

claude-windows / 47ecf23a-6312-4118-b2bd-c1af33f2eaa7

Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7.jsonl`
- SHA-256: `05f5dbbf0b546f3b959afb7c7d7101af2f3e00763144321239933c986f16f7c2`
- Source modified: `2026-06-22T09:03:02+00:00`
- Imported at: `2026-07-05T16:48:17+00:00`
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `47ecf23a-6312-4118-b2bd-c1af33f2eaa7`
```

Transcript

1. user (2026-06-12T09:16:13.456Z)

In C:\Users\avidu\Projects\badminton-highlight-indexer: config.json's `stitching.begin\_padding` / `stitching.end\_padding` are silently ignored by the live highlight compiler. `backend/main.py` (around lines 508 and 633) constructs `VideoCompiler(output\_dir, watermark=...)` WITHOUT passing the configured paddings, so the live stitcher always uses the constructor defaults in `backend/pipeline/stitcher/compiler.py:10` (begin=0.5, end=1.0). The config values are only read by `backend/pipeline/segmenters/gemini.py` (~lines 406-408) for a log-line duration estimate. This was audit finding (d) from the 2026-06-10 deep audit and was re-verified as still live on 2026-06-12. Fix: thread the typed config stitching paddings into both VideoCompiler construction sites, add a regression test asserting a config padding override changes the compiled clip boundaries. Note a cross-repo coupling: sports-data-collector `src/parsers/cvat/rally\_cvat.py` detect_issues docstring (~lines 288-294) justifies not flagging vendor boundary slack by citing the compiler's 0.5/1.0 padding ? if defaults/wiring change, update that rationale comment too. Follow repo conventions: branch off master, one PR, run `python run\_all\_tests.py`.

2. assistant (2026-06-12T09:16:24.319Z)

I'll start by reading the session handoff and the relevant code to confirm the finding, then implement the fix on a branch.

3. user (2026-06-12T09:16:26.585Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-467 of 1379 total (62639 tokens, cap 25000). Call Read with offset=468 limit=467 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4      metadata:
5          node_type: memory
6          type: project
7          originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8      ---
9
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-11 (UI-alpha **PR-2 #99 MERGED + docs-sync #100 MERGED** ? owner
authorized merge w/ due
13     diligence; master `020ff61`). **LIVE E2E AT MERGE (server up on :8123, real
artifacts):** 86MB smoke video uploaded in
14     3 sequential parts ? reassembled to **MD5 byte-identity** with the source; negatives
verified (out-of-order 409,
15     bad-ext/traversal 400, unknown-id 404; download traversal blocked at BOTH layers ?
```

handler 400 on backslash form,
 16 router 404 on %2F form); **compile of the real confirmed rally ? 3.9MB reel streamed via
 GET /api/highlights** (the
 17 fixed segment_ids path works browser-equivalent). #100 = doc due diligence: NEXT_STEPS
 (PR-2 merged + sweep recorded),
 18 UI plan PR-2 checked off w/ E2E evidence, **BACKLOG shlex item closed** (residual
 tenancy note kept). The CUJ runbook
 19 (PR2_HANDOFF \$"Tomorrow-morning") is now fully unblocked for the owner. ?? NEXT: owner
 CUJ; PR-3 identity (remember:
 20 `allowed\_video\_root` must contain `upload\_dir`). Original handoff context below:
 21 (superseded) **Checkpoint:** 2026-06-11 (UI-alpha PR-2 executed ? #99 was awaiting
 review ? per the Cowork handoff
 22 `scratch/PR2\_HANDOFF.md`; protocol = Cowork designs, Claude Code runs mechanics, **Avi
 reviews/merges ? do NOT
 23 auto-merge #99**). #88 confirmed CLOSED ? PR88_FIX skipped. Cowork's uncommitted B2+B3
 implementation verified to
 24 layer cleanly on post-sweep master (AI-7 CODE_MAP entries + pydantic fixes intact) ?
 Step 0 run here (Cowork sandbox
 25 down): suite **388 green** (15 new `tests/test\_upload\_download.py`), mypy clean; 3
 trivial ruff B904s in the new
 26 endpoints fixed per the handoff's trivial-fix rule (exception chaining only, no design
 changes). Branch
 27 `feat/upload-download`, exactly the 8 handoff files committed ? **including the two
 long-preserved doc edits**
 28 (DEFERRED_HARDENING_PLAN #16-banner + UI_ALPHA_EXECUTION_PLAN working-agreement
 rewrite), which are now COMMITTED
 29 (tree finally clean of them). CI 4/4 green on #99. ?? NEXT: owner reviews/merges #99 ?
 then the tomorrow-morning CUJ
 30 runbook in the handoff (record ? localhost:8000 ? path-paste ? index ? compile ? ?
 download; gemini default,
 31 rotated key in .env, ~\$0.05/run). PR-3 = per-user dirs + allowed_video_root?upload_dir
 (hosted-mode note).
 32
 33 **Checkpoint:** 2026-06-11 (8?9 QUALITY SWEEP ? COMPLETE ? all approved AIs landed;
 master `31069cb`). Final stretch:
 34 **AI-7 hybrid-twin collapse** = shared `segmenters/trajectory\_hybrid.py` engine
 (tracknet/wasb now ~40-line subclasses;
 35 one `\_shot\_count\_from`; `\_wsl\_bash` ? `detectors/base.py::WslCommandMixin`;
 DetectorRunner ABC now declares the
 36 parse/windowing contract; mypy quarantine ratcheted ?2; 13 equivalence tests; CODE_MAP
 2.1/2.2 updated). ? PROCESS SLIP
 37 (disclosed): AI-7 commit `2c241a8` went DIRECTLY to master (post-#96 script left
 checkout on master) ? content was
 38 pre-approved + locally verified; master CI green post-hoc; chose NOT to force-rewind
 (concurrent sessions live).
 39 - **QUALITY CHECKPOINT (owner-mandated) PASSED:** offline LOVO reproduced **EXACTLY
 0.626** (Mahadevpura Gen-0 0.744)
 40 post-sweep ? featurizer/eval edits numerically inert. **Live smoke = wasb_hybrid's
 FIRST-EVER verified E2E run**
 41 (3-min Mahadevpura slice): P1 healthcheck?P2 trajectory (9 candidates, correct
 video_id keying)?P3 Gemini (503
 42 capacity wave; 6-attempt backoff worked; 1 segment confirmed)?P4 add_play_segment +
 candidate link
 43 (`wasb-hybrid-gemini-flash-latest`). 1/9 confirms = Gemini weather, NOT code.
 44 - ? Smoke found 2 REAL pre-existing live-path bugs (masked: trajectories were always
 hand-scripted, tests mock WSL) ?
 45 **#97 MERGED**: (1) shlex.quote strangled `~` expansion in staging (the BACKLOG
 "shlex-quote breaks ~" item ? CLOSED;
 46 fix = `wsl\_tilde\_quote`, quotes only after `~/`); (2) relative `output/wasb` resolved
 against WSL cwd ? 8-min GPU run
 47 wrote CSV inside WSL invisibly (fix = abspath before /mnt translation). Regression
 tests pin both command shapes.
 48 - Merged this sweep: **#90** hygiene . **#91** pydantic model_dump . **#92** RUFF lane
 (+194 fixes; DELETED
 49 never-compiled scripts/generate_khelacharya_flyer.py ? line 39 was a literal AI-

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placeholder sentence) . **93** MYPY
50     lane (7-module quarantine w/ ratchet rule) . **94** SQLite conn-leak fix (340?4
ResourceWarnings) . **95** coverage
51     floor 70 (CI-measured 72%) . **96** config unknown-key warning . **97** WSL
tilde/abspath . (**98** Node-24
52     actions bump = owner-run chip session). Suite **373 green**; CI = 4 lanes
(smoke/lint/mypy/full+cov-floor).
53     - NEW session gotchas: `Select-Object -Last N` on a background pipe loses EVERYTHING if
the process is killed ? pipe
54     long background runs through `Out-File` instead; **sibling sessions flip branches in
THIS shared checkout** (the
55     node24 chip did, mid-session) ? re-verify `git branch --show-current` before every
commit; idle/suspend KILLS
56     background runs silently (smoke run #3 died mid-Gemini, no notification, stale report
on disk).
57     - ?? NEXT: unchanged owner-gated tracks (UI-alpha PR-2 upload/download per
`docs/UI_ALPHA_EXECUTION_PLAN.md`; corpus
58     growth; ship-gate target). Deferred-with-intent from the sweep: ruff I/UP rule groups;
yolo_hybrid P3/P4 fold-in;
59     pyproject packaging (AI-8 ? owner deselected); extra='forbid' flip (awaits gemini_
alias retirement).
60
61     **Checkpoint:** 2026-06-11 (CI Node-24 action bump ? **98 MERGED**, master `7717e15`).
GitHub forces actions onto
62     Node 24 from 2026-06-16 (Node 20 removed from runners 2026-09-16); ci.yml pinned
checkout@v4 + setup-python@v5 (Node 20).
63     Verified latest majors via `gh api releases/latest`: **checkout v6** (v6.0.3 ? NOT v5 as
the run-log warning suggested)
64     + **setup-python v6** (v6.2.0). Bumped both in all 4 lanes; all green on the PR (lint
11s / mypy 29s / smoke 15s /
65     full+cov 1m27s), squash-merged on verified rollup. PR #97 (fix/wsl-tilde-quoting) was
open + untouched throughout;
66     working tree restored to it (PR-2 doc edits intact).
67
68     **Checkpoint:** 2026-06-11 (8?9 QUALITY SWEEP IN PROGRESS ? owner approved 8 AIs via
question UI; AI-8 packaging NOT
69     selected). **MERGED on green CI: #90** (hygiene: .coveragerc repointed post-#40 +
TEST_RUN_SUMMARY ? docs/archives),
70     **91** (pydantic .dict()?model_dump, 7 sites, warnings 15?0), **92** (? RUFF LANE:
ruff.toml E/F/W/B py38 +
71     fix 194 violations + `lint` CI job; **DELETED scripts/generate_khelacharya_flyer.py ?
never compiled, line 39 was a
72     literal AI-placeholder sentence**, invisible because nothing swept scripts/), **93**
(MYPY LANE: mypy.ini lenient
73     green baseline, explicit 7-module quarantine w/ ratchet rule ? main.py+5
segmenters+registry; fixed 18 annotation-level
74     errors), **94** (? DB FIX surfaced by coverage: `with sqlite3.connect()` = transaction
scope NOT close ? all 24
75     Database call sites leaked connections; new `_connect()` CM; ResourceWarnings 340?4),
**95** (COVERAGE FLOOR:
76     full-suite lane runs --cov, **--cov-fail-under=70** vs CI-measured 72% ? ratchet-only),
**96** (config unknown-key
77     warning: dotted-path [CONFIG WARNING] for typo'd/dropped keys at all nesting levels;
repo config.json verified clean;
78     extra='forbid' still deferred). Suite 349?351 green**. CI = 4 lanes
(smoke/lint/mypy/full+cov). ?? NEXT: **AI-7
79     hybrid-twin collapse** (shared trajectory-hybrid base, fix shot_count drift, dedupe
_wsl_bash, equivalence tests) +
80     **post-AI-7 quality-eval checkpoint** (owner: verify ~0.66-class evals stick; GPU free,
Gemini key ROTATED, live calls
81     OK). Deferred-with-intent this sweep: ruff I/UP rule groups; yolo_hybrid P3/P4 fold-in
(not as follow-up).
82
83     **Checkpoint:** 2026-06-11 (PR #88 ? **89 MERGED** ? KhelSutra UI-alpha docs bundle on
master `34cf26c`). #88

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84     conflicted (branch cut from `feat/async-job-model` pre-squash of #87) ? per
`scratch/PR88_FIX.md`, recreated as
85     `docs/khelsutra-plan-2` off post-#87 master keeping 6 paths (UI_PROPOSAL, HOSTING_PLAN,
UI_ALPHA_EXECUTION_PLAN,
86     mockups/, DECISIONS **ADR-009**, NEXT_STEPS sync); #88 closed + branch deleted. CI
green-gated (statusCheckRollup),
87     squash-merged.
88     - **Dropped from #89, preserved for PR-2 as UNCOMMITTED working-tree edits on master:**
(1)
89     `docs/DEFERRED_HARDENING_PLAN.md` "#16 DONE ? MERGED (PR #87)" status hunk (also saved
as
90     `scratch/PR2_deferred_hardenening_status.patch`); (2) `docs/UI_ALPHA_EXECUTION_PLAN.md`
working-agreement rewrite
91     (Cowork designs + leaves `scratch/PRn_HANDOFF.md`; **Claude Code executes all git/PR
mechanics** ? ratified
92     2026-06-11). Don't `checkout --` these away; they commit with PR-2.
93     - ?? Next UI-alpha slice = **PR-2 (upload/download)** per
`docs/UI_ALPHA_EXECUTION_PLAN.md`.
94     - ? **SLATE CLEANED (2026-06-11):** indexer = NO open PRs, master==origin (`34cf26c`),
**all 23 stale local branches
95     pruned** (every one mapped to a MERGED PR #62?#87 before `-D`); only `master` remains.
Collector = NO open PRs, on
96     master. Intentional carryovers (NOT mess): the 2 uncommitted PR-2 doc edits (above),
untracked `scratch/` handoffs,
97     untracked `artifacts/rally_tal_gen0/0.1.0/` (Gen-0 bundle, ADR-008). ?? Collector has
a stray uncommitted
98     `.gitignore` edit (adds `artifacts/` ignore w/ ADR-008 comment) ? mega-session
leftover, owner to commit (micro-PR)
99     or fold into the next collector PR.
100
101     **Checkpoint:** 2026-06-10 (GEMINI BATTERY COMPLETE ? full coverage; owner's billing was
fine, postpay active; ALL
102     failures were burst/capacity artifacts). Merged this stretch: **#80** (chunker re-
encode/downscale, from the chip
103     session ? reviewed+merged), **#81** (rescued the chip branch's orphaned run_signals test
via cherry-pick), **#83**
104     (provider: 6 generate attempts exponential backoff+jitter ? saved chunks live), **#84**
(tracked
105     `eval_baselines/gemini_wrapper_2026-06-10.json`). Suite 336 green.
106     - **FINAL QUALITY TABLE (IoU .5 vs human golden labels):**
107     - **Mahadevpura** (clean-ish, 31 rallies): heuristic 0.657 · Gen-0 0.744 · **two-stage
WASB+Gemini refine 0.83**
108     (P .81 R .84, 32 preds) · **full-video wrapper 0.93** (P 1.0 R .87).
109     - **Kushagra** (multi-court, 32 rallies, FULL 7/7 coverage + audio): heuristic 0.157 ·
Gen-0 0.561 ·
110     **wrapper 0.66** (P .69 R .62, ±1.2s; FPS = background-game pickups + one 53s dead-
time merge ? exactly the
111     contrast-metric confound).
112     - **STRATEGY (refined):** clean footage commoditized (0.93 cheap); **multi-court drops
the wrapper to 0.66 = the
113     owned model's ship target** (Gen-0 is 0.10 behind at n=6 ? within corpus-growth
reach). Two-stage refine quality
114     (0.83) < full wrapper on short videos; its value = COST on long academy tapes (only
candidate clips uploaded).
115     Wrapper reliability is genuinely flaky (503 waves all session) ? provider-fallback
architecture is load-bearing.
116     - Artifacts: `output/mahadevpura_gemini_refined.csv`,
`output/MahadevpuraSingles_highlights.mp4` (demo reel),
117     Kushagra preds under the ORIGINAL corpus MD5 now. Total Gemini spend today: ~$1-2.
118     - PENDING/owner: rotate the chat-pasted key; ship-gate F1@tIoU decision (multi-court
target now has a number: 0.66);
119     C13 interviews + camera pilot; repo rename.
120
121     **Checkpoint:** 2026-06-10 (GEMINI BASELINE LANDED ? owner's "gradual ramp-up" theory
CORRECT). New key in gitignored

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122 `.env` (rotate later ? pasted in chat). The "credits depleted" 429 was a BURST-THROTTLE artifact: text ping + serial

123 upload worked ? **PR #79 merged**: gemini segmenter now honors `indexing.ai\_max\_workers` (was hardcoded 5; config ships

124 1=serial ? at 5 workers the same video 429'd, at 1 it worked).

125 - ? **WRAPPER BASELINE (gemini-flash-latest, ~\$0.2-0.5 total): Mahadevpura F1 0.93** (P 1.00 R 0.87, mIoU 0.80, ±0.7s;

126 3 of 4 misses = short rallies Gemini FOUND but our min_segment_duration=3.0 rejected) vs Gen-0 learned 0.744 vs

127 heuristic 0.657. **Clean footage = commoditized.**

128 - ?? **MOAT TEST (Kushagra multi-court, windowed n=12 GT): wrapper F1 0.56** (P .54 R .58 mIoU .69) ? Gen-0 learned

129 0.561, both >> heuristic 0.157. **The multi-court confound hurts Gemini too ? the wedge survives on the target

130 footage class.** Caveats: only chunks 2+4 answered (5/7 lost to GENUINE gemini-flash-latest capacity 503s ? retry

131 off-peak for full coverage); visual-only (audio stripped in downscale); 1280p re-encode (4K chunks were 1.3GB >

132 500MB upload cap ? the audit's silent-oversize-skip fired live; fix = chip task_44e72c41 ? see PR #80 checkpoint

133 below). Kushagra preds keyed under downscaled-file MD5 (`output/kushagra\_1280.mp4`).

134 - ? **DEMO ARTIFACT: `output/MahadevpuraSingles\_highlights.mp4`** ? real 27-rally reel compiled by VideoCompiler

135 (the once-broken module) from Gemini segments. The academy-pitch object exists.

136 - **STRATEGY READ:** product path = Gemini for clean footage TODAY (cheap, F1 0.93) + owned model's job = multi-court

137 (0.56 is the bar to beat there, not 0.93); 503 flakiness = real reliability cost of the wrapper (provider-fallback

138 story strengthens). PENDING: gemini_refine two-stage (503-gated, run off-peak), full Kushagra re-run, PR #80 merge.

139

140 **Checkpoint:** 2026-06-10 (4K gemini oversize fix ? **MERGED: #80 + #81 + #82; master `f01387e`, CI rollup SUCCESS**).

141 Root cause of the live "0 segments / success" on KushagraYashAviFirstVideo.MP4 (4K): gemini segmenter stream-copies 90s

142 chunks ? all 7 at 832?1299MB > provider `MAX\_UPLOAD\_MB=500` ? every chunk silently skipped. Fix (3 layers): (1) chunks

143 RE-ENCODED + downscaled to new declared knob `indexing.ai\_chunk\_scale\_width` (default 1280 = gemini_refine's

144 `--clip-scale-width`; 0 = legacy stream-copy; also fixes keyframe drift); (2) skip is LOUD ? provider

145 `metrics["oversize\_skipped"]` + segmenter aggregation (chunked + single-file) ? `logger.error` + NEW thread-safe

146 `backend/reporting/run\_signals` channel (incr/pop by video_id; popped unconditionally in `emit\_indexer\_report`);

147 (3) report flags it ? `ai\_handoff.details.oversize\_clips\_skipped` + spec.yaml ERROR rule `oversize\_clip\_skipped`.

148 Suite 325?336 green** locally (CI runs fewer: reporting tests importorskip obsreport, absent there BY DESIGN ?

149 `tests/test\_run\_signals.py` deliberately has NO importorskip = the CI-visible half of the channel). **Merge dance

150 (process note):** owner merged #80 mid-session (auto-deleting the remote branch) WHILE the follow-up test commit was

151 being pushed ? push re-created the branch orphaned; owner rescued it as #81, 39s before my cherry-picked #82 ?

152 **#82's squash `f01387e` = EMPTY commit** (harmless dup in history). Lesson: re-check `gh pr view` state before pushing

153 more commits to a PR branch ? owner + sibling sessions work concurrently (this checkpoint file too: another session

154 prepended the GEMINI-BASELINE checkpoint mid-edit). ? **OWNER/SIBLING WIP IN TREE (uncommitted, do NOT clobber):**

155 `backend/providers/gemini.py` ? generate retries 3?6 w/ exponential backoff + jitter (~3?96s) for the 503 capacity

156 waves. ?? commit/PR the 503-backoff WIP · machine-side 4K re-run (chunks should be ~tens of MB) · task-17 battery

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157     still pending Gemini billing top-up.
158
159     **Checkpoint:** 2026-06-10 (M0 GREENLIT SEQUENCE EXECUTED ? session wrap). Owner
ratified ADR-008 ("split driven by
160     productionization, not isolation") + greenlit M0.4+M0.1. **All 5 steps shipped, every PR
merged on verified-green CI:**
161     - **#73 ADR-008** (DECISIONS.md): training in-repo behind an executable wall; artifact-
bundle boundary; ship-gate
162     product-side; pre-committed split trigger ? `sports-temporal-models` at Gen-2/second-
consumer.
163     - **#74 canonical GT loader** `backend/eval/gt_loader.py::load_gt_intervals` ? closes
the two-loader fork (accepts BOTH
164     `start,end` and `start_time,end_time`); legacy loaders delegate; round-trip contract
test; LOVO 0.626 re-verified.
165     - **#75+#76 featurizer promotion + import wall** : `backend/features/rally_seq.py`
(FEATURE_SCHEMA v1, fps semantics
166     explicit) imported by train AND serve; `training/` skeleton (5 wall rules,
requirements-train.txt); smoke tests:
167     runtime no-training-in-backend.main, static sweep, featurizer purity (torch/cv2
blocked; importorskip numpy on the
168     minimal CI lane ? #76 hotfixed a red lane after #75 merged early; merges now gated on
verified SUCCESS rollout).
169     - **collector #19 ? n=6 OWNED CORPUS IN THE SoR** : all 6 golden matches via `import-
local --normalize` (262 rallies,
170     0 dropped, Proprietary, explicit fps); commercial export now 6 real matches + demo
entry.
171     - **#77 Gen-0 harness** `python -m training.gen0.harness --manifest ? --floor
eval_baselines/heuristic_lovo_n6.json
172     --version 0.1.0` : train?honest LOVO?ship-gate?artifact bundle (gitignored weights.json
+ manifest.json w/ schema pin,
173     calibration, lineage gt_hashes, attestations incl. C3, gate verdict). **First real
bundle built: LOVO 0.626, floor
174     PASSED (0.480), paired sign test 5/6 wins p=0.109 = honestly NOT significant,
ship=False** (blockers: target
175     undefined, C3, significance). Committed floor baseline
`eval_baselines/heuristic_lovo_n6.json`. Suite **325 green**.
176     - ? **Gemini baseline STILL BLOCKED after owner provided a NEW key (2026-06-10 later):**
key written to gitignored
177     `.env`, validated via `models.list` (55 models incl. `gemini-flash-latest`; **config
migrated off deprecating
178     gemini-2.5-flash ? `gemini-flash-latest` alias, PR #78 merged** ; gemini_refine already
defaulted to it). But inference
179     = 429 RESOURCE_EXHAUSTED "prepayment credits depleted" ? **the AI-Studio PROJECT
behind the key has no credits** (key
180     auth ? billing). Owner: add credits at https://ai.studio/projects (+ consider rotating
the chat-pasted key). Planned
181     battery on unblock (in task 17): wrapper Mahadevpura ? score; wrapper Kushagra (multi-
court MOAT test); two-stage
182     gemini_refine; compile a real highlight reel (academy-demo artifact). Pipeline handled
both failed runs gracefully
183     (validators pass, reports emitted, artifacts cleaned); shared-genai-client thread-
safety finding observed live.
184     - **C13 field plan (owner executing):** recruit interviewer; camera idea RATIFIED w/ 3
constraints (consent-at-capture/
185     adults-only for training; demo via Gemini-or-human-polish, NOT gated on owned model;
interviews?cameras sequencing).
186     - ?? NEXT: owner tops up Gemini billing (wrapper baseline) . owner sets the ship-gate
F1@tIoU target . grow corpus
187     (camera pilot videos ? import-local ? re-run harness) . repo rename . M0.1 event-index
contract in collector.
188
189     **Checkpoint:** 2026-06-10 (Roora PB+TT pilot review ? REPORT DELIVERED to owner; owner
emails Roora himself) .
190     ? Owner cross-checked the rally tables in VLC ? **CONFIRMED: every boundary off ?0.5s
(up to ~2s)** vs the ±0.3s bar

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191 (Brief \$4). **Root cause (my forensics): all 24 boundaries sit EXACTLY on the 30-frame
sampling grid** (~0.5s at
192 59.94fps) ? first/last box never on the true contact/dead frame ? structurally cannot
meet ±0.3s. Also decoded their
193 bogus `start/end time` attrs = **frame÷900 (decimal minutes @ assumed 15fps, ~4× off
wall-clock)**. Report findings:
194 boundary accuracy (CRITICAL), skip-rallies-in-progress-at-file-start/end convention (PB
rally#1 starts at frame 0 ?
195 shouldn't have been labeled), dead-time+warmup reminders for Phase-B, schema cut to
exactly
196 `rally\_number,shot\_count,ending\_reason,last\_shot\_outcome,note` (drop their time attrs +
scores), field quality
197 (PB outcome='0' off-vocab; TT 5/6 'other' + missing required notes; TT#5 intra-rally
other|winner). Per owner:
198 NO redelivery demand in the report (he words the email himself; thread = Jamshad@roora
on avidullu@live.com Outlook).
199 **Artifacts:** PDF `Annotation Setup\6 June 2026\Feeback\Roora Pilot Review -
Pickleball and Table Tennis
200 (2026-06-10).pdf` + md record at `Annotation Setup\Roora Pilot Review ? Pickleball &
Table Tennis (2026-06-10).md`
201 (generator: `%TEMP%\roora\make\_review\_pdf.py`). ? Local guideline docs are **DRAFT v2**
? owner mentioned "v7";
202 no v7 found on disk (possibly in the email thread; report cites v2). Spawned chip
`task\_c55f0be6`: collector
203 `rally\_cvat.py` ATTR_MAP drops `shot\_count` (present in Roora XML, empty in our CSVs) ?
OUR bug, kept out of the
204 vendor report. ?? owner sends email; Roora's badminton long-video + redelivery decisions
pending owner.
205
206 **Checkpoint:** 2026-06-10 (REPO-TOPOLGY decision panel + C13 plan ? AWAITING OWNER
RATIFICATION). Owner leaning
207 M0.4+M0.1 greenlight; asked: model in-repo vs separate repo consumed as a segmenter
(owner prefers isolation). Ran a
208 4-agent judge panel (`wf\_471a9cdd-d1d`; full output `?\tasks\wnksv545g.output`).
**Verdict (my rec): "artifact boundary
209 NOW, repo split at Gen-2"*** ? M0.4 lands as a top-level `training/` tree in the indexer
w/ executable isolation
210 (import-smoke asserts backend.main loads no training module; own requirements-train.txt
+ 3rd CI lane), featurizer
211 promoted to `backend/features/rally\_seq.py` (single source, train==serve), artifact
bundle = gitignored weights
212 (private Releases) + committed manifest (feature-schema ver, normalization, calibration,
corpus ref+gt_hash,
213 consent/license attestation incl. C3), ship-gate stays product-side (regression.py
extension); **pre-committed split
214 trigger** to a named repo (`sports-temporal-models`) when Gen-2/cloud-training or a 2nd
artifact consumer arrives.
215 Decisive evidence: Gen-0's input = the indexer's OWN featurization (mid-pipeline seam ?
WASB/Gemini video-boundary);
216 indexer not pip-installable ? split today = copy-paste = the observed drift class;
obsreport "precedent" never actually
217 exercised (pin is a commented line, repo unpublished, CI skips). **FIRST contract to
freeze: golden-label/corpus
218 contract** ? indexer carries TWO incompatible GT loaders (annotations.py `start,end` vs
calibrate_wasb `start\_time`,
219 `end\_time` ? fed to 7 modules) vs collector export `start,end`; consolidate + round-trip
test BEFORE any training epoch.
220 Panel extras: n=6 corpus's durable home = loose `Annotation Setup\` folder (neither
repo!) ? M0.1 first deliverable =
221 hash-addressed in collector; training fixtures synthetic-only until consent ledger
exists; run the Gemini-wrapper
222 baseline before/with M0.4. **C13 field plan:** owner recruiting an interviewer; my rec =
bake in the camera idea
223 (paid recordings = consented owned multi-court corpus + demo), demo served via Gemini
path or human-polish (owned model

```

224     honest 0.626 ? >0.8 yet; the comparison doubles as the wrapper baseline). ?? awaiting
owner ratification of topology +
225     M0.4/M0.1 scope; then: ADR + GT-loader consolidation + featurizer promotion + harness
scaffold.
226
227     **Checkpoint:** 2026-06-10 (CORPUS RECLASS EXECUTED + NEXT_STEPS refreshed ? session
wrap). Owner approved + I ran the
228     corpus correction: `reclassify-licenses --apply` (23 ShuttleSet broadcast rows ?
annotation_license=MIT, video=Unknown,
229     is_commercial=False; **zero annotations deleted**, 1,917 rallies intact) + `curate
export` (commercial manifest now
230     owned-data-only: 1 video/2 segments; stale ShuttleSet CSVs removed from the gitignored
eval_export; eval-only re-export
231     available via `--include-staging --include-noncommercial`). **Merged: collector #17**
(DB + .gitignore backup rule);
232     pre-change backup at `data/sports_metadata.backup-2026-06-10.db` (gitignored). Quality
signals UNAFFECTED (n=6 corpus
233     never flows through the collector DB; learned 0.626 re-verified earlier). **Indexer #72
merged:** NEXT_STEPS.md
234     refreshed (sweep banner, 0.626 vs 0.480, M0.3 partially-done with corrected
distill_local pointer, NEW prioritized PMF-
235     validation item, durable artifact paths). **Owner is now reading NEXT_STEPS to chalk out
next steps** ? open owner
236     decisions stand: greenlight M0.x (rec M0.4+M0.1), confirm collector=SoR, ship-gate
F1@tIoU, repo rename, C13 interviews
237     / Gemini-wrapper baseline. Both repos clean on master, 0 open PRs.
238
239     **Checkpoint:** 2026-06-10 (HIGH-RISK REMEDIATION COMPLETE ? owner approved "everything
A?K, open+merge PRs after CI
240     green, re-run quality evals, GPU available"). **ALL 11 PRs MERGED + CI green; both repos
clean on master.**
241     - **Indexer (badminton-highlight-indexer):** #63 compiler-syntax fix+CI (prior), **#64**
detector keying by video_id,
242     **#65** re-ingest destroy-after-confirm (add_video UPSERT + watermark/prune), **#66**
nested-config wasb typed,
243     **#67** eval gate + metric footguns (iou=0, midrank AUC, fps-normalized speed,
baseline?`eval_baselines/` exit-3),
244     **#68** API path validation, **#69** reliability (timeouts 1800s, SQLite
WAL+busy_timeout, chunk-step guard, CORS),
245     **#70** test coverage (TestClient endpoints + compiler + WasbRunner), **#71** docs
truth-sync. Suite 256?308 green**.
246     - **Collector (sports-data-collector):** **#14** licensing gate (video vs annotation
license; `reclassify-licenses`
247     found 23 conflated ShuttleSet rows) + **added collector CI**, **#15** import-local
golden-label guard (+`--force`,
248     `--normalize`, duplicate-rally guard), **#16** validate-delivery fails on
empty/corrupt/zip CVAT. Suite 124?136 green**.
249     - ? **QUALITY RE-VALIDATION (owner asked): metric fixes STRENGTHENED the conclusion.**
After midrank-AUC +
250     fps-normalized speed + occlusion-gap guard, **learned LOVO 0.583 ? 0.626** (per-video
AUC unchanged 0.83?0.92 =
251     never tie-inflated; fps-fix mainly helped the 30fps short fold 0.30?0.56). Heuristic
LOVO **0.480** unchanged (scored
252     at IoU 0.5; my iou>0 guard is a no-op there). **Learned beats heuristic by +0.146**
(was +0.10).
253     - **Eval artifacts RESCUED from `%TEMP%` ? durable `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\`**
254     (+ `roora\`); manifest `output/human_lovo_manifest.json` repointed there + reverified
0.626.
255     - **Deferred WITH INTENT (in BACKLOG): repo-rename; Gen-2 cost benchmark; ship-gate
F1@tIoU target; packaging
256     (pyproject); app-factory/lifespan + lazy torch-cv2 imports (with async slice #16); WSL
temp/frame/cache + Gemini
257     remote-file GC (retention policy); shlex-quote WSL config values (breaks `~` expansion
? non-trivial); deeper

```



```

258     validator tests + 3-hybrid contract test (partial coverage shipped).**
259     - ? **Process note:** during PR-F I briefly `git checkout --`d the owner's uncommitted
`data/sports_metadata.db`
260     change (200704b), then RECOVERED it from dangling stash `d3c2e77` and restored it to
the working tree (no loss).
261     Going forward stash-and-pop that DB for collector PRs; never `checkout --` it. The
corpus-DB reclassify (`curate
262     reclassify-licenses --apply` + `curate export`) is left for the OWNER to run
deliberately (binary SoR data).
263     - Plan doc (reference): `scratch/HIGH_RISK_REMEDIATION_PLAN.md` (untracked). Audit JSON:
`?\tasks\wf38bovts.output`
264     (batch 1) + `?\tasks\wa4d2980l.output` (full 8 auditors).
265
266     **Checkpoint:** 2026-06-10 (compiler SyntaxError FIX ? **PR #63 MERGED**, squash `master
13d4573`; owner approved merge-on-green) .
267     ? **master UNBROKEN again ? fixed the ? SyntaxError from 7fbbeb0** (= the audit's fix
chip `task_2b7a19d4` work):
268     restored the f-string quotes on the 7 mangled lines of
`backend/pipeline/stitcher/compiler.py`
269     (100/104/156/160/181/191/213; message texts verified against the pre-conversion file at
`7fbbeb0`; line 191 =
270     plain string, no placeholders) + fixed line 168 `f"\nSUMMARY]" ? `f"\n[SUMMARY]`. **NEW
regression guard
271     `tests/test_import_smoke.py`**: (a) py_compile sweep over every file in backend/
(imports nothing ? runs on
272     cv2-free machines), (b) subprocess `import backend.main` with missing cv2/torch/numpy
stubbed. Negative check:
273     the broken compiler.py fails the sweep (`SyntaxError: invalid decimal literal`, line
100).
274     ? **CI NOW EXISTS ? the audit's no-CI finding is CLOSED** (`github/workflows/ci.yml`,
runs on every PR + master
275     push): job 1 **import-smoke** = smoke tests on a GPU-stack-free runner (minimal dep set
verified empirically in a
276     fresh venv: pytest/fastapi/uvicorn/pydantic/python-dotenv/**google-genai** ? needed at
import time by
277     backend.providers); job 2 **full-suite** = requirements.txt with **CPU-only torch
wheels**
278     (`--index-url download.pytorch.org/whl/cpu`) + libgl for opencv ? `python
run_all_tests.py` (obsreport absent ?
279     reporting tests skip by design). **Both jobs GREEN on the first run** (27s / 1m29s).
Verified post-merge:
280     `import backend.main` OK on master, suite 256 green (254+2).
281     ?? Audit's **coverage-inversion** finding (wasb_hybrid/validators untested) remains open
? CI only guards the
282     import/syntax class + whatever the suite covers.
283
284     **Checkpoint:** 2026-06-10 (DEEP AUDIT session ? ultracode 18-agent audit
`wf_2a64bc88-2a8` over indexer+collector:
285     engineering/testing/reliability/docs-consistency/PMF; 3 of 8 auditors
completed+verified, 5 rate-limited).
286     - ? **CRITICAL ? master is BROKEN since 7fbbeb0 (2026-06-07 print?logger sweep):**
287     `backend/pipeline/stitcher/compiler.py` has **7 unquoted f-string lines
(100/104/156/160/181/191/213)** ? SyntaxError ?
288     `import backend.main` fails ? FastAPI server + `/api/compile` dead. Triple-verified +
reproduced first-hand
289     (`py_compile` exit 1). The **254-green suite never noticed** (zero tests import
backend.main / compiler / wasb_hybrid /
290     motion / any concrete validator; **no CI exists at all**). Spawned fix chip
`task_2b7a19d4` (fix + import/compileall
291     smoke test). Coverage is INVERTED vs production risk (tracknet twin 8 tests vs live
wasb_hybrid 0; TrackNetRunner 21
292     vs WasbRunner 1). Suite itself healthy where it exists (254 green / 19.9s / faithful
mocks / low flake risk); "fully
293     mocked runs anywhere" is false on cv2-free machines (bare `import cv2` at collection
time, only obsreport importorskips).

```

294 - ****Doc drift (verified, one-directional ? newest decisions not back-propagated):**** (1)
licensing "Apache-2.0 only" still
295 in CLAUDE.md:33 / PLATFORM:269,301 / COMMERCIALIZATION C14 vs ratified
****MIT/Apache/BSD**** (a literal CLAUDE.md reader
296 rejects the plan's own MIT ActionFormer/ASFormer heads); (2) ****Gemma-4 still**
"approved"** in those same docs vs
297 AMBER/bench-only in the plan; (3) ****M0.3 purge pointer WRONG**** ?
`training.label\_candidates` is a pure GT-agnostic fn;
298 the live Gemini-silver training violation = `backend/eval/distill\_local.py` (+
calibrate_local thresholds) + gitignored
299 `output/local\_rally\_model.json`; (4) ****C3 escape hatch broken**** ? "train-own weights
(Phase-4 roadmap)" cites a phase
300 that contains no such plan and ADR-002 says corpus labels are temporal-only (CAN'T
train a detector) ? own-weights is
301 an unscoped new workstream (coordinate labels or detector swap); (5)
OWNED_MODEL_IMPLEMENTATION_PLAN + NEXT_STEPS are
302 linked from NO entry point (CLAUDE.md "read first" / docs/README index) and NEXT_STEPS
is branch-only; (6) README
303 "yolo_hybrid by default" vs config.json `gemini` vs CODE_MAP "wasb = LIVE default";
(7) ROADMAP still says audio
304 "gated on golden labels" (trigger now satisfied!) vs plan "parked, not in reserve";
(8) Roora multisport CSVs live ONLY
305 in `%TEMP%` (OS-clearable) + n=6 manifest gitignored ? headline
TT-0.909/pickleball-0.308 evidence is one cleanup from
306 loss; (9) TT-on-WASB idea silently walks into the C3 ship-gate ("multiplies per
sport"); (10) ship-gate F1@tIoU still
307 undefined while 0.44/0.54/0.480 floors circulate; repo-rename "ASAP" tracked in no
tracker (BACKLOG lacks it).
308 - ****PMF (web-grounded, sources in audit output):**** pricing/cost benchmarks VERIFIED
(SwingVision \$179.99/yr ~20k paying
309 subs rev \$2.75M +12% YoY; PB Vision \$99/yr + ****\$8/hr-of-video partner API****; Gemini
COGS ~\$0.30?0.60/hr ? fat margin);
310 but ****"India badminton whitespace" FALSIFIED** ? Bangalore is the most contested
badminton-AI city**: Machaxi (\$1.5M
311 Rainmatter+Prakash Padukone), VISIST.AI (70+ Bengaluru academies, Padukone School),
Game Theory (Gopichand-backed smart
312 courts w/ auto-highlights), Stupa (\$3.3M, BWF-approved, TT?badminton) ? NONE in our
2026-06-07 competitive table.
313 Surviving wedge (narrow, real): ****software-only rally-precise indexing of EXISTING**
messy multi-court BYO footage**
314 (market solves it only with per-court hardware) sold per-hour as engine/API, possibly
THROUGH those players
315 (PB Vision×PodPlay template). Fastest falsification: C13 interviews (still undone) +
****Gemini-wrapper baseline vs the**
316 **n=6 LOVO corpus**** (if wrapper ? learned 0.583, owned-model is premature) + 1 paid
academy pilot.
317 - ****AUDIT NOW COMPLETE** (8/8 auditors, resumed `wa4d2980l` after Max-20x refresh; 40
agents). **** New high/critical findings**
318 beyond the first batch:
319 - ? ****CRITICAL (collector) ? LICENSING GUARDRAIL VIOLATION:**** `src/cli/ingest.py:12`
defaults `--license MIT`; the
320 ShuttleSet ****annotation**** license is stamped onto ****YouTube broadcast video**** (e.g.
shuttleSet_21 = a YONEX Thailand
321 Open broadcast, `is\_commercial=true, license\_type=MIT`). ****24 videos / 1915 segments**
currently exported as
322 "commercially-safe" on unsound grounds** ? directly violates the "training data must
be commercial-clean" guardrail.
323 - ****HIGH (indexer, all verified):**** (a) ****nested config sections silently drop unknown**
keys** ? only `AppConfig` has
324 `extra='allow'`, so the ENTIRE `indexing.wasb` block
(weights/conda_env/velocity_threshold) + `yolo\_prefetch\_workers`
325 + `debug\_duration` are unconfigurable via config.json; meanwhile
`yolo\_parallel\_chunks/yolo\_chunk\_workers` are read
326 nowhere. (b) ****detector artifacts keyed by file BASENAME not video_id**** ? two
`match.mp4` files reuse stale frames ?

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327         a trajectory + candidate_segments computed from the WRONG video (silent training-
data poisoning). (c) **re-ingest is
328         destroy-before-confirm** ? `add_video('processed'/'failed')` INSERT-OR-REPLACE
cascade-deletes prior good segments
329         BEFORE the new run; a failed/empty re-run leaves zero. (d) **stitching padding
config not wired** ? API reels always
330         end_padding=1.0 despite config 0.5. (e) **`/api/process` arbitrary server-file read
+ **`/api/compile` path-traversal/
331         absolute write** via output_filename. (f) **run_regression "CI gate" not in CI +
silently passes** on missing baseline;
332         baseline omits iou/detector/GT-hash. (g) **metrics bug**: `match_intervals` at
iou_threshold=0.0 matches
333         non-overlapping intervals (P=R=F1=1.0 for garbage); **rally_seq_proto AUC has no tie
handling** (order-dependent;
334         all-zero dead-time rows guarantee ties) + speed features fps-inconsistent within one
vector.
335         - **HIGH (collector)** silent DELETE-then-INSERT replace of golden labels (no
warn/backup, filename-derived video_id);
336         validate-delivery **PASSES on a zero-rally CVAT export**; not yet SoR (no
provenance/maturity/labeled_window cols,
337         import-local hardcodes CURATED); `int(float())` rally_number crashes mid-import
leaving a committed video row w/
338         rolled-back annotations (normalize_annotations guard is opt-in, NOT wired into
import-local).
339         - **MEDIUM batch** WSL `timeout_sec=0` = hang-forever (both runners + provider
PROCESSING poll); `uvicorn backend.main:app`
340         leaves globals None (init only in main()); SQLite no WAL/no busy_timeout + swallow-
on-lock; chunk_overlap?duration =
341         infinite loop; WSL temp/frame/cache + Gemini remote-file leaks; CORS
**+credentials; sport-profile relevance config
342         dead under typed config (bites first non-badminton sport ? the current branch!).
343         - **VERDICT NUANCE** verifier DOWNGRADED 2 "high"?not-real (Result-contract gap + DB-
swallow are DOCUMENTED CODE_MAP
344         gotchas/deliberate-incremental, medium at most; DB "no busy_timeout" claim factually
wrong ? sqlite3 default
345         timeout=5s). Eng-pipeline/eval/collector strengths confirmed: registry seams,
DetectorRunner ABC, WASB crash-cache,
346         honest fit-on-self labeling, leakage-free LOVO, frame-true CVAT conversion all
genuinely good.
347         - ? **SPOOF-CHECK PASSED ? quality iterations are REAL (re-ran the evals first-hand on
this box, pure numpy/scipy):**
348         all 6 videos + 6 human label CSVs + 6 trajectories exist; traj row-counts match docs
exactly. **`rally_seq_proto` LOVO
349         reproduced: mean 0.583** (AUC 0.877/0.851/0.916/0.832/0.730/0.902 vs doc
0.879/0.853/0.915/0.833/0.726/0.902).
350         **`calibrate_local` heuristic LOVO reproduced: 0.480 ± 0.242** (per-fold Adarsh
0.751/Kushagra 0.157/doubles 0.737/
351         gBaaddy 0.309/testlarge 0.267/mahadevpura 0.657 ? exact). **Contrast metric recomputed
independently from raw CSVs:**
352         3.7/1.4/3.7/1.9/1.4/2.2× vs doc 3.9/1.4/3.6/1.9/1.4/2.2. Opus 4.8 did NOT spoof. ? one
nit: gBaaddy CSV has 48 rallies
353         (doc says 47). Caveat: trajectories were WASB-GPU-generated (can't regenerate here)
but sizes/frame-counts/visibility all sane.
354         - **HIGH-RISK REMEDIATION PLAN written for owner approval:**
`scratch/HIGH_RISK_REMEDIATION_PLAN.md` (12 PRs, sequenced;
355         PR-1 compiler fix already DONE via #63). Awaiting owner scope + push/PR decision
before execution. Full audit JSON:
356         `%LOCALAPPDATA%\Temp\claude\...\tasks\wa4d2980l.output` (8 auditors) +
`wf38bovts.output` (first batch).
357         Full audit JSON: `%LOCALAPPDATA%\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\45ea3a91-654d-4972-9cd3-65f35db54caf\tasks\wf38bovts.output`.
358
359         **Checkpoint** 2026-06-09 (Roora MULTISPORT recon ? for FRESH next session) . ??
**Pickleball + Tabletennis golden labels
360         arrived; INGESTION = SOLVED, but the "merged model" experiment is BLOCKED on per-sport

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trajectory detectors (validates the
361     roadmap blocker).** Owner left a recon-only ask (deliberately did NOT drain context /
spin heavy jobs ? owner wants me fresh).
362     - **Deliverables:** `Annotation Setup/6 June 2026/{pickleball,tabletennis}/*.zip` (3
CVAT export FORMATS each: COCO
363     `instances_default.json`, **CVAT-XML `annotations.xml`** ? use this, Pascal-VOC).
Videos: `~/6 June 2026/Annotation Videos/
364     Video 2 - Pickleball.mp4` (4K, 59.94fps, 97s, 5737fr) + `Video 3 - Tabletennis.mp4`
(4K, 59.94fps, 58s, 3490fr). Short clips.
365     - **What Roora annotated:** label class "rally" as **per-frame BOUNDING BOXES on sampled
frames (~every 30fr)** ? box geometry
366     is just a CARRIER; the boxes carry the FULL rally schema as CVAT **attributes**
(rally_number, start/end time, shot_count,
367     score_before/after(A-B), ending_reason, last_shot_outcome [+note for TT]). 456/513
attr instances. Boxed frames cluster into
368     rallies (runs of ~30-spaced frames = a rally; big gaps = dead time).
369     - **INGESTION = YES, cleanly:** the collector's `src/parsers/cvat/rally_cvat.py`
(`curate convert-cvat`) is PURPOSE-BUILT for
370     exactly this ? ignores box geometry, **recomputes rally interval from the frame
range** (`sec=frame/fps` at true fps, ignoring
371     vendor start/end times which are unreliable), groups boxes by rally_number, maps
attrs?canonical. Minor: verify the attr-name
372     mapping handles spaces/"(A-B)" (`ATTR_MAP`). Collector already has pickleball +
tabletennis schemas.
373     - ? **"MERGED LEARNT MODEL" across sports is BLOCKED:** `rally_seq_proto` features come
from a WASB **shuttle** trajectory =
374     BADMINTON-ONLY; pickleball (wiffle ball) + TT (tiny fast ball on a table) are out-of-
distribution ? no clean trajectory yet.
375     This EMPIRICALLY validates the roadmap's binding multisport blocker (clean per-sport
MIT/Apache ball detectors not identified).
376     - **PROGRESS (owner asked to complete 1/2/3 before badminton):**
377     - ? ** (1) convert-cvat DONE ? INGESTION PROVEN.** `python -m src.cli.curate convert-
cvat --input <annotations.xml> --out
378     <csv> --fps 59.94` ? **pickleball 6 rallies + tabletennis 6 rallies**, sane
intervals, attrs mapped. CSVs at
379     `%TEMP%\roora\{pickleball,tabletennis}_rallies.csv` (+ extracted XMLs
`%TEMP%\roora\{pb,tt}\annotations.xml`). ? Roora data
380     nits (non-blocking): pickleball `last_shot_outcome='0'` (off-vocab in/n_a/net/out);
TT `ending_reason` mostly 'other' with the
381     real outcome in `notes` ("out of table"/"net") ? ending_reason taxonomy mapping
could improve. Both clips SHORT (97s/58s).
382     - ? ** (2) WASB-transfer test DONE (`%TEMP%\{pickleball,tabletennis}_traj.csv`; log
`~/clips/multisport_run.log`):** WASB
383     (badminton shuttle detector) detects SOMETHING on both ? **pickleball 28% /
tabletennis 39% visibility** (NOT near-zero;
384     comparable to badminton 40?67%). So it's not blind.
385     - ? ** (3) DONE ? SURPRISE RESULT (directional, n=6 each, TINY so noisy):**
`eval_partial_labels` on (WASB-transfer traj + Roora
386     6-rally CSV, fps 59.94, fw 1920): **TABLE TENNIS best F1 0.909** (tuned+gate2:
P1.0/R0.833, 5/6 matched, boundary err 0.49s ?
387     **WASB transfers to TT remarkably well!); **PICKLEBALL best F1 0.308** (poor ?
wiffle ball too out-of-distribution).
388     ? **NUANCES the roadmap blocker: per-sport detector transfer is SPORT-DEPENDENT ? TT
may be servable with WASB short-term;
389     pickleball needs a proper detector.** ? BIG caveat: 6 rallies each = could be partly
luck ? owner sending more videos to
390     confirm. A "merged" prototype LOVO across badminton+pickleball+TT is the natural
next experiment once n grows per sport.
391     Roora LABELS ingest cleanly + are corpus-gold regardless. **Great honest surprise to
greet owner with.**
392
393     **Checkpoint:** 2026-06-09 (PR #61 MERGED ? roadmap SET, session wrap) . ? **Owner
agreed + merged #61 (`master 4d04a73`)).
394     ALL CLEAN: master, 0 uncommitted, 0 open PRs, branches pruned.** The owned-model
**ROADMAP is now set + agreed** ?

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395 authoritative doc = `**`docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md`**` (full verified decisions in the "(PR review)" checkpoint

396 just below: ms-swift + Qwen3-VL, TAL-head-first, conversational-QA seam-now-model-later, multisport one-head/base, Python

397 everywhere, MIT/Apache/BSD ratified, "envision now build later"). `[[session-handoff]]` rewritten to this phase.

398 - ?? `**NEXT SESSION` (awaiting owner greenlight ? NO build until then):`**` greenlight a Phase-0 milestone (rec `**M0.1`

399 `corpus+event-index**` ? `**M0.4` Gen-0 TAL harness`**`, both \$0/local); confirm `**collector` = system-of-record`**`; define the

400 `**ship-gate F1@tIoU target**` (floor = beat heuristic LOVO 0.480); `**rename repo off "badminton"**`; M0.3 purge Gemini

401 training labels + WASB C3 (own-weights/clean swap). Corpus = n=6 (``output/human_lovo_manifest.json``); learned prototype

402 is the Gen-0 seed. Committed next PLATFORM slice stays the async job model (single-tenant). Owner flagged context ~85% ? this is the wrap checkpoint.

403

404 `**Checkpoint:**` 2026-06-09 (PR review) . ? `**OWNER REVIEWED PR #61 ? REFRAMED ROADMAP` (important checkpoint).`**` Owner left

405 ~12 inline comments (mostly state-updates + recommendations wanted). `**Key roadmap shifts to internalize:**`

406 - `**Immediate goal REFRAMED:**` not "on-device model" but a `**TRAINING FRAMEWORK + HARNESS` to produce a VERY HIGH

407 precision/recall model that extracts HIGHLIGHTS + RALLIES from a clip.`**` Quality first. On-device = LATER (compression

408 for power users to save upload cost), definitely on roadmap but not now.

409 - `**MULTISPORT FROM THE START**` ? all net-separated racquet sports ? eventually all sports; a SINGLE sports-agnostic model.

410 Collector + indexer already sport-generic by design. ? `**ACTION: rename the repo off "badminton" ASAP**` (owner emphatic).

411 - `**CONVERSATIONAL VIDEO-QA` = north-star feature:`**` upload video ? ask contextual questions ("longest rally", "make a clip

412 of all service faults", "rally >20 shots"). Needs per-video STATE/bookkeeping ? design the framework for it. Owner asks:

413 embark now or punt + revisit after quality? (My lean: design the bookkeeping/event-store layer now, punt the heavy

414 conversational model ? pending the strategy workflow.)

415 - `**LICENSING GUARDRAIL` (owner: "never corrupted"): `**EVERY` dependency ? code, data, AND weights ? must be usable in

416 PRIVATE PROPRIETARY COMMERCIAL software. `**MIT / Apache-2.0 / BSD only.**` Ratify (was "Apache-2.0 only"). Enforce religiously.

417 - Owner wants a `**coherent recommendation: local Gemma 4 vs ms-swift starting point**` (multisport, single, conversational,

418 commercial). NB: ms-swift = training FRAMEWORK, Gemma 4 = BASE model ? orthogonal; clarify in the reply.

419 - Owner: `**upgrade WASB C3` (own-weights/provenance) to a first-class action item now.`**` M0.2 (grow n) = practically DONE

420 (n=6); owner will add 1 more varied multi-court-club video soon ? Phase-0 ready to start.

421 - `**LANGUAGE QUESTION` (owner callout, no preference): `**` is Python right for the upcoming work, or brainstorm alternatives?

422 - ? `**STRATEGY WORKFLOW` done (``wf_f9aa1d1c-c83``, 10 agents, verified+critiqued) ? VERIFIED DECISIONS (now in the rewritten

423 ``docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md``, the authoritative roadmap; pushed to PR #61 + a consolidated PR comment reply):`**`

424 - `**Framework = ms-swift` (Apache-2.0)`**` (only forge w/ native video+audio+timestamp-grounding+video-GRP0); unsloth = 8GB-local

425 SFT/compression tool (avoid its AGPL Studio). `**Base = Qwen3-VL` (Apache-2.0, timestamp emission)`**` primary; `**Gemma-4 STAYS`

426 AMBER/bench-only`**` (critic corrected the workflow's "clear AMBER" ? posture unconfirmed + 60s/no-timestamp dispute);

427 InternVL3 (MIT) fallback. ms-swift?Gemma4 (framework vs base ? clarified for owner).

428 - `**START WITH THE TAL HEAD, NOT THE VLM**` (VLMs miss strict temporal boundaries; boundary accuracy = the product; head

429 de-risked AUC 0.83?0.92, \$0 local, doubles as teacher+baseline). Head now ? VLM

later (two-stage: VLM proposes, head refines).

430 - **Conversational-QA**: build the SEAM now (per-video event store + structured text-to-SQL + ffmpeg clip-cut), PUNT the NL

431 model** until quality clears the bar. Queries ("longest rally"/"service faults"/">20 shots") = structured-numeric ? SQL, not

432 a VLM. Reserve fault/event taxonomy in M0.1. ? real DB delta (new cols + a `highlights` table; play_segments lacks them today).

433 - **Multisport**: one shared head + one base; scale DATA + per-sport clean trajectory detectors (the binding blocker ? none

434 identified for tennis/TT/pickleball/padel yet); WASB C3 multiplies. Single-model temporal generalization = OPEN HYPOTHESIS

435 (RacketVision evidence REFUTED = ball-tracking not temporal). Repo rename off "badminton" = action item.**

436 - **Language**: Python everywhere** (training Python-locked; platform I/O-bound, no hot path ? reject Go/Rust split); on-device =

437 deferred EXPORTED runtime shell (ONNX/CoreML/GGUF), doesn't dictate dev language. Constraint NOW: **train export-clean ops**.

438 - **Honest gaps**:** cost-to-Gen-2 UNQUANTIFIED (biggest risk); ship-gate "very high P/R" target undefined (floor = beat heuristic

439 LOVO 0.480 ? set explicit F1@tIoU); per-sport detectors TBD; base PRETRAINING provenance unauditable ? owner risk-acceptance;

440 **Gemini-silver TRAINING** labels = LIVE violation ? hard-blocker purge M0.3.**

441 - **Owner principle recorded**: "envision now, build later"*** ? design every deferred seam (event store, export boundary, sport-

442 agnostic head) up front; punt the feature, never the seam.

443 - ?? **NEXT**:** owner reviews the reframed plan + decisions (greenlight a Phase-0 milestone ? rec M0.1 corpus+event-index then M0.4

444 harness; confirm collector=system-of-record). Then start the greenlit milestone. Repo rename. Define the ship-gate F1@tIoU target.

445

446 **Checkpoint**:** 2026-06-09 . ? **n=6: MahadevpuraSingles added while owner reviews PR #61.** Owner moved ALL golden videos

447 + CSVs to `Annotation Setup/Collect/Golden Labelled/` (manifest label paths updated; trajectories unaffected in `%TEMP%`).

448 New: **MahadevpuraSingles**** (singles, 1080p, 29.97fps, 31 rallies; traj `%TEMP%mahadevpura\_traj.csv`, script

449 `run\_wasb\_mahadevpura.sh`). **Borderline multi-court** (contrast 2.2x)** ? heuristic best 0.667, learned held-out **AUC**

450 0.902/F1 0.712** (clean generalization to an unseen match). **n=6 LOVO: heuristic 0.480 ± 0.242 vs learned 0.583** (learned

451 holds ~+0.10 since n=3). Contrast metric now predicts heuristic F1 MONOTONICALLY across 5 matches (Mahadevpura's 2.2x?0.67

452 lands on the curve). **Decoupling headline**: learned held-out AUC 0.83?0.92 regardless of contrast; heuristic swings 0.16?0.75

453 with it.** Committed to PR #61 (n=6 doc update). ? GOTCHA: don't combine inner `&` with run_in_background (orphans the job);

454 calibrate_local buffers output to end + is slow (~17?20min). "Ingested" = added to EVAL corpus (manifest); collector

455 import-local (system-of-record) still = M0.1, plan-only pending greenlight. ?? owner reviewing PR #61 (4 plan decisions pending).

456

457 **Checkpoint**:** 2026-06-08 (later+4, SESSION WRAP) . ? **n=5 corpus + MULTI-COURT root cause + heuristic discovery + SINGLE PR #61 for offline review.**

458 Owner tagged gBaaddy (47, 1080p, multi-court) + TestLargeVideo (6, short) ? corpus **n=5** (`output/human\_lovo\_manifest.json`).

459 - **HEURISTIC DISCOVERY** (user asked "discover more heuristics like audio"):** tested 2 label-free trajectory pre-filters

460 for multi-court (measured on all 5): foreground spatial-density gate = **net ?0.04** (helps multi-court +0.03, hurts clean

461 ?0.12); track-continuity/jump filter = **no-op**. ? simple single-cue trajectory heuristics CAN'T isolate the foreground

462 court. **AUDIO-foreground idea PROTOTYPED** (no more videos needed) ? WEAK ?:** on the 2 multi-court videos + clean

463 control, audio onset/RMS energy AUC only **0.53?0.65** for in-rally vs dead-time; HF bands (8?22kHz @48kHz) raise

```

464     CONTRAST (Kushagra 1.1?1.7x, gBaaddy 1.3?1.9x, confirming distance-attenuation
physics) but AUC stays ~0.6. Fails:
465     dead-time isn't quiet ? foreground court loud during dead-time too
(footwork/talk/pickup) + gym reverb ? audio loudness
466     ? court occupancy, not rally (same wall as E1 ~2.2x). **Audio alone ? multi-court
gate.** Remaining heuristic lever =
467     real court-region detection (player-cluster/court-line, ROADMAP #2); robust path = the
learned visual model (AUC 0.85).

```

4. user (2026-06-12T09:16:26.270Z)

```

1     import os
2     import subprocess
3     import tempfile
4     import logging
5     from typing import Any, Dict, List, Optional, Tuple, Union
6
7     logger = logging.getLogger(__name__)
8
9     class VideoCompiler:
10         def __init__(self, output_dir: str, end_padding: float = 1.0, begin_padding: float =
0.5,
11                     watermark: Optional[Any] = None):
12             self.output_dir = output_dir
13             self.end_padding = end_padding
14             self.begin_padding = begin_padding
15             # Optional branding overlay. Accepts a WatermarkConfig model or a plain dict;
16             # normalized to a dict (or None) so this module stays config-package-agnostic.
17             self.watermark = self._normalize_watermark(watermark)
18             os.makedirs(output_dir, exist_ok=True)
19
20         @staticmethod
21         def _normalize_watermark(watermark: Optional[Any]) -> Optional[Dict[str, Any]]:
22             if watermark is None:
23                 return None
24             if hasattr(watermark, "model_dump"):
25                 return watermark.model_dump()
26             if isinstance(watermark, dict):
27                 return dict(watermark)
28             return None
29
30         @staticmethod
31         def _ff_quote(path: str) -> str:
32             """Make a filesystem path safe to embed inside single quotes in an ffmpeg
33             filtergraph: forward slashes (Windows-friendly) + escaped single quotes."""
34             return path.replace("\\", "/").replace("'", r"\'")
35
36         @staticmethod
37         def _parse_time(val: Union[int, float, str]) -> float:
38             """Normalizes a timestamp value to float seconds.
39             Handles: float/int (pass-through), plain numeric strings ("12.4"),
40             and colon-separated strings ("MM:SS", "HH:MM:SS", "H:MM:SS.ms").
41             This mirrors the parse_time logic used in gemini.py to ensure
42             the stitcher can always consume whatever format the indexer produced."""
43             if isinstance(val, (int, float)):
44                 return float(val)
45             if isinstance(val, str):
46                 val = val.strip()
47                 if ":" in val:
48                     # Handle MM:SS, HH:MM:SS, etc.
49                     parts = val.split(":")
50                     parts.reverse()
51                     total = 0.0
52                     for i, p in enumerate(parts):
53                         try:

```

```

54         total += float(p) * (60 ** i)
55     except ValueError:
56         pass
57     return total
58     try:
59         return float(val)
60     except ValueError:
61         logger.warning(f"Could not parse timestamp value '{val}' as a number.
Defaulting to 0.0.")
62         return 0.0
63     logger.warning(f"Unexpected timestamp type {type(val).__name__} for value
'{val}'. Defaulting to 0.0.")
64     return 0.0
65
66 def _get_video_duration(self, video_path: str) -> float:
67     """Probes the video file to get its total duration in seconds.
68     Returns 0.0 if it cannot be determined."""
69     try:
70         result = subprocess.run(
71             [
72                 "ffprobe", "-v", "error",
73                 "-show_entries", "format=duration",
74                 "-of", "default=noprint_wrappers=1:nokey=1",
75                 video_path
76             ],
77             capture_output=True, text=True, check=True
78         )
79         return float(result.stdout.strip())
80     except Exception as e:
81         logger.warning(f"Could not determine video duration via ffprobe: {e}")
82         return 0.0
83
84 def _watermark_filter(self, tmp_dir: str) -> Optional[str]:
85     """Build the ffmpeg -vf filtergraph that burns the configured watermark (text
86     and/or logo PNG) into a clip during the per-clip re-encode. Returns None when
87     watermarking is disabled or has nothing to draw. The concat stage stays -c copy
88     because the mark is already baked into every clip identically."""
89     wm = self.watermark
90     if not wm or not wm.get("enabled"):
91         return None
92
93     text = (wm.get("text") or "").strip()
94     logo_path = (wm.get("logo_path") or "").strip()
95     if logo_path and not os.path.exists(logo_path):
96         logger.warning(f"[watermark] logo not found: {logo_path} ? skipping logo
overlay.")
97     logo_path = ""
98     if not text and not logo_path:
99         return None
100
101     margin = int(wm.get("margin", 24))
102     opacity = float(wm.get("opacity", 0.85))
103     position = str(wm.get("position", "bottom-right"))
104     # drawtext uses lowercase w/h (main) + tw/th (text); overlay uses W/H (main) +
w/h (logo).
105     dt_xy = {
106         "bottom-right": f"x=w-tw-{margin}:y=h-th-{margin}",
107         "bottom-left": f"x={margin}:y=h-th-{margin}",
108         "top-right": f"x=w-tw-{margin}:y={margin}",
109         "top-left": f"x={margin}:y={margin}",
110     }.get(position, f"x=w-tw-{margin}:y=h-th-{margin}")
111     ov_xy = {
112         "bottom-right": f"W-w-{margin}:H-h-{margin}",
113         "bottom-left": f"{margin}:H-h-{margin}",
114         "top-right": f"W-w-{margin}:{margin}",

```



```

115         "top-left": f"{margin}:{margin}",
116     }.get(position, f"W-w-{margin}:H-h-{margin}")
117
118     drawtext = ""
119     if text:
120         ratio = float(wm.get("font_size_ratio") or 0.035)
121         divisor = max(1, round(1.0 / ratio))
122         # textfile= sidesteps the notorious drawtext text= escaping for arbitrary
strings.
123         tf_path = os.path.join(tmp_dir, "watermark.txt")
124         with open(tf_path, "w", encoding="utf-8") as fh:
125             fh.write(text)
126         font_path = (wm.get("font_path") or "").strip()
127         if font_path:
128             font_opt = f"fontfile='{self._ff_quote(font_path)}'"
129         else:
130             font_opt = f"font='{wm.get('font', 'Sans')}'"
131         box = ""
132         if wm.get("box", True):
133             box = f":box=1:boxcolor=black@{min(opacity, 0.5):.2f}:boxborderw=10"
134         drawtext = (
135             f"drawtext=textfile='{self._ff_quote(tf_path)}':{font_opt}"
136             f":fontcolor=white@{opacity:.2f}:fontsize=h/{divisor}:{dt_xy}{box}"
137         )
138
139         if logo_path and drawtext:
140             return
141         f"movie='{self._ff_quote(logo_path)}'[wm];[in][wm]overlay={ov_xy},{drawtext}"
142         if logo_path:
143             return f"movie='{self._ff_quote(logo_path)}'[wm];[in][wm]overlay={ov_xy}"
144         return drawtext
145
146     def compile_highlights(self, raw_video_path: str, segments: List[Tuple[float,
float]], output_filename: str) -> str:
147         """
148         Extracts selected segments from a raw video and stitches them together
149         into a seamless highlights file, applying end_padding to prevent abrupt endings.
150         """
151         if not segments:
152             raise ValueError("No highlight segments provided to compile.")
153
154         if not os.path.exists(raw_video_path):
155             raise FileNotFoundError(f"[COMPILER ERROR] Source video file not found:
{raw_video_path}")
156
157         # Defense-in-depth: never let the output name escape output_dir (the API also
158         # validates this at the request boundary).
159         output_path = os.path.join(self.output_dir, os.path.basename(output_filename))
160
161         # Probe video duration so we can detect out-of-range segments
162         video_duration = self._get_video_duration(raw_video_path)
163         if video_duration > 0:
164             logger.info(f"Source video duration: {video_duration:.2f}s")
165         else:
166             logger.warning("Video duration unknown. Cannot validate segment boundaries
against video length.")
167
168         # We will create temporary clips and stitch them using FFmpeg concat demuxer
169         temp_clips = []
170         skipped_segments = []
171
172         # Create a temp directory within output_dir
173         with tempfile.TemporaryDirectory(dir=self.output_dir) as tmp_dir:
174             logger.info(f"Trimming {len(segments)} segments
(begin_padding={self.begin_padding}s, end_padding={self.end_padding}s)...")

```

```

174         # Build the (optional) watermark filtergraph once; it is applied during
every
175         # per-clip re-encode so the mark is baked into each clip identically.
176         watermark_filter = self._watermark_filter(tmp_dir)
177         if watermark_filter:
178             logger.info("[watermark] applying branding overlay to each clip.")
179         for idx, (raw_start, raw_end) in enumerate(segments):
180             # Normalize timestamps to float seconds ? handles float, int,
181             # plain numeric strings, and colon-separated formats like "MM:SS"
182             start = self._parse_time(raw_start)
183             end = self._parse_time(raw_end)
184             segment_label = f"Segment #{idx+1}/{len(segments)} [{start:.2f}s -
{end:.2f}s]"
185
186             # Validate the segment times
187             if start < 0:
188                 logger.warning(f"{segment_label}: start_time is negative
({start:.2f}s). Clamping to 0.")
189                 start = 0.0
190
191             if start >= end:
192                 logger.error(f"{segment_label}: start_time ({start:.2f}s) >=
end_time ({end:.2f}s). Skipping invalid segment.")
193                 skipped_segments.append((idx, start, end, "start >= end"))
194                 continue
195
196             # Check if segment extends beyond video duration
197             if video_duration > 0:
198                 if start >= video_duration:
199                     logger.error(f"{segment_label}: start_time ({start:.2f}s) is
beyond the video duration ({video_duration:.2f}s). "
200                                 f"This segment cannot be extracted ? the video ran
out. Skipping.")
201                     skipped_segments.append((idx, start, end, "start beyond video
end"))
202                     continue
203
204                 if end > video_duration:
205                     original_end = end
206                     end = video_duration
207                     logger.warning(f"{segment_label}: end_time ({original_end:.2f}s)
exceeds video duration ({video_duration:.2f}s). "
208                                 f"Clamping end_time to {video_duration:.2f}s. The
segment may be truncated.")
209
210                 clip_path = os.path.join(tmp_dir, f"clip_{idx}.mp4")
211
212             # Precise sub-second trim using fast re-encoding to preserve stream
headers
213             padded_start = max(0.0, start - self.begin_padding)
214             padded_end = end + self.end_padding
215
216             # Clamp padded_end to video duration if known
217             if video_duration > 0 and padded_end > video_duration:
218                 padded_end = video_duration
219                 logger.info(f"{segment_label}: Padded end ({end +
self.end_padding:.2f}s) exceeds video duration. "
220                             f"Clamping to {video_duration:.2f}s (end padding
partially applied).")
221
222             trim_duration = padded_end - padded_start
223             logger.info(f"Trimming {segment_label} -> [{padded_start:.3f}s -
{padded_end:.3f}s] (duration: {trim_duration:.2f}s)")
224
225             base_cmd = [

```

```

226         "ffmpeg", "-y",
227         "-ss", str(round(padded_start, 3)),
228         "-to", str(round(padded_end, 3)),
229         "-i", raw_video_path,
230         "-c:v", "libx264",
231         "-preset", "superfast",
232         "-crf", "22",
233         "-c:a", "aac",
234         "-b:a", "128k",
235     ]
236     vf_args = ["-vf", watermark_filter] if watermark_filter else []
237
238     # Try with the watermark first; if that encode fails (e.g. a bad
font/logo
239     # path), retry once WITHOUT it so a branding misconfig can never nuke
the
240     # whole reel. Clips stay codec-identical either way, so concat -c copy
holds.
241     attempts = [vf_args, []] if vf_args else [[]]
242     produced = False
243     last_exc: Optional[subprocess.CalledProcessError] = None
244     cmd = base_cmd + [clip_path]
245     for attempt_vf in attempts:
246         cmd = base_cmd + attempt_vf + [clip_path]
247         try:
248             subprocess.run(cmd, capture_output=True, check=True)
249         except subprocess.CalledProcessError as e:
250             last_exc = e
251             if attempt_vf and vf_args:
252                 logger.warning(f"{segment_label}: encode with watermark
failed; retrying without it.")
253                 continue
254                 break
255             if os.path.exists(clip_path) and os.path.getsize(clip_path) > 0:
256                 produced = True
257                 if not attempt_vf and vf_args:
258                     logger.warning(f"{segment_label}: encoded WITHOUT the
watermark (the watermark filter failed).")
259                     break
260                 last_exc = None # ffmpeg succeeded but produced nothing
261                 break
262
263     if produced:
264         temp_clips.append(clip_path)
265     elif last_exc is not None:
266         stderr_msg = last_exc.stderr.decode(errors='replace').strip()
267         logger.error(f"{segment_label}: FFmpeg trim failed.")
268         print(f"  Command: {' '.join(cmd)}")
269         print(f"  FFmpeg stderr: {stderr_msg[-500:]}") # Last 500 chars to
avoid flooding
270         skipped_segments.append((idx, start, end, f"ffmpeg error:
{stderr_msg[-200:]}"))
271         continue
272     else:
273         logger.error(f"{segment_label}: FFmpeg exited successfully but
output clip is missing or empty. Skipping.")
274         skipped_segments.append((idx, start, end, "empty output clip"))
275         continue
276
277     # Report summary of skipped segments
278     if skipped_segments:
279         logger.info(f"\n[SUMMARY] {len(skipped_segments)} out of {len(segments)}
segments were SKIPPED:")
280         for seg_idx, seg_start, seg_end, reason in skipped_segments:
281             logger.info(f"  - Segment #{seg_idx+1} [{seg_start:.2f}s -

```

```

{seg_end:.2f}s]: {reason}")
282         logger.info("")
283
284     if not temp_clips:
285         raise RuntimeError(
286             f"[COMPILER ERROR] Failed to generate ANY valid highlight clips. "
287             f"All {len(segments)} segments were skipped. "
288             f"Check the errors above for details. "
289             f"Common causes: video file is corrupted, segments are out of range,
or ffmpeg is not installed."
290         )
291
292     logger.info(f"Successfully trimmed {len(temp_clips)}/{len(segments)} clips.
Preparing to stitch...")
293
294     # Create a list file for FFmpeg concat demuxer
295     list_file_path = os.path.join(tmp_dir, "clips.txt")
296     with open(list_file_path, "w") as f:
297         for clip in temp_clips:
298             # Escape single quotes in path
299             safe_path = clip.replace("'", "'\\'")
300             f.write(f"file '{safe_path}'\n")
301
302     logger.info("Stitching clips together...")
303     # Stitch them together without re-encoding (since they were encoded with
identical settings)
304     concat_cmd = [
305         "ffmpeg", "-y",
306         "-f", "concat",
307         "-safe", "0",
308         "-i", list_file_path,
309         "-c", "copy",
310         output_path
311     ]
312
313     try:
314         subprocess.run(concat_cmd, capture_output=True, check=True)
315
316         # Final verification
317         if not os.path.exists(output_path) or os.path.getsize(output_path) == 0:
318             raise RuntimeError(
319                 f"[COMPILER ERROR] FFmpeg concatenation appeared to succeed, "
320                 f"but output file is missing or empty: {output_path}"
321             )
322
323         output_size_mb = os.path.getsize(output_path) / (1024 * 1024)
324         logger.info(f"Dynamic highlight reel saved: {output_path}
({output_size_mb:.1f} MB)")
325         return output_path
326     except subprocess.CalledProcessError as e:
327         stderr_msg = e.stderr.decode(errors='replace').strip()
328         raise RuntimeError(
329             f"[COMPILER ERROR] FFmpeg concatenation failed.\n"
330             f"Command: {' '.join(concat_cmd)}\n"
331             f"FFmpeg stderr: {stderr_msg[-500:]}"
332         ) from e
333

```

5. user (2026-06-12T09:16:27.731Z)

```

C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py-127-
run_process_and_stitch.py-128-     if not time_pairs:
C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py-129-
print("No play segments passed the filtering criteria. Cannot compile highlights.")
C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py-130-
return

```

```

C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py-131-
run_process_and_stitch.py:132:     compiler = VideoCompiler(output_dir, end_padding=end_padding,
begin_padding=begin_padding,
C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py-133-
watermark=cfg.stitching.watermark)
run_process_and_stitch.py-134-     try:
C:\Users\avidu\Projects\badminton-highlight-indexer\run_process_and_stitch.py-135-
out_path = compiler.compile_highlights(video_path, time_pairs, output_filename)
run_process_and_stitch.py-136-         print(f"Stitched highlight video saved to: {out_path}",
flush=True)
run_process_and_stitch.py-137-     except Exception as e:
--
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-503-         time_pairs =
[(s["start_time"], s["end_time"]) for s in matched_segments]
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-504-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-505-         # Run compiler (with
the optional configured branding watermark, if enabled)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-506-         output_dir =
getattr(getattr(global_config, "output", None), "output_dir", "output") or "output"
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-507-         watermark =
getattr(getattr(global_config, "stitching", None), "watermark", None)
backend\main.py:508:     compiler = VideoCompiler(output_dir, watermark=watermark)
backend\main.py-509-     try:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-510-         out_path =
compiler.compile_highlights(video["filepath"], time_pairs, out_name)
backend\main.py-511-         return {"status": "success", "filepath": out_path}
backend\main.py-512-     except Exception as e:
backend\main.py-513-         raise HTTPException(status_code=500, detail=f"Highlight stitching
failed: {str(e)}") from e
--
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-628-
os.makedirs(out_dir, exist_ok=True)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-629-         print(f"[CLI]
Extracting debug segment from {args.trim_start}s to {args.trim_end}s from {args.video_path}...")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-630-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-631-         # global_config
isn't initialized this early; load the typed config so the trim
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-632-         # clip carries
the same configured watermark as a real highlight (default-on).
backend\main.py:633:     compiler = VideoCompiler(out_dir,
watermark=load_app_config().stitching.watermark)
backend\main.py-634-     try:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py-635-         res_path =
compiler.compile_highlights(args.video_path, [(args.trim_start, args.trim_end)], out_name)
backend\main.py-636-         print(f"[SUCCESS] Trimmed segment saved to: {res_path}")
backend\main.py-637-     except Exception as e:
backend\main.py-638-         print(f"[ERROR] Failed to compile trimmed segment: {e}")
--
tests\test_compiler.py-17-     assert f("1:00:00") == 3600.0
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-18-     assert
f("garbage") == 0.0
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-19-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-20-
tests\test_compiler.py-21-def test_compile_basenames_output_and_stitches(tmp_path):
tests\test_compiler.py:22:     comp = VideoCompiler(str(tmp_path / "out"))
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-23-     raw = tmp_path
/ "raw.mp4"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-24-
raw.write_bytes(b"x")
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-25-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-26-     calls = []
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-27-
--
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-44-     # Per-clip
trims + one concat = at least 3 ffmpeg invocations (plus an ffmpegprobe).

```

```

C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-45-    assert sum(1
for c in calls if c[0] == "ffmpeg") >= 3
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-46-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-47-
tests\test_compiler.py-48-def test_compile_raises_on_no_segments(tmp_path):
tests\test_compiler.py:49:     comp = VideoCompiler(str(tmp_path / "out"))
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-50-     raw = tmp_path
/ "raw.mp4"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-51-
raw.write_bytes(b"x")
tests\test_compiler.py-52-     with pytest.raises(ValueError):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-53-
comp.compile_highlights(str(raw), [], "reel.mp4")
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-54-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-55-
tests\test_compiler.py-56-def test_compile_raises_when_source_missing(tmp_path):
tests\test_compiler.py:57:     comp = VideoCompiler(str(tmp_path / "out"))
tests\test_compiler.py-58-     with pytest.raises(FileNotFoundError):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_compiler.py-59-
comp.compile_highlights(str(tmp_path / "nope.mp4"), [(0.0, 5.0)], "reel.mp4")
--
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md-148-### 2.3a Audio (E1
probe; NOT adopted for fusion ? ADR-007)
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md-149-[Omitted long context
line]
docs\CODE_MAP.md-150-- `@backend\eval\audio\e1_probe.py` ? E1 GO\NO-GO diagnostic (no
training\fusion). `::cluster_hits` (groups thwacks; ? `min_hits>=2` so a lone service-feed hit
isn't a rally), `::run` (discrimination ratio + recall recovery + audio-only P/R/F1), `::main`.
reuses `hit_detector` + `calibrate_wasb.{load_golden_gts,make_predict_fn,BASELINE}` + `metrics`.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md-151-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md-152-### 2.4 Stitcher /
tools / utils
docs\CODE_MAP.md:153:[Omitted long matching line]
docs\CODE_MAP.md-154-- `@backend\tools\rally_slicer.py` ? standalone CLI.
`::compute_clip_windows` (pure, unit-tested), `::load_rally_labels` (golden schema; ? silently
skips degenerate rows), `::slice_rallies`, `::ClipWindow` (rally:str),
`::BEGIN_PADDING=0.5`/`::END_PADDING=1.0` (? duplicate compiler's by copy). `read_time ->
backend/eval/annotations.parse_timestamp`.
docs\CODE_MAP.md-155-- `@backend\utils\video.py` ? `::get_video_duration` (ffprobe; ? 0.0 on
failure = "unknown"), `::extract_video_chunk(...,reencode=False,scale_width=None)` ? copy mode
cuts at nearest keyframe (drift) ? pass `reencode=True` for short/accurate clips; `scale_width`
requires `reencode=True`.
docs\CODE_MAP.md-156-- `@backend\utils\geometry.py` ? `::get_velocity_normalised` (fraction of
frame-width/s; ? raw-px fallback if width<=0; used by yolo_hybrid), `::calculate_iou` (? +1 px
convention inflates IoU), `::get_center/get_distance/get_velocity`.
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md-157-
C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md-158-### 2.5 Validators ?
--
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-38-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-39-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-40-# --- filtergraph
builder -----
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-41-
tests\test_watermark.py-42-def test_disabled_or_none_builds_no_filter(tmp_path):
tests\test_watermark.py:43:     assert VideoCompiler(str(tmp_path / "a"),
watermark=None)._watermark_filter(str(tmp_path)) is None
tests\test_watermark.py:44:     off = VideoCompiler(str(tmp_path / "b"), watermark={"enabled":
False})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-45-     assert
off._watermark_filter(str(tmp_path)) is None
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-46-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-47-
tests\test_watermark.py-48-def
test_text_filter_writes_configurable_string_and_builds_drawtext(tmp_path):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-49-     custom = "My

```

Academy 2026 ? Khelsutra.guru"

```
tests\test_watermark.py:50:    comp = VideoCompiler(str(tmp_path / "o"), watermark={"enabled":
True, "text": custom})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-51-    graph =
comp._watermark_filter(str(tmp_path))
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-52-    assert graph
is not None and graph.startswith("drawtext=")
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-53-    # The
arbitrary, configurable string is written verbatim to the textfile (no escaping pain).
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-54-    assert
(tmp_path / "watermark.txt").read_text(encoding="utf-8") == custom
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-55-    assert
"textfile=" in graph
--
tests\test_watermark.py-58-    assert "x=w-tw-24:y=h-th-24" in graph          # bottom-right
default, margin 24
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-59-    assert
"box=1" in graph
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-60-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-61-
tests\test_watermark.py-62-def test_position_and_margin_map_to_corner(tmp_path):
tests\test_watermark.py:63:    comp = VideoCompiler(str(tmp_path / "o"),
tests\test_watermark.py-64-                          watermark={"enabled": True, "text": "X",
"position": "top-left", "margin": 12})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-65-    graph =
comp._watermark_filter(str(tmp_path))
tests\test_watermark.py-66-    assert "x=12:y=12" in graph
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-67-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-68-
tests\test_watermark.py-69-def test_font_path_uses_fontfile_else_fontconfig(tmp_path):
tests\test_watermark.py:70:    with_font = VideoCompiler(str(tmp_path / "o1"),
tests\test_watermark.py-71-                              watermark={"enabled": True, "text":
"X", "font_path": "/fonts/Brand.ttf"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-72-    g1 =
with_font._watermark_filter(str(tmp_path))
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-73-    assert
"fontfile='/fonts/Brand.ttf'" in g1
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-74-
tests\test_watermark.py:75:    no_font = VideoCompiler(str(tmp_path / "o2"),
tests\test_watermark.py-76-                            watermark={"enabled": True, "text": "X",
"font": "DejaVu Sans"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-77-    g2 =
no_font._watermark_filter(str(tmp_path))
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-78-    assert
"font='DejaVu Sans'" in g2 and "fontfile=" not in g2
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-79-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-80-
tests\test_watermark.py-81-def test_logo_plus_text_overlays_then_draws(tmp_path):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-82-    logo =
tmp_path / "logo.png"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-83-
logo.write_bytes(b"\x89PNG\r\n")
tests\test_watermark.py:84:    comp = VideoCompiler(str(tmp_path / "o"),
tests\test_watermark.py-85-                          watermark={"enabled": True, "text": "X",
"logo_path": str(logo)})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-86-    graph =
comp._watermark_filter(str(tmp_path))
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-87-    assert
graph.startswith("movie=")
tests\test_watermark.py-88-    assert "overlay=W-w-24:H-h-24" in graph
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-89-    assert
"drawtext=" in graph
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-90-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py-91-
tests\test_watermark.py-92-def test_missing_logo_is_skipped_gracefully(tmp_path):
```

```

tests\test_watermark.py:93:    comp = VideoCompiler(str(tmp_path / "o"),
tests\test_watermark.py:94-                                watermark={"enabled": True, "text": "X",
"logo_path": str(tmp_path / "nope.png")})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:95-    graph =
comp._watermark_filter(str(tmp_path))
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:96-    assert
"movie=" not in graph and graph.startswith("drawtext=") # text still drawn
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:97-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:98-
--
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:113-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:114-
tests\test_watermark.py:115-def
test_compile_applies_watermark_vf_to_each_clip_not_concat(tmp_path):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:116-    raw =
tmp_path / "raw.mp4"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:117-
raw.write_bytes(b"x")
tests\test_watermark.py:118:    comp = VideoCompiler(str(tmp_path / "out"),
tests\test_watermark.py:119-                                watermark={"enabled": True, "text":
"Generated by Khelsutra.guru"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:120-    calls = []
tests\test_watermark.py:121-    with patch.object(subprocess, "run",
side_effect=_fake_run_factory(calls)):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:122-
comp.compile_highlights(str(raw), [(0.0, 5.0), (10.0, 15.0)], "reel.mp4")
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:123-
--
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:132-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:133-
tests\test_watermark.py:134-def test_compile_without_watermark_adds_no_vf(tmp_path):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:135-    raw =
tmp_path / "raw.mp4"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:136-
raw.write_bytes(b"x")
tests\test_watermark.py:137:    comp = VideoCompiler(str(tmp_path / "out")) # no watermark
passed
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:138-    calls = []
tests\test_watermark.py:139-    with patch.object(subprocess, "run",
side_effect=_fake_run_factory(calls)):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:140-
comp.compile_highlights(str(raw), [(0.0, 5.0)], "reel.mp4")
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:141-    assert
all("-vf" not in c for c in calls)
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:142-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:143-
tests\test_watermark.py:144-def
test_watermark_failure_falls_back_to_unwatermarked_clip(tmp_path):
tests\test_watermark.py:145-    """A bad font\logo must not nuke the reel: the clip re-encodes
without the watermark."""
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:146-    raw =
tmp_path / "raw.mp4"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:147-
raw.write_bytes(b"x")
tests\test_watermark.py:148:    comp = VideoCompiler(str(tmp_path / "out"),
tests\test_watermark.py:149-                                watermark={"enabled": True, "text":
"Generated by Khelsutra.guru"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:150-    calls = []
tests\test_watermark.py:151-    with patch.object(subprocess, "run",
side_effect=_fake_run_factory(calls, fail_on_vf=True)):
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:152-    out_path
= comp.compile_highlights(str(raw), [(0.0, 5.0)], "reel.mp4")
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py:153-

```



```

config.json:73:         "begin_padding": 0.5,
config.json:74:         "end_padding": 0.5,
README.md:28:*   **High-Speed FFmpeg Stitching** with configurable `begin_padding`,
`end_padding`, and `min_shot_count` filtering, plus an optional **branding watermark** burned
into every reel (**on by default** ? `"Generated by Khelsutra.guru"`, fully configurable, with
an off switch).
README.md:104:`run_process_and_stitch.py` is an interactive convenience runner: it MD5-hashes a
hardcoded video path, reuses cached indexed segments if available (offering to re-index from
scratch), and then prompts for stitching parameters (`begin_padding`, `end_padding`,
`min_shot_count`) before compiling the final highlight reel. Useful when iterating on stitching
settings against a fixed large recording without re-running expensive segmentation.
README.md:149:         "begin_padding": 0.5,
README.md:150:         "end_padding": 0.5,
run_process_and_stitch.py:85:         end_padding = cfg.stitching.end_padding
run_process_and_stitch.py:86:         begin_padding = cfg.stitching.begin_padding
run_process_and_stitch.py:92:         bpad_input = input(f"Enter begin_padding in seconds
(default {begin_padding}): ").strip()
run_process_and_stitch.py:94:         begin_padding = float(bpad_input)
run_process_and_stitch.py:96:         pad_input = input(f"Enter end_padding in seconds
(default {end_padding}): ").strip()
run_process_and_stitch.py:98:         end_padding = float(pad_input)
run_process_and_stitch.py:106:     print(f"\nCompiling highlights (with
begin_padding={begin_padding}s, end_padding={end_padding}s,
min_shot_count={min_shot_count})...", flush=True)
run_process_and_stitch.py:132:     compiler = VideoCompiler(output_dir, end_padding=end_padding,
begin_padding=begin_padding,
backend\config\models.py:142:         begin_padding: float = 0.5
backend\config\models.py:143:         end_padding: float = 0.5
tests\test_config.py:162:     assert cfg.get("begin_padding") == 0.5           # lives inside
`stitching`
docs\CODE_MAP.md:100:[Omitted long matching line]
docs\CODE_MAP.md:153:[Omitted long matching line]
docs\CODE_MAP.md:275:| Change stitching padding | `config.json
stitching.{begin_padding,end_padding}`; impl
`@backend/pipeline/stitcher/compiler.py::VideoCompiler.compile_highlights`; mirror
`@backend/tools/rally_slicer.py::BEGIN_PADDING/END_PADDING` |
backend\pipeline\stitcher\compiler.py:10:     def __init__(self, output_dir: str, end_padding:
float = 1.0, begin_padding: float = 0.5,
backend\pipeline\stitcher\compiler.py:13:         self.end_padding = end_padding
backend\pipeline\stitcher\compiler.py:14:         self.begin_padding = begin_padding
backend\pipeline\stitcher\compiler.py:148:         into a seamless highlights file, applying
end_padding to prevent abrupt endings.
backend\pipeline\stitcher\compiler.py:173:         logger.info(f"Trimming {len(segments)}
segments (begin_padding={self.begin_padding}s, end_padding={self.end_padding}s)...")
backend\pipeline\stitcher\compiler.py:213:         padded_start = max(0.0, start -
self.begin_padding)
backend\pipeline\stitcher\compiler.py:214:         padded_end = end + self.end_padding
backend\pipeline\stitcher\compiler.py:219:         logger.info(f"{segment_label}:
Padded end ({end + self.end_padding:.2f}s) exceeds video duration. ")
backend\pipeline\segmenters\gemini.py:407:         begin_pad = stitch_cfg.get("begin_padding",
0.5)
backend\pipeline\segmenters\gemini.py:408:         end_pad = stitch_cfg.get("end_padding", 0.5)

```

7. user (2026-06-12T09:16:51.117Z)

```

460     try:
461         path = resolve_under_root(path, output_dir)
462     except ValueError as e:
463         raise HTTPException(status_code=400, detail=str(e)) from e
464     if not os.path.isfile(path):
465         raise HTTPException(status_code=404,
466                             detail=f"No compiled file named '{name}'. Compile first via
/api/compile.")
467     media = "video/quicktime" if name.lower().endswith(".mov") else "video/mp4"
468     return FileResponse(path, media_type=media, filename=name)

```

```

469
470
471 @app.post("/api/query")
472 def query_play_segments(req: QueryRequest):
473     filters = {
474         "server": req.server,
475         "receiver": req.receiver,
476         "duration_min": req.duration_min,
477         "event_type": req.event_type
478     }
479     segments = db_instance.query_play_segments(req.video_id, filters)
480     return {"play_segments": segments}
481
482 @app.post("/api/compile")
483 def compile_highlights(req: CompileRequest):
484     video = db_instance.get_video(req.video_id)
485     if not video:
486         raise HTTPException(status_code=404, detail="Indexed video record not found.")
487
488     # Retrieve matching play segments from db
489     all_segments = db_instance.query_play_segments(req.video_id, {})
490     matched_segments = [s for s in all_segments if s["id"] in req.segment_ids]
491
492     if not matched_segments:
493         raise HTTPException(status_code=400, detail="No matching play segments found to
494 stitch.")
495
496     # Reject path traversal / absolute output names (os.path.join would otherwise let an
497     # absolute or '..'-laden output_filename write anywhere on disk).
498     try:
499         out_name = safe_output_filename(req.output_filename)
500     except ValueError as e:
501         raise HTTPException(status_code=400, detail=str(e)) from e
502
503     # Extract start/end time pairs
504     time_pairs = [(s["start_time"], s["end_time"]) for s in matched_segments]
505
506     # Run compiler (with the optional configured branding watermark, if enabled)
507     output_dir = getattr(getattr(global_config, "output", None), "output_dir", "output")
508     or "output"
509     watermark = getattr(getattr(global_config, "stitching", None), "watermark", None)
510     compiler = VideoCompiler(output_dir, watermark=watermark)
511     try:
512         out_path = compiler.compile_highlights(video["filepath"], time_pairs, out_name)
513         return {"status": "success", "filepath": out_path}
514     except Exception as e:
515         raise HTTPException(status_code=500, detail=f"Highlight stitching failed:
516 {str(e)}") from e
517
518 # --- CLI Debug Utility & Orchestration ---
519 def run_cli_diagnose_clip(video_path: str, timestamp: float):
520     """CLI debugging utility to examine segment properties around a timestamp."""
521     print("=" * 60)
522     print(f"DEBUGGING VIDEO CLIP: {video_path} at {timestamp}s")
523     print("=" * 60)
524
525     cfg = load_config() # legacy wrapper still works, returns dict for now
526
527     # 1. Validation check diagnostics
528     print("[DIAGNOSTIC] Running validation framework...")
529     results = registry.run_all(video_path, cfg)
530     for r in results:
531         status_symbol = "[PASS]" if r.passed else "[FAIL]"
532         print(f" {status_symbol} {r.validator_name}: {r.message}")
533         if r.details:

```

```

531         print(f"          Details: {r.details}")
532
533     # 2. Local segment analysis around the timestamp (sampling +/- 5 seconds)
534     print("-" * 60)
535     print("[DIAGNOSTIC] Analyzing local motion around the timestamp...")
536     import cv2
537     cap = cv2.VideoCapture(video_path)
538     if not cap.isOpened():
539         print("[FAIL] OpenCV could not open video.")

```

8. user (2026-06-12T09:16:52.525Z)

```

125     """Branding overlay burned into each highlight clip during the per-clip re-encode
126     (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
127     (under `stitching.watermark` in config.json) to turn it off. The text is fully
128     configurable. The concat stage stays stream-copy (-c copy)."""
129     enabled: bool = True
130     text: str = "Generated by Khelsutra.guru"
131     logo_path: str = ""          # optional transparent PNG overlay; "" = no logo
132     font_path: str = ""         # optional .ttf; "" = use the fontconfig `font` family
below
133     font: str = "Sans"          # fontconfig family used when font_path is empty
134     font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5) # text height as a
fraction of frame height
135     opacity: float = Field(default=0.85, ge=0.0, le=1.0)
136     box: bool = True           # faint backing box behind the text for legibility
137     margin: int = Field(default=24, ge=0) # px inset from the chosen corner
138     position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = "bottom-
right"
139
140
141     class StitchingConfig(BaseModel):
142         begin_padding: float = 0.5
143         end_padding: float = 0.5
144         use_cache: bool = False
145         min_shot_count: int = 1
146         watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
147
148
149     class SecurityConfig(BaseModel):
150         # When set, /api/process video_path must resolve under this directory (hosted/tenant
151         # mode). Default None preserves the single-user local 'any local file' workflow.
152         allowed_video_root: Optional[str] = None
153
154         # --- Chunked upload (B2, PR-2) ---
155         # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
156         # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
157         # contain upload_dir or /api/process will reject the uploaded paths.
158         upload_dir: str = "output/uploads"
159         # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
160         # free-tier edge proxies cap request bodies there (HOSTING_PLAN §2).
161         max_upload_mb: int = 4096
162         max_part_mb: int = 95
163         allowed_upload_extensions: List[str] = ["mp4", "mov"]
164

```

9. user (2026-06-12T09:16:53.062Z)

```

275         frame_max=fmax,
276         n_boxes=len(frames),
277         vendor_start_time=attrs.get("start_time"),
278         vendor_end_time=attrs.get("end_time"),
279     ))
280     return records
281

```

```

282
283 def detect_issues(records: List[RallyRecord],
284                   stop_frame: Optional[int] = None,
285                   allowed_reasons: Optional[set] = None,
286                   allowed_outcomes: Optional[set] = None,
287                   max_gap_s: float = 15.0) -> List[Issue]:
288     """Find the defects that block or degrade a clean ground-truth import.
289
290     Note: we deliberately do NOT flag the sampling density (step/fps). A boundary
291     slack of ~0.5s is acceptable for this pipeline because the downstream
292     highlight compiler pads every clip on output (badminton-highlight-indexer,
293     backend/stitcher/compiler.py: begin_padding=0.5s, end_padding=1.0s), which
294     absorbs the sampling uncertainty when highlights are trimmed.
295     """
296     allowed = allowed_reasons if allowed_reasons is not None else
DEFAULT_ALLOWED_REASONS
297     allowed_out = allowed_outcomes if allowed_outcomes is not None else
DEFAULT_ALLOWED_OUTCOMES
298     issues: List[Issue] = []
299
300     ordered = sorted(records, key=lambda r: r.rally_number)
301     for i, r in enumerate(ordered):
302         if r.end_time <= r.start_time:
303             issues.append(Issue(
304                 "blocker", "zero_duration",
305                 f"Rally {r.rally_number}: end <= start "
306                 f"({frames {r.frame_min}-{r.frame_max}}); single-sample or stray box.",
307                 r.rally_number))
308
309         if r.ending_reason is not None and r.ending_reason not in allowed:

```

10. user (2026-06-12T09:17:16.350Z)

```

1     import os
2     import sys
3     import argparse
4     import logging
5     import threading
6     import traceback
7     import uuid
8     from concurrent.futures import ThreadPoolExecutor
9     import uvicorn
10    from dotenv import load_dotenv
11
12    load_dotenv()
13
14    # Centralized logging setup for the indexer (hot paths use logger, CLI output uses print
for direct user-facing messages).
15    logging.basicConfig(
16        level=logging.INFO,
17        format='%(asctime)s [%(levelname)s] %(name)s: %(message)s',
18        datefmt='%H:%M:%S'
19    )
20    logger = logging.getLogger(__name__)
21    from fastapi import FastAPI, HTTPException, Request
22    from fastapi.middleware.cors import CORSMiddleware
23    from pydantic import BaseModel
24    from typing import List, Dict, Any, Optional
25
26    from fastapi.staticfiles import StaticFiles
27    from fastapi.responses import FileResponse, HTMLResponse, JSONResponse
28    from backend.pipeline.validators.base import registry
29    from backend.infrastructure.database import Database
30    from backend.pipeline.segmenters.base import segmenter_registry
31    from backend.pipeline.stitcher.compiler import VideoCompiler

```

```

32     from backend.utils.hashing import compute_video_id # canonical file-identity hashing
33     from backend.utils.paths import resolve_under_root, safe_output_filename,
validate_input_path
34     from backend.config import load_config as load_app_config, AppConfig # new typed config
35     from backend.results import Failure # standardized error/result contract
36     from backend.reporting import emit_indexer_report
37
38     app = FastAPI(title="Sports Highlight Indexer API")
39
40     # Enable CORS for the local web interface. allow_origins='' with allow_credentials=True
is
41     # rejected by browsers per the fetch spec and is an unsafe default to carry into the
42     # auth/tenancy work; this tool uses no cookies, so credentials stay off.
43     app.add_middleware(
44         CORSMiddleware,
45         allow_origins=["*"],
46         allow_credentials=False,
47         allow_methods=["*"],
48         allow_headers=["*"],
49     )
50
51     # Serves static frontend files directly (now under api/ per approved structure)
52     os.makedirs("backend/api/static", exist_ok=True)
53     app.mount("/static", StaticFiles(directory="backend/api/static"), name="static")
54
55     @app.get("/", response_class=HTMLResponse)
56     def serve_index():
57         index_path = "backend/api/static/index.html"
58         if os.path.exists(index_path):
59             with open(index_path, "r", encoding="utf-8") as f:
60                 return f.read()
61         return "<h3>Sports Highlight Indexer Server is Running. Please create index.html in
backend/api/static/</h3>"
62
63     # Global variables initialized at startup
64     db_instance: Optional[Database] = None
65     global_config: Optional[AppConfig] = None # now typed (Pydantic model)
66
67     # Async job executor (DEFERRED_HARDENING_PLAN #16). max_workers=1 on purpose: a single
68     # self-hosted box serializes GPU work, and the Gemini provider is rate-fragile under
69     # parallel uploads (2026-06-10 ops learning: serial uploads + backoff). The segmenters
70     # already parallelize their AI handoff internally via their own pools.
71     _job_executor: Optional[ThreadPoolExecutor] = None
72
73     def _get_executor() -> ThreadPoolExecutor:
74         """Lazy module-global executor; tests monkeypatch this for inline execution."""
75         global _job_executor
76         if _job_executor is None:
77             _job_executor = ThreadPoolExecutor(max_workers=1,
thread_name_prefix="jobworker")
78         return _job_executor
79
80     # Legacy thin wrapper for any remaining direct calls during transition.

```

11. user (2026-06-12T09:17:17.745Z)

```

C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-109-   client,
db, tmp_path, mp = env
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-110-   db.add_video("vid", str(_video(tmp_path)), "processed", [])
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-111-   sid =
db.add_play_segment("vid", 1.0, 4.0)
tests\test_api_endpoints.py:112:   r = client.post("/api/compile", json={
tests\test_api_endpoints.py-113-       "video_id": "vid", "segment_ids": [sid],
"output_filename": "../evil.mp4"})

```

```

C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-114-    assert
r.status_code == 400
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-115-
--
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-127-
return f"output/{name}"
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-128-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-129-
mp setattr(m, "VideoCompiler", _FakeCompiler)
tests\test_api_endpoints.py:130:     r = client.post("/api/compile", json={
tests\test_api_endpoints.py-131-         "video_id": "vid", "segment_ids": [sid],
"output_filename": "out.mp4"})
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-132-    assert
r.status_code == 200
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py-133-    assert
r.json()["filepath"] == "output/out.mp4"
--
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py-119-
tests\test_config.py-120-def test_output_dir_field_default_and_override():
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py-121-    """output_dir
now lives on OutputConfig (was a write-only runtime hack), so
tests\test_config.py:122:     /api/compile and the CLI resolve the same value. Default + file
override + the
tests\test_config.py-123-     runtime mutation that backend\main.py::main performs for --output-
dir."""
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py-124-    cfg =
AppConfig()
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_config.py-125-    assert
cfg.output.output_dir == "output" # built-in default
--
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_import_smoke.py-33-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_import_smoke.py-34-
tests\test_import_smoke.py-35-def test_backend_main_imports():
tests\test_import_smoke.py:36:     """backend.main (FastAPI app, /api/compile) must be
importable.
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_import_smoke.py-37-
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_import_smoke.py-38-    Runs in a
subprocess so stubbing missing heavy deps cannot leak into this
C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_import_smoke.py-39-    pytest
process; cwd=repo root puts the package on sys.path.

```

12. user (2026-06-12T09:17:24.930Z)

```

1     """TestClient coverage for the FastAPI seam (previously zero ? the compiler SyntaxError
2     that broke `import backend.main` shipped green because nothing imported it)."""
3     import pytest
4
5     pytest.importorskip("httpx") # TestClient needs httpx; skip cleanly where absent
6     from fastapi.testclient import TestClient
7
8     import backend.main as m
9     from backend.infrastructure.database import Database
10    from backend.config.models import AppConfig
11    from backend.pipeline.validators.base import ValidationResult
12
13
14    class _InlineExecutor:
15        """submit() runs the fn synchronously ? async jobs complete before the response
16        is inspected (full lifecycle coverage lives in test_async_jobs.py)."""
17
18        def submit(self, fn, *args, **kwargs):
19            fn(*args, **kwargs)
20
21
22    @pytest.fixture

```

```

23     def env(tmp_path, monkeypatch):
24         db = Database(str(tmp_path / "t.db"))
25         cfg = AppConfig()
26         monkeypatch.setattr(m, "db_instance", db)
27         monkeypatch.setattr(m, "global_config", cfg)
28         monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
29         client = TestClient(m.app)
30         return client, db, tmp_path, monkeypatch
31
32
33     def _video(tmp_path, name="m.mp4"):
34         p = tmp_path / name
35         p.write_bytes(b"not-a-real-video-but-hashable")
36         return p
37
38
39     def _pass():
40         return [ValidationResult(validator_name="x", passed=True, message="ok", details={})]
41
42
43     def _fail():
44         return [ValidationResult(validator_name="blur", passed=False, message="blurry",
45 details={})]
46
47     class _FakeSeg:
48         def __init__(self, db, cfg):
49             self.db = db
50
51         def process_video(self, video_id, video_path, *a, **k):
52             sid = self.db.add_play_segment(video_id, 0.0, 5.0)
53             return [{"id": sid, "start_time": 0.0, "end_time": 5.0, "duration": 5.0,
54 "metadata": {}}]
55
56     def test_api_config_ok(env):
57         client = env[0]
58         r = client.get("/api/config")
59         assert r.status_code == 200
60         assert r.json()["sport"] == "badminton"
61
62
63     def test_process_success_with_mocked_segmenter(env):
64         """#16: submit returns 202 + job_id; the (inline) job carries the old response."""
65         client, db, tmp_path, mp = env
66         vid_file = _video(tmp_path)
67         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {}))
68         mp.setattr(m.segmenter_registry, "get", lambda name: _FakeSeg)
69         r = client.post("/api/process", json={"video_path": str(vid_file), "segmenter":
70 "x"})
71         assert r.status_code == 202
72         job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
73         assert job["status"] == "done"
74         assert job["result"]["status"] == "success"
75         assert len(job["result"]["play_segments"]) == 1
76
77     def test_process_validation_failure_preserves_prior_segments(env):
78         client, db, tmp_path, mp = env
79         vid_file = _video(tmp_path)
80         vid = m.compute_video_id(str(vid_file))
81         db.add_video(vid, str(vid_file), "processed", [])
82         db.add_play_segment(vid, 1.0, 2.0) # prior good result
83         mp.setattr(m.registry, "run_all_with_context", lambda *a, **k: (_fail(), {}))
84         r = client.post("/api/process", json={"video_path": str(vid_file)})

```

```

85     assert r.status_code == 202
86     job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
87     assert job["status"] == "done" # worker finished; the PIPELINE verdict failed
88     assert job["result"]["status"] == "failed"
89     # PR-C: a failed re-validation must NOT cascade-delete prior segments.
90     assert len(db.query_play_segments(vid, {})) == 1
91
92
93     def test_process_rejects_null_byte_path(env):
94         client = env[0]
95         r = client.post("/api/process", json={"video_path": "a\x00b.mp4"})
96         assert r.status_code == 400
97
98
99     def test_query_returns_segments(env):
100         client, db, tmp_path, mp = env
101         db.add_video("vid", "/x.mp4", "processed", [])
102         db.add_play_segment("vid", 1.0, 4.0)
103         r = client.post("/api/query", json={"video_id": "vid"})
104         assert r.status_code == 200
105         assert len(r.json()["play_segments"]) == 1
106
107
108     def test_compile_rejects_traversal_output_filename(env):
109         client, db, tmp_path, mp = env
110         db.add_video("vid", str(_video(tmp_path)), "processed", [])
111         sid = db.add_play_segment("vid", 1.0, 4.0)
112         r = client.post("/api/compile", json={
113             "video_id": "vid", "segment_ids": [sid], "output_filename": "../evil.mp4"})
114         assert r.status_code == 400
115
116
117     def test_compile_success_with_mocked_compiler(env):
118         client, db, tmp_path, mp = env
119         db.add_video("vid", str(_video(tmp_path)), "processed", [])
120         sid = db.add_play_segment("vid", 1.0, 4.0)
121
122         class _FakeCompiler:
123             def __init__(self, out_dir, watermark=None, **kwargs):
124                 pass
125
126             def compile_highlights(self, raw, pairs, name):
127                 return f"output/{name}"
128
129         mp.setattr(m, "VideoCompiler", _FakeCompiler)
130         r = client.post("/api/compile", json={
131             "video_id": "vid", "segment_ids": [sid], "output_filename": "out.mp4"})
132         assert r.status_code == 200
133         assert r.json()["filepath"] == "output/out.mp4"
134

```

13. user (2026-06-12T09:18:03.679Z)

```

1     """Configuration package.
2
3     Public API:
4         from backend.config import load_config, AppConfig, Config
5
6     This package is the canonical place for all configuration access.
7     """
8
9     from .loader import load_config, save_default_config
10    from .models import AppConfig, Config, IndexingConfig, ValidationConfig # re-export
common ones
11

```



```

12     __all__ = [
13         "load_config",
14         "save_default_config",
15         "AppConfig",
16         "Config",
17         "IndexingConfig",
18         "ValidationConfig",
19     ]
20

```

14. user (2026-06-12T09:18:04.014Z)

```

1     """VideoCompiler coverage (previously zero ? the user-facing highlight-reel output path,
2     where the f-string SyntaxError lived undetected). ffmpeg/ffprobe are mocked."""
3     import os
4     import subprocess
5     from unittest.mock import patch, MagicMock
6
7     import pytest
8
9     from backend.pipeline.stitcher.compiler import VideoCompiler
10
11
12     def test_parse_time_formats():
13         f = VideoCompiler._parse_time
14         assert f(12.5) == 12.5
15         assert f("12.5") == 12.5
16         assert f("1:30") == 90.0
17         assert f("1:00:00") == 3600.0
18         assert f("garbage") == 0.0
19
20
21     def test_compile_basenames_output_and_stitches(tmp_path):
22         comp = VideoCompiler(str(tmp_path / "out"))
23         raw = tmp_path / "raw.mp4"
24         raw.write_bytes(b"x")
25
26         calls = []
27
28         def fake_run(cmd, *a, **k):
29             calls.append(cmd)
30             # Create whatever output file ffmpeg/ffprobe was told to write so existence
31             checks pass.
32             if cmd[0] == "ffprobe":
33                 return MagicMock(stdout="100.0\n", returncode=0)
34             out = cmd[-1]
35             if isinstance(out, str) and out.endswith(("mp4",)):
36                 open(out, "wb").write(b"clip")
37                 return MagicMock(returncode=0, stdout=b"", stderr=b"")
38
39         with patch.object(subprocess, "run", side_effect=fake_run):
40             # Traversal in the name must be stripped (defense-in-depth basename).
41             out_path = comp.compile_highlights(str(raw), [(0.0, 5.0), (10.0, 15.0)],
42             "../reel.mp4")
43
44             assert os.path.basename(out_path) == "reel.mp4"
45             assert os.path.dirname(out_path).endswith("out")
46             # Per-clip trims + one concat = at least 3 ffmpeg invocations (plus an ffprobe).
47             assert sum(1 for c in calls if c[0] == "ffmpeg") >= 3
48
49     def test_compile_raises_on_no_segments(tmp_path):
50         comp = VideoCompiler(str(tmp_path / "out"))
51         raw = tmp_path / "raw.mp4"
52         raw.write_bytes(b"x")

```

```

52         with pytest.raises(ValueError):
53             comp.compile_highlights(str(raw), [], "reel.mp4")
54
55
56     def test_compile_raises_when_source_missing(tmp_path):
57         comp = VideoCompiler(str(tmp_path / "out"))
58         with pytest.raises(FileNotFoundError):
59             comp.compile_highlights(str(tmp_path / "nope.mp4"), [(0.0, 5.0)], "reel.mp4")
60

```

15. user (2026-06-12T09:18:06.648Z)

```

master
?? arial.ttf
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
master
M .gitignore

```

16. user (2026-06-12T09:18:53.360Z)

```

96     - `@backend/config/models.py` ? Pydantic v2 models; **single source of truth for
defaults**.
97     - `::AppConfig` ? root. Scalars `sport='badminton'`, `ai_provider='gemini'`,
`ai_model='gemini-2.5-flash'`; sections `validation`, `indexing`, `output`, `stitching`
(`default_factory`). `model_config = {extra:'allow', validate_assignment:True}`. ?
`extra='allow'` -> unknown/typo'd config.json keys silently kept (legacy tolerance).
98     - ? `::AppConfig.get` ? legacy dict-compat shim: returns nested **sections as plain
dicts** (`model_dump(mode='json')`, NOT the model) so chained
`config.get("indexing",{}).get(...)` keeps working; falls back to searching a leaf key in any
section (sections discovered dynamically from `model_fields`). ? "bit every validator on real
runs" per docstring ? returning the model breaks `get`. `::AppConfig.__getitem__` raises
`KeyError` for unknown top-level keys else delegates to `get`.
99     - `::IndexingConfig` ? segmenter/detector knobs (`motion_threshold=0.08`,
`min_segment_duration=3.0`, `dead_time_merge_gap=2.0`, `chunk_duration=90.0`,
`chunk_overlap=10.0`, `yolo_velocity_threshold=0.15`, `yolo_frame_skip=3`,
`ai_handoff_padding=2.0`, `ai_max_workers=5`, `log/store_yolo_candidates=True`, nested
`tracknet`). ? `default_segmenter='yolo_hybrid'` HERE but config.json overrides to `gemini`. ?
`::IndexingConfig.segmenter_must_be_registered` `@field_validator` is a **no-op pass-through** ?
a typo'd segmenter validates fine, only fails later at `segmenter_registry.get(...)`
100     - `::ValidationConfig` (`min_resolution_width=1280`, `min_resolution_height=720`, `min_fps=29.9`, `min_blur_threshold=20.0`, `max_scene_cuts_per_minute=0.5`, nested `relevance`);
`::ValidationRelevanceConfig` (court HSV gates, `court_pixels_min_ratio=0.05`).
`::TrackNetConfig` (WSL invocation contract: `wsl_distro='Ubuntu'`, `repo_dir`,
`conda_env='TrackNetV4'`, `weights_path='', `queue_length=5`, `velocity_threshold=0.02`).
`::OutputConfig` (`default_highlights_name`, `db_name='sports_indexer.db'`,
`output_dir='output'` ? CLI `--output-dir` overrides it on the typed model in `main()`).
`::StitchingConfig`
(`begin_padding=0.5`, `end_padding=0.5`, `use_cache=False`, `min_shot_count=1`). `::Sport` Literal
of 5 sports. `::Config = AppConfig` alias.
101     - `@backend/config/loader.py` ? load/save with defaults-merge.
102     - `::DEFAULT_CONFIG_PATH = Path("config.json")` (? CWD-relative, NOT repo-root).
`_get_built_in_defaults` = `AppConfig().model_dump(...)`
103     - `::load_config` ? precedence built-in-defaults -> file overrides. ? **shallow top-
level merge** (`{**defaults, **file_data}`): a section in config.json REPLACES the whole
defaults dict for that section (Pydantic then re-applies field defaults for missing leaves).
Missing/unreadable/parse-error file -> fully-defaulted AppConfig + `[CONFIG WARNING]` (non-
fatal).

```

17. user (2026-06-12T09:18:54.720Z)

```
152     ### 2.4 Stitcher / tools / utils
153     - `@backend/pipeline/stitcher/compiler.py::VideoCompiler` ? `compile_highlights(raw,
segments:[(start,end)], out_name)`. Per-clip re-encode -> `-f concat -c copy`. `::_parse_time`
(mirrors gemini.parse_time; ? unparseable -> 0.0), `::_get_video_duration`. ? if duration probe
fails (0.0) NO boundary validation; `begin_padding`(0.5)/`end_padding`(1.0) defaults; temp clips
in a TemporaryDirectory under output_dir; raises ValueError/FileNotFoundError/RuntimeError; uses
`print()`. ? two-stage codec contract: clips re-encoded with IDENTICAL settings precisely so
final concat can stream-copy ? change a preset and concat may break. ? optional **branding
watermark** (`stitching.watermark`, default-on): `::_watermark_filter` builds a `-vf` (drawtext
via `textfile=` + optional logo `movie/overlay`) applied to the per-clip re-encode only (concat
stays `-c copy`); a failed watermark encode retries the clip WITHOUT it (never nukes the reel).
Pass `watermark=` to `VideoCompiler(...)` (a `WatermarkConfig` model or dict; `None` = off).
154     - `@backend/tools/rally_slicer.py` ? standalone CLI. `::compute_clip_windows` (pure,
unit-tested), `::load_rally_labels` (golden schema; ? silently skips degenerate rows),
`::slice_rallies`, `::ClipWindow` (rally:str), `::BEGIN_PADDING=0.5`/`::END_PADDING=1.0` (?
duplicate compiler's by copy). `read_time` -> backend/eval/annotations.parse_timestamp`.
155     - `@backend/utls/video.py` ? `::get_video_duration` (ffprobe; ? 0.0 on failure =
"unknown"), `::extract_video_chunk(...,reencode=False,scale_width=None)`. ? copy mode cuts at
nearest keyframe (drift) ? pass `reencode=True` for short/accurate clips; `scale_width` requires
`reencode=True`.
```

18. user (2026-06-12T09:18:55.244Z)

```
272     | Read config in new code | `from backend.config import load_config, AppConfig`
(`@backend/config/__init__.py`); attribute access, NOT raw `json.load` |
273     | Tune candidate windowing (velocity/merge/min-dur) |
`@backend/pipeline/detectors/tracknet_runner.py::trajectory_to_action_windows`; cfg `config.json
indexing.{wasb,tracknet}.velocity_threshold`, `dead_time_merge_gap`,
`min_action_window_duration` |
274     | Add a segmenter | subclass `@backend/pipeline/segmenters/base.py::BasePlaySegmenter`;
`segmenter_registry.register(...)` at module bottom; import in
`@backend/pipeline/segmenters/__init__.py`; if AI-based add to
`@backend/reporting/harvest.py::AI_SEGMENTERS` |
275     | Change stitching padding | `config.json stitching.{begin_padding,end_padding}`; impl
`@backend/pipeline/stitcher/compiler.py::VideoCompiler.compile_highlights`; mirror
`@backend/tools/rally_slicer.py::BEGIN_PADDING/END_PADDING` |
276     | Add a sport | subclass `@backend/sports/base.py::SportProfile`;
`sport_registry.register(...)` in `@backend/sports/__init__.py`; define prompt+HSV in new file |
277     | Add an AI provider | subclass `@backend/providers/base.py::AIVideoProvider`;
`provider_registry.register('name', Cls)` |
```

19. user (2026-06-12T09:19:31.763Z)

Switched to a new branch 'fix/stitching-padding-config'

20. user (2026-06-12T09:19:38.872Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

21. user (2026-06-12T09:19:37.718Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config__init__.py has been updated successfully. (file state is current in your context ? no need to Read it back)

22. assistant (2026-06-12T09:19:44.025Z)

Now the two construction sites:

23. user (2026-06-12T09:19:47.819Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated

successfully. (file state is current in your context ? no need to Read it back)

24. user (2026-06-12T09:19:50.315Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

25. user (2026-06-12T09:19:58.878Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)

26. user (2026-06-12T09:20:07.863Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)

27. user (2026-06-12T09:20:16.756Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

28. user (2026-06-12T09:20:19.339Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\docs\CODE_MAP.md has been updated successfully. (file state is current in your context ? no need to Read it back)

29. assistant (2026-06-12T09:20:22.857Z)

Now run the full offline suite:

30. user (2026-06-12T09:21:11.438Z)

```
=====
Full test suite: C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
-q
obsreport (optional reporting dep): available
=====
..... [ 17%]
..... [ 35%]
..... [ 53%]
..... [ 71%]
..... [ 89%]
..... [100%]
401 passed in 41.99s
=====
RESULT: PASS - all tests green
=====
```

31. assistant (2026-06-12T09:21:18.204Z)

Suite is green ? 401 passed, including the new regression test. CI also has ruff and mypy lanes, so let me check those on the changed files before committing:

32. user (2026-06-12T09:21:25.044Z)

All checks passed!

backend\sports__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

backend\providers__init__.py:7: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs [annotation-unchecked]

Success: no issues found in 2 source files

33. user (2026-06-12T09:21:33.917Z)

```
fix/stitching-padding-config
M backend/config/__init__.py
M backend/main.py
M docs/CODE_MAP.md
M tests/test_api_endpoints.py
?? arial.ttf
?? artifacts/
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
```

34. user (2026-06-12T09:21:48.597Z)

```
[fix/stitching-padding-config 30e4899] fix(api): thread config stitching paddings into live
VideoCompiler sites
 4 files changed, 54 insertions(+), 9 deletions(-)
remote:
remote: Create a pull request for 'fix/stitching-padding-config' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/stitching-
padding-config
remote:
branch 'fix/stitching-padding-config' set up to track 'origin/fix/stitching-padding-config'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      fix/stitching-padding-config -> fix/stitching-padding-config
```

35. user (2026-06-12T09:22:09.333Z)

```
Warning: 8 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/102
```

36. user (2026-06-12T09:22:18.747Z)

```
M .gitignore
Switched to a new branch 'docs/compiler-padding-rationale'
False
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

37. user (2026-06-12T09:22:27.556Z)

```
The file C:\Users\avidu\Projects\sports-data-collector\src\parsers\cvat\rally_cvat.py has been
updated successfully. (file state is current in your context ? no need to Read it back)
```

38. user (2026-06-12T09:22:37.798Z)

```
..... [ 52%]
..... [100%]
137 passed in 2.30s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

39. user (2026-06-12T09:22:51.292Z)

```
docs/compiler-padding-rationale
M .gitignore
M src/parsers/cvat/rally_cvat.py
[docs/compiler-padding-rationale 4ce9e01] docs(cvat): update detect_issues padding rationale to
config-driven values
 1 file changed, 2 insertions(+), 1 deletion(-)
```

```
remote:
remote: Create a pull request for 'docs/compiler-padding-rationale' on GitHub by visiting:
remote:   https://github.com/avidullu/sports-data-collector/pull/new/docs/compiler-padding-
rationale
remote:
branch 'docs/compiler-padding-rationale' set up to track 'origin/docs/compiler-padding-
rationale'.
To https://github.com/avidullu/sports-data-collector.git
* [new branch]      docs/compiler-padding-rationale -> docs/compiler-padding-rationale
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

40. user (2026-06-12T09:23:05.447Z)

```
Warning: 1 uncommitted change
https://github.com/avidullu/sports-data-collector/pull/20
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

41. assistant (2026-06-12T09:23:13.695Z)

Both PRs are open. Now verifying CI goes green on the indexer PR (4 lanes):

42. user (2026-06-12T09:24:00.025Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    22s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27406851871/job/80998008861
Full test suite (offline, mocked)                pending 0      https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27406851871/job/80998008701
Full test suite (offline, mocked)                pass     1m28s   https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27406851871/job/80998008701
Mypy (lenient baseline) pass     31s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27406851871/job/80998008713
Ruff lint      pass     14s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27406851871/job/80998008705
Syntax sweep + import smoke (no GPU stack)      pass     22s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27406851871/job/80998008861
Full test suite (offline, mocked)                pass     1m28s   https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27406851871/job/80998008701
Mypy (lenient baseline) pass     31s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27406851871/job/80998008713
Ruff lint      pass     14s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27406851871/job/80998008705
Syntax sweep + import smoke (no GPU stack)      pass     22s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27406851871/job/80998008861
```

43. assistant (2026-06-12T09:24:07.415Z)

Indexer CI is 4/4 green. Checking the collector PR's CI:

44. user (2026-06-12T09:24:10.910Z)

```
test      pass     15s      https://github.com/avidullu/sports-data-
collector/actions/runs/27406898883/job/80998164484
test      pass     15s      https://github.com/avidullu/sports-data-
collector/actions/runs/27406898883/job/80998164484
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

45. user (2026-06-12T09:24:33.385Z)

```
Your branch is up to date with 'origin/master'.
Switched to branch 'master'
M       .gitignore
Your branch is up to date with 'origin/master'.
Switched to branch 'master'
M       .gitignore
```

46. user (2026-06-12T09:24:52.733Z)

<tool_use_error>File has not been read yet. Read it first before writing to it.</tool_use_error>

47. user (2026-06-12T09:24:57.896Z)

```
1      ---
2      name: current-state
3      description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
4      only on request). What just landed, what's next, open decisions."
5      metadata:
6      node_type: memory
7      type: project
8      originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
9      ---
10     # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12     **Checkpoint:** 2026-06-12 (Roora ANNOTATION-PROXY assessment delivered ? AWAITING OWNER
13     DECISION, no code).
14     Owner asked: re-encode videos to a Roora-friendly profile before handoff? Researched via
15     workflow `wr6mnel6c`
16     (3 sweeps + adversarial verify; full JSON
17     `%LOCALAPPDATA%\Temp\claude\?\tasks\wr6mnel6c.output`). **Facts:** root
18     cause of the ?0.5s boundaries = 30-frame CVAT sampling grid (NOT playback delay;
19     direction never measured ? docs say
```

48. assistant (2026-06-12T09:25:03.611Z)

A sibling session prepended a new checkpoint since my first read ? I'll insert mine above it without disturbing it:

49. user (2026-06-12T09:25:14.194Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

50. assistant (2026-06-12T09:25:28.990Z)

Done. The fix is shipped as two PRs, both CI-green and awaiting your review ? I did not merge anything.

```
**[badminton-highlight-indexer #102](https://github.com/avidullu/badminton-highlight-indexer/pull/102)** ? the actual fix:
- Both `VideoCompiler` construction sites in [backend/main.py](backend/main.py) (`/api/compile` around line 508 and the CLI trim path around line 633) now pass `begin_padding`/`end_padding` from the typed `StitchingConfig`, alongside the watermark they already passed. The API site keeps the existing defensive idiom by falling back to `StitchingConfig()` when `global_config` is unset; `StitchingConfig` is now re-exported from `backend.config`.
- New regression test in [tests/test_api_endpoints.py](tests/test_api_endpoints.py) drives the **real** compiler through `/api/compile` with mocked ffmpeg/ffprobe and asserts that a config override of 2.0/3.0 actually moves the trim boundaries to `-ss 8.0 / -to 17.0` for a 10?14s segment ? the previously unwired defaults would have produced 9.5/15.0, so this test fails on the old code.
- [docs/CODE_MAP.md](docs/CODE_MAP.md) truth-synced: the stitcher entry now distinguishes the constructor defaults (0.5/1.0) from the config-driven live call sites (0.5/0.5), and the padding recipe row points at the `main.py` wiring.
- Verified: `python run_all_tests.py` ? **401 passed**; ruff and mypy clean on touched files; all 4 CI lanes green.
```

One behavior change worth knowing before merge: live API reels currently get 1.0s of end padding; with the shipped `config.json` they'll get the configured **0.5s** after this lands.

****[sports-data-collector #20](https://github.com/avidullu/sports-data-collector/pull/20)**** ? the cross-repo coupling you flagged: the `detect\_issues` docstring in `rally\_cvat.py` now cites the config-driven 0.5/0.5 paddings instead of the hardcoded 0.5/1.0 (and the corrected module path). Docstring-only; collector suite 137 passed, CI green. Your uncommitted `.gitignore` edit there was left untouched.

Both checkouts are back on `master`, and I recorded a checkpoint in the memory current-state file (noting the padding-rationale interaction with the in-flight Roora boundary-slack thread, since that rationale now rests on 0.5s rather than 1.0s of end padding). One observation: an untracked `arial.ttf` appeared at the indexer repo root ? not from this work; I left it alone, but it may be a stray from a watermark-font experiment.

51. user (2026-06-12T09:45:30.912Z)

Can you update the global skill/memory/reminder for yourself to **always git pull the repo before even starting to read it** because I have an occupational hazard of ensuring I always close/leave a session with a "clean slate" but often I have multiple parallel slates, so a git pull would ensure the local starts where remote truly is.

52. assistant (2026-06-12T09:46:17.784Z)

I'll add this to your global `~/.claude/CLAUDE.md` (the instructions loaded in every project session), since the memory directory is per-project and you asked for a global rule.

53. user (2026-06-12T09:46:17.997Z)

```
1      # Global instructions (all projects)
2
3      ## Session-handoff convention
4      Maintain a living **session handoff** for each project and use it to resume context
quickly
5      across windows.
6
7      - **Where it lives:**
8      - If the project has an auto-memory dir (`~/.claude/projects/<project>/memory/`), keep
the
9      handoff as `session-handoff.md` there, and list it under a ***? Start here*** entry
at the
10     top of that project's `MEMORY.md`.
11     - Otherwise, keep a `SESSION_HANDOFF.md` at the repo root.
12     - **What it contains** (keep it to ~1 page ? it POINTS to detailed artifacts, never
duplicates them):
13     1. **You are here** ? current state.
14     2. **Next steps / open threads.**
15     3. **Ramp-up kit** ? the exact files/docs to read to get current.
16     4. **Key decisions** ? so they aren't relitigated.
17     - **At session start:** if a handoff exists, read it before starting substantive work,
and use it
18     (plus its ramp-up kit) to get up to speed instead of replaying past conversation.
19     - **Updating it:** keep it current **automatically** ? refresh at meaningful checkpoints
(a PR is
20     pushed/merged, a key decision is made, work shifts phase, or a session is wrapping
up), so a
21     restart can ramp up without losing context. Do **not** rewrite it every turn ? only
when
22     something worth resuming from changed. The phrases ***update the handoff*** /
***refresh the
23     handoff*** force an immediate full rewrite on demand.
24     - **Keep it honest:** it reflects real, shipped state ? never aspirational. (See each
project's
25     attribution/working-agreement note where present.)
26
```

54. user (2026-06-12T09:46:28.513Z)

The file C:\Users\avidu\.claude\CLAUDE.md has been updated successfully. (file state is current in your context ? no need to Read it back)

55. assistant (2026-06-12T09:46:35.795Z)

The rule is saved. Applying it right now to both repos as a first exercise:

56. user (2026-06-12T09:46:44.873Z)

From <https://github.com/avidullu/badminton-highlight-indexer>

```
* [new branch]      claude/watermark-bundled-font -> origin/claude/watermark-bundled-font
```

Already up to date.

Already up to date.

57. assistant (2026-06-12T09:46:59.126Z)

Done. The rule now lives at the top of your global `~/.claude/CLAUDE.md`, so it loads in every session across all projects:

- ****Always `git pull` before starting to read a repo**** ? at session start for the working directory, and again for any sibling repo (like `sports-data-collector`) the moment work touches it.
- ****Safe form only:**** `git pull --ff-only` ? never auto-merge, rebase, or discard local state to make a pull succeed.
- ****If it can't fast-forward**** (diverged branch, dirty tree): report the state and proceed read-only on local, flagged as possibly stale ? uncommitted changes are treated as a sibling session's or your WIP and never clobbered.

I also captured your *why* in the rule itself ? parallel "clean slates" mean master moves between sessions ? so future sessions understand the intent, not just the mechanic.

And it paid off immediately: running the pull on both repos just now, both masters were already current, but the fetch surfaced a new remote branch `claude/watermark-bundled-font` from another session ? which neatly explains the stray `arial.ttf` I'd flagged at the indexer root earlier (it's that session's watermark-font experiment, left alone).

One option if you ever want this enforced mechanically rather than by instruction: a `SessionStart` hook in `settings.json` could run the `--ff-only` pull automatically before I even wake up. I'd lean against it ? a hook pulls blindly without first checking for a dirty tree or a sibling session mid-work on a feature branch, whereas the instruction lets me look before pulling ? but say the word and I'll set it up.

58. user (2026-06-12T16:42:42.470Z)

Can you run the highlight indexer? Using the GPU? end to end. and show me highlights for the video that I'm attaching. Only for rallies where the number of shots were more than 7. You will have to. update the configs and make sure that The number of shots config is picked up. Let me know if that is not possible. "F:\GoPro Hero Black 12\Khe1Sutra\Badminton 12 June 2026\GX010125.MP4"

59. user (2026-06-12T16:43:37.630Z)

Chapter marked: "E2E GPU highlight run (GX010125)". Continue your current work.

60. assistant (2026-06-12T16:43:43.842Z)

I'll first verify the video exists, the repo is fresh, and then trace exactly where `min\_shot\_count` is (or isn't) honored before running anything.

61. user (2026-06-12T16:43:48.791Z)

From <https://github.com/avidullu/badminton-highlight-indexer>

```
* [new branch]      claude/indexer-doc-review-XVPTH -> origin/claude/indexer-doc-review-XVPTH
ffdfde0..2a8d1ea  master      -> origin/master
```

Already up to date.

FullName
Length LastWriteTime

F:\GoPro Hero Black 12\KhelSutra\Badminton 12 June 2026\GX010125.MP4 11522892112 6/12/2026
7:25:06 AM

62. user (2026-06-12T16:43:48.808Z)

```
1      import os
2      import json
3      import logging
4      from dotenv import load_dotenv
5
6      load_dotenv()
7
8      # Configure root logger ? all backend modules use logging.getLogger(__name__)
9      logging.basicConfig(
10         level=logging.INFO,
11         format="%(asctime)s [%(name)s] %(levelname)s: %(message)s",
12         datefmt="%H:%M:%S",
13     )
14
15     from backend.infrastructure.database import Database
16     from backend.config import load_config
17     from backend.utils.hashing import compute_video_id
18     from backend.pipeline.segmenters import segmenter_registry
19     from backend.pipeline.stitcher.compiler import VideoCompiler
20
21     def main():
22         video_path = r"C:\Users\avidu\Videos\Testing\KushagraYashAviFirstVideo.mp4"
23         output_dir = r"C:\Users\avidu\Videos\Testing"
24         output_filename = "Final_2_OutputOfKushagraYashAviFirstVideo.mp4"
25
26         print("Loading config...")
27         cfg = load_config() # typed AppConfig ? single source of defaults
28
29         print("Verifying files exist...")
30         if not os.path.exists(video_path):
31             print(f"Error: Video file not found at {video_path}")
32             return
33
34         os.makedirs(cfg.output.output_dir, exist_ok=True)
35         db_path = os.path.join(cfg.output.output_dir, cfg.output.db_name)
36         db = Database(db_path)
37
38         # Calculate MD5 of the video to uniquely identify it
39         import time
40         print("Calculating video MD5 (this might take a few seconds on large files)...",
41             flush=True)
42         t0 = time.time()
43         video_id = compute_video_id(video_path)
44         print(f"Calculated MD5: {video_id} (took {time.time()-t0:.1f}s)", flush=True)
45
46         segments = []
47         cached_video = db.get_video(video_id)
48         cache_hit = False
49
50         if cached_video and cached_video.get("status") == "processed":
51             segments = db.query_play_segments(video_id, {})
52             if len(segments) > 0:
53                 print(f"\n[CACHE] Found cached indexed video in database (MD5: {video_id})
54 with {len(segments)} parsed play segments.", flush=True)
```

```

53         choice = input("Would you like to (1) Run stitching on the cached segments,
or (2) Reprocess the video with Gemini from scratch? [1/2]: ").strip()
54         if choice == '1':
55             print("[CACHE] Using cached play segments. Skipping segmenter API
call.", flush=True)
56             cache_hit = True
57         else:
58             print("[CACHE] Ignoring cache...", flush=True)
59         else:
60             print("[CACHE] Video exists in DB but contains 0 play segments. Treating as
cache miss.", flush=True)
61
62         if not cache_hit:
63             print("[CACHE] Initializing clean indexing...", flush=True)
64             db.clear_video_data(video_id)
65
66             print("Bypassing OpenCV validations for performance on massive files...",
flush=True)
67             db.add_video(video_id, video_path, "processed", [])
68
69             print("Running segmenter & indexing...", flush=True)
70             seg_name = cfg.indexing.default_segmenter
71             segmenter_cls = segmenter_registry.get(seg_name)
72             if not segmenter_cls:
73                 print(f"[FAIL] Segmenter '{seg_name}' not found.")
74                 return
75             print(f"Using segmenter: {seg_name}", flush=True)
76             segmenter = segmenter_cls(db, cfg)
77             segments = segmenter.process_video(video_id, video_path)
78             print(f"Successfully indexed {len(segments)} play segments.", flush=True)
79
80             if not segments:
81                 print("No play segments detected. Cannot compile highlights.")
82                 return
83
84
85             end_padding = cfg.stitching.end_padding
86             begin_padding = cfg.stitching.begin_padding
87             min_shot_count = cfg.stitching.min_shot_count
88
89             if cache_hit:
90                 print("\n--- Stitching Customizations ---")
91                 try:
92                     bpad_input = input(f"Enter begin_padding in seconds (default
{begin_padding}): ").strip()
93                     if bpad_input:
94                         begin_padding = float(bpad_input)
95
96                     pad_input = input(f"Enter end_padding in seconds (default {end_padding}):
").strip()
97                     if pad_input:
98                         end_padding = float(pad_input)
99
100                     shot_input = input(f"Enter min_shot_count to filter by (default
{min_shot_count}): ").strip()
101                     if shot_input:
102                         min_shot_count = int(shot_input)
103                     except ValueError:
104                         print("[WARNING] Invalid input detected, falling back to defaults from
config.json.")
105
106                 print(f"\nCompiling highlights (with begin_padding={begin_padding}s,
end_padding={end_padding}s, min_shot_count={min_shot_count})...", flush=True)
107
108                 # Filter segments where shot_count >= min_shot_count

```

```

109         time_pairs = []
110         skipped_count = 0
111         for s in segments:
112             meta = s.get("metadata", {})
113             if isinstance(meta, str):
114                 try:
115                     meta = json.loads(meta)
116                 except Exception:
117                     meta = {}
118             shot_count = meta.get("shot_count", 0)
119             if shot_count >= min_shot_count:
120                 time_pairs.append((s["start_time"], s["end_time"]))
121             else:
122                 skipped_count += 1
123
124         if skipped_count > 0:
125             print(f"[FILTER] Excluded {skipped_count} play segments with fewer than
126 {min_shot_count} shots.", flush=True)
127         print(f"[FILTER] Retaining {len(time_pairs)} play segments for stitching.",
128 flush=True)
129
130         if not time_pairs:
131             print("No play segments passed the filtering criteria. Cannot compile
132 highlights.")
133         return
134
135         compiler = VideoCompiler(output_dir, end_padding=end_padding,
136 begin_padding=begin_padding,
137 watermark=cfg.stitching.watermark)
138
139         try:
140             out_path = compiler.compile_highlights(video_path, time_pairs, output_filename)
141             print(f"Stitched highlight video saved to: {out_path}", flush=True)
142         except Exception as e:
143             print(f"Error during stitching: {e}", flush=True)
144
145     if __name__ == "__main__":
146         main()
147
148

```

63. user (2026-06-12T16:44:44.482Z)

85abe26 docs(platform): video-locality mental model - the 10GB question (owner-requested)
1e327d4 docs(platform): video-locality mental model - the 10GB question (owner-requested)
ffdfde0 Merge pull request #103 from avidullu:claude/watermark-bundled-font
112e75d feat(stitcher): bundle Apache-2.0 default watermark font + log fallback reason
82e479f Merge pull request #102 from avidullu:fix/stitching-padding-config
30e4899 fix(api): thread config stitching paddings into live VideoCompiler sites
?? arial.ttf
?? artifacts/
?? docs/STORAGE_SHARING_MODEL.md
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardenening_status.patch
?? scratch/PR88_FIX.md

64. user (2026-06-12T16:44:44.497Z)

```

1      {
2          "sport": "badminton",
3          "ai_provider": "gemini",
4          "ai_model": "gemini-flash-latest",
5          "validation": {
6              "min_resolution_width": 1280,

```

```

7         "min_resolution_height": 720,
8         "min_fps": 29.9,
9         "min_blur_threshold": 20.0,
10        "max_scene_cuts_per_minute": 0.5,
11        "relevance": {
12            "court_green_lower": [
13                35,
14                40,
15                40
16            ],
17            "court_green_upper": [
18                85,
19                255,
20                255
21            ],
22            "court_pixels_min_ratio": 0.05
23        }
24    },
25    "indexing": {
26        "default_segmenter": "gemini",
27        "use_gpu": true,
28        "motion_threshold": 0.08,
29        "min_segment_duration": 3.0,
30        "dead_time_merge_gap": 2.0,
31        "chunk_duration": 90,
32        "chunk_overlap": 10,
33        "detector_model": "fasterrcnn_resnet50_fpn",
34        "detector_score_threshold": 0.5,
35        "yolo_velocity_threshold": 0.15,
36        "yolo_frame_skip": 3,
37        "yolo_tracker": "bytetrack.yaml",
38        "yolo_prefetch_workers": 2,
39        "min_action_window_duration": 2.0,
40        "log_yolo_candidates": true,
41        "store_yolo_candidates": true,
42        "ai_handoff_padding": 2.0,
43        "ai_max_workers": 1,
44        "tracknet": {
45            "wsl_distro": "Ubuntu",
46            "repo_dir": "~/models/TrackNetV4",
47            "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
48            "conda_env": "TrackNetV4",
49            "weights_path": "",
50            "queue_length": 5,
51            "stage_in_wsl": true,
52            "wsl_stage_dir": "~/clips",
53            "timeout_sec": 1800,
54            "velocity_threshold": 0.02
55        },
56        "wasb": {
57            "wsl_distro": "Ubuntu",
58            "repo_dir": "~/models/WASB-SBDT",
59            "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
60            "conda_env": "wasb",
61            "weights_path": "~/models/WASB-
62            SBDT/pretrained_weights/wasb_badminton_best.pth.tar",
63            "sport": "badminton",
64            "wsl_stage_dir": "~/clips",
65            "timeout_sec": 1800,
66            "velocity_threshold": 0.02
67        }
68    },
69    "output": {
70        "default_highlights_name": "highlights.mp4",
71        "db_name": "sports_indexer.db"

```

```

71     },
72     "stitching": {
73         "begin_padding": 0.5,
74         "end_padding": 0.5,
75         "use_cache": false,
76         "min_shot_count": 1,
77         "watermark": {
78             "enabled": true,
79             "text": "Generated by Khelsutra.guru",
80             "position": "bottom-right",
81             "opacity": 0.85,
82             "logo_path": "",
83             "font_path": ""
84         }
85     }
86 }

```

65. user (2026-06-12T16:44:55.455Z)

```

backend\config\models.py-141-class StitchingConfig(BaseModel):
backend\config\models.py-142-    begin_padding: float = 0.5
backend\config\models.py-143-    end_padding: float = 0.5
backend\config\models.py-144-    use_cache: bool = False
backend\config\models.py:145-    min_shot_count: int = 1
backend\config\models.py-146-    watermark: WatermarkConfig =
Field(default_factory=WatermarkConfig)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py-147-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py-148-
backend\config\models.py-149-class SecurityConfig(BaseModel):
--
backend\sports\badminton.py-31-Pre-serve Rituals: Do not start the timestamp during the players'
setup or stance. Wait for physical racket-to-shuttle contact.
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-32-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-33-Shot Tracking
& Data Confidence
backend\sports\badminton.py-34-Shot Definition: Count every successful traverse of the
shuttlecock across the net (serves, clears, smashes, drops, drives). Do not count the final
grounding or pre-net faults.
backend\sports\badminton.py:35:Confidence Scoring: Because the shuttlecock moves at extreme
velocities that may fall between sampled frames, provide a shot_count_confidence metric (Low,
Medium, High). Base your estimate on player biomechanics, positioning, and racket swings if the
shuttle itself is blurred.
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-36-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-37-Strict Output
Schema
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-38-Output
strictly in the following JSON array format. Do not include markdown formatting, introductory
text, conversational filler, or explanations. Any deviation from this schema will break the
downstream pipeline.
backend\sports\badminton.py-39-Remember: start_time and end_time are TOTAL SECONDS (decimal),
not MM:SS.
--
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-42-{
backend\sports\badminton.py-43-"start_time": 74.0,
backend\sports\badminton.py-44-"end_time": 81.3,
backend\sports\badminton.py-45-"shuttle_exchange_count": 11,
backend\sports\badminton.py:46:"shot_count_confidence": "High"
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-47-},
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-48-{
backend\sports\badminton.py-49-"start_time": 150.5,
backend\sports\badminton.py-50-"end_time": 174.2,
backend\sports\badminton.py-51-"shuttle_exchange_count": 24,
backend\sports\badminton.py:52:"shot_count_confidence": "Medium"
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-53-}
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-54-]

```

```

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\sports\badminton.py-55-"""
backend\sports\badminton.py-56-         if chunk_duration > 0.0:
--
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-72-     }
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-73-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-74-
backend\tools\export.py-75-def segments_to_csv(segments: Iterable[Dict[str, Any]],
include_header: bool = True) -> str:
backend\tools\export.py:76:     """Return a CSV string
(start,end,duration,shot_count,detector,...) .
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-77-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-78-     Compatible
with many editors and the existing golden/slicer schemas
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-79-     (extra
columns are harmless).
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-80-     """
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-81-     segs =
_to_list(segments)
backend\tools\export.py-82-     lines: List[str] = []
backend\tools\export.py-83-     if include_header:
backend\tools\export.py:84:
lines.append("start_time,end_time,duration,shot_count,shot_count_confidence,detector,id")
backend\tools\export.py-85-     for s in segs:
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-86-         md =
s.get("metadata") or {}
backend\tools\export.py:87:         shot = md.get("shot_count", "")
backend\tools\export.py:88:         conf = md.get("shot_count_confidence", "")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-89-         det =
md.get("detector", "")
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-90-
lines.append(
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-91- f"{s.get('start_time',0)},{s.get('end_time',0)},{s.get('duration',0)},"
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\tools\export.py-92- f"{shot},{conf},{det},{s.get('id','')}"
--
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-424-
end = c["end"]
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-425-
segment = c["raw"]
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-426-
dur = end - start
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-427-
backend\pipeline\segmenters\gemini.py:428:         # Extract shot_count from Gemini or
compute estimate if not present
backend\pipeline\segmenters\gemini.py:429:         shot_count =
segment.get("shuttle_exchange_count")
backend\pipeline\segmenters\gemini.py:430:         if shot_count is None:
backend\pipeline\segmenters\gemini.py:431:             shot_count = max(3, int(dur / 0.8)) #
1 shot every 0.8 seconds as fallback estimate
backend\pipeline\segmenters\gemini.py-432-         else:
backend\pipeline\segmenters\gemini.py-433-             try:
shot_count = int(shot_count)
backend\pipeline\segmenters\gemini.py-434-             except Exception:
shot_count = max(3, int(dur /
0.8))
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-437-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-438-
# Only store metrics we actually know ? no fabricated data
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-439-
metadata = {
backend\pipeline\segmenters\gemini.py:440:         "shot_count": shot_count,
backend\pipeline\segmenters\gemini.py:441:         "shot_count_confidence":
segment.get("shot_count_confidence", "Unknown"),
backend\pipeline\segmenters\gemini.py-442-         "detector": model_name

```

```

C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-443-
}
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-444-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\gemini.py-445-
segment_id = self.db.add_play_segment(video_id, start, end, None, None, metadata)
--
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py-164-
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py-165-
# Formulate dynamic rich tags
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py-166-
dur = end - start
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py-167-
metadata = {
backend\pipeline\segmenters\motion.py:168:         "shot_count": random.randint(4, 25),
backend\pipeline\segmenters\motion.py-169-         "intensity": "high" if dur > 12 else
"medium",
backend\pipeline\segmenters\motion.py-170-         "avg_speed_kmh":
round(random.uniform(40, 85), 1)
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py-171-
}
C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\segmenters\motion.py-172-
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-63-         cap.release()
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-64-         return (fps if fps > 0 else
30.0, width if width > 0 else 1280.0)
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-65-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-66-     @staticmethod
backend\pipeline\segmenters\trajectory_hybrid.py:67:     def _shot_count_from(segment: Dict[str,
Any], duration: float) -> int:
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-68-         """Provider-reported
exchange count, else a duration-based estimate.
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-69-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-70-         One canonical implementation
? the twins had drifted (wasb relied on
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-71-         ``int(None)`` raising into
the fallback; same result, by accident).
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-72-         """
backend\pipeline\segmenters\trajectory_hybrid.py:73:         shot_count =
segment.get("shuttle_exchange_count")
backend\pipeline\segmenters\trajectory_hybrid.py:74:         if shot_count is None:
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-75-             return max(3,
int(duration / 0.8))
backend\pipeline\segmenters\trajectory_hybrid.py-76-         try:
backend\pipeline\segmenters\trajectory_hybrid.py:77:             return int(shot_count)
backend\pipeline\segmenters\trajectory_hybrid.py-78-         except (ValueError, TypeError):
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-79-             return max(3,
int(duration / 0.8))
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-80-
backend\pipeline\segmenters\trajectory_hybrid.py-81-     def process_video(self, video_id: str,
video_path: str, # type: ignore[override]
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-242-         start =

```



```

float(segment["start_time"])
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-243-
float(segment["end_time"])
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-244-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-245-
backend\pipeline\segmenters\trajectory_hybrid.py:246:
self._shot_count_from(segment, dur),
backend\pipeline\segmenters\trajectory_hybrid.py:247:
"shot_count_confidence": segment.get("shot_count_confidence", "Unknown"),
backend\pipeline\segmenters\trajectory_hybrid.py-248-
f"{self.family}-hybrid-{model_name}",
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-249-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\trajectory_hybrid.py-250-
self.db.add_play_segment(video_id, start, end, None, None, metadata)
backend\pipeline\segmenters\trajectory_hybrid.py-251-
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-534-
float(segment["start_time"])
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-535-
float(segment["end_time"])
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-536-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-537-
backend\pipeline\segmenters\yolo_hybrid.py:538:
segment.get("shuttle_exchange_count")
backend\pipeline\segmenters\yolo_hybrid.py:539:
backend\pipeline\segmenters\yolo_hybrid.py:540:
0.8))
backend\pipeline\segmenters\yolo_hybrid.py-541-
backend\pipeline\segmenters\yolo_hybrid.py-542-
backend\pipeline\segmenters\yolo_hybrid.py:543:
int(shot_count)
backend\pipeline\segmenters\yolo_hybrid.py-544-
TypeError):
backend\pipeline\segmenters\yolo_hybrid.py:545:
int(dur / 0.8))
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-546-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-547-
backend\pipeline\segmenters\yolo_hybrid.py:548:
backend\pipeline\segmenters\yolo_hybrid.py:549:
segment.get("shot_count_confidence", "Unknown"),
backend\pipeline\segmenters\yolo_hybrid.py-550-
hybrid-{model_name}"
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-551-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-552-
C:\Users\avidu\Projects\badminton-highlight-
indexer\backend\pipeline\segmenters\yolo_hybrid.py-553-
self.db.add_play_segment(video_id, start, end, None, None, metadata)

```

```

end =
dur = end - start
metadata = {
    "shot_count":
}
segment_id =
if segment_id:
start =
end =
dur = end - start
shot_count =
if shot_count is None:
    shot_count = max(3, int(dur /
else:
    try:
        shot_count =
    except (ValueError,
        shot_count = max(3,
metadata = {
    "shot_count": shot_count,
    "shot_count_confidence":
    "detector": f"yolo-
}
segment_id =

```

66. user (2026-06-12T16:44:56.318Z)

```

backend\pipeline\segmenters\gemini.py:459:segmenter_registry.register(GeminiPlaySegmenter)
backend\pipeline\segmenters\motion.py:227:segmenter_registry.register(MotionPlaySegmenter)

```

```
backend\pipeline\segmenters\tracknet_hybrid.py:42:segmenter_registry.register(TrackNetHybridSegmenter)
backend\pipeline\segmenters\yolo_hybrid.py:574:segmenter_registry.register(HybridYoloSegmenter)
backend\pipeline\segmenters\wasb_hybrid.py:41:segmenter_registry.register(WasbHybridSegmenter)
```

67. user (2026-06-12T16:45:28.764Z)

```
1      """WASB-based hybrid play segmenter.
2
3      Phase 2 uses **WASB** (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via
4      WSL2 (backend/pipeline/detectors/wasb_runner.py). WASB ships pretrained
5      badminton weights and runs today, so this is the live shuttle-motion substrate.
6
7      For the per-video *calibration* goal we mainly use Phase 2 (trajectory ->
8      candidate windows); the windowing thresholds (velocity_thresh / merge_gap /
9      min_window_duration) are what get tuned against golden rally labels.
10
11     The pipeline engine (healthcheck -> trajectory -> windows -> AI handoff -> DB)
12     lives in trajectory_hybrid.TrajectoryHybridSegmenter; this class only wires the
13     WASB runner and identity.
14     """
15     from typing import Any, Dict, Tuple
16
17     from backend.pipeline.segmenters.base import segmenter_registry
18     from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter
19     from backend.pipeline.detectors.base import DetectorRunner
20     from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
21
22
23     class WasbHybridSegmenter(TrajectoryHybridSegmenter):
24         family = "wasb"
25         display_label = "WASB"
26
27         @property
28         def name(self) -> str:
29             return "wasb_hybrid"
30
31         @property
32         def description(self) -> str:
33             return ("Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained
badminton "
34                     "weights) for precise candidate rallies + AI provider for semantic
boundaries.")
35
36         def _build_runner(self, idx_cfg: Dict[str, Any]) -> Tuple[DetectorRunner, Dict[str,
Any]]:
37             cfg = WasbConfig.from_indexing_cfg(idx_cfg)
38             return WasbRunner(cfg), {"weights_path": cfg.weights_path}
39
40
41     segmenter_registry.register(WasbHybridSegmenter)
42
```

68. user (2026-06-12T16:45:34.041Z)

```
cv2 4.13.0 numpy 2.4.6
width=3840
height=2160
r_frame_rate=60000/1001
duration=768.768000
size=11522892112
bit_rate=119910215
True
True
```

69. user (2026-06-12T16:45:48.596Z)

```
NVIDIA GeForce RTX 2070 Super with Max-Q Design, 8192 MiB, 1548 MiB
/home/avidullu/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar
wasb /home/avidullu/miniconda3/envs/wasb
/dev/sdc 1007G 126G 831G 14% /
```

70. user (2026-06-12T16:46:00.058Z)

[Request interrupted by user]

71. user (2026-06-12T16:46:00.066Z)

make sure to sync to head and run!

72. assistant (2026-06-12T16:46:51.771Z)

All preflight checks pass ? RTX 2070 Super visible in WSL, WASB weights + conda env present, 831G free. Syncing to head once more and launching.

73. user (2026-06-12T16:46:55.582Z)

Already up to date.
85abe26 docs(platform): video-locality mental model - the 10GB question (owner-requested)

74. user (2026-06-12T16:47:02.360Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\config.json has been updated successfully. (file state is current in your context ? no need to Read it back)

75. user (2026-06-12T16:47:03.534Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\config.json has been updated successfully. (file state is current in your context ? no need to Read it back)

76. assistant (2026-06-12T16:47:25.758Z)

"""One-off non-interactive E2E runner for GX010125.MP4 (modeled on run_process_and_stitch.py).

Every knob ? segmenter, min_shot_count, paddings, watermark ? comes from config.json.
Nothing is overridden here, so the run demonstrates the config is actually picked up.
"""

```
import json
import logging
import os
import time

from dotenv import load_dotenv

load_dotenv()

logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s [%(name)s] %(levelname)s: %(message)s",
    datefmt="%H:%M:%S",
)

from backend.infrastructure.database import Database
from backend.config import load_config
from backend.utils.hashing import compute_video_id
from backend.pipeline.segmenters import segmenter_registry
from backend.pipeline.stitcher.compiler import VideoCompiler

VIDEO_PATH = r"F:\GoPro Hero Black 12\KhelSutra\Badminton 12 June 2026\GX010125.MP4"
```

```
OUTPUT_FILENAME = "GX010125_highlights.mp4"
```

```
def main():
    cfg = load_config()
    print(f"[CONFIG] segmenter={cfg.indexing.default_segmenter} | "
          f"min_shot_count={cfg.stitching.min_shot_count} | "
          f"begin_padding={cfg.stitching.begin_padding}s
end_padding={cfg.stitching.end_padding}s | "
          f"watermark={'on' if cfg.stitching.watermark.enabled else 'off'}", flush=True)

    if not os.path.exists(VIDEO_PATH):
        raise SystemExit(f"video not found: {VIDEO_PATH}")

    os.makedirs(cfg.output.output_dir, exist_ok=True)
    db = Database(os.path.join(cfg.output.output_dir, cfg.output.db_name))

    t0 = time.time()
    print("[HASH] computing MD5 of the source video (11.5 GB ? takes a while)...", flush=True)
    video_id = compute_video_id(VIDEO_PATH)
    print(f"[HASH] video_id={video_id} ({time.time() - t0:.0f}s)", flush=True)

    segments = []
    cached = db.get_video(video_id)
    if cached and cached.get("status") == "processed":
        segments = db.query_play_segments(video_id, {})
        if segments:
            print(f"[CACHE] reusing {len(segments)} cached segments for {video_id}", flush=True)

    if not segments:
        print("[INDEX] clean indexing (validators bypassed for the massive file, "
              "as in run_process_and_stitch.py)...", flush=True)
        db.clear_video_data(video_id)
        db.add_video(video_id, VIDEO_PATH, "processed", [])
        seg_name = cfg.indexing.default_segmenter
        segmenter_cls = segmenter_registry.get(seg_name)
        if not segmenter_cls:
            raise SystemExit(f"segmenter '{seg_name}' not registered")
        segmenter = segmenter_cls(db, cfg)
        segments = segmenter.process_video(video_id, VIDEO_PATH)
        print(f"[INDEX] {len(segments)} play segments indexed "
              f"(total {time.time() - t0:.0f}s).", flush=True)

    if not segments:
        raise SystemExit("no play segments detected ? nothing to stitch")

    min_shots = cfg.stitching.min_shot_count
    time_pairs, skipped = [], 0
    for s in segments:
        meta = s.get("metadata") or {}
        if isinstance(meta, str):
            try:
                meta = json.loads(meta)
            except Exception:
                meta = {}
        sc = meta.get("shot_count", 0)
        line = (f" segment id={s.get('id')} "
                f"[{float(s['start_time']):8.2f}s - {float(s['end_time']):8.2f}s] "
                f"shot_count={sc} confidence={meta.get('shot_count_confidence', '?')}")
        if sc >= min_shots:
            time_pairs.append((s["start_time"], s["end_time"]))
            print(line + " -> KEEP", flush=True)
        else:
            skipped += 1
            print(line + " -> drop", flush=True)
```

```

print(f"[FILTER] config min_shot_count={min_shots}: "
      f"kept {len(time_pairs)}, dropped {skipped}", flush=True)
if not time_pairs:
    raise SystemExit(f"no segments passed the min_shot_count={min_shots} filter")

compiler = VideoCompiler(cfg.output.output_dir,
                          begin_padding=cfg.stitching.begin_padding,
                          end_padding=cfg.stitching.end_padding,
                          watermark=cfg.stitching.watermark)
out = compiler.compile_highlights(VIDEO_PATH, time_pairs, OUTPUT_FILENAME)
print(f"[DONE] highlight reel: {os.path.abspath(out)} (total {time.time() - t0:.0f}s)",
      flush=True)

if __name__ == "__main__":
    main()

```

77. user (2026-06-12T16:47:25.776Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\run_gx010125.py (file state is current in your context ? no need to Read it back)

78. user (2026-06-12T16:47:33.889Z)

Command running in background with ID: bmr9aws10. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\bmr9aws10.output. You will be notified when it completes. To check interim output, use Read on that file path.

79. user (2026-06-12T16:47:39.078Z)

<tool_use_error>Blocked: Start-Sleep 45 followed by: Get-Content C:\Users\avidu\Projects\badminton-highlight-indexer\output\gx010125_run.log -Tail 15. To wait for a condition, use Monitor with an until-loop (e.g. `until <check>; do sleep 2; done` ? Monitor runs bash). To wait for a command you started, use run_in_background: true. Do not chain shorter sleeps to work around this block.</tool_use_error>

80. assistant (2026-06-12T16:47:43.817Z)

The run failed at startup ? checking the log:

81. user (2026-06-12T16:47:44.195Z)

```

1      Traceback (most recent call last):
2          File "C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\run_gx010125.py",
line 21, in <module>
3          from backend.infrastructure.database import Database
4      ModuleNotFoundError: No module named 'backend'
5

```

82. assistant (2026-06-12T16:47:50.873Z)

Path issue ? the script lives in `scratch/` so the repo root isn't on `sys.path`. Fixing and relaunching:

83. user (2026-06-12T16:47:52.935Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\run_gx010125.py has been updated successfully. (file state is current in your context ? no need to Read it back)

84. user (2026-06-12T16:47:59.327Z)

Command running in background with ID: bwqevfnlg. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\bwqevfnlg.output. You will be notified when it completes. To check interim output, use Read on that file path.

85. user (2026-06-12T16:48:27.193Z)

```
93:         logger.error(str(e))
103:         logger.error(
112:         logger.error(str(e))
117:         logger.error("Invalid video duration.")
143:         logger.error("%s WSL environment not ready: %s", label, msg)
145:         logger.info("%s WSL env OK: %s", label, msg)
154:         logger.error("%s produced no trajectory; aborting.", label)
158:         logger.info("Parsed %d trajectory points (%d visible).",
166:         logger.info("Phase 2 (%s) completed in %.1fs. Candidate windows: %d",
171:         logger.info(f"[{label} rally] {_fmt_ts(cw['start'])}-{_fmt_ts(cw['end'])} ")
198:         logger.info("Phase 3: %s validation on %d candidate windows...",
219:         logger.warning("Dropping segment with unparseable timestamps: %s ? %s",
seg, e)
236:         logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
len(ai_segments))
260:         logger.warning("Skipping invalid segment struct: %s", e)
```

86. assistant (2026-06-12T16:48:40.532Z)

Arming a monitor on the run log for phase milestones and failures:

87. user (2026-06-12T16:48:44.564Z)

Monitor started (task b0emrus25, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

88. assistant (2026-06-12T16:49:01.064Z)

The end-to-end run is **launched and being monitored**. Here's the setup and what to expect:

What's running (background job `bwqevfnlg`, log at output/gx010125_run.log):

- MD5 hash** of the 11.5 GB source (this is the `video\_id`) ? a few minutes.
- WASB shuttle-trajectory detection on the GPU** (RTX 2070 Super via WSL2 Ubuntu, pretrained badminton weights ? verified present) over the full 12.8-minute 4K60 video ? candidate rally windows.
- Gemini confirmation** of each candidate window (serial uploads with backoff, re-encoded 1280p chunks) ? confirmed rallies **with per-rally `shot\_count`** (Gemini's shuttle-exchange count; duration-based estimate only if the model omits it).
- Filter + stitch**: only rallies with `shot\_count >= 8` (your ">7 shots") survive, then the compiler cuts and concatenates them with the configured paddings and watermark into `output/GX010125\_highlights.mp4`.

Your config question ? yes, it's possible, with one nuance. **`stitching.min\_shot\_count`** is honored by the offline runner flow (`run\_process\_and\_stitch.py` lineage), which is what I'm using ? I set it to **8** in config.json and the runner reads it from config with no override, logging each segment's shot count and KEEP/drop decision so you can audit the filter. The nuance: the **API path** (`/api/compile`) does **not** apply **`min\_shot\_count`** ? it stitches exactly the **`segment\_ids`** the caller picks (the UI is expected to filter). If you want the config filter enforced on the API path too, that's a small PR ? say the word.

Config changes made (local, uncommitted): **`indexing.default\_segmenter`**: gemini ? wasb_hybrid (the GPU path you asked for) and **`stitching.min\_shot\_count`**: 1 ? 8. Also synced to head first ? your merge of the padding fix (#102) means this run uses the now config-driven 0.5s paddings.

Expect roughly 30?60+ minutes total (GPU inference on ~46k frames is the long pole). The monitor

will report each phase as it lands; I'll summarize the kept rallies and the reel path when it finishes.

89. user (2026-06-12T16:49:02.686Z)

```
<task-notification>
<task-id>b0emrus25</task-id>
<summary>Monitor event: "GX010125 E2E run milestones (phases, filter, errors)"</summary>
<event>[CONFIG] segmenter=wasb_hybrid | min_shot_count=8 | begin_padding=0.5s end_padding=0.5s |
watermark=on
[HASH] computing MD5 of the source video (11.5 GB ? takes a while)...
[HASH] video_id=c794a7af282ab54fc8ebab23830fcdb7 (25s)
[INDEX] clean indexing (validators bypassed for the massive file, as in
run_process_and_stitch.py)...
22:18:31 [backend.pipeline.segmenters.trajectory_hybrid] INFO: WASB WSL env OK: torch
1.11.0+cu113 cuda True</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

90. assistant (2026-06-12T16:49:13.497Z)

Early milestones are clean ? and the config pickup is proven right at startup:

```
- `[CONFIG] segmenter=wasb_hybrid | min_shot_count=8 | begin_padding=0.5s end_padding=0.5s |
watermark=on` ? all four knobs read from `config.json`.
- MD5 hashing the 11.5 GB took only 25s (`video_id=c794a7af?`).
- GPU healthcheck passed: WSL env OK, `torch 1.11.0+cu113, cuda True`.
```

WASB trajectory inference is now running on the GPU ? the long pole. Next milestone will be the trajectory-points/candidate-windows line.

91. user (2026-06-12T16:51:42.076Z)

I dont see GPU being used. can you check

92. assistant (2026-06-12T16:51:49.076Z)

Likely the run is still in the staging step (copying the 11.5 GB file into WSL before inference), but let me verify directly:

93. user (2026-06-12T16:51:54.056Z)

```
39 %, 1512 MiB
-rw-r--r-- 1 avidullu avidullu   95K Jun  9 12:43 tabletennis_traj.csv
drwxr-xr-x 2 avidullu avidullu  4.0K Jun  9 12:43 tabletennis_traj_wasbcache
-rw-r--r-- 1 avidullu avidullu   95K Jun  6 16:50 testlarge_traj.csv
drwxr-xr-x 2 avidullu avidullu  4.0K Jun  6 16:50 testlarge_traj_wasbcache
-rw-r--r-- 1 avidullu avidullu   4.7K Jun  6 10:52 wasb_traj_test.csv
[CONFIG] segmenter=wasb_hybrid | min_shot_count=8 | begin_padding=0.5s end_padding=0.5s |
watermark=on
[HASH] computing MD5 of the source video (11.5 GB ? takes a while)...
[HASH] video_id=c794a7af282ab54fc8ebab23830fcdb7 (25s)
[INDEX] clean indexing (validators bypassed for the massive file, as in
run_process_and_stitch.py)...
22:18:31 [backend.pipeline.segmenters.trajectory_hybrid] INFO: WASB WSL env OK: torch
1.11.0+cu113 cuda True
22:19:36 [backend.pipeline.detectors.wasb_runner] INFO: Running WASB inference on GX010125.MP4
...
```

94. user (2026-06-12T16:52:26.091Z)

```
avidullu 120169 137 5.4 2608260 893880 pts/2 Rsl+ 16:49 3:30 python wasb_infer.py --video
/home/avidullu/clips/GX010125__c794a7af282a.MP4 --frames_out_dir
```

```
/home/avidullu/clips/GX010125__c794a7af282a_frames --weights /home/avidullu/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar --sport badminton --out /mnt/c/Users/avidu/Projects/badminton-highlight-indexer/output/wasb/GX010125__c794a7af282a_wasb.csv
-rwxr-xr-x 1 avidullu avidullu 11G Jun 12 16:49 GX010125__c794a7af282a.MP4
drwxr-xr-x 2 avidullu avidullu 20K Jun 12 16:52 GX010125__c794a7af282a_frames
5 %, 1505 MiB
21 %, 1506 MiB
17 %, 1504 MiB
```

95. user (2026-06-12T16:52:47.166Z)

Exit code 1
715

96. assistant (2026-06-12T16:53:02.948Z)

715 frames extracted so far, trajectory CSV not started yet. Let me measure the extraction rate to give you a real ETA:

97. user (2026-06-12T16:53:09.089Z)

Exit code 1
wc: The term 'wc' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
wc: The term 'wc' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
b-a: The term 'b-a' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
rate: The term 'rate' is not recognized as a name of a cmdlet, function, script file, or executable program.
Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

98. assistant (2026-06-12T16:53:14.247Z)

PowerShell ate the `` ? quoting fix:

99. user (2026-06-12T16:53:39.027Z)

frames: -> (rate 0/min)
16:53:38

100. user (2026-06-12T16:54:06.786Z)

1015
mahadevpura_smoke_180s__05e0e577029d_wasb.csv
mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache
16:54:06

101. user (2026-06-12T16:55:00.287Z)

Your questions have been answered: "The WASB trajectory pass on the 4K60 original will take ~2-3 hours (decode-bound). How do you want to proceed?"="Downscale first (Recommended)". You can now continue with these answers in mind.

102. user (2026-06-12T16:55:39.834Z)


```
{"message":"Successfully stopped task: bwqevfnlg (cd C:\\Users\\avidu\\Projects\\badminton-highlight-indexer && python -u scratch\\run_gx010125.py 2>&1 | Out-File -Encoding utf8 output\\gx010125_run.log)","task_id":"bwqevfnlg","task_type":"local_bash","command":"cd C:\\Users\\avidu\\Projects\\badminton-highlight-indexer && python -u scratch\\run_gx010125.py 2>&1 | Out-File -Encoding utf8 output\\gx010125_run.log"}
```

103. user (2026-06-12T16:55:39.873Z)

```
{"message":"Successfully stopped task: b0emrus25 (tail -n +1 -f \"/c/Users/avidu/Projects/badminton-highlight-indexer/output/gx010125_run.log\" | grep -E --line-buffered '\\\\[CONFIG\\\\]|\\\\[HASH\\\\]|\\\\[INDEX\\\\]|\\\\[CACHE\\\\]|\\\\[FILTER\\\\]|\\\\[DONE\\\\]|Traceback|ERROR|Exception|WSL env|trajectory points|Phase 2|Phase 3|confirmed|oversize|SUMMARY|COMPILE R\")","task_id":"b0emrus25","task_type":"local_bash","command":"tail -n +1 -f \"/c/Users/avidu/Projects/badminton-highlight-indexer/output/gx010125_run.log\" | grep -E --line-buffered '\\\\[CONFIG\\\\]|\\\\[HASH\\\\]|\\\\[INDEX\\\\]|\\\\[CACHE\\\\]|\\\\[FILTER\\\\]|\\\\[DONE\\\\]|Traceback|ERROR|Exception|WSL env|trajectory points|Phase 2|Phase 3|confirmed|oversize|SUMMARY|COMPILER\""}}
```

104. user (2026-06-12T16:55:48.741Z)

V....D av1_nvenc	NVIDIA NVENC av1 encoder (codec av1)
V....D h264_nvenc	NVIDIA NVENC H.264 encoder (codec h264)
V....D hevc_nvenc	NVIDIA NVENC hevc encoder (codec hevc)

105. user (2026-06-12T16:56:00.145Z)

no wasb_infer running
staged files cleaned

106. assistant (2026-06-12T16:56:06.307Z)

Clean. Now checking the source codec to build the right GPU transcode, then making the 1080p proxy:

107. user (2026-06-12T16:56:09.834Z)

```
codec_name=hevc  
profile=Main 10  
pix_fmt=yuv420p10le
```

108. user (2026-06-12T16:56:36.949Z)

Command running in background with ID: b5edsetb3. Output is being written to:
C:\\Users\\avidu\\AppData\\Local\\Temp\\claude\\C--Users-avidu-Projects-badminton-highlight-indexer\\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\\tasks\\b5edsetb3.output. You will be notified when it completes. To check interim output, use Read on that file path.

109. assistant (2026-06-12T16:56:43.332Z)

While that runs, updating the runner for the proxy-index / 4K-stitch split:

110. user (2026-06-12T16:56:47.863Z)

The file C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\scratch\\run_gx010125.py has been updated successfully. (file state is current in your context ? no need to Read it back)

111. user (2026-06-12T16:56:56.758Z)

The file C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\scratch\\run_gx010125.py has been updated successfully. (file state is current in your context ? no need to Read it back)

112. user (2026-06-12T16:57:00.524Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\run_gx010125.py has been updated successfully. (file state is current in your context ? no need to Read it back)

113. user (2026-06-12T16:57:02.350Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\run_gx010125.py has been updated successfully. (file state is current in your context ? no need to Read it back)

114. user (2026-06-12T16:57:09.929Z)

frame= 7968 fps=253 q=22.0 size= 396288KiB time=00:02:12.88 bitrate=24430.5kbits/s speed=4.23x elapsed=0:00:31.44
frame= 8130 fps=254 q=22.0 size= 404224KiB time=00:02:15.58 bitrate=24423.0kbits/s speed=4.24x elapsed=0:00:31.97

115. assistant (2026-06-12T16:57:18.280Z)

The GPU transcode is ripping along at ****254 fps (4.2x realtime)**** ? the proxy will be ready in ~3 minutes. The runner is already updated for the split flow: index the 1080p proxy, stitch the final reel from the original 4K (timestamps transfer 1:1 ? same timeline). I'll relaunch the pipeline the moment the transcode lands.

116. user (2026-06-12T16:57:18.336Z)

and show me a log file which when refreshed I can see the tail
You become unresponsive and I want to see the logs
Can you keep merging the PRs which you identify as good engineering fixes. like the API option. definitely
But make sure to stitch the video from the original and not the compressed. And also check if this is the behaviour and fix if not

117. assistant (2026-06-12T16:57:41.116Z)

Got it on all three: (1) I'll give you a live-tailable log path, (2) merge authorization for good engineering fixes noted ? and I'll ship the API ``min_shot_count`` enforcement now, (3) the stitch ****is**** from the 4K original ? I'll show you the exact line and verify the output resolution when the reel lands.

First, checking the transcode and relaunching the pipeline:

118. user (2026-06-12T16:57:45.309Z)

frame=18892 fps=282 q=22.0 size= 930048KiB time=00:05:15.13 bitrate=24177.1kbits/s speed= 4.7x elapsed=0:01:07.03
frame=19054 fps=282 q=22.0 size= 938496KiB time=00:05:17.83 bitrate=24189.2kbits/s speed= 4.7x elapsed=0:01:07.56

```
Length LastWriteTime
-----
962068532 6/12/2026 10:27:45 PM
```

119. assistant (2026-06-12T16:58:33.902Z)

While the transcode finishes (~1.5 min left), let me check how the DB returns segment metadata so the API filter handles it correctly:

120. user (2026-06-12T16:58:35.345Z)

```
57-         )
58-         """)
59-
60-         # Play Segments Table (Extensible metadata JSON)
61-         cursor.execute("""
```

```

62-         CREATE TABLE IF NOT EXISTS play_segments (
63-             id INTEGER PRIMARY KEY AUTOINCREMENT,
64-             --
65-             duration REAL NOT NULL,
66-             server TEXT,
67-             receiver TEXT,
68-             metadata TEXT,
69-             FOREIGN KEY (video_id) REFERENCES videos(id) ON DELETE CASCADE
70-         )
71-     """
72-
73-
74-
75-
76-
77-
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79-
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```

121. user (2026-06-12T16:59:16.883Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\main.py has been updated successfully. (file state is current in your context ? no need to Read it back)

122. assistant (2026-06-12T16:59:23.642Z)

Transcode done ? relaunching the pipeline on the proxy immediately:

123. user (2026-06-12T16:59:25.352Z)

Command running in background with ID: b4ez47ehk. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\b4ez47ehk.output. You will be notified when it completes. To check interim output, use Read on that file path.

124. user (2026-06-12T16:59:32.071Z)

Monitor started (task bwiwctgbr, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

125. user (2026-06-12T16:59:48.075Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_api_endpoints.py has been updated successfully. (file state is current in your context ? no need to Read it back)

126. user (2026-06-12T17:00:27.060Z)

```
..... [ 88%]
..... [100%]
407 passed in 27.45s
=====
RESULT: PASS - all tests green
=====
```

127. assistant (2026-06-12T17:00:35.466Z)

Green ? 407 passed. Lint check, then branch/PR:

128. user (2026-06-12T17:00:39.873Z)

```
All checks passed!
M      backend/main.py
M      config.json
M      tests/test_api_endpoints.py
Switched to a new branch 'feat/api-min-shot-count'
M      backend/main.py
M      config.json
M      tests/test_api_endpoints.py
?? arial.ttf
?? artifacts/
?? docs/STORAGE_SHARING_MODEL.md
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
?? scratch/run_gx010125.py
```

129. assistant (2026-06-12T17:00:46.726Z)

Only the two intended files staged (`config.json` run-config stays local). Committing and opening the PR:

130. user (2026-06-12T17:01:02.943Z)

```
[feat/api-min-shot-count bfa3cde] feat(api): enforce stitching.min_shot_count on /api/compile
2 files changed, 66 insertions(+), 2 deletions(-)
remote:
remote: Create a pull request for 'feat/api-min-shot-count' on GitHub by visiting:
remote:   https://github.com/avidullu/badminton-highlight-indexer/pull/new/feat/api-min-shot-
count
remote:
branch 'feat/api-min-shot-count' set up to track 'origin/feat/api-min-shot-count'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch]      feat/api-min-shot-count -> feat/api-min-shot-count
Warning: 11 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/107
```

131. user (2026-06-12T17:02:54.518Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    30s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27430516406/job/81079339547
Full test suite (offline, mocked)                pass   1m32s  https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27430516406/job/81079339627
Mypy (lenient baseline) pass    29s  https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27430516406/job/81079339579
Ruff lint          pass    9s  https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27430516406/job/81079339763
Syntax sweep + import smoke (no GPU stack)      pass    30s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27430516406/job/81079339547
```

132. user (2026-06-12T17:03:11.857Z)

```
["SUCCESS"]
From https://github.com/avidullu/badminton-highlight-indexer
 * branch          master      -> FETCH_HEAD
 2a8d1ea..3461898  master      -> origin/master
Updating ffdfe0..3461898
Fast-forward
 backend/main.py          | 26 +++-
 docs/COMMERCIALIZATION.md |  3 +-
 docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md | 89 -----
 docs/FIELD_VISIT_ANSWER_SHEET.md | 104 ++++++++
 docs/FIELD_VISIT_PLAYBOOK.md | 116 ++++++++
 docs/I18N_PLAN.md | 229 ++++++++
 docs/NEXT_STEPS.md |  9 +-
 docs/PLATFORM_ARCHITECTURE.md |  5 +
 docs/PMF_FIELD_SIGNALS.md | 169 ++++++++
 docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md | 84 ++++++++
 docs/README.md |  1 +
 tests/test_api_endpoints.py | 42 +++++
12 files changed, 783 insertions(+), 94 deletions(-)
delete mode 100644 docs/FIELD_RESEARCH_ASSOCIATE_FLYER.md
create mode 100644 docs/FIELD_VISIT_ANSWER_SHEET.md
create mode 100644 docs/FIELD_VISIT_PLAYBOOK.md
create mode 100644 docs/I18N_PLAN.md
create mode 100644 docs/PMF_FIELD_SIGNALS.md
create mode 100644 docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md
M      config.json
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
3461898 feat(api): enforce stitching.min_shot_count on /api/compile (#107)
```

133. assistant (2026-06-12T17:03:36.589Z)

```
---
name: working-agreement-merging
description: "Owner authorization (2026-06-12): Claude may MERGE its own PRs that are good
engineering fixes, on verified-green CI; product/strategy/licensing changes stay owner-gated."
metadata:
  type: feedback
---
```

****Owner (2026-06-12, GX010125 E2E session):**** "Can you keep merging the PRs which you identify as good engineering fixes. like the API option. definitely."

****Why:**** owner runs multiple parallel sessions and doesn't want to be the merge bottleneck for clear-cut engineering fixes (bug fixes, config wiring, parity gaps, test coverage, hardening).

****How to apply:****

- Scope = engineering fixes Claude itself identified and verified (full suite + ruff/mypy + CI).

First exercised on PR #107 (API min_shot_count parity, merged 2026-06-12).

- Merge ONLY on a verified-green rollup (``gh pr view --json statusCheckRollup` ? all SUCCESS`), squash-merge, delete branch ? same dance the owner uses.
- STILL owner-gated: product/strategy/roadmap decisions, licensing-guardrail or training-data policy changes, schema/data deletions, and anything Claude is unsure belongs in "engineering fix". When in doubt, open the PR and leave it for review.
- Supersedes the earlier "do NOT auto-merge" default from the UI-alpha protocol (see `[[working-agreement-checkpoints]]`).

134. user (2026-06-12T17:03:36.609Z)

File created successfully at: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\working-agreement-merging.md (file state is current in your context ? no need to Read it back)

135. user (2026-06-12T17:03:39.725Z)

```
[HASH] video_id=89605e80ed012ca259a640b7ec06cd80 (3s)
[INDEX] clean indexing (validators bypassed for the massive file, as in
run_process_and_stitch.py)...
22:29:35 [backend.pipeline.segmenters.trajectory_hybrid] INFO: WASB WSL env OK: torch
1.11.0+cu113 cuda True
22:29:48 [backend.pipeline.detectors.wasb_runner] INFO: Running WASB inference on
GX010125_1080p.mp4 ...
0
```

136. user (2026-06-12T17:04:12.370Z)

```
GX010125_1080p__89605e80ed01.mp4
GX010125_1080p__89605e80ed01_frames
frames to in 20s
```

137. assistant (2026-06-12T17:04:22.819Z)

Quoting gremlins again ? plain counts instead:

138. user (2026-06-12T17:04:57.197Z)

```
4436
4951
```

139. assistant (2026-06-12T17:05:17.145Z)

Status on all three of your asks, plus the run:

****1. Live log**** ? open or tail this file; ``-Wait`` streams updates so you don't even need to refresh:

```
```powershell
Get-Content -Wait -Tail 30 C:\Users\avidu\Projects\badminton-highlight-
indexer\output\gx010125_run.log
```
```

Every phase, every per-rally KEEP/drop decision with its shot count, and the final reel path all land there. If I ever go quiet, that file is the source of truth (the WASB trajectory CSV also appears under ``output\wasb\`` when phase 2 completes).

****2. Merging good engineering fixes ? done and recorded.**** The API option is shipped: [PR #107](<https://github.com/avidullu/badminton-highlight-indexer/pull/107>) (``/api/compile`` now enforces ``stitching.min_shot_count``; segments with a known below-floor shot count are dropped, segments lacking shot-count metadata pass so explicitly chosen ids can't silently vanish; all-below-floor ? 400). CI 4/4 green, squash-merged to master. I also saved the new working agreement to memory ? future sessions will merge verified engineering fixes on green CI, while product/strategy/licensing changes stay yours.

****3. Stitch from the original ? verified, it's the behavior.**** The runner indexes the proxy but

compiles from the 4K source: `compile\_highlights(SOURCE\_PATH, ?)` where `SOURCE\_PATH` is your `F:\?\GX010125.MP4` ? the proxy is never touched at stitch time. Timestamps transfer 1:1 since both share the same timeline. As a hard check, I'll `ffprobe` the finished reel and confirm it's 3840x2160 HEVC-sourced footage before calling it done.

****Run progress:**** WASB is extracting/inferring at ~17 fps on the 1080p proxy (vs ~5 fps on 4K) ? ~4,950 of ~46,000 frames done. ETA: trajectory pass ~40 more minutes, then Gemini confirmation of the candidate rallies, then the >7-shot filter and the 4K stitch. I'll report when the kept-rally list and the reel land.

140. user (2026-06-12T17:29:49.701Z)

```
<task-notification>
<task-id>b4ez47ehk</task-id>
<tool-use-id>toolu_01Kv2v4JzknkahB3fGLEyaN1</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\b4ez47ehk.output</output-file>
<status>failed</status>
<summary>Background command "Run E2E pipeline on 1080p proxy, stitch from 4K original" failed with exit code 1</summary>
</task-notification>
```

141. assistant (2026-06-12T17:29:59.956Z)

The run died ? checking the log:

142. user (2026-06-12T17:30:03.443Z)

```
[CONFIG] segmenter=wasb_hybrid | min_shot_count=8 | begin_padding=0.5s end_padding=0.5s | watermark=on
[HASH] computing MD5 of the 1080p proxy...
[HASH] video_id=89605e80ed012ca259a640b7ec06cd80 (3s)
[INDEX] clean indexing (validators bypassed for the massive file, as in run_process_and_stitch.py)...
22:29:35 [backend.pipeline.segmenters.trajectory_hybrid] INFO: WASB WSL env OK: torch 1.11.0+cu113 cuda True
22:29:48 [backend.pipeline.detectors.wasb_runner] INFO: Running WASB inference on GX010125_1080p.mp4 ...
22:59:48 [backend.pipeline.detectors.wasb_runner] ERROR: WASB inference timed out after 1800s.
22:59:48 [backend.pipeline.segmenters.trajectory_hybrid] ERROR: WASB produced no trajectory; aborting.
[INDEX] 0 play segments indexed (total 1819s).
no play segments detected ? nothing to stitch
```

143. user (2026-06-12T17:30:24.562Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\config.json has been updated successfully. (file state is current in your context ? no need to Read it back)

144. user (2026-06-12T17:30:27.121Z)

no lingering process
30087

145. assistant (2026-06-12T17:30:34.427Z)

30,087 frames (65%) are already cached, no zombie process. The crash-cache means the re-run resumes rather than starting over. Relaunching with the 2-hour budget:

146. user (2026-06-12T17:30:35.757Z)

Command running in background with ID: bshx6hglsa. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\bshx6hglsa.output. You will be notified when

it completes. To check interim output, use Read on that file path.

147. assistant (2026-06-12T17:30:50.063Z)

Relaunched (config now `timeout\_sec: 7200`). The cached 30k frames mean the trajectory pass should finish in roughly **20?25 minutes** instead of 45. Same log file for your live view ? it restarted from the top:

```
```powershell
Get-Content -Wait -Tail 30 C:\Users\avidu\Projects\badminton-highlight-
indexer\output\gx010125_run.log
```
```

What happened, for the record: the 30-min WASB ceiling (`indexing.wasb.timeout\_sec: 1800`) was tuned for the shorter eval corpus videos ? a 12.8-min 4K60 GoPro file is a new envelope for this pipeline. The timeout killed inference cleanly, the pipeline correctly reported "0 segments" rather than corrupting anything, and the crash-cache preserved all extracted frames. I'll report at the next milestone (trajectory ? candidate windows ? Gemini confirmations ? filter ? 4K stitch).

148. user (2026-06-12T17:30:50.116Z)

```
<task-notification>
<task-id>bwiwctgbr</task-id>
<summary>Monitor event: "GX010125 proxy-run milestones (phases, kept rallies, errors)"</summary>
<event>[CONFIG] segmenter=wasb_hybrid | min_shot_count=8 | begin_padding=0.5s end_padding=0.5s |
watermark=on
[HASH] computing MD5 of the 1080p proxy...</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

149. assistant (2026-06-12T17:30:58.412Z)

Run restarted cleanly ? hashing the proxy now. Next meaningful milestone will be the WASB trajectory completing (~20?25 min). I'll stay quiet until then unless something breaks.

150. user (2026-06-12T17:30:58.519Z)

```
<task-notification>
<task-id>bwiwctgbr</task-id>
<summary>Monitor event: "GX010125 proxy-run milestones (phases, kept rallies, errors)"</summary>
<event>[HASH] video_id=89605e80ed012ca259a640b7ec06cd80 (3s)
[INDEX] clean indexing (validators bypassed for the massive file, as in
run_process_and_stitch.py)...</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

151. assistant (2026-06-12T17:31:05.247Z)

Routine ? same video_id as before (cache will be reused), indexing phase starting. Waiting for the trajectory milestone.

152. user (2026-06-12T17:31:06.862Z)

```
<task-notification>
<task-id>bwiwctgbr</task-id>
<summary>Monitor event: "GX010125 proxy-run milestones (phases, kept rallies, errors)"</summary>
<event>23:00:45 [backend.pipeline.segmenters.trajectory_hybrid] INFO: WASB WSL env OK: torch
1.11.0+cu113 cuda True</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```


153. assistant (2026-06-12T17:31:13.276Z)

GPU healthcheck passed again ? WASB inference is resuming over the cached frames. Next event should be the trajectory-points line.

154. user (2026-06-13T08:14:29.223Z)

hi

155. assistant (2026-06-13T08:14:30.436Z)

There's an issue with the selected model (claude-fable-5). It may not exist or you may not have access to it.

156. user (2026-06-13T08:14:41.976Z)

any update? can you rerun if needed?

157. assistant (2026-06-13T08:14:42.849Z)

There's an issue with the selected model (claude-fable-5). It may not exist or you may not have access to it.

158. user (2026-06-13T08:15:04.735Z)

<local-command-caveat>Caveat: The messages below were generated by the user while running local commands. DO NOT respond to these messages or otherwise consider them in your response unless the user explicitly asks you to.</local-command-caveat>

159. user (2026-06-13T08:15:04.735Z)

```
<command-name>/model</command-name>
      <command-message>model</command-message>
      <command-args>claude-opus-4-8</command-args>
```

160. user (2026-06-13T08:15:04.735Z)

<local-command-stdout>Set model to claude-opus-4-8</local-command-stdout>

161. user (2026-06-13T08:21:53.174Z)

hello

162. assistant (2026-06-13T08:21:59.534Z)

Let me check on the run's current state.

163. user (2026-06-13T08:22:02.346Z)

```
[CONFIG] segmenter=wasb_hybrid | min_shot_count=8 | begin_padding=0.5s end_padding=0.5s |
watermark=on
[HASH] computing MD5 of the 1080p proxy...
[HASH] video_id=89605e80ed012ca259a640b7ec06cd80 (3s)
[INDEX] clean indexing (validators bypassed for the massive file, as in
run_process_and_stitch.py)...
23:00:45 [backend.pipeline.segmenters.trajectory_hybrid] INFO: WASB WSL env OK: torch
1.11.0+cu113 cuda True
23:01:05 [backend.pipeline.detectors.wasb_runner] INFO: Running WASB inference on
GX010125_1080p.mp4 ...
```

164. user (2026-06-13T08:22:03.697Z)

NO wasb_infer process running

165. assistant (2026-06-13T08:22:13.642Z)

The WASB process is gone but the log shows neither completion nor timeout ? something's off. Let me investigate thoroughly before deciding whether to rerun.

166. user (2026-06-13T08:22:18.216Z)

```
NOW: 06/13/2026 13:52:18
--- Background python (parent) ---
--- Run log full ---
[CONFIG] segmenter=wasb_hybrid | min_shot_count=8 | begin_padding=0.5s end_padding=0.5s |
watermark=on
[HASH] computing MD5 of the 1080p proxy...
[HASH] video_id=89605e80ed012ca259a640b7ec06cd80 (3s)
[INDEX] clean indexing (validators bypassed for the massive file, as in
run_process_and_stitch.py)...
23:00:45 [backend.pipeline.segmenters.trajectory_hybrid] INFO: WASB WSL env OK: torch
1.11.0+cu113 cuda True
23:01:05 [backend.pipeline.detectors.wasb_runner] INFO: Running WASB inference on
GX010125_1080p.mp4 ...
```

167. user (2026-06-13T08:22:22.715Z)

```
--- GX frames cached ---
30087
--- trajectory CSV ---
no CSV yet
--- wasbcache dir ---
--- any python in wsl ---
260 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-
signal
--- GPU ---
22 %, 1608 MiB
```

168. assistant (2026-06-13T08:23:07.042Z)

The diagnosis is clear: it's now **13:52 on June 13** but the run froze at **23:01 last night (June 12)** ? the log never advanced past "Running WASB inference," the Python parent is gone, the WSL process is gone, and the frame cache is frozen at 30,087 (exactly where it stalled). This is the **overnight machine-suspend kill** ? a known failure mode in this project's history ("idle/suspend KILLS background runs silently"). The machine slept overnight and took both processes with it.

The good news: the 30,087-frame crash-cache survived. Let me confirm the suspend theory and understand the runner's resume behavior before relaunching robustly.

169. user (2026-06-13T08:23:12.551Z)

--- Sleep/resume events (last 18h) ---

TimeCreated	Id	Msg
-----	--	---
6/13/2026 9:53:15 AM	1	Activating keepalive handle from process 3464.
6/13/2026 9:53:15 AM	1	Opened keepalive file from process 3464.
6/13/2026 9:53:15 AM	1	Opened file with user mode dispatch blocked: "__drive_fs_keepalive"
6/13/2026 9:53:15 AM	1	Handle created before IOCTL_EVENT_WAIT for file "(null)"
6/13/2026 9:53:15 AM	1	Handle created before IOCTL_EVENT_WAIT for file "(null)"
6/13/2026 9:53:15 AM	1	Finished event start with status 1 and flags: 0
6/13/2026 9:53:15 AM	1	Returning actual mount point G
6/13/2026 9:53:15 AM	1	Mounting successfully done.
6/13/2026 9:53:15 AM	1	Link name matches the current one.
6/13/2026 9:53:15 AM	1	Link created: "\\DosDevices\\G:"
6/13/2026 9:53:15 AM	1	Link created: "\\??\\Volume{7bf6c51f-66c0-11f1-a10f-182649797673}"
6/13/2026 9:53:15 AM	1	Returning suggested name: "\\DosDevices\\G:"

6/13/2026 9:53:15 AM 1 Mount manager is querying for desired drive letter.
ForceDriveLetterAutoAssignme
6/13/2026 9:53:15 AM 1 Starting FCB garbage collector with 2000 ms interval.
6/13/2026 9:53:15 AM 1 Mount entry found: \DosDevices\G: -> \Device\Volume{f5ae2bcb-da00-3bf2-8935-4081
6/13/2026 9:53:15 AM 1 Found entry with matching device name: \Device\Volume{f5ae2bcb-da00-3bf2-8935-40
6/13/2026 9:53:15 AM 1 Finding mount entry; lockGlobal = 1; mount point = .
6/13/2026 9:53:15 AM 1 5-408102040811}"
6/13/2026 9:53:15 AM 1 Mounting volume using MountPoint "\DosDevices\G:" device
"\Device\Volume{f5ae2bc
6/13/2026 9:53:15 AM 1 Event start using mount ID: 2; device name: \Volume{f5ae2bcb-da00-3bf2-8935-4081
6/13/2026 9:53:15 AM 1 Inserted new mount entry.
6/13/2026 9:53:15 AM 1 ume{f5ae2bcb-da00-3bf2-8935-408102040811}" created
6/13/2026 9:53:15 AM 1 SymbolicLink: "\DosDevices\Global\Volume{f5ae2bcb-da00-3bf2-8935-408102040811}"
6/13/2026 9:53:15 AM 1 ; type: 8
6/13/2026 9:53:15 AM 1 \DosDevices\Global\Volume{f5ae2bcb-da00-3bf2-8935-408102040811}"; mount
point: "
6/13/2026 9:53:15 AM 1 disk device name: "\Device\Volume{f5ae2bcb-da00-3bf2-8935-408102040811}"; symbol
6/13/2026 9:53:15 AM 1 Set volume security descriptor successfully.
6/13/2026 9:53:15 AM 1 Creating disk device; mount point = G; mount ID = 21
6/13/2026 9:53:15 AM 1 No mount entry found.
6/13/2026 9:53:15 AM 1 Finding mount entry; lockGlobal = 0; mount point = \DosDevices\G:.
6/13/2026 9:53:15 AM 1 Entered event start.
6/13/2026 9:53:11 AM 1 Ignoring keepalive close because unmount is already in progress.
6/13/2026 9:53:05 AM 1 Finished event release.
6/13/2026 9:53:05 AM 1 Finished deleting device.
6/13/2026 9:53:05 AM 1 Deleting volume device object.
6/13/2026 9:53:05 AM 1 Removing mount entry.
6/13/2026 9:53:05 AM 1 Mount entry found: \DosDevices\G: -> \Device\Volume{f5ae2bcb-da01-3bf2-8935-4081
6/13/2026 9:53:05 AM 1 Found entry with matching device name: \Device\Volume{f5ae2bcb-da01-3bf2-8935-40
6/13/2026 9:53:05 AM 1 Finding mount entry; lockGlobal = 1; mount point = .
6/13/2026 9:53:05 AM 1 Deleting device object.
6/13/2026 9:53:05 AM 1 Cleaning up all change notification waiters.
6/13/2026 9:53:05 AM 1 Stopping FCB garbage collector.
6/13/2026 9:53:05 AM 1 Starting unmount for device "\Device\Volume{f5ae2bcb-da01-3bf2-8935-408102040811
6/13/2026 9:53:05 AM 1 Mount manager is disabled for this device or dokan initiated deletion;
ignoring.
6/13/2026 9:53:05 AM 1 Link deleted: "\DosDevices\G:"
6/13/2026 9:53:05 AM 1 Mount manager is disabled for this device or dokan initiated deletion;
ignoring.
6/13/2026 9:53:05 AM 1 Link deleted: "?\Volume{b128ed38-666c-11f1-a10f-182649797673}"
6/13/2026 9:53:05 AM 1 40811}"
6/13/2026 9:53:05 AM 1 Issuing a clean mount manager delete for device
"\Device\Volume{f5ae2bcb-da01-3b
6/13/2026 9:53:05 AM 1 Entered event release.
6/12/2026 9:53:08 PM 1 Opened file with user mode dispatch blocked: "\drive_fs_notification"
6/12/2026 9:53:04 PM 1 Activating keepalive handle from process 35148.
6/12/2026 9:53:04 PM 1 Opened keepalive file from process 35148.
6/12/2026 9:53:04 PM 1 Opened file with user mode dispatch blocked: "__drive_fs_keepalive"
6/12/2026 9:53:04 PM 1 Handle created before IOCTL_EVENT_WAIT for file
"\$Extend\$Reparse:\$R:\$INDEX_ALL
6/12/2026 9:53:04 PM 1 Handle created before IOCTL_EVENT_WAIT for file "(null)"
6/12/2026 9:53:04 PM 1 Handle created before IOCTL_EVENT_WAIT for file "(null)"
6/12/2026 9:53:04 PM 1 Finished event start with status 1 and flags: 0
6/12/2026 9:53:04 PM 1 Returning actual mount point G
6/12/2026 9:53:04 PM 1 Mounting successfully done.
6/12/2026 9:53:04 PM 1 Link name matches the current one.
6/12/2026 9:53:04 PM 1 Link created: "\DosDevices\G:"

```

6/12/2026 9:53:04 PM 1 Link created: "\\??\\Volume{b128ed38-666c-11f1-a10f-182649797673}"
6/12/2026 9:53:04 PM 1 Returning suggested name: "\\DosDevices\\G:"
6/12/2026 9:53:04 PM 1 Mount manager is querying for desired drive letter.
ForceDriveLetterAutoAssignme
6/12/2026 9:53:04 PM 1 Starting FCB garbage collector with 2000 ms interval.
6/12/2026 9:53:04 PM 1 Mount entry found: \\DosDevices\\G: -> \\Device\\Volume{f5ae2bcb-
da01-3bf2-8935-4081
6/12/2026 9:53:04 PM 1 Found entry with matching device name: \\Device\\Volume{f5ae2bcb-
da01-3bf2-8935-40
6/12/2026 9:53:04 PM 1 Finding mount entry; lockGlobal = 1; mount point = .
6/12/2026 9:53:04 PM 1 5-408102040811}"
6/12/2026 9:53:04 PM 1 Mounting volume using MountPoint "\\DosDevices\\G:" device
"\\Device\\Volume{f5ae2bc
6/12/2026 9:53:04 PM 1 Event start using mount ID: 1; device name: \\Volume{f5ae2bcb-
da01-3bf2-8935-4081
6/12/2026 9:53:04 PM 1 Inserted new mount entry.
6/12/2026 9:53:04 PM 1 ume{f5ae2bcb-da01-3bf2-8935-408102040811}" created
6/12/2026 9:53:04 PM 1 SymbolicLink: "\\DosDevices\\Global\\Volume{f5ae2bcb-
da01-3bf2-8935-408102040811}"
6/12/2026 9:53:04 PM 1 ; type: 8
6/12/2026 9:53:04 PM 1 \\DosDevices\\Global\\Volume{f5ae2bcb-da01-3bf2-8935-408102040811}"; mount
point: "
6/12/2026 9:53:04 PM 1 disk device name: "\\Device\\Volume{f5ae2bcb-
da01-3bf2-8935-408102040811}"; symbol
6/12/2026 9:53:04 PM 1 Set volume security descriptor successfully.
6/12/2026 9:53:04 PM 1 Creating disk device; mount point = G; mount ID = 11
6/12/2026 9:53:04 PM 1 No mount entry found.
6/12/2026 9:53:04 PM 1 Finding mount entry; lockGlobal = 0; mount point = \\DosDevices\\G:.
6/12/2026 9:53:04 PM 1 Entered event start.

```

--- Current power scheme sleep timeouts ---

```

Current AC Power Setting Index: 0x00000000
Current DC Power Setting Index: 0x00000000

```

170. user (2026-06-13T08:23:12.587Z)

```

1      """"WASB shuttle-trajectory runner (WSL2 subprocess adapter).
2
3      Windows-side counterpart to backend/detectors/wasb_infer.py. It:
4          1. stages the video into the WSL filesystem (fast local reads, not /mnt),
5          2. copies wasb_infer.py into the WASB repo's src/ dir,
6          3. runs it in the WSL `wasb` conda env -> a Frame,Visibility,X,Y trajectory CSV,
7          4. reuses the (model-agnostic) trajectory->action-window logic from tracknet_runner.
8
9      So the only WASB-specific bits live here; the windowing + CSV parsing are shared
10     with the TrackNet path. Verified inference core: wasb_infer.py (see its docstring).
11     """"
12     import os
13     import shlex
14     import logging
15     import subprocess
16     from dataclasses import dataclass
17     from typing import Dict, Any, Optional, Tuple
18
19     # Reuse the model-agnostic helpers ? trajectory shape + windowing are identical.
20     # parse_trajectory_csv / trajectory_to_action_windows are @staticmethods on
TrackNetRunner.
21     from backend.pipeline.detectors.tracknet_runner import to_wsl_mnt_path, TrackNetRunner
22     from backend.pipeline.detectors.base import DetectorRunner, WslCommandMixin,
wsl_tilde_quote
23
24     logger = logging.getLogger(__name__)
25
26     # Windows path to the inference wrapper we stage into WSL.

```

```

27     _INFER_WIN = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
28
29
30     @dataclass
31     class WasbConfig:
32         """Resolved settings for a WASB WSL run (built from config.json ->
indexing.wasb)."""
33         distro: str = "Ubuntu"
34         repo_dir: str = "~/models/WASB-SBDT"
35         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
36         conda_env: str = "wasb"
37         weights_path: str = "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
38         sport: str = "badminton"
39         wsl_stage_dir: str = "~/clips"
40         timeout_sec: int = 1800 # 30 min; a hung GPU/conda call is killed (0 = no timeout)
41
42         @classmethod
43         def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "WasbConfig":
44             w = (idx_cfg or {}).get("wasb", {}) or {}
45             return cls(
46                 distro=w.get("wsl_distro", "Ubuntu"),
47                 repo_dir=w.get("repo_dir", "~/models/WASB-SBDT"),
48                 conda_sh=w.get("conda_sh", "~/miniconda3/etc/profile.d/conda.sh"),
49                 conda_env=w.get("conda_env", "wasb"),
50                 weights_path=w.get("weights_path",
51                                     "~/models/WASB-
SBDT/pretrained_weights/wasb_badminton_best.pth.tar"),
52                 sport=w.get("sport", "badminton"),
53                 wsl_stage_dir=w.get("wsl_stage_dir", "~/clips"),
54                 timeout_sec=int(w.get("timeout_sec", 1800)),
55             )
56
57
58     class WasbRunner(WslCommandMixin, DetectorRunner):
59         def __init__(self, cfg: WasbConfig):
60             self.cfg = cfg
61
62         def healthcheck(self) -> Tuple[bool, str]:
63             """Verify the WSL `wasb` env imports torch and sees a GPU. Returns (ok, msg)."""
64             cmd = (f"conda activate {self.cfg.conda_env} && "
65                 f"python -c \"import torch; print('torch', torch.__version__, "
66                 f"'cuda', torch.cuda.is_available())\"")
67             try:
68                 res = self._wsl_bash(cmd)
69             except FileNotFoundError:
70                 return False, "`wsl` not found ? Windows + WSL2 required."
71             except subprocess.TimeoutExpired:
72                 return False, "WSL healthcheck timed out."
73             out = (res.stdout or "").strip()
74             if res.returncode != 0:
75                 return False, f"WSL `wasb` env check failed: {(res.stderr or out).strip()}"
76             return True, out
77
78         def _stage_video(self, video_win_path: str, dest_name: str) -> Optional[str]:
79             src_mnt = to_wsl_mnt_path(video_win_path)
80             dest = f"{self.cfg.wsl_stage_dir}/{dest_name}"
81             cmd = f"mkdir -p {self.cfg.wsl_stage_dir} && cp {shlex.quote(src_mnt)}
{wsl_tilde_quote(dest)} && echo OK"
82             try:
83                 res = self._wsl_bash(cmd)
84             except subprocess.TimeoutExpired:
85                 logger.error("Staging video into WSL timed out.")
86                 return None
87             if res.returncode != 0 or "OK" not in (res.stdout or ""):

```

```

88         logger.error("Failed to stage video into WSL: %s", (res.stderr or
res.stdout).strip())
89         return None
90         return dest
91
92     def _stage_infer_script(self) -> bool:
93         """Copy wasb_infer.py into the WASB repo src/ so it can import WASB modules."""
94         src_mnt = to_wsl_mnt_path(_INFER_WIN)
95         cmd = f"cp {shlex.quote(src_mnt)} {self.cfg.repo_dir}/src/wasb_infer.py && echo
OK"
96
97         res = self._wsl_bash(cmd)
98         return res.returncode == 0 and "OK" in (res.stdout or "")
99
100     def run_predict(self, video_win_path: str, output_win_dir: str,
101                     video_id: Optional[str] = None) -> Optional[str]:
102         """Run WASB on a video. Returns the Windows path to the trajectory CSV, or None.
103
104         All WSL/cache/output artifacts are namespaced by ``video_id`` (the file MD5) so
105         two videos that share a filename (e.g. two ``match.mp4`` from different folders)
106         can never reuse each other's staged frames, resumable cache, or CSV.
107
108         # Resolve relative dirs (the hybrids pass "output/wasb") BEFORE the /mnt
109         # translation: to_wsl_mnt_path passes relative paths through unchanged, so
110         # WSL resolved them against the inference cwd (~/.models/WASB-SBDT/src) and
111         # the CSV landed inside WSL while Windows polled the repo-relative path.
112         output_win_dir = os.path.abspath(output_win_dir)
113         os.makedirs(output_win_dir, exist_ok=True)
114         # Normalize for cross-platform basename (tests use Windows paths on Linux).
115         video_win_path = str(video_win_path).replace("\\", "/")
116         base = os.path.basename(video_win_path)
117         stem, ext = os.path.splitext(base)
118         # Unique, content-derived key: same filename + different bytes -> different key.
119         key = f"{stem}__{video_id[:12]}" if video_id else stem
120         wsl_out_dir = to_wsl_mnt_path(output_win_dir)
121         expected_csv_win = os.path.join(output_win_dir, f"{key}_wasb.csv")
122
123         if not self._stage_infer_script():
124             logger.error("Could not stage wasb_infer.py into WSL.")
125             return None
126         staged_video = self._stage_video(video_win_path, f"{key}{ext}")
127         if staged_video is None:
128             return None
129         frames_dir = f"{self.cfg.wsl_stage_dir}/{key}_frames"
130
131         cmd = (
132             f"conda activate {self.cfg.conda_env} && cd {self.cfg.repo_dir}/src && "
133             f"python wasb_infer.py "
134             f"--video {wsl_tilde_quote(staged_video)} "
135             f"--frames_out_dir {wsl_tilde_quote(frames_dir)} "
136             f"--weights {wsl_tilde_quote(self.cfg.weights_path)} "
137             f"--sport {shlex.quote(self.cfg.sport)} "
138             f"--out {wsl_tilde_quote(wsl_out_dir + '/' + key + '_wasb.csv')}"
139         )
140         logger.info("Running WASB inference on %s ...", base)
141         try:
142             res = self._wsl_bash(cmd)
143         except subprocess.TimeoutExpired:
144             logger.error("WASB inference timed out after %ss.", self.cfg.timeout_sec)
145             return None
146         if res.returncode != 0:
147             logger.error("WASB inference failed:\n%s", (res.stderr or
res.stdout)[-2000:])
148             return None
149         if not os.path.exists(expected_csv_win):
150             logger.error("WASB finished but expected CSV not found at %s",

```

```

expected_csv_win)
150         return None
151         logger.info("WASB trajectory CSV: %s", expected_csv_win)
152         return expected_csv_win
153
154     # Trajectory parsing + windowing are inherited from tracknet_runner (model-
agnostic).
155     parse_trajectory_csv = staticmethod(TrackNetRunner.parse_trajectory_csv)
156     trajectory_to_action_windows =
staticmethod(TrackNetRunner.trajectory_to_action_windows)
157

```

171. assistant (2026-06-13T08:23:59.763Z)

Key finding: power settings show ****sleep is disabled**** (`STANDBYIDLE = 0` on both AC/DC), yet the WSL filesystem remounted at 9:53 AM today (a fresh WSL VM start ? consistent with an overnight reboot or WSL teardown). Python orchestrator gone with no timeout logged = abrupt kill, not a clean failure. Regardless of exact cause (reboot/hibernate/WSL teardown), the fix is the same: rerun, but first I need to understand the runner's resume behavior so I don't waste 45 minutes re-extracting frames. Let me read the inference wrapper.

172. user (2026-06-13T08:24:00.822Z)

```

1     """WASB shuttle-trajectory inference on an arbitrary frame sequence (runs in WSL).
2
3     WASB-SBDT only ships *dataset* evaluation; this is a thin wrapper that reuses its
4     own building blocks (ImageDataset transform + detector + online tracker) to run on
5     any folder of extracted frames and emit a trajectory CSV in the exact shape the
6     indexer's `tracknet_runner.parse_trajectory_csv` already consumes:
7
8         Frame,Visibility,X,Y
9
10    It must run INSIDE the WSL `wasb` conda env, from the WASB-SBDT `src/` dir (it
11    imports WASB's `detectors`/`trackers`/`dataloaders`). The Windows-side adapter
12    (`wasb_runner.py`) stages this file + the frames into WSL and invokes it.
13
14    Resumable across reboots
15    -----
16    The GPU detector pass (one forward per sliding window ? ~21k for a 6-min clip) is
17    the slow, expensive part; the CPU tracker pass that turns detections into a
18    trajectory is cheap and *stateful* (it must see every frame in order, so it can't
19    resume mid-stream). We exploit that split:
20
21    * The detector pass writes its per-window raw outputs incrementally to
22      ``<cache>/det_raw.jsonl`` (flushed+fsync'd every ``--flush-every`` windows) and
23      records progress in ``<cache>/manifest.json`` (atomic write). A reboot loses at
24      most ``--flush-every`` windows of GPU work instead of the whole pass.
25    * On restart we replay the cached windows, skip the DataLoader ahead to the first
26      un-cached window, and continue. Once every window is cached, the detector pass
27      is skipped entirely and we just re-run the (cheap, deterministic) tracker over
28      the full ordered cache -> trajectory CSV. So a lost trajectory CSV costs seconds,
29      not the GPU hour.
30
31    The cache is keyed by the output path (``<out>_wasbcache/`` by default), lives next
32    to the output (NOT in %TEMP%), and is invalidated automatically if the frames dir,
33    weights, sport, window size, or frame count change.
34
35    Usage (in WSL, from ~/models/WASB-SBDT/src):
36        python wasb_infer.py --frames_dir /path/frames --weights
../pretrained_weights/wasb_badminton_best.pth.tar \
37        --out /path/traj.csv [--sport badminton] [--limit N] [--cache-dir DIR] [--flush-
every 1000] [--fresh]
38    """
39    import os
40    import os.path as osp

```

```

41     import csv
42     import glob
43     import json
44     import argparse
45     import logging
46     from collections import defaultdict
47
48     logger = logging.getLogger(__name__)
49
50     CACHE_VERSION = 1
51     _DET_FILE = "det_raw.jsonl"
52     _MANIFEST_FILE = "manifest.json"
53
54
55     def _build_cfg(weights: str, sport: str):
56         """Compose the WASB eval config via Hydra, overriding model/weights/device."""
57         from hydra import compose, initialize
58         with initialize(config_path="configs", version_base=None):
59             cfg = compose(config_name="eval", overrides=[
60                 f"dataset={sport}",
61                 "model=wasb",
62                 f"detector.model_path={weights}",
63                 "runner.device=cuda",
64                 "runner.gpus=[0]",          # this box has a single GPU; default config assumes
more
65             ])
66         return cfg
67
68
69     def _frame_id(path: str) -> int:
70         """Best-effort integer frame id from a frame filename (e.g. frame_000180.png ->
180)."""
71         stem = osp.splitext(osp.basename(path))[0]
72         digits = "".join(ch for ch in stem if ch.isdigit())
73         return int(digits) if digits else -1
74
75
76     # ----- #
77     # Resumable detector cache (pure-Python, no torch/numpy ? unit-testable)
78     # ----- #
79     def _cache_paths(cache_dir: str):
80         return osp.join(cache_dir, _DET_FILE), osp.join(cache_dir, _MANIFEST_FILE)
81
82
83     def default_cache_dir(out_csv: str) -> str:
84         """Stable cache dir next to the trajectory output (survives reboots; not %TEMP%)."""
85         d = osp.dirname(osp.abspath(out_csv))
86         stem = osp.splitext(osp.basename(out_csv))[0]
87         return osp.join(d, stem + "_wasbcache")
88
89
90     def _atomic_write_text(path: str, text: str) -> None:
91         """Write then rename so a crash mid-write can't leave a half-written file."""
92         tmp = path + ".tmp"
93         with open(tmp, "w") as f:
94             f.write(text)
95             f.flush()
96             os.fsync(f.fileno())
97         os.replace(tmp, path)
98
99
100     def _det_plain(det) -> list:
101         """A detector candidate dict {'xy': np.array([x,y]), 'score', 'scale'} -> JSON-able
list."""
102         xy = det["xy"]

```



```

103         return [float(xy[0]), float(xy[1]), float(det["score"]), int(det["scale"])]
104
105
106     def _dets_to_tracker(plain_list):
107         """Inverse of _det_plain: rebuild the dicts the tracker expects (xy MUST be a numpy
array,
108         the tracker does vector math / np.linalg.norm on it)."""
109         import numpy as np
110         return [{"xy": np.array([p[0], p[1]], dtype=float), "score": p[2], "scale": p[3]}
111                 for p in plain_list]
112
113
114     def write_manifest(cache_dir: str, manifest: dict) -> None:
115         _, mpath = _cache_paths(cache_dir)
116         _atomic_write_text(mpath, json.dumps(manifest, indent=2))
117
118
119     def read_manifest(cache_dir: str):
120         _, mpath = _cache_paths(cache_dir)
121         if not osp.exists(mpath):
122             return None
123         try:
124             with open(mpath) as f:
125                 return json.load(f)
126         except (json.JSONDecodeError, OSError):
127             return None
128
129
130     def manifest_compatible(manifest: dict, *, frames_dir: str, weights: str, sport: str,
131                             frames_in: int, frames_total: int) -> bool:
132         """A cache is reusable only if the run that produced it matches this run's
inputs."""
133         if not manifest or manifest.get("version") != CACHE_VERSION:
134             return False
135         return (
136             manifest.get("frames_dir") == osp.abspath(frames_dir)
137             and manifest.get("weights") == weights
138             and manifest.get("sport") == sport
139             and manifest.get("frames_in") == frames_in
140             and manifest.get("frames_total") == frames_total
141         )
142
143
144     def reset_cache(cache_dir: str) -> None:
145         """Drop any existing cache so the next run starts clean."""
146         det_jsonl, mpath = _cache_paths(cache_dir)
147         for p in (det_jsonl, mpath):
148             if osp.exists(p):
149                 os.remove(p)
150
151
152     def load_and_clean_cache(cache_dir: str):
153         """Read the longest *contiguous* (w == line index) valid prefix of det_raw.jsonl,
154         rebuild the per-frame detection accumulation, and rewrite the file to exactly that
155         prefix so a trailing half-written line from a crash can't corrupt future appends.
156
157         Returns (windows_done, det_by_fid) where det_by_fid maps frame_id -> list of
158         [x, y, score, scale] candidates (accumulated across every window the frame is in,
159         in window order ? identical to a non-resumed run).
160         """
161         det_jsonl, _ = _cache_paths(cache_dir)
162         det_by_fid = defaultdict(list)
163         valid: list = []
164         if osp.exists(det_jsonl):
165             with open(det_jsonl, "r") as f:

```

```

166         for line in f:
167             s = line.strip()
168             if not s:
169                 continue
170             try:
171                 obj = json.loads(s)
172             except json.JSONDecodeError:
173                 break # truncated trailing line (crash mid-
flush)
174             if obj.get("w") != len(valid): # non-contiguous -> stop at the gap
175                 break
176             valid.append(s)
177             for fid, plain in obj["f"]:
178                 det_by_fid[int(fid)].extend([list(p) for p in plain])
179             # Guarantee a clean append boundary by rewriting only the good prefix.
180             _atomic_write_text(det_jsonl, ("\n".join(valid) + "\n") if valid else "")
181         return len(valid), det_by_fid
182
183
184 def _append_windows(fh, objs) -> None:
185     """Append window records as JSONL and durably flush (bounded loss on crash)."""
186     for o in objs:
187         fh.write(json.dumps(o, separators=(",", ":")) + "\n")
188     fh.flush()
189     os.fsync(fh.fileno())
190
191
192 def _atomic_write_trajectory(out_csv: str, rows) -> None:
193     os.makedirs(osp.dirname(osp.abspath(out_csv)), exist_ok=True)
194     tmp = out_csv + ".tmp"
195     with open(tmp, "w", newline="") as f:
196         w = csv.writer(f)
197         w.writerow(["Frame", "Visibility", "X", "Y"])
198         w.writerows(rows)
199         f.flush()
200         os.fsync(f.fileno())
201     os.replace(tmp, out_csv)
202
203
204 # ----- #
205 # Inference
206 # ----- #
207 def run(frames_dir: str, weights: str, out_csv: str, sport: str = "badminton",
208         limit: int = 0, batch_size: int = 8, num_workers: int = 4,
209         log_every_batches: int = 50, cache_dir: "str | None" = None,
210         flush_every: int = 1000, fresh: bool = False) -> str:
211     import time
212     import torch
213     from torch.utils.data import DataLoader
214     from detectors import build_detector
215     from trackers import build_tracker
216     from dataloaders import build_img_transforms, build_seq_transforms
217     from dataloaders.dataset_loader import ImageDataset
218     from utils import Center
219
220     cfg = _build_cfg(weights, sport)
221     frames_in = int(cfg["model"]["frames_in"])
222     input_wh = (int(cfg["model"]["inp_width"]), int(cfg["model"]["inp_height"]))
223     output_wh = (int(cfg["model"]["out_width"]), int(cfg["model"]["out_height"]))
224
225     frames = sorted(glob.glob(osp.join(frames_dir, "*.png"))) +
glob.glob(osp.join(frames_dir, "*.jpg")))
226     if limit:
227         frames = frames[:limit]
228     if len(frames) < frames_in:

```

```

229         raise SystemExit(f"need >= {frames_in} frames, found {len(frames)} in
{frames_dir}")
230
231     # Sliding windows of `frames_in` consecutive frames (the unit of GPU work +
checkpoint).
232     samples = []
233     for i in range(len(frames) - frames_in + 1):
234         window = frames[i:i + frames_in]
235         annos = [{"frame_path": p, "center": Center(is_visible=False, x=-1.0, y=-1.0)}
for p in window]
236         samples.append({"frames": window, "annos": annos})
237     windows_total = len(samples)
238
239     # ---- cache setup / resume decision ----- #
240     if cache_dir is None:
241         cache_dir = default_cache_dir(out_csv)
242         os.makedirs(cache_dir, exist_ok=True)
243         det_jsonl, _ = _cache_paths(cache_dir)
244         manifest = None if fresh else read_manifest(cache_dir)
245         resume = manifest_compatible(
246             manifest or {}, frames_dir=frames_dir, weights=weights, sport=sport,
247             frames_in=frames_in, frames_total=len(frames),
248         )
249         if resume:
250             windows_done, det_by_fid = load_and_clean_cache(cache_dir)
251             logger.info(f"resuming from cache: {windows_done}/{windows_total} windows
already detected")
252         else:
253             if manifest is not None:
254                 logger.info("cache incompatible with this run -> starting fresh")
255                 reset_cache(cache_dir)
256                 windows_done, det_by_fid = 0, defaultdict(list)
257
258     base_manifest = {
259         "version": CACHE_VERSION,
260         "frames_dir": osp.abspath(frames_dir),
261         "weights": weights,
262         "sport": sport,
263         "frames_in": frames_in,
264         "frames_total": len(frames),
265         "windows_total": windows_total,
266         "windows_done": windows_done,
267     }
268     write_manifest(cache_dir, base_manifest)
269
270     # ---- detector pass over the un-cached windows ----- #
271     remaining = samples[windows_done:]
272     if remaining:
273         _, transform_test = build_img_transforms(cfg)
274         try:
275             _, seq_transform_test = build_seq_transforms(cfg)
276         except Exception:
277             seq_transform_test = None
278         ds = ImageDataset(cfg, remaining, input_wh, output_wh,
279                         transform=transform_test, seq_transform=seq_transform_test,
is_train=False)
280         loader = DataLoader(ds, batch_size=batch_size, shuffle=False,
num_workers=num_workers)
281         detector = build_detector(cfg)
282
283         t0 = time.time()
284         done = windows_done
285         win_idx = windows_done
286         log_every = max(1, log_every_batches)
287         flush_n = max(1, flush_every)

```

```

288         pending = []
289         logger.info(f"detecting on {len(remaining)} remaining windows "
290                    f"(total {windows_total}, batch={batch_size},
workers={num_workers})...")
291         det_f = open(det_jsonl, "a")
292         try:
293             with torch.no_grad():
294                 for bi, batch in enumerate(loader):
295                     imgs, _hms, trans = batch[0], batch[1], batch[2]
296                     img_paths = [list(t) for t in batch[-1]] # [frame_in][batch] ->
path
297                     batch_results, _ = detector.run_tensor(imgs, trans)
298                     nb = imgs.shape[0]
299                     for ib in range(nb):
300                         frames_payload = []
301                         for ie in sorted(batch_results[ib].keys()):
302                             p = img_paths[ie][ib]
303                             fid = _frame_id(p)
304                             plain = [_det_plain(d) for d in batch_results[ib][ie]]
305                             frames_payload.append([fid, plain])
306                             det_by_fid[fid].extend(plain)
307                         pending.append({"w": win_idx, "f": frames_payload})
308                         win_idx += 1
309                     done += nb
310                     if len(pending) >= flush_n or done >= windows_total:
311                         _append_windows(det_f, pending)
312                         pending = []
313                         base_manifest["windows_done"] = done
314                         write_manifest(cache_dir, base_manifest)
315                     if (bi + 1) % log_every == 0 or done >= windows_total:
316                         el = time.time() - t0
317                         new_done = done - windows_done
318                         rate = new_done / el if el > 0 else 0.0
319                         eta = (windows_total - done) / rate if rate > 0 else 0.0
320                         logger.info(f"{done}/{windows_total} "
321                                   f"({100 * done // max(1, windows_total)}%) |
{rate:.1f} win/s | "
322                                   f"elapsed {el:.0f}s | ETA {eta:.0f}s")
323                     if pending:
324                         _append_windows(det_f, pending)
325                         base_manifest["windows_done"] = done
326                         write_manifest(cache_dir, base_manifest)
327                 finally:
328                     det_f.close()
329                 logger.info(f"detection done in {time.time() - t0:.0f}s; running tracker...")
330             else:
331                 logger.info("detector cache already complete -> tracker-only re-run")
332
333             # ---- tracker re-run over the full ordered cache (cheap, deterministic) ---- #
334             tracker = build_tracker(cfg)
335             tracker.refresh()
336             rows = []
337             for fid in sorted(det_by_fid.keys()):
338                 r = tracker.update(_dets_to_tracker(det_by_fid[fid]))
339                 rows.append((fid, 1 if r.get("visi") else 0, r.get("x", -1), r.get("y", -1)))
340
341             _atomic_write_trajectory(out_csv, rows)
342             visible = sum(1 for _, v, *_ in rows if v)
343             logger.info(f"wrote {len(rows)} rows -> {out_csv}")
344             logger.info(f"visible-shuttle frames: {visible}/{len(rows)}")
345             return out_csv
346
347
348 def extract_frames(video_path: str, frames_dir: str) -> str:
349     """Decode a video to PNG frames named by source frame index (00000000.png ...).

```

```

350
351 Idempotent: if the dir already holds the expected number of frames (matching the
352 container's reported frame count) we skip re-decoding ? re-extraction after a
353 reboot is wasteful. NOTE: cv2 PNG extraction is slow; for big clips prefer
354 extracting with ffmpeg on the host and passing --frames_dir.
355 """
356 import cv2
357 os.makedirs(frames_dir, exist_ok=True)
358 cap = cv2.VideoCapture(video_path)
359 if not cap.isOpened():
360     raise SystemExit(f"cannot open video: {video_path}")
361 expected = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
362 existing = len(glob.glob(osp.join(frames_dir, "*.png")))
363 if expected > 0 and existing >= expected:
364     cap.release()
365     logger.info(f"reusing {existing} already-extracted frames -> {frames_dir}")
366     return frames_dir
367 i = 0
368 while True:
369     ok, frame = cap.read()
370     if not ok:
371         break
372     cv2.imwrite(osp.join(frames_dir, f"{i:08d}.png"), frame)
373     i += 1
374 cap.release()
375 logger.info(f"extracted {i} frames -> {frames_dir}")
376 return frames_dir
377
378
379 def main():
380     ap = argparse.ArgumentParser()
381     ap.add_argument("--frames_dir", help="folder of pre-extracted frames")
382     ap.add_argument("--video", help="video file; frames are extracted internally (in
WSL)")
383     ap.add_argument("--frames_out_dir", help="where to extract frames when --video is
used")
384     ap.add_argument("--weights", required=True)
385     ap.add_argument("--out", required=True)
386     ap.add_argument("--sport", default="badminton")
387     ap.add_argument("--limit", type=int, default=0, help="cap #frames (for quick
tests)")
388     ap.add_argument("--batch-size", type=int, default=8)
389     ap.add_argument("--num-workers", type=int, default=4)
390     ap.add_argument("--log-every-batches", type=int, default=50, help="progress log
cadence")
391     ap.add_argument("--cache-dir", default=None,
392                     help="resumable detector cache dir (default: <out>_wasbcache next to
--out)")
393     ap.add_argument("--flush-every", type=int, default=1000,
394                     help="fsync the detector cache every N windows (bounds reboot
loss)")
395     ap.add_argument("--fresh", action="store_true", help="ignore any existing cache and
start over")
396     args = ap.parse_args()
397
398     frames_dir = args.frames_dir
399     if args.video:
400         frames_dir = extract_frames(args.video, args.frames_out_dir or (args.video +
"_frames"))
401     if not frames_dir:
402         raise SystemExit("provide --frames_dir or --video")
403     run(frames_dir, args.weights, args.out, args.sport, args.limit,
404         batch_size=args.batch_size, num_workers=args.num_workers,
405         log_every_batches=args.log_every_batches, cache_dir=args.cache_dir,
406         flush_every=args.flush_every, fresh=args.fresh)

```

```
407
408
409     if __name__ == "__main__":
410         main()
411
```

173. user (2026-06-13T08:24:07.445Z)

Exit code 1

--- Last boot time ---

Thursday, June 11, 2026 12:43:17 AM

--- Power events 42(sleep)/1(resume)/107(resume) via Kernel-Power ---

TimeCreated	Id	Msg
-----	--	----
6/13/2026 1:32:46 PM	105	Power source change.
6/13/2026 1:32:42 PM	566	The system session has transitioned from 61 to 63.
		Reason AcDcBurst
6/13/2026 1:32:41 PM	105	Power source change.
6/13/2026 1:15:25 PM	566	The system session has transitioned from 60 to 61.
		Reason SessionUn
6/13/2026 12:34:43 PM	566	The system session has transitioned from 58 to 60.
		Reason InputHid
6/13/2026 8:38:51 AM	566	The system session has transitioned from 57 to 58.
		Reason SessionUn
6/13/2026 8:23:51 AM	566	The system session has transitioned from 55 to 57.
		Reason InputHid
6/13/2026 7:45:32 AM	566	The system session has transitioned from 54 to 55.
		Reason AcDcBurst
6/13/2026 7:45:26 AM	105	Power source change.
6/13/2026 7:45:22 AM	566	The system session has transitioned from 52 to 54.
		Reason AcDcBurst
6/13/2026 7:45:21 AM	105	Power source change.
6/12/2026 10:53:52 PM	566	The system session has transitioned from 51 to 52.
		Reason SessionUn
6/12/2026 9:51:26 PM	566	The system session has transitioned from 49 to 51.
		Reason InputHid
6/12/2026 8:43:31 PM	566	The system session has transitioned from 48 to 49.
		Reason SessionUn
6/12/2026 8:22:10 PM	566	The system session has transitioned from 46 to 48.
		Reason InputHid
6/12/2026 8:13:27 PM	566	The system session has transitioned from 45 to 46.
		Reason SessionUn

--- Unexpected shutdown (6008) ---

174. assistant (2026-06-13T08:25:53.465Z)

Now I have the real root cause. Let me lay out what the evidence shows:

- ****No reboot**** (last boot June 11) and ****no sleep (S3) events**** ? but a chain of `Kernel-Power`

****session transitions**** ("transitioned from 51 to 52", reasons `InputHid`/`SessionUn`) around 22:53 last night. That's ****Modern Standby (connected standby)****: when the machine sits idle with the screen off, Windows enters S0 low-power and the ****Desktop Activity Moderator** suspends Win32 processes and pauses the WSL2 VM**. That froze the Python orchestrator and the WSL inference mid-run ? no clean exit, no timeout logged, frame cache frozen at exactly 30,087. `standby-timeout=0` doesn't prevent this; Modern Standby is screen-off-triggered.

The crucial good news from reading `wasb\_infer.py`: the ****GPU detector pass** is checkpointed and resumable** (`det\_raw.jsonl` + manifest, fsync'd every 1000 windows), and frame extraction is idempotent once complete. The killer was purely the machine idling ? not a code bug.

Let me gather the exact numbers I need to choose the fastest extraction path, and install a keep-awake so this can't happen again.

175. user (2026-06-13T08:25:58.890Z)

Command running in background with ID: b97ebue57. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\b97ebue57.output. You will be notified when it completes. To check interim output, use Read on that file path.

176. user (2026-06-13T08:26:03.176Z)

```
Exit code 1
ParserError:
Line |
    1 |   ? 10125_1080p__89605e80ed01.mp4');
print('cv2_expected_frames', int(cap ?
    |
~
    | Missing expression
after ', '.
```

177. user (2026-06-13T08:26:17.367Z)

```
1      SetThreadExecutionState returned: 2147483648 (nonzero = success)
2      --- powercfg requests (who is holding wake locks) ---
3      This command requires administrator privileges and must be executed from an elevated
command prompt.
4
```

178. assistant (2026-06-13T08:27:25.202Z)

The wake-lock call ****succeeded**** (`SetThreadExecutionState` returned the prior state) ? but the job then exited, which releases the lock, and `powercfg /requests` tripped the exit code by needing admin. I need a ***persistent*** process to hold the lock. Let me fix that, and re-probe the frame metrics cleanly (the comma broke PowerShell's parser). Writing a small probe script avoids the nested-quote hell.

179. assistant (2026-06-13T08:27:32.317Z)

Holds a wake lock so Modern Standby (connected standby) can't suspend the long GPU run + WSL VM mid-flight. ES_CONTINUOUS persists only while THIS thread is alive, so we keep the process alive and re-assert periodically.

```
$sig = @'
using System;
using System.Runtime.InteropServices;
public class Caffeine {
    [DllImport("kernel32.dll", SetLastError = true)]
    public static extern uint SetThreadExecutionState(uint esFlags);
}
'@
```

```
Add-Type -TypeDefinition $sig
```

ES_CONTINUOUS(0x80000000) | ES_SYSTEM_REQUIRED(0x1) | ES_DISPLAY_REQUIRED

```
$FLAGS = [uint32]"0x80000003"
```

```
$deadlineMin = 300 # ~5h ceiling; covers the whole pipeline with margin
```

```
$start = [DateTime]::UtcNow
```

```
while (([DateTime]::UtcNow - $start).TotalMinutes -lt $deadlineMin) {
```

```
    [void][Caffeine]::SetThreadExecutionState($FLAGS)
```

```
    Write-Output ("[keepawake] asserted ES_SYSTEM_REQUIRED|ES_DISPLAY_REQUIRED at " +
```

```
[DateTime]::Now.ToString("HH:mm:ss"))
```

```
    Start-Sleep -Seconds 60
```

```
}
```

Release on clean exit.

```
[void][Caffeine]::SetThreadExecutionState([uint32]"0x80000000")
```

```
Write-Output "[keepawake] released"
```

180. user (2026-06-13T08:27:32.342Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-

indexer\scratch\keep_awake.ps1 (file state is current in your context ? no need to Read it back)

181. user (2026-06-13T08:27:37.208Z)

Command running in background with ID: brysrtrco. Output is being written to:

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-

indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\brysrtrco.output. You will be notified when

it completes. To check interim output, use Read on that file path.

182. assistant (2026-06-13T08:27:43.491Z)

Keep-awake is now holding the lock persistently. Now the frame metrics ? using `ffprobe`

(Windows) for the exact count and simple WSL commands (no commas to trip the parser):

183. user (2026-06-13T08:29:47.754Z)

Command running in background with ID: b0ak89bhi. Output is being written to:

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-

indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\b0ak89bhi.output. You will be notified when

it completes. To check interim output, use Read on that file path.

184. user (2026-06-13T08:29:49.213Z)

```
PNG_COUNT
```

```
0
```

```
FRAMES_DIR_SIZE
```

```
63G /home/avidullu/clips/GX010125_1080p__89605e80ed01_frames
```

```
FREE_SPACE
```

```
/dev/sdc 1007G 191G 766G 20% /
```

```
SAMPLE_FRAME
```

```
-rw-r--r-- 1 avidullu avidullu 2282728 Jun 12 17:31
```

```
/home/avidullu/clips/GX010125_1080p__89605e80ed01_frames/00000100.png
```

185. assistant (2026-06-13T08:30:28.939Z)

PNG_COUNT=0 is a false reading ? `ls \*.png` with 30k files overflows ARG_MAX; the dir is **63 GB

of ~2.3 MB PNGs** (?30,087 frames, confirmed by the sample file). Free space 766 G ? plenty. So

the prior run died **during cv2 frame extraction, before GPU inference ever started** (no

`det\_raw.jsonl` cache existed) ? extraction is the long pole, exactly as `wasb\_infer.py`'s own

docstring warns ("cv2 PNG extraction is slow; for big clips prefer ffmpeg").

Let me get cv2's exact expected count (the number `extract\_frames` checks against) so I can pre-extract with ffmpeg safely, and check the ffprobe result:

186. user (2026-06-13T08:30:30.364Z)

```
1      --- ffmpeg exact decoded frame count (proxy) ---
2
```

187. assistant (2026-06-13T08:30:31.622Z)

```
import sys
import cv2
cap = cv2.VideoCapture(sys.argv[1])
print("CAP_PROP_FRAME_COUNT", int(cap.get(cv2.CAP_PROP_FRAME_COUNT)))
print("FPS", cap.get(cv2.CAP_PROP_FPS))
cap.release()
```

188. user (2026-06-13T08:30:31.635Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\probe_cv2_count.py (file state is current in your context ? no need to Read it back)

189. assistant (2026-06-13T08:30:40.233Z)

ffmpeg is still counting (it decodes every frame). Let me probe cv2's count directly in the WSL `wasb` env ? that's the exact number `extract\_frames` will check against, on the staged video:

190. user (2026-06-13T08:30:44.756Z)

```
CAP_PROP_FRAME_COUNT 46080
FPS 59.94005994005994
```

191. assistant (2026-06-13T08:31:51.613Z)

cv2 expects exactly **46080 frames** (and the math is clean: $768.768s \times 59.94 \text{ fps} = 46080$ exactly, CFR ? so ffmpeg will produce precisely that count, perfectly aligned with cv2's indexing). That makes ffmpeg pre-extraction safe and removes the long pole: I'll extract all 46080 PNGs with ffmpeg natively inside WSL (fast ext4 disk, ~8 min) so the pipeline's slow cv2 extraction is skipped entirely, going straight to GPU inference (which is itself resumable).

Let me wipe the partial 30k and extract fresh:

192. user (2026-06-13T08:31:55.468Z)

Command running in background with ID: bpbtssydt. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\bpbtssydt.output. You will be notified when it completes. To check interim output, use Read on that file path.

193. assistant (2026-06-13T08:32:38.027Z)

While ffmpeg extracts (~8?12 min), let me verify it started cleanly and confirm the stitch-from-original wiring once more:

194. user (2026-06-13T08:32:39.042Z)

```
1      -vsync is deprecated. Use -fps_mode
2      Passing a number to -vsync is deprecated, use a string argument as described in the
manual.
3      [swscaler @ 0x5558fb368680] deprecated pixel format used, make sure you did set range
correctly
4      [swscaler @ 0x5558faf6e8c0] deprecated pixel format used, make sure you did set range
correctly
5      frame=    0 fps=0.0 q=0.0 size=          0kB time=N/A bitrate=N/A speed=N/A
frame=   16 fps=0.0 q=-0.0 size=N/A time=00:00:00.26 bitrate=N/A speed=0.469x
```

frame= 41 fps= 38 q=-0.0 size=N/A time=00:00:00.68 bitrate=N/A speed=0.632x
frame= 73 fps= 46 q=-0.0 size=N/A time=00:00:01.21 bitrate=N/A speed=0.767x
frame= 98 fps= 45 q=-0.0 size=N/A time=00:00:01.63 bitrate=N/A speed=0.758x
frame= 132 fps= 49 q=-0.0 size=N/A time=00:00:02.18 bitrate=N/A speed=0.819x
frame= 157 fps= 49 q=-0.0 size=N/A time=00:00:02.61 bitrate=N/A speed=0.82x
frame= 185 fps= 50 q=-0.0 size=N/A time=00:00:03.08 bitrate=N/A speed=0.834x
frame= 211 fps= 49 q=-0.0 size=N/A time=00:00:03.52 bitrate=N/A speed=0.814x
frame= 241 fps= 50 q=-0.0 size=N/A time=00:00:04.02 bitrate=N/A speed=0.831x
frame= 267 fps= 50 q=-0.0 size=N/A time=00:00:04.47 bitrate=N/A speed=0.835x
frame= 293 fps= 50 q=-0.0 size=N/A time=00:00:04.88 bitrate=N/A speed=0.833x
frame= 324 fps= 51 q=-0.0 size=N/A time=00:00:05.40 bitrate=N/A speed=0.848x
frame= 344 fps= 50 q=-0.0 size=N/A time=00:00:05.73 bitrate=N/A speed=0.834x
frame= 371 fps= 50 q=-0.0 size=N/A time=00:00:06.17 bitrate=N/A speed=0.832x
frame= 400 fps= 50 q=-0.0 size=N/A time=00:00:06.67 bitrate=N/A speed=0.842x
frame= 427 fps= 50 q=-0.0 size=N/A time=00:00:07.12 bitrate=N/A speed=0.839x
frame= 458 fps= 51 q=-0.0 size=N/A time=00:00:07.64 bitrate=N/A speed=0.849x
frame= 485 fps= 51 q=-0.0 size=N/A time=00:00:08.09 bitrate=N/A speed=0.851x
frame= 511 fps= 51 q=-0.0 size=N/A time=00:00:08.52 bitrate=N/A speed=0.852x
frame= 538 fps= 51 q=-0.0 size=N/A time=00:00:08.95 bitrate=N/A speed=0.852x
frame= 566 fps= 51 q=-0.0 size=N/A time=00:00:09.44 bitrate=N/A speed=0.857x
frame= 594 fps= 51 q=-0.0 size=N/A time=00:00:09.90 bitrate=N/A speed=0.858x
frame= 625 fps= 52 q=-0.0 size=N/A time=00:00:10.42 bitrate=N/A speed=0.865x
frame= 652 fps= 52 q=-0.0 size=N/A time=00:00:10.87 bitrate=N/A speed=0.866x
frame= 685 fps= 52 q=-0.0 size=N/A time=00:00:11.42 bitrate=N/A speed=0.871x
frame= 716 fps= 52 q=-0.0 size=N/A time=00:00:11.94 bitrate=N/A speed=0.873x
frame= 740 fps= 52 q=-0.0 size=N/A time=00:00:12.34 bitrate=N/A speed=0.87x
frame= 773 fps= 53 q=-0.0 size=N/A time=00:00:12.89 bitrate=N/A speed=0.877x
frame= 801 fps= 53 q=-0.0 size=N/A time=00:00:13.36 bitrate=N/A speed=0.879x
frame= 826 fps= 53 q=-0.0 size=N/A time=00:00:13.78 bitrate=N/A speed=0.877x
frame= 855 fps= 53 q=-0.0 size=N/A time=00:00:14.26 bitrate=N/A speed=0.88x
frame= 882 fps= 53 q=-0.0 size=N/A time=00:00:14.71 bitrate=N/A speed=0.88x
frame= 910 fps= 53 q=-0.0 size=N/A time=00:00:15.18 bitrate=N/A speed=0.881x
frame= 938 fps= 53 q=-0.0 size=N/A time=00:00:15.64 bitrate=N/A speed=0.882x
frame= 959 fps= 52 q=-0.0 size=N/A time=00:00:15.99 bitrate=N/A speed=0.876x
frame= 988 fps= 53 q=-0.0 size=N/A time=00:00:16.49 bitrate=N/A speed=0.879x
frame= 1018 fps= 53 q=-0.0 size=N/A time=00:00:16.98 bitrate=N/A speed=0.88x
frame= 1041 fps= 52 q=-0.0 size=N/A time=00:00:17.36 bitrate=N/A speed=0.874x
frame= 1072 fps= 53 q=-0.0 size=N/A time=00:00:17.86 bitrate=N/A speed=0.877x
frame= 1096 fps= 52 q=-0.0 size=N/A time=00:00:18.26 bitrate=N/A speed=0.874x
frame= 1118 fps= 52 q=-0.0 size=N/A time=00:00:18.65 bitrate=N/A speed=0.871x
frame= 1148 fps= 52 q=-0.0 size=N/A time=00:00:19.13 bitrate=N/A speed=0.871x
frame= 1170 fps= 52 q=-0.0 size=N/A time=00:00:19.51 bitrate=N/A speed=0.867x
frame= 1203 fps= 52 q=-0.0 size=N/A time=00:00:20.07 bitrate=N/A speed=0.869x
frame= 1227 fps= 52 q=-0.0 size=N/A time=00:00:20.45 bitrate=N/A speed=0.866x
frame= 1256 fps= 52 q=-0.0 size=N/A time=00:00:20.95 bitrate=N/A speed=0.868x
frame= 1280 fps= 52 q=-0.0 size=N/A time=00:00:21.35 bitrate=N/A speed=0.865x
frame= 1306 fps= 52 q=-0.0 size=N/A time=00:00:21.78 bitrate=N/A speed=0.864x
frame= 1327 fps= 51 q=-0.0 size=N/A time=00:00:22.13 bitrate=N/A speed=0.858x
frame= 1351 fps= 51 q=-0.0 size=N/A time=00:00:22.53 bitrate=N/A speed=0.856x
frame= 1378 fps= 51 q=-0.0 size=N/A time=00:00:22.98 bitrate=N/A speed=0.857x
frame= 1407 fps= 51 q=-0.0 size=N/A time=00:00:23.47 bitrate=N/A speed=0.857x
frame= 1437 fps= 51 q=-0.0 size=N/A time=00:00:23.97 bitrate=N/A speed=0.859x
frame= 1462 fps= 51 q=-0.0 size=N/A time=00:00:24.39 bitrate=N/A speed=0.858x
frame= 1494 fps= 52 q=-0.0 size=N/A time=00:00:24.90 bitrate=N/A speed=0.86x
frame= 1520 fps= 52 q=-0.0 size=N/A time=00:00:25.35 bitrate=N/A speed=0.86x
frame= 1551 fps= 52 q=-0.0 size=N/A time=00:00:25.87 bitrate=N/A speed=0.863x
frame= 1575 fps= 52 q=-0.0 size=N/A time=00:00:26.27 bitrate=N/A speed=0.861x
frame= 1604 fps= 52 q=-0.0 size=N/A time=00:00:26.76 bitrate=N/A speed=0.862x
frame= 1630 fps= 52 q=-0.0 size=N/A time=00:00:27.19 bitrate=N/A speed=0.861x
frame= 1659 fps= 52 q=-0.0 size=N/A time=00:00:27.67 bitrate=N/A speed=0.862x
frame= 1690 fps= 52 q=-0.0 size=N/A time=00:00:28.17 bitrate=N/A speed=0.864x
frame= 1716 fps= 52 q=-0.0 size=N/A time=00:00:28.62 bitrate=N/A speed=0.864x
frame= 1740 fps= 52 q=-0.0 size=N/A time=00:00:29.02 bitrate=N/A speed=0.863x
frame= 1772 fps= 52 q=-0.0 size=N/A time=00:00:29.56 bitrate=N/A speed=0.865x
frame= 1797 fps= 52 q=-0.0 size=N/A time=00:00:29.97 bitrate=N/A speed=0.864x

```

frame= 1828 fps= 52 q=-0.0 size=N/A time=00:00:30.49 bitrate=N/A speed=0.866x
frame= 1852 fps= 52 q=-0.0 size=N/A time=00:00:30.89 bitrate=N/A speed=0.865x
frame= 1879 fps= 52 q=-0.0 size=N/A time=00:00:31.34 bitrate=N/A speed=0.866x
frame= 1908 fps= 52 q=-0.0 size=N/A time=00:00:31.83 bitrate=N/A speed=0.867x
frame= 1931 fps= 52 q=-0.0 size=N/A time=00:00:32.21 bitrate=N/A speed=0.865x
frame= 1963 fps= 52 q=-0.0 size=N/A time=00:00:32.74 bitrate=N/A speed=0.867x
frame= 1994 fps= 52 q=-0.0 size=N/A time=00:00:33.26 bitrate=N/A speed=0.868x
frame= 2023 fps= 52 q=-0.0 size=N/A time=00:00:33.75 bitrate=N/A speed=0.868x
frame= 2051 fps= 52 q=-0.0 size=N/A time=00:00:34.21 bitrate=N/A speed=0.867x

```

195. user (2026-06-13T08:32:40.008Z)

```

95         time_pairs.append((s["start_time"], s["end_time"]))
96         print(line + " -> KEEP", flush=True)
97     else:
98         skipped += 1
99         print(line + " -> drop", flush=True)
100
101     print(f"[FILTER] config min_shot_count={min_shots}: "
102           f"kept {len(time_pairs)}, dropped {skipped}", flush=True)
103     if not time_pairs:
104         raise SystemExit(f"no segments passed the min_shot_count={min_shots} filter")
105
106     compiler = VideoCompiler(cfg.output.output_dir,
107                              begin_padding=cfg.stitching.begin_padding,
108                              end_padding=cfg.stitching.end_padding,
109                              watermark=cfg.stitching.watermark)
110     out = compiler.compile_highlights(SOURCE_PATH, time_pairs, OUTPUT_FILENAME) # cut
from the 4K original
111     print(f"[DONE] highlight reel: {os.path.abspath(out)} (total {time.time() -
t0:.0f}s)", flush=True)
112
113
114     if __name__ == "__main__":

```

196. assistant (2026-06-13T08:33:12.278Z)

Good ? ffmpeg is extracting cleanly at ~52 fps, and I've confirmed the stitch line cuts from the ****4K original**** (`SOURCE\_PATH`), not the proxy. Here's the full picture:

What happened overnight

It wasn't a code bug ? it was the ****machine, twice****. The Kernel-Power event log shows ****Modern Standby (connected standby)**** transitions around 22:53 last night. When the machine sits idle with the screen off, Windows enters S0 low-power and ****suspends Win32 processes and pauses the WSL2 VM****. That froze the Python orchestrator + WSL inference mid-run, with no clean exit (which is exactly why you saw no error ? the log just stopped at "Running WASB inference" and the frame cache froze at 30,087). Note `standby-timeout=0` does ***not*** prevent this; Modern Standby is screen-off-triggered.

I also discovered ***why it kept dying at the same place***: the run ****never even reached the GPU**** ? it was still in ****cv2 frame extraction**** (the slow part ? `wasb\_infer.py`'s own docstring warns "cv2 PNG extraction is slow; for big clips prefer ffmpeg"). 46,080 frames at ~20 fps via cv2 = ~38 min of fragile single-threaded extraction, and standby kept killing it before inference began.

Fixes applied

1. ****Root cause ? keep-awake.**** A persistent wake-lock process (`SetThreadExecutionState(ES\_SYSTEM\_REQUIRED|ES\_DISPLAY\_REQUIRED)`, re-asserted every 60s) is now holding the machine in S0 working state so standby can't suspend the run.
2. ****The bottleneck ? ffmpeg pre-extraction.**** Extracting all 46,080 PNGs with ffmpeg natively inside WSL right now (~15 min, running). The frame count is provably exact (768.768s × 59.94fps = 46,080, CFR ? matches cv2's count precisely), so the pipeline's slow cv2 step is ****skipped**

entirely** and it goes straight to GPU inference (which is itself checkpointed/resumable).
3. **Stitch-from-original ? confirmed.** The runner indexes the 1080p proxy but cuts the final reel from `F:\?\GX010125.MP4` (the 4K original) ? same timeline, so timestamps transfer 1:1. I'll `ffprobe` the output to confirm it's 3840x2160 before calling it done.

Also: the **API `min\_shot\_count` fix you authorized is merged**
([#107](https://github.com/avidullu/badminton-highlight-indexer/pull/107), squashed to master on green CI).

What's next (automatic)

When extraction finishes (~15 min), I'll launch the pipeline ? it'll skip extraction ? **GPU WASB inference** ? candidate windows ? **Gemini rally confirmation** (serial, your API key's in `.env`) ? **filter to shot_count > 7** ? **stitch from 4K** . ETA after extraction: ~30?50 min. The live log stays at:

```
```powershell
Get-Content -Wait -Tail 40 C:\Users\avidu\Projects\badminton-highlight-
indexer\output\gx010125_run.log
```
```

I'll report at each milestone and won't go idle on you.

197. user (2026-06-13T08:37:05.081Z)

```
<task-notification>
<task-id>b0ak89bhi</task-id>
<tool-use-id>toolu_01PZEujMX5yRKubGxSBv8kSU</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\b0ak89bhi.output</output-file>
<status>completed</status>
<summary>Background command "Get exact decoded frame count of the 1080p proxy" completed (exit
code 0)</summary>
</task-notification>
```

198. user (2026-06-13T08:37:09.977Z)

```
1      --- ffprobe exact decoded frame count (proxy) ---
2      avg_frame_rate=60000/1001
3      nb_read_frames=46080
4
```

199. assistant (2026-06-13T08:37:14.174Z)

Confirmed: ffprobe's exact decoded count is **46080** ? matching cv2's `CAP\_PROP\_FRAME\_COUNT` precisely. So the ffmpeg pre-extraction will produce exactly the count the pipeline expects, and the slow cv2 step will be skipped. Extraction is running (~15 min). I'll launch the pipeline the moment it completes.

200. user (2026-06-13T08:46:02.675Z)

```
<task-notification>
<task-id>bpbtsyydt</task-id>
<tool-use-id>toolu_01SKK1dnHyEAEdLVvfnE45xE</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\bpbtsyydt.output</output-file>
<status>completed</status>
<summary>Background command "Extract all proxy frames to PNG natively in WSL" completed (exit
code 0)</summary>
</task-notification>
```

201. assistant (2026-06-13T08:46:08.857Z)

Extraction done. Let me verify the count is exactly 46080 before launching the pipeline:

202. user (2026-06-13T08:46:13.315Z)

```
46080
FIRST
00000000.png
LAST
00046079.png
FREE
/dev/sdc      1007G   220G   736G   24% /
```

203. assistant (2026-06-13T08:46:26.630Z)

Perfect: ****46080 PNGs****, named `00000000.png`?`00046079.png` ? exact match to cv2's expected count and naming. The pipeline will skip extraction and go straight to GPU inference. Launching now (stopping the stale monitor first):

204. user (2026-06-13T08:46:26.632Z)

<tool_use_error>No task found with ID: bwiwctgbr</tool_use_error>

205. assistant (2026-06-13T08:46:30.649Z)

Old monitor already gone. Launching the pipeline and a fresh monitor:

206. user (2026-06-13T08:46:33.039Z)

Command running in background with ID: b09gv8s9s. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\b09gv8s9s.output. You will be notified when it completes. To check interim output, use Read on that file path.

207. user (2026-06-13T08:46:40.824Z)

Monitor started (task b53194x5t, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

208. assistant (2026-06-13T08:46:47.531Z)

Pipeline relaunched (`b09gv8s9s`) with a fresh milestone monitor. This run should sail past the extraction step (frames pre-staged, count-verified 46080) straight into ****GPU WASB inference**** ? the part that never got to run before. The keep-awake lock is holding the machine in working state so standby can't interrupt it.

```
Live log as always:
```powershell
Get-Content -Wait -Tail 40 C:\Users\avidu\Projects\badminton-highlight-
indexer\output\gx010125_run.log
```
```

I'll report when inference progress appears (the `win/s` / `% | ETA` lines from the detector pass), then at trajectory completion, Gemini confirmation, the >7-shot filter, and the final 4K stitch.

209. user (2026-06-13T08:46:47.674Z)

```
<task-notification>
<task-id>b53194x5t</task-id>
<summary>Monitor event: "GX010125 pipeline milestones (inference progress, phases, filter,
errors)"</summary>
<event>[CONFIG] segmenter=wasb_hybrid | min_shot_count=8 | begin_padding=0.5s end_padding=0.5s |
watermark=on
[HASH] computing MD5 of the 1080p proxy...</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
```

output doesn't need one.
</task-notification>

210. assistant (2026-06-13T08:46:53.505Z)

Routine startup. Watching for the "reusing 46080 already-extracted frames" line ? that's the confirmation the extraction-skip worked ? followed by inference progress.

211. user (2026-06-13T08:46:57.621Z)

```
<task-notification>
<task-id>b53194x5t</task-id>
<summary>Monitor event: "GX010125 pipeline milestones (inference progress, phases, filter,
errors)"</summary>
<event>[HASH] video_id=89605e80ed012ca259a640b7ec06cd80 (4s)
[INDEX] clean indexing (validators bypassed for the massive file, as in
run_process_and_stitch.py)...</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

212. assistant (2026-06-13T08:47:02.703Z)

Indexing started. Waiting for the GPU healthcheck and the extraction-skip confirmation.

213. user (2026-06-13T08:47:04.134Z)

```
<task-notification>
<task-id>b53194x5t</task-id>
<summary>Monitor event: "GX010125 pipeline milestones (inference progress, phases, filter,
errors)"</summary>
<event>14:16:49 [backend.pipeline.segmenters.trajectory_hybrid] INFO: WASB WSL env OK: torch
1.11.0+cu113 cuda True</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

214. assistant (2026-06-13T08:47:33.085Z)

GPU healthcheck passed. Important heads-up about visibility: the WASB inference runs as a *subprocess inside WSL*, so its progress logs are captured by the subprocess and **won't appear in the live run log until it finishes** ? that's precisely the "goes unresponsive" window you flagged. But `wasb\_infer.py` writes incremental progress to a checkpoint manifest on disk. Let me set up a monitor that polls that manifest so we both get live GPU inference progress:

215. user (2026-06-13T08:47:38.319Z)

Monitor started (task bnl9fplws, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

216. assistant (2026-06-13T08:47:42.685Z)

Now I have two monitors running: the run-log milestone watcher (`b53194x5t`) and the GPU detector-progress watcher (`bnl9fplws`) polling the checkpoint manifest. Between them, you'll see live progress through the whole inference phase ? no more silent gaps. I'll report as the detector windows tick up and at each phase transition.

217. user (2026-06-13T08:47:42.804Z)

```
<task-notification>
<task-id>bnl9fplws</task-id>
<summary>Monitor event: "WASB GPU detector progress via checkpoint manifest"</summary>
```

<event>WASB detector: 0/46078 windows (0%)</event>

If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.

</task-notification>

218. assistant (2026-06-13T08:47:53.231Z)

Inference is **live on the GPU** ? and this confirms the optimization worked:

`windows\_total=46078` with `windows\_done=0` means extraction was skipped entirely (frames_in=3, so 46080-2=46078 sliding windows). The run went straight to the detector pass that never got to start in any prior attempt.

The checkpoint updates every 1000 windows, so the next ticks will let me estimate the rate and give you a real ETA. The GPU phase is now visible in real time via the manifest monitor.

219. user (2026-06-13T08:47:59.156Z)

<task-notification>

<task-id>bnl9fppls</task-id>

<summary>Monitor event: "WASB GPU detector progress via checkpoint manifest"</summary>

<event>WASB detector: 1000/46078 windows (2%)</event>

If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.

</task-notification>

220. assistant (2026-06-13T08:48:18.517Z)

Progress is flowing (2%). Let me grab one timestamped anchor so I can compute the actual throughput and give you a real ETA (the per-window rate isn't in the manifest):

221. user (2026-06-13T08:48:24.027Z)

now=06/13/2026 14:18:23 manifest_mtime=06/13/2026 14:18:20 windows_done=2000 windows_total=46078
78 %, 5188 MiB, 73.88 W

222. assistant (2026-06-13T08:48:43.472Z)

Solid anchor: **2000 windows by 14:18:20**, GPU at 78% util / 5.2 GB VRAM / 74 W ? healthy and steady. That's roughly **25 windows/sec**, so the detector pass (46,078 windows) should finish in **~30 minutes, around 14:47-14:50**.

After that the sequence is quick ? variable:

- **Tracker pass** (CPU, deterministic) ? seconds ? writes the trajectory CSV.
- **Candidate windowing** ? instant ? turns the trajectory into candidate rallies.
- **Gemini confirmation** ? the next variable phase: each candidate rally gets uploaded to Gemini (serial, `ai\_max\_workers=1`) for semantic boundary + shot-count confirmation. Length depends on how many candidates WASB finds.
- **Filter to shot_count > 7**, then **stitch from the 4K original**, then I **ffprobe** the reel to confirm 3840x2160.

The detector progresses in the chat via the manifest monitor, so you'll see it tick toward 46,078. I'll flag the trajectory-complete and Gemini phases as they hit.

223. user (2026-06-13T08:48:43.679Z)

<task-notification>

<task-id>bnl9fppls</task-id>

<summary>Monitor event: "WASB GPU detector progress via checkpoint manifest"</summary>

<event>WASB detector: 2000/46078 windows (4%)</event>

If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.

</task-notification>

224. assistant (2026-06-13T08:48:51.637Z)

Tracking as expected (4%). I'll go quiet through the routine detector ticks ? you'll see them stream in the chat ? and speak up at trajectory completion, the Gemini phase, or if anything looks off.

225. user (2026-06-13T08:48:59.925Z)

```
<task-notification>
<task-id>bnl9fplws</task-id>
<summary>Monitor event: "WASB GPU detector progress via checkpoint manifest"</summary>
<event>WASB detector: 3000/46078 windows (6%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

226. assistant (2026-06-13T08:49:06.238Z)

(6% ? routine; holding for the next milestone.)

227. user (2026-06-13T08:49:40.447Z)

```
<task-notification>
<task-id>bnl9fplws</task-id>
<summary>Monitor event: "WASB GPU detector progress via checkpoint manifest"</summary>
<event>WASB detector: 4000/46078 windows (8%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

228. assistant (2026-06-13T08:49:55.943Z)

These per-1000 ticks are more notification noise than signal. Let me make the monitor coarser (every ~10%) so it reports meaningfully without spamming, while still catching completion and errors:

229. user (2026-06-13T08:49:56.288Z)

```
{"message":"Successfully stopped task: bnl9fplws (cache=\"/c/Users/avidu/Projects/badminton-highlight-indexer/output/wasb/GX010125_1080p__89605e80ed01_wasb_wasbcache/manifest.json\";
csv=\"/c/Users/avidu/Projects/badminton-highlight-indexer/output/wasb/GX010125_1080p__89605e80ed01_wasb.csv\"; prev=\"\"; waited=0; while true; do
if [ -f \"$csv\" ]; then echo \"WASB trajectory CSV WRITTEN -> detector+tracker done\"; break;
fi; if [ -f \"$cache\" ]; then cur=$(grep -o '\"windows_done\"[: ]*[0-9]*' \"$cache\" | grep -o '[0-9]*$'); tot=$(grep -o '\"windows_total\"[: ]*[0-9]*' \"$cache\" | grep -o '[0-9]*$'); if [ -n \"$cur\" ] && [ \"$cur\" != \"$prev\" ]; then pct=$(( cur*100/(tot>0?tot:1) )); echo \"WASB detector: $cur/$tot windows (${pct}%)\"; prev=$cur; fi; else waited=$((waited+20)); if [ $((waited%120)) -eq 0 ]; then echo \"waiting for detector cache to appear (${waited}s)...\"; fi; fi; sleep 20; done)\",\"task_id\":\"bnl9fplws\",\"task_type\":\"local_bash\",\"command\":\"cache=\"/c/Users/avidu/Projects/badminton-highlight-indexer/output/wasb/GX010125_1080p__89605e80ed01_wasb_wasbcache/manifest.json\";
csv=\"/c/Users/avidu/Projects/badminton-highlight-indexer/output/wasb/GX010125_1080p__89605e80ed01_wasb.csv\"; prev=\"\"; waited=0; while true; do
if [ -f \"$csv\" ]; then echo \"WASB trajectory CSV WRITTEN -> detector+tracker done\"; break;
fi; if [ -f \"$cache\" ]; then cur=$(grep -o '\"windows_done\"[: ]*[0-9]*' \"$cache\" | grep -o '[0-9]*$'); tot=$(grep -o '\"windows_total\"[: ]*[0-9]*' \"$cache\" | grep -o '[0-9]*$'); if [ -n \"$cur\" ] && [ \"$cur\" != \"$prev\" ]; then pct=$(( cur*100/(tot>0?tot:1) )); echo \"WASB detector: $cur/$tot windows (${pct}%)\"; prev=$cur; fi; else waited=$((waited+20)); if [ $((waited%120)) -eq 0 ]; then echo \"waiting for detector cache to appear (${waited}s)...\"; fi; fi; sleep 20; done\"}
```

230. user (2026-06-13T08:50:03.218Z)

Monitor started (task bq5a7u7db, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

231. assistant (2026-06-13T08:50:06.670Z)

Coarser monitor in place (reports? ~10% plus completion/errors). I'll surface the next update around the 30?40% mark, then at trajectory completion when the Gemini phase begins. Everything's healthy and on track for detector completion ~14:47?14:50.

232. user (2026-06-13T08:50:08.128Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 4000/46078 windows (0%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

233. assistant (2026-06-13T08:50:16.770Z)

(Detector in the 0?10% bucket; coarser monitor working. Holding for the 10% mark.)

234. user (2026-06-13T08:50:34.115Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 5000/46078 windows (10%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

235. assistant (2026-06-13T08:50:38.755Z)

(10% ? on pace. Next update around the 40?50% mark or trajectory completion.)

236. user (2026-06-13T08:53:05.297Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 10000/46078 windows (20%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

237. assistant (2026-06-13T08:53:07.756Z)

(20% ? steady. Holding.)

238. user (2026-06-13T08:55:06.156Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 14000/46078 windows (30%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

239. assistant (2026-06-13T08:55:12.154Z)

(30% ? tracking steady, no issues. ETA refresh at the halfway point.)

240. user (2026-06-13T08:58:07.668Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 19000/46078 windows (40%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

241. assistant (2026-06-13T08:58:20.479Z)

We're at 40% and the next phase (Gemini confirmation) is historically the flakier one ? past sessions hit 429/503 waves. Let me use this GPU wait productively and ****pre-flight the Gemini path now****, so we don't discover a dead key or empty credits ***after*** 30 minutes of inference. Checking how the provider authenticates and testing connectivity:

242. user (2026-06-13T08:58:23.063Z)

```
18: def __init__(self, api_key: str, model_name: str):
19:     super().__init__(api_key, model_name)
26:     self._client = genai.Client(api_key=self.api_key)
```

243. user (2026-06-13T08:58:26.966Z)

env keys present: GEMINI_API_KEY

244. assistant (2026-06-13T08:58:35.152Z)

`GEMINI\_API\_KEY` is present and the provider uses `genai.Client`. Let me verify it actually authenticates and has quota ? a free `models.list` call (no generation cost), via a small script to avoid quoting issues:

245. assistant (2026-06-13T08:58:38.505Z)

```
"""Lightweight Gemini connectivity + quota preflight (no generation cost)."""
import os
import sys

sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
from dotenv import load_dotenv

load_dotenv()
key = os.environ.get("GEMINI_API_KEY")
if not key:
    raise SystemExit("GEMINI_API_KEY not set")
print(f"key present: ...{key[-6:]} (len {len(key)})")

from google import genai

client = genai.Client(api_key=key)
models = list(client.models.list())
flash = [m.name for m in models if "flash-latest" in m.name]
print(f"models.list OK: {len(models)} models")
print(f"gemini-flash-latest available: {bool(flash)} -> {flash[:2]}")
```

246. user (2026-06-13T08:58:38.522Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\probe_gemini.py (file state is current in your context ? no need to Read it back)

247. user (2026-06-13T08:58:47.858Z)

```
key present: ...ZHex2Q (len 53)
models.list OK: 55 models
```

gemini-flash-latest available: True -> ['models/gemini-flash-latest']

248. assistant (2026-06-13T08:59:02.105Z)

****Gemini preflight passed**** ? key authenticates (55 models visible), `gemini-flash-latest` available. So the confirmation phase won't hit a dead key; if it meets the usual 503 capacity waves, the provider already has 6-attempt backoff. That de-risks the next phase.

Detector is ~40% and on pace ? trajectory completion still looks like ****~14:48****. After it: tracker ? candidate windows ? Gemini confirmation (now verified ready) ? >7-shot filter ? 4K stitch. Holding until ~50% or trajectory completion.

249. user (2026-06-13T09:00:38.814Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 24000/46078 windows (50%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

250. assistant (2026-06-13T09:00:43.537Z)

(50% ? halfway, on schedule for ~14:48. Holding for trajectory completion.)

251. user (2026-06-13T09:03:10.063Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 28000/46078 windows (60%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

252. assistant (2026-06-13T09:03:12.604Z)

(60% ? steady.)

253. user (2026-06-13T09:05:41.193Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 33000/46078 windows (70%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

254. assistant (2026-06-13T09:05:43.670Z)

(70% ? on track for ~14:48.)

255. user (2026-06-13T09:08:12.415Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 37000/46078 windows (80%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

256. assistant (2026-06-13T09:08:22.926Z)

(80% ? ~6 min to trajectory completion. The Gemini phase logs run in the main process, so they'll stream live; I'll report the candidate-window count when it lands, which sets the Gemini-phase ETA.)

257. user (2026-06-13T09:11:13.763Z)

```
<task-notification>
<task-id>bq5a7u7db</task-id>
<summary>Monitor event: "WASB detector progress (every ~10%) + completion"</summary>
<event>WASB detector: 42000/46078 windows (90%)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

258. assistant (2026-06-13T09:11:16.702Z)

(90% ? trajectory completion imminent.)

259. user (2026-06-13T09:13:02.535Z)

```
<task-notification>
<task-id>b53194x5t</task-id>
<summary>Monitor event: "GX010125 pipeline milestones (inference progress, phases, filter,
errors)"</summary>
<event>14:43:01 [backend.pipeline.segmenters.trajectory_hybrid] INFO: Parsed 46080 trajectory
points (29879 visible).
14:43:01 [backend.pipeline.segmenters.trajectory_hybrid] INFO: Phase 2 (WASB) completed in
1572.2s. Candidate windows: 18
14:43:01 [backend.pipeline.segmenters.trajectory_hybrid] INFO: Phase 3: gemini validation on 18
candidate windows...</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

260. assistant (2026-06-13T09:13:20.011Z)

****Trajectory complete**** ? and the signal quality is good:

- ****46,080 points parsed, 29,879 visible (64.8% shuttle visibility)**** ? squarely in the healthy badminton range, so WASB tracked the shuttle well on this footage.
- Phase 2 (WASB GPU) took ~26 min.
- ****18 candidate rallies**** detected.
- ****Phase 3: Gemini confirmation on 18 windows is now running**** (serial).

18 serial Gemini calls (clip extract + upload + confirm each) ? ****~15?25 min****, plus any 503 backoff. Let me check whether Phase 3 logs per-window progress or goes quiet ? if quiet, I'll add a DB-poll monitor so we keep visibility through this phase too:

261. user (2026-06-13T09:13:21.514Z)

```
196         handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
197         os.makedirs(handoff_dir, exist_ok=True)
198         logger.info("Phase 3: %s validation on %d candidate windows...",
199                     provider_name, len(all_candidate_windows))
200
201     def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
202         w_start = max(0, window["start"] - handoff_padding)
203         w_end = min(video_duration, window["end"] + handoff_padding)
204         w_dur = w_end - w_start
205         w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
206         if not extract_video_chunk(video_path, w_start, w_dur, w_path,
reencode=True):
207             return []
```

```

208         prompt = sport_profile.get_prompt(chunk_duration=w_dur)
209         segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
210                                             label=f"Handoff {wi+1}")
211         valid = []
212         for seg in segments:
213             try:
214                 seg["start_time"] = float(seg["start_time"]) + w_start
215                 seg["end_time"] = float(seg["end_time"]) + w_start
216                 seg["_candidate_id"] = candidate_ids[wi]
217                 valid.append(seg)
218             except (ValueError, TypeError, KeyError) as e:
219                 logger.warning("Dropping segment with unparseable timestamps: %s ?
%s", seg, e)
220             try:
221                 os.remove(w_path)
222             except OSError:
223                 pass
224         return valid
225
226     ai_segments: List[dict] = []
227     with concurrent.futures.ThreadPoolExecutor(max_workers=ai_max_workers) as
executor:
228         futures = [executor.submit(process_candidate_window, wi, cw)
229                     for wi, cw in enumerate(all_candidate_windows)]
230         for future in futures:
231             ai_segments.extend(future.result())
232     try:
233         os.rmdir(handoff_dir)
234     except OSError:
235         pass
236     logger.info("Phase 3 completed. AI confirmed %d semantic play segments.",
len(ai_segments))
237
238     # --- Phase 4: DB population + candidate linking ---
239     segments_data = []
240     for segment in ai_segments:
241         try:
242             start = float(segment["start_time"])
243             end = float(segment["end_time"])
244             dur = end - start
245             metadata = {
246                 "shot_count": self._shot_count_from(segment, dur),
247                 "shot_count_confidence": segment.get("shot_count_confidence",
"Unknown"),
248                 "detector": f"{self.family}-hybrid-{model_name}",
249             }
250             segment_id = self.db.add_play_segment(video_id, start, end, None, None,
metadata)
251             if segment_id:
252                 candidate_id = segment.get("_candidate_id")
253                 if candidate_id is not None:
254                     self.db.link_candidate_to_segment(candidate_id, segment_id)
255                 segments_data.append({
256                     "id": segment_id, "start_time": start, "end_time": end,
257                     "duration": dur, "metadata": metadata,
258                 })
259             except Exception as e:
260                 logger.warning("Skipping invalid segment struct: %s", e)
261
262     return segments_data
263

```

262. assistant (2026-06-13T09:13:49.702Z)

Trajectory CSV is written (detector monitor closed cleanly). Now the key observability insight

for Phase 3: looking at the code (lines 226?236), **segments are only written to the DB in Phase 4, *after* all 18 Gemini calls finish** ? so the DB won't show incremental progress. The reliable live signal is the per-window handoff clip: each candidate extracts `handoff\_<id>\_<wi>.mp4`, processes it, deletes it (lines 205?223). Let me watch that directory so we see Gemini tick through the 18 windows:

263. user (2026-06-13T09:13:53.824Z)

Monitor started (task b9hizl8yk, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

264. assistant (2026-06-13T09:14:02.588Z)

Visibility for Phase 3 is now wired up. Two monitors active:

- **Per-window Gemini progress** (`b9hizl8yk`) ? watches the handoff clips, reports "confirming window N/18" as it goes.
- **Run-log milestones** (`b53194x5t`) ? catches "Phase 3 completed", the shot-count filter (KEEP/drop per rally), the final stitch, and any 503/error.

So you'll see Gemini tick through the 18 candidates rather than a silent gap. When Phase 3 finishes you'll get the per-rally shot counts and the >7 filter decisions, then the 4K stitch. Expect ~15?25 min for the Gemini pass.

265. user (2026-06-13T09:14:04.525Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 1/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

266. assistant (2026-06-13T09:14:13.782Z)

(Window 1/18 ? Gemini phase underway. Holding; will report progress and any 503 backoff.)

267. user (2026-06-13T09:14:15.321Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 2/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

268. assistant (2026-06-13T09:14:27.233Z)

(2/18 ? moving well.)

269. user (2026-06-13T09:16:11.158Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 3/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

270. assistant (2026-06-13T09:16:17.447Z)

(3/18.)

271. user (2026-06-13T09:16:56.927Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 4/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

272. assistant (2026-06-13T09:17:04.836Z)

(4/18 ? steady, no backoff so far.)

273. user (2026-06-13T09:17:57.478Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 5/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

274. assistant (2026-06-13T09:18:05.808Z)

(5/18.)

275. user (2026-06-13T09:18:12.927Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 6/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

276. assistant (2026-06-13T09:18:17.757Z)

(6/18 ? a third done, all clean.)

277. user (2026-06-13T09:19:28.653Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 7/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

278. assistant (2026-06-13T09:19:32.211Z)

(7/18.)

279. user (2026-06-13T09:19:58.967Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
```

<event>Gemini confirming candidate window 8/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

280. assistant (2026-06-13T09:20:01.579Z)

(8/18 ? approaching halfway.)

281. user (2026-06-13T09:20:29.269Z)

<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 9/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

282. assistant (2026-06-13T09:20:32.950Z)

(9/18 ? halfway through the Gemini pass, no 503s. On track for the filter + 4K stitch in ~8?12 min.)

283. user (2026-06-13T09:20:44.727Z)

<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 10/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

284. assistant (2026-06-13T09:20:47.346Z)

(10/18.)

285. user (2026-06-13T09:21:15.170Z)

<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 11/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

286. assistant (2026-06-13T09:21:18.311Z)

(11/18.)

287. user (2026-06-13T09:21:45.848Z)

<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 12/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

288. assistant (2026-06-13T09:21:48.563Z)

(12/18 ? two-thirds done.)

289. user (2026-06-13T09:22:31.524Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 13/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

290. assistant (2026-06-13T09:22:36.028Z)

(13/18.)

291. user (2026-06-13T09:23:47.412Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 14/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

292. assistant (2026-06-13T09:23:50.443Z)

(14/18.)

293. user (2026-06-13T09:24:17.915Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 15/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

294. assistant (2026-06-13T09:24:20.293Z)

(15/18 ? home stretch.)

295. user (2026-06-13T09:25:33.652Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 16/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

296. assistant (2026-06-13T09:25:37.006Z)

(16/18.)

297. user (2026-06-13T09:26:04.273Z)

```
<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
```

<event>Gemini confirming candidate window 17/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

298. assistant (2026-06-13T09:26:08.408Z)

(17/18 ? last one, then Phase 3 wraps and the shot-count filter + 4K stitch follow.)

299. user (2026-06-13T09:27:20.356Z)

<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Gemini confirming candidate window 18/18</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

300. assistant (2026-06-13T09:27:27.871Z)

(18/18 done ? Phase 3 wrapping. Next: DB populate ? my runner prints each rally's shot count with KEEP/drop at the >7 threshold ? stitch from the 4K original. Watching for those.)

301. user (2026-06-13T09:27:50.023Z)

<task-notification>
<task-id>b53194x5t</task-id>
<summary>Monitor event: "GX010125 pipeline milestones (inference progress, phases, filter, errors)"</summary>
<event>14:57:49 [backend.pipeline.segmenters.trajectory_hybrid] INFO: Phase 3 completed. AI confirmed 52 semantic play segments.</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

302. assistant (2026-06-13T09:27:57.821Z)

****Phase 3 done ? Gemini confirmed 52 rallies**** from the 18 candidate windows (windows get split into individual rallies). Now the DB populates and my runner applies the ****shot_count > 7 filter**** (``min_shot_count=8``), printing each rally's count with KEEP/drop, then stitches the survivors from the 4K original. Watching for the filter breakdown and stitch.

303. user (2026-06-13T09:27:59.352Z)

<task-notification>
<task-id>b9hizl8yk</task-id>
<summary>Monitor event: "Gemini Phase 3 per-window progress via handoff dir"</summary>
<event>Phase 3 handoff dir gone -> Gemini phase finishing</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign output doesn't need one.
</task-notification>

304. assistant (2026-06-13T09:28:05.012Z)

(Handoff monitor closed ? Phase 4 DB write, then the filter + stitch. Standing by for the KEEP/drop breakdown and the reel.)

305. user (2026-06-13T09:28:07.511Z)

<task-notification>
<task-id>b9hizl8yk</task-id>
<tool-use-id>toolu_016CawjkX1WvH91UjV2YrWFM</tool-use-id>

```
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-
highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\b9hizl8yk.output</output-file>
<status>completed</status>
<summary>Monitor "Gemini Phase 3 per-window progress via handoff dir" stream ended</summary>
</task-notification>
```

306. assistant (2026-06-13T09:28:21.228Z)

(Phase 3 fully closed. Awaiting the >7-shot filter result ? that's the headline number ? then the 4K stitch and `[DONE]`.)

307. user (2026-06-13T09:28:23.075Z)

```
<task-notification>
<task-id>b53194x5t</task-id>
<summary>Monitor event: "GX010125 pipeline milestones (inference progress, phases, filter,
errors)"</summary>
<event>[INDEX] 52 play segments indexed (total 2468s).
segment id=288 [ 6.80s - 16.50s] shot_count=10 confidence=Medium -&gt; KEEP
segment id=289 [ 26.00s - 36.00s] shot_count=10 confidence=Medium -&gt; KEEP
segment id=290 [ 42.00s - 48.00s] shot_count=6 confidence=Medium -&gt; drop
segment id=291 [ 53.00s - 57.00s] shot_count=4 confidence=Medium -&gt; drop
segment id=292 [ 67.09s - 85.09s] shot_count=14 confidence=Medium -&gt; KEEP
segment id=293 [ 88.59s - 110.09s] shot_count=16 confidence=Medium -&gt; KEEP
segment id=294 [ 128.59s - 132.09s] shot_count=3 confidence=High -&gt; drop
segment id=295 [ 141.09s - 145.09s] shot_count=4 confidence=High -&gt; drop
segment id=296 [ 180.09s - 192.59s] shot_count=11 confidence=Medium -&gt; KEEP
segment id=297 [ 202.35s - 205.35s] shot_count=3 confidence=High -&gt; drop
segment id=298 [ 211.15s - 215.85s] shot_count=4 confidence=High -&gt; drop
segment id=299 [ 221.35s - 225.35s] shot_count=4 confidence=High -&gt; drop
segment id=300 [ 230.85s - 237.35s] shot_count=6 confidence=High -&gt; drop
segment id=301 [ 246.25s - 249.75s] shot_count=4 confidence=High -&gt; drop
segment id=302 [ 255.75s - 259.25s] shot_count=4 confidence=High -&gt; drop
segment id=303 [ 266.25s - 269.25s] shot_count=4 confidence=High -&gt; drop
segment id=304 [ 275.75s - 282.75s] shot_count=8 confidence=High -&gt; KEEP
segment id=305 [ 288.47s - 290.97s] shot_count=2 confidence=High -&gt; drop
segment id=306 [ 296.41s - 298.71s] shot_count=3 confidence=High -&gt; drop
segment id=307 [ 305.01s - 316.01s] shot_count=9 confidence=High -&gt; KEEP
segment id=308 [ 322.51s - 330.01s] shot_count=6 confidence=High -&gt; drop
segment id=309 [ 336.51s - 340.01s] shot_count=2 confidence=High -&gt; drop
segment id=310 [ 347.60s - 351.30s] shot_count=3 confidence=High -&gt; drop
segment id=311 [ 357.80s - 365.80s] shot_count=6 confidence=High -&gt; drop
segment id=312 [ 371.24s - 374.04s] shot_count=4 confidence=High -&gt; drop
segment id=313 [ 381.54s - 389.54s] shot_count=6 confidence=High -&gt; drop
segment id=314 [ 390.73s - 398.13s] shot_count=8 confidence=High -&gt; KEEP
segment id=315 [ 403.80s - 405.80s] shot_count=3 confidence=High -&gt; drop
segment id=316 [ 410.00s - 413.20s] shot_count=4 confidence=High -&gt; drop
segment id=317 [ 418.94s - 422.54s] shot_count=4 confidence=High -&gt; drop
segment id=318 [ 431.24s - 434.94s] shot_count=4 confidence=High -&gt; drop
segment id=319 [ 441.21s - 451.71s] shot_count=11 confidence=High -&gt; KEEP
segment id=320 [ 458.21s - 460.71s] shot_count=3 confidence=High -&gt; drop
segment id=321 [ 481.12s - 492.62s] shot_count=7 confidence=High -&gt; drop
segment id=322 [ 501.12s - 504.62s] shot_count=3 confidence=High -&gt; drop
segment id=323 [ 512.12s - 516.62s] shot_count=3 confidence=High -&gt; drop
segment id=324 [ 523.62s - 528.12s] shot_count=3 confidence=High -&gt; drop
segment id=325 [ 541.12s - 548.62s] shot_count=5 confidence=High -&gt; drop
segmen
...(truncated)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

308. assistant (2026-06-13T09:28:34.310Z)

The filter is working exactly as specified ? and note ****segment 321 (shot_count=7) was dropped****, confirming "more than 7" correctly excludes 7 (threshold is 8). The event was truncated at 52 segments; let me read the full log to get the complete KEEP list and the filter summary:

309. user (2026-06-13T09:28:43.743Z)

```
--- KEEP lines ---
segment id=288 [ 6.80s - 16.50s] shot_count=10 confidence=Medium -> KEEP
segment id=289 [ 26.00s - 36.00s] shot_count=10 confidence=Medium -> KEEP
segment id=292 [ 67.09s - 85.09s] shot_count=14 confidence=Medium -> KEEP
segment id=293 [ 88.59s - 110.09s] shot_count=16 confidence=Medium -> KEEP
segment id=296 [ 180.09s - 192.59s] shot_count=11 confidence=Medium -> KEEP
segment id=304 [ 275.75s - 282.75s] shot_count=8 confidence=High -> KEEP
segment id=307 [ 305.01s - 316.01s] shot_count=9 confidence=High -> KEEP
segment id=314 [ 390.73s - 398.13s] shot_count=8 confidence=High -> KEEP
segment id=319 [ 441.21s - 451.71s] shot_count=11 confidence=High -> KEEP
segment id=330 [ 615.43s - 627.43s] shot_count=9 confidence=High -> KEEP
segment id=332 [ 648.93s - 661.43s] shot_count=12 confidence=High -> KEEP
--- FILTER / DONE / errors ---
[FILTER] config min_shot_count=8: kept 11, dropped 41
14:57:49 [backend.pipeline.stitcher.compiler] INFO: Source video duration: 768.77s
14:57:49 [backend.pipeline.stitcher.compiler] INFO: Trimming 11 segments (begin_padding=0.5s,
end_padding=0.5s)...
14:57:49 [backend.pipeline.stitcher.compiler] INFO: [watermark] applying branding overlay to
each clip.
14:57:49 [backend.pipeline.stitcher.compiler] INFO: Trimming Segment #1/11 [6.80s - 16.50s] ->
[6.300s - 17.000s] (duration: 10.70s)
14:57:49 [backend.pipeline.stitcher.compiler] WARNING: Segment #1/11 [6.80s - 16.50s]: watermark
encode failed; retrying without it. ffmpeg: nd/assets/fonts/RobotoSlab-Regular.ttf':fontcolor=wh
ite@0.85;fontsize=h/29;x=w-tw-24;y=h-th-24:box=1:boxcolor=black@0.50:boxborderw=10' around:
Error opening output file C:\Users\avidu\Projects\badminton-highlight-
indexer\output\tmp98nqyb02\clip_0.mp4.
Error opening output files: Invalid argument
14:58:17 [backend.pipeline.stitcher.compiler] WARNING: Segment #1/11 [6.80s - 16.50s]: encoded
WITHOUT the watermark (the watermark filter failed).
14:58:17 [backend.pipeline.stitcher.compiler] INFO: Trimming Segment #2/11 [26.00s - 36.00s] ->
[25.500s - 36.500s] (duration: 11.00s)
14:58:17 [backend.pipeline.stitcher.compiler] WARNING: Segment #2/11 [26.00s - 36.00s]:
watermark encode failed; retrying without it. ffmpeg: nd/assets/fonts/RobotoSlab-Regular.ttf':fo
ntcolor=white@0.85;fontsize=h/29;x=w-tw-24;y=h-th-24:box=1:boxcolor=black@0.50:boxborderw=10'
around:
Error opening output file C:\Users\avidu\Projects\badminton-highlight-
indexer\output\tmp98nqyb02\clip_1.mp4.
Error opening output files: Invalid argument
```

310. assistant (2026-06-13T09:29:27.028Z)

We have the answer. The ****>7-shot filter kept 11 of 52 rallies**** (``min_shot_count=8``, so "more than 7" correctly excludes 7 ? e.g. segment 321 at exactly 7 shots was dropped):

#	Time (4K source)	Shots	Confidence
1	6.8?16.5s	10	Medium
2	26.0?36.0s	10	Medium
3	67.1?85.1s	14	Medium
4	88.6?110.1s	16	Medium
5	180.1?192.6s	11	Medium
6	275.8?282.8s	8	High
7	305.0?316.0s	9	High
8	390.7?398.1s	8	High
9	441.2?451.7s	11	High
10	615.4?627.4s	9	High
11	648.9?661.4s	12	High

****Stitching is running now from the 4K original**** (11 clips, ~28s each to re-encode ? ~5 min).
The padding fix is visibly working too (`begin\_padding=0.5s, end\_padding=0.5s` from config, e.g. 6.8?6.3 / 16.5?17.0).

?? ****One real bug surfaced:**** the default watermark fails on every clip ?
`fontfile=...assets/fonts/RobotoSlab-Regular.ttf ... Invalid argument` (the bundled-font path from PR #103 doesn't resolve / isn't escaped for ffmpeg on Windows). The compiler's fallback is working correctly ? it re-encodes each clip ****without**** the watermark, so your reel is unaffected ? but the on-by-default branding is silently disabled. Since you didn't ask for branding this doesn't block the result; I'll investigate and propose a fix as a follow-up once the reel lands. Waiting for `[DONE]`, then I'll `ffprobe` to confirm it's true 3840×2160.

311. user (2026-06-13T09:33:38.573Z)

```
<task-notification>
<task-id>b09gv8s9s</task-id>
<tool-use-id>toolu_017XvPXnnbRWDGNbkoBKhhau</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\b09gv8s9s.output</output-file>
<status>completed</status>
<summary>Background command "Launch E2E pipeline (frames pre-extracted, GPU inference next)"
completed (exit code 0)</summary>
</task-notification>
```

312. assistant (2026-06-13T09:33:41.618Z)

The pipeline exited 0. Let me confirm the reel and verify it's true 4K from the original:

313. user (2026-06-13T09:33:47.126Z)

```
15:03:37 [backend.pipeline.stitcher.compiler] INFO: Dynamic highlight reel saved:
output\GX010125_highlights.mp4 (1151.9 MB)
[DONE] highlight reel: C:\Users\avidu\Projects\badminton-highlight-indexer\output\GX010125_highlights.mp4 (total 2816s)
--- REEL PROBE ---
codec_name=h264
width=3840
height=2160
r_frame_rate=60000/1001
duration=143.326733
size=1207870728
size_MB=1151.9
```

314. assistant (2026-06-13T09:34:03.122Z)

****Done ? and verified true 4K.**** The reel probes at ****3840×2160, h264, 59.94fps, 143.3s, 1151.9 MB**** ? the 4K resolution confirms it was cut from the original `F:\GX010125.MP4`, not the proxy. Let me release the keep-awake lock, stop the monitors, and notify you:

315. user (2026-06-13T09:34:07.501Z)

```
{"message": "Successfully stopped task: brysrtrco (powershell -NoProfile -ExecutionPolicy Bypass -File C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\scratch\\keep_away.ps1)", "task_id": "brysrtrco", "task_type": "local_bash", "command": "powershell -NoProfile -ExecutionPolicy Bypass -File C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\scratch\\keep_away.ps1"}
```

316. user (2026-06-13T09:34:08.325Z)

```
{"message": "Successfully stopped task: b53194x5t (tail -n +1 -f \"/c/Users/avidu/Projects/badminton-highlight-indexer/output/gx010125_run.log\" | grep -E --line-buffered \"\\[\\[CONFIG\\]\\[\\[HASH\\]\\[\\[INDEX\\]\\[\\[CACHE\\]\\[\\[FILTER\\]\\[\\[DONE\\]]|Traceback|ERROR|Exception|WSL env|reusing|already-extracted|detecting|win/s|detection done|tracker|wrote .* rows|visible-shuttle|trajectory points|Phase 2|Phase 3|Candidate|confirmed|oversize|SUMMARY|COMPILER|KEEP|drop|SystemExit|no play|no
```

```
segments\\"", "task_id": "b53194x5t", "task_type": "local_bash", "command": "tail -n +1 -f  
\\\"/c/Users/avidu/Projects/badminton-highlight-indexer/output/gx010125_run.log\\\" | grep -E  
--line-buffered \\\"\\\"[CONFIG\\\"]\\\"[HASH\\\"]\\\"[INDEX\\\"]\\\"[CACHE\\\"]\\\"[FILTER\\\"]\\\"[DONE\\\"]|Traceb  
ack|ERROR|Exception|WSL env|reusing|already-extracted|detecting|win/s|detection  
done|tracker|wrote .* rows|visible-shuttle|trajectory points|Phase 2|Phase  
3|Candidate|confirmed|oversize|SUMMARY|COMPILER|KEEP|drop|SystemExit|no play|no segments\\\"\"}
```

317. assistant (2026-06-13T09:34:09.069Z)

GX010125 highlights done: 11 rallies >7 shots, stitched from 4K (3840x2160, 2:23).
output\\GX010125_highlights.mp4

318. user (2026-06-13T09:34:09.072Z)

Mobile push not sent (Remote Control inactive).

319. assistant (2026-06-13T09:34:34.385Z)

Cleanup done; the run is fully complete. Here's the bottom line, then I'll dig into that watermark bug.

? Delivered: `output\\GX010125\_highlights.mp4`

- ****11 rally highlights, every one with more than 7 shots**** (8?16 shots), drawn from 52 detected rallies in your 12.8-min match.
- ****True 4K****: 3840x2160, h264, 59.94 fps, ****2:23 long****, 1.15 GB ? stitched from the ****original**** `F:\\GX010125.MP4` (the 4K resolution is the proof), using config paddings (0.5s/0.5s).
- ****GPU end-to-end****: WASB shuttle-tracking ran on the RTX 2070 Super via WSL (64.8% shuttle visibility), Gemini confirmed rally boundaries + shot counts, the config `min\_shot\_count=8` filter selected ">7".

****On reliability**** the two overnight failures were Modern Standby suspending the machine, not the code ? the keep-awake lock fixed that, and ffmpeg pre-extraction (replacing the slow cv2 path that was actually where it kept dying) got the full run through in one pass.

Now let me diagnose that watermark font bug properly ? it's a real defect worth a fix under your "merge good engineering fixes" standing:

320. user (2026-06-13T09:34:34.677Z)

```
84         capture_output=True, text=True, check=True
85     )
86     return float(result.stdout.strip())
87 except Exception as e:
88     logger.warning(f"Could not determine video duration via ffprobe: {e}")
89     return 0.0
90
91 def _watermark_filter(self, tmp_dir: str) -> Optional[str]:
92     """Build the ffmpeg -vf filtergraph that burns the configured watermark (text
93     and/or logo PNG) into a clip during the per-clip re-encode. Returns None when
94     watermarking is disabled or has nothing to draw. The concat stage stays -c copy
95     because the mark is already baked into every clip identically."""
96     wm = self.watermark
97     if not wm or not wm.get("enabled"):
98         return None
99
100     text = (wm.get("text") or "").strip()
101     logo_path = (wm.get("logo_path") or "").strip()
102     if logo_path and not os.path.exists(logo_path):
103         logger.warning(f"[watermark] logo not found: {logo_path} ? skipping logo
104         overlay.")
105         logo_path = ""
106     if not text and not logo_path:
```

```

106         return None
107
108     margin = int(wm.get("margin", 24))
109     opacity = float(wm.get("opacity", 0.85))
110     position = str(wm.get("position", "bottom-right"))
111     # drawtext uses lowercase w/h (main) + tw/th (text); overlay uses W/H (main) +
w/h (logo).
112     dt_xy = {
113         "bottom-right": f"x=w-tw-{margin}:y=h-th-{margin}",
114         "bottom-left": f"x={margin}:y=h-th-{margin}",
115         "top-right": f"x=w-tw-{margin}:y={margin}",
116         "top-left": f"x={margin}:y={margin}",
117     }.get(position, f"x=w-tw-{margin}:y=h-th-{margin}")
118     ov_xy = {
119         "bottom-right": f"W-w-{margin}:H-h-{margin}",
120         "bottom-left": f"{margin}:H-h-{margin}",
121         "top-right": f"W-w-{margin}:{margin}",
122         "top-left": f"{margin}:{margin}",
123     }.get(position, f"W-w-{margin}:H-h-{margin}")
124
125     drawtext = ""
126     if text:
127         ratio = float(wm.get("font_size_ratio") or 0.035)
128         divisor = max(1, round(1.0 / ratio))
129         # textfile= sidesteps the notorious drawtext text= escaping for arbitrary
strings.
130         tf_path = os.path.join(tmp_dir, "watermark.txt")
131         with open(tf_path, "w", encoding="utf-8") as fh:
132             fh.write(text)
133         # Font precedence: explicit font_path ? bundled Apache-2.0 default (renders
134         # without fontconfig) ? fontconfig family (last-resort, environment-
dependent).
135         font_path = (wm.get("font_path") or "").strip()
136         if not font_path and _BUNDLED_FONT.exists():
137             font_path = str(_BUNDLED_FONT)
138         if font_path:
139             font_opt = f"fontfile='{self._ff_quote(font_path)}'"
140         else:
141             font_opt = f"font='{wm.get('font', 'Sans')}'"
142         box = ""
143         if wm.get("box", True):
144             box = f"box=1:boxcolor=black@{min(opacity, 0.5):.2f}:boxborderw=10"
145         drawtext = (
146             f"drawtext=textfile='{self._ff_quote(tf_path)}':{font_opt}"
147             f":fontcolor=white@{opacity:.2f}:fontsize=h/{divisor}:{dt_xy}{box}"
148         )
149
150     if logo_path and drawtext:
151         return
152     f"movie='{self._ff_quote(logo_path)}'[wm];[in][wm]overlay={ov_xy},{drawtext}"
153     if logo_path:
154         return f"movie='{self._ff_quote(logo_path)}'[wm];[in][wm]overlay={ov_xy}"
155     return drawtext
156
157     def compile_highlights(self, raw_video_path: str, segments: List[Tuple[float,
float]], output_filename: str) -> str:
158         """
159         Extracts selected segments from a raw video and stitches them together

```

321. assistant (2026-06-13T09:35:21.820Z)

The root cause is clear from the code. `textfile=` uses a path under `tmp_dir` (relative `output/tmp?` ? no drive letter), which is why text never errored. But `_BUNDLED_FONT` (added in PR #103) is an **absolute** path `C:\?\RobotoSlab-Regular.ttf` ? `_ff_quote` turns it into `C:/?` and the **drive-letter colon** breaks ffmpeg's drawtext option parsing** (``` separates

filter options). Let me confirm `\_BUNDLED\_FONT` and reproduce/verify the fix empirically before touching code:

322. user (2026-06-13T09:35:23.356Z)

```
14:_BUNDLED_FONT = Path(__file__).resolve().parents[2] / "assets" / "fonts" / "RobotoSlab-
Regular.ttf"
136:         if not font_path and _BUNDLED_FONT.exists():
137:             font_path = str(_BUNDLED_FONT)
```

323. user (2026-06-13T09:35:26.950Z)

```
bundled font exists: True
size: 251880 bytes
```

324. assistant (2026-06-13T09:35:44.603Z)

Confirmed: `\_BUNDLED\_FONT` is absolute (`C:\?`) and exists. Now let me reproduce the failure and verify the fix empirically ? mirroring exactly how the compiler invokes ffmpeg (subprocess list, no shell), testing the raw `C:/` path vs. a colon-escaped one:

325. assistant (2026-06-13T09:35:52.646Z)

"""Reproduce the watermark drawtext failure + verify the colon-escape fix.

Mirrors VideoCompiler exactly: ffmpeg invoked via subprocess list (no shell), so only ffmpeg filter-level escaping matters, not shell escaping.

"""

```
import subprocess
import tempfile
import os
```

```
FONT = r"C:\Users\avidu\Projects\badminton-highlight-indexer\backend\assets\fonts\RobotoSlab-
Regular.ttf"
```

```
def ff_quote_current(path: str) -> str:
    # current VideoCompiler._ff_quote
    return path.replace("\\", "/").replace("'", r"\'")
```

```
def ff_quote_fixed(path: str) -> str:
    # proposed: also escape the drive-letter (and any) colon for ffmpeg filter parsing
    return path.replace("\\", "/").replace("'", r"\'").replace(":", r"\:")
```

```
def try_filter(label: str, font_quoted: str) -> None:
    out = os.path.join(tempfile.gettempdir(), f"wm_{label}.mp4")
    vf = (f"drawtext=text='hi':fontfile='{font_quoted}'"
f":fontcolor=white@0.85:fontsize=20:x=10:y=10:box=1:boxcolor=black@0.50:boxborderw=10")
    cmd = ["ffmpeg", "-y", "-f", "lavfi", "-i", "testsrc=size=320x240:duration=1:rate=10",
        "-vf", vf, "-frames:v", "3", out]
    r = subprocess.run(cmd, capture_output=True)
    ok = r.returncode == 0 and os.path.exists(out) and os.path.getsize(out) > 0
    print(f"[{label}] vf={vf}")
    print(f"[{label}] returncode={r.returncode} ok={ok}")
    if not ok:
        print(f"[{label}] stderr tail: {r.stderr.decode(errors='replace')[-300:]}")
    print()
```

```
print("CURRENT _ff_quote (raw C:/ colon):")
try_filter("current", ff_quote_current(FONT))
print("FIXED _ff_quote (escaped C\\:/ colon):")
```



```
try_filter("fixed", ff_quote_fixed(FONT))
```

326. user (2026-06-13T09:35:52.666Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\repro_drawtext_font.py (file state is current in your context ? no need to Read it back)

327. user (2026-06-13T09:35:58.252Z)

```
CURRENT _ff_quote (raw C:/ colon):
[current] vf=drawtext=text='hi':fontfile='C:/Users/avidu/Projects/badminton-highlight-indexer/backend/assets/fonts/RobotoSlab-Regular.ttf':fontcolor=white@0.85:fontsize=20:x=10:y=10:box=1:boxcolor=black@0.50:boxborderw=10
[current] returncode=4294967274 ok=False
[current] stderr tail: /Projects/badminton-highlight-indexer/backend/assets/fonts/RobotoSlab-Regular.ttf':fontcolor=white@0.85:fontsize=20:x=10:y=10:box=1:boxcolor=black@0.50:boxborderw=10'
around:
```

Error opening output file C:\Users\avidu\AppData\Local\Temp\wm_current.mp4.

Error opening output files: Invalid argument

```
FIXED _ff_quote (escaped C\:/ colon):
[fixed] vf=drawtext=text='hi':fontfile='C\:/Users/avidu/Projects/badminton-highlight-indexer/backend/assets/fonts/RobotoSlab-Regular.ttf':fontcolor=white@0.85:fontsize=20:x=10:y=10:box=1:boxcolor=black@0.50:boxborderw=10
[fixed] returncode=0 ok=True
```

328. assistant (2026-06-13T09:36:36.583Z)

```
**Root cause + fix both confirmed empirically:**
- Current (`fontfile='C:/?`'): fails with the *exact* production error (`Invalid argument`,
"around:").
- Fixed (`fontfile='C\:/`'): returncode 0, works.
```

The drive-letter colon breaks ffmpeg's drawtext option parsing even inside single quotes. The fix is to escape colons in `\_ff\_quote` (which also feeds `textfile=` and the logo `movie=` source). Let me verify the `movie=` operand behaves the same so the shared-helper fix is safe everywhere:

329. user (2026-06-13T09:36:40.142Z)

logo png: True C:\Users\avidu\AppData\Local\Temp\wm_logo.png

330. assistant (2026-06-13T09:36:49.313Z)

```
"""Verify the colon-escape fix is also correct for the logo `movie=` filter source."""
```

```
import subprocess
import tempfile
import os
```

```
LOGO = os.path.join(tempfile.gettempdir(), "wm_logo.png") # created beforehand
```

```
def ff_quote_current(p):
    return p.replace("\\", "/").replace("'", r"\'")
```

```
def ff_quote_fixed(p):
    return p.replace("\\", "/").replace("'", r"\'").replace(":", r"\:")
```

```

def try_movie(label, logo_quoted):
    out = os.path.join(tempfile.gettempdir(), f"mv_{label}.mp4")
    vf = f"movie='{logo_quoted}'[wm];[in][wm]overlay=W-w-24:H-h-24"
    cmd = ["ffmpeg", "-y", "-f", "lavfi", "-i", "testsrc=size=320x240:duration=1:rate=10",
           "-vf", vf, "-frames:v", "3", out]
    r = subprocess.run(cmd, capture_output=True)
    ok = r.returncode == 0 and os.path.exists(out) and os.path.getsize(out) > 0
    print(f"[{label}] ok={ok} rc={r.returncode}")
    if not ok:
        print(f"[{label}] stderr tail: {r.stderr.decode(errors='replace')[-200:]}")

print("movie= CURRENT (raw colon):")
try_movie("current", ff_quote_current(LOGO))
print("movie= FIXED (escaped colon):")
try_movie("fixed", ff_quote_fixed(LOGO))

```

331. user (2026-06-13T09:36:49.326Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\repro_movie_overlay.py (file state is current in your context ? no need to Read it back)

332. user (2026-06-13T09:36:54.748Z)

```

movie= CURRENT (raw colon):
[current] ok=False rc=4294967294
[current] stderr tail: 'C'

[AVFilterGraph @ 000002180ff5de80] Error initializing filters

Error opening output file C:\Users\avidu\AppData\Local\Temp\mv_current.mp4.

Error opening output files: No such file or directory

```

```

movie= FIXED (escaped colon):
[fixed] ok=True rc=0

```

333. assistant (2026-06-13T09:37:33.510Z)

Confirmed ? the `movie=` logo source has the **same** colon bug (fails with parse error on `C`), and the same escape fixes it. So one fix in the shared `\_ff\_quote` helper correctly repairs all three operands (drawtext `fontfile`, `textfile`, and logo `movie=`). This means absolute logo paths were silently broken on Windows too. Let me read the current exact helper text and apply the fix:

334. user (2026-06-13T09:37:34.048Z)

```

1     import os
2     import subprocess
3     import tempfile
4     import logging
5     from pathlib import Path
6     from typing import Any, Dict, List, Optional, Tuple, Union
7
8     logger = logging.getLogger(__name__)
9
10    # Bundled, commercially-clean default watermark font (Apache-2.0). Shipped so the
overlay
11    # renders out-of-the-box without a system fontconfig setup (a common gap on Windows,
where
12    # `font='Sans` fails). Overridable via stitching.watermark.font_path. This module lives
at

```

```

13     # backend/pipeline/stitcher/, so parents[2] == backend/.
14     _BUNDLED_FONT = Path(__file__).resolve().parents[2] / "assets" / "fonts" / "RobotoSlab-
Regular.ttf"
15
16     class VideoCompiler:
17         def __init__(self, output_dir: str, end_padding: float = 1.0, begin_padding: float =
0.5,
18                 watermark: Optional[Any] = None):
19             self.output_dir = output_dir
20             self.end_padding = end_padding
21             self.begin_padding = begin_padding
22             # Optional branding overlay. Accepts a WatermarkConfig model or a plain dict;
23             # normalized to a dict (or None) so this module stays config-package-agnostic.
24             self.watermark = self._normalize_watermark(watermark)
25             os.makedirs(output_dir, exist_ok=True)
26
27         @staticmethod
28         def _normalize_watermark(watermark: Optional[Any]) -> Optional[Dict[str, Any]]:
29             if watermark is None:
30                 return None
31             if hasattr(watermark, "model_dump"):
32                 return watermark.model_dump()
33             if isinstance(watermark, dict):
34                 return dict(watermark)
35             return None
36
37         @staticmethod
38         def _ff_quote(path: str) -> str:
39             """Make a filesystem path safe to embed inside single quotes in an ffmpeg
40             filtergraph: forward slashes (Windows-friendly) + escaped single quotes."""

```

335. user (2026-06-13T09:37:37.849Z)

```

37         @staticmethod
38         def _ff_quote(path: str) -> str:
39             """Make a filesystem path safe to embed inside single quotes in an ffmpeg
40             filtergraph: forward slashes (Windows-friendly) + escaped single quotes."""
41             return path.replace("\\", "/").replace("'", r"\'")
42
43         @staticmethod
44         def _parse_time(val: Union[int, float, str]) -> float:

```

336. user (2026-06-13T09:37:45.298Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\stitcher\compiler.py has been updated successfully. (file state is current in your context ? no need to Read it back)

337. assistant (2026-06-13T09:37:49.042Z)

Now the regression tests. Let me check the end of the watermark test file:

338. user (2026-06-13T09:37:50.001Z)

```

1     """Tests for the configurable highlight watermark (stitching.watermark).
2
3     Covers the config defaults, the filtergraph builder (text + logo, positions, font,
4     the configurable string), and the end-to-end wiring through compile_highlights with
5     ffmpeg/ffprobe mocked ? including the graceful fallback when the watermark encode fails.
6     """
7     import os
8     import subprocess
9     from pathlib import Path
10    from unittest.mock import MagicMock, patch

```

```

11
12 import pytest
13 from pydantic import ValidationError
14
15 from backend.config import AppConfig
16 from backend.config.models import WatermarkConfig
17 from backend.pipeline.stitcher.compiler import VideoCompiler
18
19
20 # --- config -----
21
22 def test_watermark_config_defaults_on_with_brand_text():
23     wm = AppConfig().stitching.watermark
24     assert wm.enabled is True # default-on, with an off switch
25     assert wm.text == "Generated by Khelsutra.guru"
26     assert wm.position == "bottom-right"
27
28
29 def test_watermark_can_be_disabled_via_config():
30     cfg = AppConfig.model_validate({"stitching": {"watermark": {"enabled": False}}})
31     assert cfg.stitching.watermark.enabled is False
32
33
34 def test_font_size_ratio_and_opacity_bounds_enforced():
35     with pytest.raises(ValidationError):
36         WatermarkConfig(opacity=1.5)
37     with pytest.raises(ValidationError):
38         WatermarkConfig(font_size_ratio=0.0)
39
40
41 # --- filtergraph builder -----
42
43 def test_disabled_or_none_builds_no_filter(tmp_path):
44     assert VideoCompiler(str(tmp_path / "a"),
45 watermark=None)._watermark_filter(str(tmp_path)) is None
46     off = VideoCompiler(str(tmp_path / "b"), watermark={"enabled": False})

```

339. user (2026-06-13T09:37:57.325Z)

```

62
63 def test_position_and_margin_map_to_corner(tmp_path):
64     comp = VideoCompiler(str(tmp_path / "o"),
65 watermark={"enabled": True, "text": "X", "position": "top-
left", "margin": 12})
66     graph = comp._watermark_filter(str(tmp_path))
67     assert "x=12:y=12" in graph
68
69
70 def test_explicit_font_path_uses_fontfile(tmp_path):
71     comp = VideoCompiler(str(tmp_path / "o1"),
72 watermark={"enabled": True, "text": "X", "font_path":
"/fonts/Brand.ttf"})
73     g = comp._watermark_filter(str(tmp_path))
74     assert "fontfile='/fonts/Brand.ttf'" in g
75
76
77 def test_default_uses_bundled_apache_font(tmp_path):
78     # No font_path ? the bundled Apache-2.0 Roboto Slab (renders without fontconfig).
79     comp = VideoCompiler(str(tmp_path / "o2"), watermark={"enabled": True, "text": "X"})
80     g = comp._watermark_filter(str(tmp_path))
81     assert "fontfile=" in g and "RobotoSlab-Regular.ttf" in g
82     assert "font=" not in g # not the fontconfig path
83
84
85 def test_bundled_font_actually_ships_and_is_apache():

```

```

86         import backend.pipeline.stitcher.compiler as cmod
87         assert cmod._BUNDLED_FONT.exists(), "bundled default watermark font must ship in the
repo"
88         with open(cmod._BUNDLED_FONT, "rb") as fh:
89             assert fh.read(4) in (b"\x00\x01\x00\x00", b"OTTO", b"true", b"ttcf") # valid
sfnt
90             lic = cmod._BUNDLED_FONT.parent / "LICENSE-Apache-2.0.txt"
91             assert lic.exists() and "Apache License" in lic.read_text(encoding="utf-8",
errors="replace")
92
93
94     def test_fontconfig_fallback_when_bundled_missing(tmp_path, monkeypatch):
95         import backend.pipeline.stitcher.compiler as cmod
96         monkeypatch.setattr(cmod, "_BUNDLED_FONT", Path("/no/such/font.ttf"))
97         comp = VideoCompiler(str(tmp_path / "o3"),
98                               watermark={"enabled": True, "text": "X", "font": "DejaVu
Sans"})
99         g = comp._watermark_filter(str(tmp_path))
100         assert "font='DejaVu Sans'" in g and "fontfile=" not in g
101

```

340. assistant (2026-06-13T09:38:16.532Z)

Now I'll add cross-platform regression tests (using `\_ff\_quote` directly and an explicit colon font_path, so they're deterministic on Linux CI too):

341. user (2026-06-13T09:38:27.128Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test_watermark.py has been updated successfully. (file state is current in your context ? no need to Read it back)

342. user (2026-06-13T09:39:05.466Z)

```

..... [ 88%]
..... [100%]
409 passed in 29.82s
=====
RESULT: PASS - all tests green
=====

```

343. assistant (2026-06-13T09:39:08.712Z)

All 409 pass (+2 new). Lint/type check, then branch and PR:

344. user (2026-06-13T09:39:12.504Z)

All checks passed!
Success: no issues found in 1 source file

345. user (2026-06-13T09:39:21.365Z)

```

M      backend/pipeline/stitcher/compiler.py
M      config.json
M      tests/test_watermark.py
Switched to a new branch 'fix/watermark-windows-colon'
M      backend/pipeline/stitcher/compiler.py
M      config.json
M      tests/test_watermark.py
?? arial.ttf
?? artifacts/
?? docs/STORAGE_SHARING_MODEL.md
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md

```

```
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
?? scratch/keep_awake.ps1
?? scratch/probe_cv2_count.py
?? scratch/probe_gemini.py
?? scratch/repro_drawtext_font.py
?? scratch/repro_movie_overlay.py
?? scratch/run_gx010125.py
```

346. assistant (2026-06-13T09:39:25.442Z)

Only the two fix files staged (config.json and scratch stay local). Committing, pushing, and opening the PR:

347. user (2026-06-13T09:39:48.005Z)

```
[fix/watermark-windows-colon a1b8cfb] fix(stitcher): escape drive-letter colon in ffmpeg
filtergraph paths
 2 files changed, 33 insertions(+), 2 deletions(-)
remote:
remote: Create a pull request for 'fix/watermark-windows-colon' on GitHub by visiting:
remote:      https://github.com/avidullu/badminton-highlight-indexer/pull/new/fix/watermark-
windows-colon
remote:
branch 'fix/watermark-windows-colon' set up to track 'origin/fix/watermark-windows-colon'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      fix/watermark-windows-colon -> fix/watermark-windows-colon
Warning: 16 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/108
```

348. user (2026-06-13T09:41:19.438Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27463151486/job/81180603911
Full test suite (offline, mocked)                pass    1m25s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27463151486/job/81180603962
Mypy (lenient baseline) pass    26s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27463151486/job/81180603933
Ruff lint      pass    12s      https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27463151486/job/81180603924
Syntax sweep + import smoke (no GPU stack)      pass    19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27463151486/job/81180603911
```

349. user (2026-06-13T09:41:32.763Z)

```
["SUCCESS"]
From https://github.com/avidullu/badminton-highlight-indexer
 * branch      master      -> FETCH_HEAD
 3461898..6de70aa master      -> origin/master
Updating 3461898..6de70aa
Fast-forward
 backend/pipeline/stitcher/compiler.py | 11 ++++++--
 tests/test_watermark.py               | 24 ++++++
 2 files changed, 33 insertions(+), 2 deletions(-)
M      config.json
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
6de70aa fix(stitcher): escape drive-letter colon in ffmpeg filtergraph paths (#108)
```

350. assistant (2026-06-13T09:41:57.044Z)

Both PRs merged. Let me record the durable lessons (Modern Standby killing long runs is a

reusable gotcha for this box) and update the checkpoint:

351. user (2026-06-13T09:41:57.966Z)

```
1      # Memory Index
2
3      ## ? Start here
4      - [Current state](current-state.md) ? **live micro-checkpoint** (updated every
checkpoint): open PRs, what just landed, next steps, open decisions. **Read this first.**
5      - [Session handoff](session-handoff.md) ? current state · next steps · open decisions ·
ramp-up kit. AUTO-refreshed at meaningful checkpoints (PR merged, key decision, phase shift,
session wrap) ? see working-agreement-checkpoints staleness tripwire.
6      - [Code map artifact](code-map-artifact.md) ? for CODE/architecture ramp-up, read
`docs/CODE_MAP.md` FIRST (dense anchor-rich map; replaces a file crawl). How to regenerate it.
7
8      - [Working agreement: attribution](working-agreement-attribution.md) ? never assert
fabricated facts; NEVER invent a provenance (a discussion/review/source that didn't happen). The
silver-bullet rule.
9      - [Working agreement: checkpoints](working-agreement-checkpoints.md) ? user prefers
frequent small PRs + a Claude-consumable current-state file updated at every micro-checkpoint.
10     - [Feedback: tail long-job logs](feedback-tail-long-job-logs.md) ? proactively tail
progress/summaries from long-running background jobs while they run, not just the final result.
11
12     ## Project: rally segmentation pipeline
13     - [Candidate segments & fine-tuning](candidate-segments-finetuning.md) ? why raw
detector candidates are persisted (detector-agnostic table); golden-eval-set goal
14     - [TrackNetV4/WASB via WSL2](tracknet-wasb-wsl2-decision.md) ? self-hosted shuttle
trajectory models (not LLM APIs); WASB/TrackNetV3 = MIT, commercial-clear
15     - [Collector?indexer bridge](collector-indexer-bridge.md) ? 4-phase plan to train/eval a
generic temporal rally segmenter; Phases 1?2 done, 3 in progress, 4 next
16     - [Golden-set vendoring](golden-set-vendoring.md) ? generate golden rally labels via
3rd-party vendors (Bangalore-local) + extensible regression flow; Roora selected for a 3-sport
pilot, guideline + collector ingestion built, GS-1/2/4 landed, GS-3 + gold key pending
17     - [Roora guidelines Drive doc](roora-guidelines-drive-doc.md) ? the CANONICAL vendor-
facing Brief+Guide Google Doc (supersedes v3/v5/v6/v7 Drive copies; v7 anomaly RESOLVED
2026-06-12); TODO markers still to fill before sending
18     - [Audio (E1) findings](audio-e1-findings.md) ? quantitative gains/losses of the
classical-onset audio cue (recall 100% but discrimination ~2.2x, band-pass null) + reusable
lessons for adding the NEXT signal. Audio NOT adopted for fusion; detector kept (PR #35).
19
20     ## Monetization (hosted API)
21     - [Monetization plan](monetization-plan.md) ? hosted video?highlights API on a GPU VM;
licensing & per-video cost are first-class; user is Bangalore-based
22     - [Monetization research status](monetization-research-status.md) ? audit done:
Ultralytics AGPL swapped out (PR #8); MonoTrack/YOLO-NAS weights avoided; codec=watch; full
report in docs/
23
24     ## Forward-looking
25     - [Sibling internship repo](sibling-internship-repo.md) ? khelacharya shot-intelligence
layer builds ON TOP of our rally segmentation; kept off master at branch level (NOT a separate
GitHub repo ? corrected 2026-06-06)
26
```

352. assistant (2026-06-13T09:42:25.459Z)

name: gotcha-long-runs-gpu-wsl

description: "Running the WASB GPU pipeline E2E on a big 4K video: Modern Standby kills long runs; cv2 frame extraction is the bottleneck; the proxy+keepawake+ffmpeg-preextract recipe."

metadata:

type: project

Hard-won recipe for running the indexer E2E on the GPU over a LONG/large video (e.g. the 12.8-min 4K60 GoPro `GX010125.MP4`, 11.5 GB). First proven 2026-06-13.

****Three traps, in the order they bite:****

1. ****Modern Standby (connected standby) silently kills long runs.**** This box enters S0 low-power when idle/screen-off and the Desktop Activity Moderator SUSPENDS Win32 processes + pauses the WSL2 VM ? the Python orchestrator and WSL inference freeze with NO clean exit, no timeout logged (the log just stops). `powercfg standby-timeout=0` does NOT prevent it (it's screen-off-triggered, not idle-timeout). Confirm via `Get-WinEvent ... Kernel-Power` session transitions. ****Fix:**** hold a wake lock for the whole run ? a background PowerShell loop calling
`SetThreadExecutionState(ES\_CONTINUOUS|ES\_SYSTEM\_REQUIRED|ES\_DISPLAY\_REQUIRED)` and staying alive (the flag releases when the thread dies). Saved as `scratch/keep\_awake.ps1`. This is the same class as the older "idle/suspend kills background runs" note in `[[current-state]]`.
2. ****cv2 PNG frame extraction is the real bottleneck, NOT the GPU.**** `wasb\_infer.extract\_frames` writes 46k PNGs single-threaded (~20 fps ? ~38 min for a 12.8-min clip) BEFORE any GPU work. The GPU detector pass is fast (~25 win/s, ~26 min for 46k windows) and is checkpointed/resumable
(`<out>\_wasbcache/{det\_raw.jsonl,manifest.json}`, fsync every 1000 windows). ****Fix:**** pre-extract
with ffmpeg natively inside WSL (ext4, not /mnt) to PNG `%08d.png -start\_number 0`; the count is provably exact for CFR (duration*fps = 46080) so `extract\_frames` sees `existing>=expected` and skips its slow path entirely. wasb_infer's docstring literally recommends this but the runner never wired a `--frames\_dir` passthrough.
3. ****4K is decode-bound for WASB.**** Downscale to a 1080p PROXY first (GPU `h264\_nvenc`, ~4.7x realtime), index the proxy, but STITCH the final reel from the 4K ORIGINAL ? timestamps transfer
1:1 (identical timeline). The runner I used: `scratch/run\_gx010125.py` (INDEX_PATH=proxy, SOURCE_PATH=4K). 1080p is also the resolution all golden evals used.

****Observability while it runs**** (the user explicitly hates silent gaps): WASB inference is a WSL subprocess so its logs are captured, NOT in the live run log ? poll the cache `manifest.json` windows_done/windows_total instead. Phase-3 Gemini progress: watch
`output/chunks\_gemini\_handoff/`
(one `handoff\_<id>\_<wi>.mp4` per candidate, deleted after). DB segments only land in Phase 4 (after ALL Gemini calls), so the DB shows no incremental Phase-3 progress.

****Result of the proven run:**** 64.8% shuttle visibility, 18 candidate windows ? 52 Gemini-confirmed rallies ? 11 with shot_count>7 ? 2:23 4K reel. Preflight Gemini with a free `models.list` before the long GPU pass so a dead key doesn't waste 30 min.

353. user (2026-06-13T09:42:25.486Z)

File created successfully at: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\gotcha-long-runs-gpu-wsl.md (file state is current in your context ? no need to Read it back)

354. user (2026-06-13T09:42:33.359Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

355. user (2026-06-13T09:42:37.189Z)

```
9
10      # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12      **Checkpoint:** 2026-06-12 (latest ? PAID COMPONENT REMOVED + VIDEO-LOCALITY MODEL; **PR
```


#104 updated `9c84fea` +

356. user (2026-06-13T09:42:56.580Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

357. assistant (2026-06-13T09:43:18.438Z)

Everything's complete and recorded. Here's the wrap-up.

? Your highlights: `output\GX010125\_highlights.mp4`

****11 rally highlights, every one with more than 7 shots**** (8, 8, 9, 9, 10, 10, 11, 11, 12, 14, 16 shots), selected from 52 rallies detected across your 12.8-minute match.

- ****True 4K, from the original****: 3840×2160 h264 59.94fps, 2:23, 1.15 GB ? cut from `F:\?GX010125.MP4`, not the proxy (the 4K resolution is the proof). Config paddings applied (0.5s/0.5s).
- ****GPU end-to-end****: WASB shuttle-tracking on your RTX 2070 Super via WSL (64.8% visibility) ? 18 candidate windows ? Gemini confirmed 52 rallies with shot counts ? config `min\_shot\_count=8` filter. A 7-shot rally was correctly dropped, confirming ">7" semantics.

The config is picked up ? and now enforced everywhere

`stitching.min\_shot\_count` drove the filter (you can see each rally's KEEP/drop in `output\gx010125\_run.log`). I also ****merged [#107](https://github.com/avidullu/badminton-highlight-indexer/pull/107)**** so the ****API path**** (`/api/compile`) enforces the same floor ? previously it ignored the config and stitched whatever segment IDs it was given.

Two things fixed along the way (both merged on green CI, per your standing authorization)

1. ****[#107](https://github.com/avidullu/badminton-highlight-indexer/pull/107)**** ? API `min\_shot\_count` parity (above).
2. ****[#108](https://github.com/avidullu/badminton-highlight-indexer/pull/108)**** ? a real bug the run surfaced: the on-by-default watermark (bundled font from #103) was ****silently failing on Windows**** ? the absolute font path's drive-letter colon (`C:/`) broke ffmpeg's drawtext parser, so every clip fell back to no-watermark. Fixed by escaping the colon in `\_ff\_quote` (also repaired absolute logo paths). Verified against real ffmpeg, with regression tests.

Why the earlier attempts died (not a code bug)

****Modern Standby**** suspended the machine overnight twice, freezing the run mid-inference. Fixed with a keep-awake lock. I also found the actual bottleneck was ****cv2 frame extraction**** (~38 min, where it kept dying ? ****before**** the GPU even started), so I pre-extract with ffmpeg now (count-exact for CFR) and the full run completed in one ~47-min pass. The whole recipe is saved to memory for next time.

Two notes

- ****`config.json` has local uncommitted changes**** from this run (`default\_segmenter=wasb\_hybrid`, `min\_shot\_count=8`, `wasb.timeout\_sec=7200`). Tell me if you want them reverted to the defaults (gemini / 1 / 1800).
- ****The delivered reel has no watermark**** (it stitched ****before**** #108 landed). If you'd like a branded version, I can re-stitch ? the 52 rallies are cached in the DB, so it skips all inference/Gemini and just re-cuts the 11 clips (~5 min) now that the watermark works. Want me to?

358. user (2026-06-13T14:32:18.079Z)

Can you reuse the info and make me a new video with shot_count >= 5 and also before that can you

tell me whether we used the 4K video to upload to gemini or 1080p? I also want to see a "flowchart of execution" if you may, where was wasb involved, how did we decide to "find the rally end" etc.

359. assistant (2026-06-13T14:33:20.211Z)

Good questions ? let me confirm the upload resolution from the code/DB precisely before answering, and verify the cached data is still there for the new reel.

360. user (2026-06-13T14:33:21.640Z)

```
180             "peak_velocity": cw["peak_velocity"],
181             "mean_velocity": cw["mean_velocity"],
182             "active_frame_count": cw["active_frame_count"],
183             "track_id_count": cw["track_id_count"],
184         }
185         candidate_ids[i] = self.db.add_candidate_segment(
186             video_id, detector=self.name,
187             start_time=cw["start"], end_time=cw["end"],
188             chunk_index=cw["chunk_index"], confidence=min(1.0,
189 cw["peak_velocity"]),
190             stats=stats, detection_params=detection_params,
191         )
192     if not all_candidate_windows:
193         return []
194
195     # --- Phase 3: AI provider handoff over padded candidate clips ---
196     handoff_dir = os.path.join("output", f"chunks_{provider_name}_handoff")
197     os.makedirs(handoff_dir, exist_ok=True)
198     logger.info("Phase 3: %s validation on %d candidate windows...",
199                 provider_name, len(all_candidate_windows))
200
201     def process_candidate_window(wi: int, window: Dict[str, Any]) -> List[dict]:
202         w_start = max(0, window["start"] - handoff_padding)
203         w_end = min(video_duration, window["end"] + handoff_padding)
204         w_dur = w_end - w_start
205         w_path = os.path.join(handoff_dir, f"handoff_{video_id}_{wi}.mp4")
206         if not extract_video_chunk(video_path, w_start, w_dur, w_path,
reencode=True):
207             return []
208         prompt = sport_profile.get_prompt(chunk_duration=w_dur)
209         segments, _ = provider.analyze_video(w_path, prompt, chunk_duration=w_dur,
```

361. user (2026-06-13T14:33:27.502Z)

video row status: processed | filepath: C:\Users\avidu\Projects\badminton-highlight-indexer\output\GX010125_1080p.mp4
cached segments: 52
shot_count>=5: 24 | <5: 28

362. user (2026-06-13T14:33:28.833Z)

proxy exists: True size_MB=2243.2

363. assistant (2026-06-13T14:34:32.691Z)

Both confirmed from the code and DB:

- **Gemini received 1080p, not 4K.** Phase 3 extracts each candidate clip from the proxy that was passed to the segmenter (`extract\_video\_chunk(video\_path=?1080p proxy?, reencode=True)`, no further downscale) ? so the clips uploaded to Gemini were 1080p. The 4K original was touched **only** at the final stitch.
- **Cache is intact** for reuse: video_id `89605e80?`, status `processed`, **52 rallies

cached**, of which **24 have shot_count ? 5** (so the new reel will be 24 rallies). The 4K source is untouched, so I can re-stitch without re-running WASB or Gemini.

Let me build the flowchart, then kick off the ?5 reel.

364. user (2026-06-13T14:34:33.706Z)

<persisted-output>

Output too large (61.5KB). Full output saved to: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tool-results\toolu_01ThyJioAbxybnszG7THJnhM.json

Preview (first 2KB):

```
[
  {
    "type": "text",
    "text": "# Imagine ? Visual Creation Suite\n\n## Modules\n\nCall read_me again with the\nmodules parameter to load detailed guidance:\n- `diagram` ? SVG flowcharts, structural diagrams,\nillustrative diagrams\n- `mockup` ? UI mockups, forms, cards, dashboards\n- `interactive` ?\ninteractive explainers with controls\n- `chart` ? charts, data analysis, geographic maps\n(Char.js, D3 choropleth)\n- `art` ? illustration and generative art\n\nPick the closest fit. The\nmodule includes all relevant design guidance.\n\n**Complexity budget ? hard limits:**\n- Box\nsubtitles: ?5 words. Detail goes in click-through (`sendPrompt`) or the prose below ? not the\nbox.\n- Colors: ?2 ramps per diagram. If colors encode meaning (states, tiers), add a 1-line\nlegend. Otherwise use one neutral ramp.\n- Horizontal tier: ?4 boxes at full width (~140px\neach). 5+ boxes ? shrink to ?110px OR wrap to 2 rows OR split into overview + detail\ndiagrams.\n\nIf you catch yourself writing \"click to learn more\" in prose, the diagram itself\nmust ACTUALLY be sparse. Don't promise brevity then front-load everything.\n\n**Accessibility:**\nFor HTML widgets, begin with a visually-hidden `<h2 class=\"sr-only\">` containing a one-\nsentence summary of the visualization for screen-reader users. (SVG widgets use `role=\"img\"`\nwith `<title>` and `<desc>` instead ? see SVG setup.)\n\nYou create rich visual content ? SVG\ndiagrams/illustrations and HTML interactive widgets ? that renders inline in conversation. The\nbest output feels like a natural extension of the chat.\n\n## Core Design System\n\nThese rules\napply to ALL use cases.\n\n### Philosophy\n- **Seamless**: Users shouldn't notice where\nclaude.ai ends and your widget begins.\n- **Flat**: No gradients, mesh backgrounds, noise\ntextures, or decorative effects. Clean flat surfaces.\n- **Compact**: Show the essential inline.\nExplain the rest in text.\n- **Text goes in your response, visuals go in the tool** ? All\nexplanatory text, descripti\n...\n</persisted-output>
```

365. assistant (2026-06-13T14:34:37.732Z)

Let me read the full diagram guidance:

366. user (2026-06-13T14:34:39.769Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-2 of 6 total (27116 tokens, cap 25000). Call Read with offset=3 limit=2 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```
1      [
2      {
```

367. user (2026-06-13T14:34:46.758Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.

368. user (2026-06-13T14:34:50.977Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.

369. user (2026-06-13T14:35:00.253Z)

4:[Omitted long matching line]

370. user (2026-06-13T14:35:10.028Z)

4:--font-serif
4:--color-text-primary
4:--border-radius-md
4:--color-background-primary
4:--color-text-primary
4:--color-border-tertiary
4:--font-sans
4:--font-serif
4:--font-mono
4:--border-radius-md
4:--border-radius-lg
4:--border-radius-xl
4:--color-text-primary
4:--color-text-secondary
4:#EEDFE
4:#CECBF6
4:#AFA9EC
4:#7F77DD
4:#534AB7
4:#3C3489
4:#26215C
4:#E1F5EE
4:#9FE1CB
4:#5DCAA5
4:#1D9E75
4:#0F6E56
4:#085041
4:#04342C
4:#FAECE7
4:#F5C4B3
4:#F0997B
4:#D85A30
4:#993C1D
4:#712B13
4:#4A1B0C
4:#FBEAF0
4:#F4C0D1
4:#ED93B1
4:#D4537E
4:#993556
4:#72243E
4:#4B1528
4:#F1EFE8
4:#D3D1C7
4:#B4B2A9
4:#888780
4:#5F5E5A
4:#444441
4:#2C2C2A
4:#E6F1FB
4:#B5D4F4
4:#85B7EB
4:#378ADD
4:#185FA5
4:#0C447C
4:#042C53
4:#EAF3DE
4:#C0DD97
4:#97C459

4:#639922

[Showing results with pagination = limit: 60]

371. assistant (2026-06-13T14:37:13.651Z)

Now the execution flowchart. Let me render it:

372. user (2026-06-13T14:37:54.183Z)

Content rendered and shown to the user. Please do not duplicate the shown content in text because it's already visually represented.

373. assistant (2026-06-13T14:38:06.276Z)

""Re-stitch ONLY (no re-inference): reuse the 52 cached Gemini-confirmed rallies for GX010125 and build a new reel with shot_count >= 5, cut from the 4K original.

Pulls segments straight from the DB by the known video_id, so it does not touch WASB / Gemini / the proxy at all ? pure reuse of the cached info.

""

```
import json
import logging
import os
import sys
import time
```

```
sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
from dotenv import load_dotenv
```

```
load_dotenv()
logging.basicConfig(level=logging.INFO,
                    format="%(asctime)s [%(name)s] %(levelname)s: %(message)s",
                    datefmt="%H:%M:%S")
```

```
from backend.infrastructure.database import Database
from backend.config import load_config
from backend.pipeline.stitcher.compiler import VideoCompiler
```

```
VIDEO_ID = "89605e80ed012ca259a640b7ec06cd80" # the 1080p-proxy MD5 we indexed
SOURCE_PATH = r"F:\GoPro Hero Black 12\Khe\Sutra\Badminton 12 June 2026\GX010125.MP4" # 4K original
OUTPUT_FILENAME = "GX010125_highlights_min5.mp4"
MIN_SHOTS = 5 # user request: shot_count >= 5
```

```
def main():
    cfg = load_config()
    db = Database(os.path.join(cfg.output.output_dir, cfg.output.db_name))
    segs = db.query_play_segments(VIDEO_ID, {})
    print(f"[REUSE] {len(segs)} cached rallies for video_id={VIDEO_ID}", flush=True)
    if not segs:
        raise SystemExit("no cached segments ? nothing to reuse")
    if not os.path.exists(SOURCE_PATH):
        raise SystemExit(f"4K source not found: {SOURCE_PATH}")

    t0 = time.time()
    time_pairs, kept, dropped = [], 0, 0
    for s in sorted(segs, key=lambda x: x["start_time"]):
        meta = s.get("metadata") or {}
        if isinstance(meta, str):
            try:
                meta = json.loads(meta)
            except Exception:
```

```

        meta = {}
        sc = meta.get("shot_count", 0)
        tag = f"  [{float(s['start_time']):8.2f}s - {float(s['end_time']):8.2f}s]
shot_count={sc}"
        if sc >= MIN_SHOTS:
            time_pairs.append((s["start_time"], s["end_time"]))
            kept += 1
            print(tag + "  -> KEEP", flush=True)
        else:
            dropped += 1
            print(tag + "  -> drop", flush=True)
        print(f"[FILTER] shot_count>={MIN_SHOTS}: kept {kept}, dropped {dropped}", flush=True)
        if not time_pairs:
            raise SystemExit("no rallies passed the filter")

        compiler = VideoCompiler(cfg.output.output_dir,
                                begin_padding=cfg.stitching.begin_padding,
                                end_padding=cfg.stitching.end_padding,
                                watermark=cfg.stitching.watermark)
        out = compiler.compile_highlights(SOURCE_PATH, time_pairs, OUTPUT_FILENAME) # cut from 4K
        print(f"[DONE] {os.path.abspath(out)} ({kept} rallies, {time.time()-t0:.0f}s)", flush=True)

if __name__ == "__main__":
    main()

```

374. user (2026-06-13T14:38:06.292Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\stitch_min5.py (file state is current in your context ? no need to Read it back)

375. assistant (2026-06-13T14:38:10.691Z)

Now let me launch the ?5 re-stitch (reuses the 24 cached rallies, cuts from 4K ? no WASB/Gemini re-run) and monitor it:

376. user (2026-06-13T14:38:12.928Z)

Command running in background with ID: boilngexb. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\boilngexb.output. You will be notified when it completes. To check interim output, use Read on that file path.

377. user (2026-06-13T14:38:18.817Z)

Monitor started (task bd1qgo1zh, persistent ? runs until TaskStop or session end). You will be notified on each event. Keep working ? do not poll or sleep. Events may arrive while you are waiting for the user ? an event is not their reply.

378. assistant (2026-06-13T14:38:37.089Z)

Both your questions answered, the flowchart is above, and the ?5 reel is now building (reusing cache ? no WASB/Gemini re-run).

1. Resolution: Gemini saw ****1080p****, not 4K

The pipeline deliberately splits resolution by stage:

- ****WASB + Gemini operated on the 1080p proxy.**** The candidate clips uploaded to Gemini were ``extract_video_chunk(<1080p proxy>, ?, reencode=True)`` with no further downscaling ? so 1080p reached Gemini. This is the right trade: shot-counting and boundary detection don't need 4K, and it keeps WASB decode and Gemini upload fast/cheap.
- ****The 4K original was used only for the final stitch**** ? that's why your reel came out true 3840×2160 while the analysis ran on 1080p.

2. Flowchart ? where WASB sits, how "rally end" is decided

The diagram above traces it, but the key idea is ****rally boundaries are decided in two passes**** (the two purple/pink-flagged nodes):

- ****WASB (coarse, motion-based)**** ? runs on the GPU over the 1080p frames and produces a per-frame shuttle trajectory `(x, y, visible)`. `trajectory\_to\_action\_windows` turns that into candidate rally windows: ****shuttle moving = in-rally****; short pauses are fused (`dead\_time\_merge\_gap`), and a long stretch of stillness/invisibility ends the window. That's the ***approximate*** "where could a rally be" ? it found 18 windows.
- ****Gemini (fine, semantic)**** ? each candidate window is cut to a padded 1080p clip and sent to Gemini, which watches the actual play and returns the ****exact**** rally start/end (start = racket-to-shuttle contact, end = grounding/fault) plus the `shot\_count`. One WASB window can contain several real rallies ? that's how 18 windows became 52 confirmed rallies.

So: ****WASB proposes, Gemini disposes.**** The shot count that your filter uses is Gemini's `shuttle\_exchange\_count`.

3. The ?5 reel is building now

`output\GX010125\_min5\_stitch.log` is the live log. It's reusing the ****24 cached rallies with shot_count ? 5**** (vs 11 at >7) straight from the DB ? ****no WASB or Gemini re-run**** ? and cutting from the 4K original. Output will be `output\GX010125\_highlights\_min5.mp4`. ETA ~8?12 min (24 clips re-encoded at 4K).

One note: this reel ****will have the watermark**** ("Generated by Khelsutra.guru"), because the bundled-font fix (#108) is now in place and the config default is on ? so it also doubles as live confirmation that fix works on real 4K. Say the word if you'd prefer it without and I'll regenerate watermark-free in a couple minutes. I'll confirm the count, duration, and 4K resolution when it lands.

379. user (2026-06-13T14:38:39.147Z)

```
<task-notification>
<task-id>bd1qgo1zh</task-id>
<summary>Monitor event: "?5 re-stitch milestones (filter, stitch, done, errors)"</summary>
<event>[REUSE] 52 cached rallies for video_id=89605e80ed012ca259a640b7ec06cd80
[FILTER] shot_count>=5: kept 24, dropped 28
20:08:13 [backend.pipeline.stitcher.compiler] INFO: Trimming 24 segments (begin_padding=0.5s,
end_padding=0.5s)...</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

380. assistant (2026-06-13T14:38:48.892Z)

Cache reuse confirmed ? ****24 of 52 rallies have shot_count ? 5**** (28 dropped), no re-inference. Now trimming 24 clips from the 4K original (~8?12 min). I'll confirm when it lands, and check whether the watermark rendered cleanly this time (validating #108 on real 4K).

381. user (2026-06-13T14:44:21.633Z)

where di this trajectory_to_action_windows come from and can you explain more about this? How did it detect the "trajectory", code pointer on github will also help

382. assistant (2026-06-13T14:44:43.116Z)

Great question ? let me pull the actual implementation so I explain it precisely, and get the GitHub permalink info.

383. user (2026-06-13T14:44:44.129Z)

1 ""TrackNet shuttle-trajectory runner (WSL2 subprocess adapter).

```

2
3     This bridges the Windows-side indexer to a TensorFlow TrackNetV4 install living
4     inside WSL2 (ADR-001 in docs/DECISIONS.md). The flow is:
5
6         1. Translate the Windows video path to its WSL-visible form (and optionally
7            stage it into the WSL filesystem to avoid slow /mnt/c per-frame reads).
8         2. Invoke ``src/predict.py`` inside the TrackNetV4 conda env via ``wsl``.
9         3. Read back the per-frame ``Frame,Visibility,X,Y`` CSV it writes.
10        4. Convert that trajectory into high-motion "action windows" ? the same
11           candidate-rally shape HybridYoloSegmenter produces ? so the rest of the
12           pipeline (AI handoff, DB population) is unchanged.
13
14     Nothing here imports TensorFlow; all model code stays in WSL.
15     """
16
17     import os
18     import csv
19     import shlex
20     import logging
21     import subprocess
22     from dataclasses import dataclass
23     from typing import List, Dict, Any, Optional, Tuple
24
25     from backend.pipeline.detectors.base import DetectorRunner, WslCommandMixin,
wsl_tilde_quote
26
27     logger = logging.getLogger(__name__)
28
29
30     @dataclass
31     class TrackNetConfig:
32         """Resolved settings for a TrackNet WSL run (built from config.json ->
indexing.tracknet)."""
33         distro: str = "Ubuntu"
34         repo_dir: str = "~/models/TrackNetV4" # WSL path to the cloned repo
35         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
36         conda_env: str = "TrackNetV4"
37         weights_path: str = "" # WSL path to the .h5/.keras weights
(REQUIRED)
38         queue_length: int = 5
39         stage_in_wsl: bool = True # copy video into WSL fs first
(perf)
40         wsl_stage_dir: str = "~/clips" # where staged videos/outputs live
in WSL
41         timeout_sec: int = 1800 # 30 min; kills a hung call (0 = no
timeout)
42
43     @classmethod
44     def from_indexing_cfg(cls, idx_cfg: Dict[str, Any]) -> "TrackNetConfig":
45         tn = (idx_cfg or {}).get("tracknet", {}) or {}
46         return cls(
47             distro=tn.get("wsl_distro", "Ubuntu"),
48             repo_dir=tn.get("repo_dir", "~/models/TrackNetV4"),
49             conda_sh=tn.get("conda_sh", "~/miniconda3/etc/profile.d/conda.sh"),
50             conda_env=tn.get("conda_env", "TrackNetV4"),
51             weights_path=tn.get("weights_path", ""),
52             queue_length=int(tn.get("queue_length", 5)),
53             stage_in_wsl=bool(tn.get("stage_in_wsl", True)),
54             wsl_stage_dir=tn.get("wsl_stage_dir", "~/clips"),
55             timeout_sec=int(tn.get("timeout_sec", 1800)),
56         )
57
58
59     def to_wsl_mnt_path(win_path: str) -> str:
60         """Translate a Windows path (C:\\Users\\x) to its WSL /mnt form (/mnt/c/Users/x).

```



```

61
62     Already-POSIX paths (starting with / or ~) are returned unchanged so the
63     same helper is safe to call on either kind of path.
64     """
65     p = str(win_path)
66     if p.startswith("/") or p.startswith("~"):
67         return p
68     p = p.replace("\\", "/")
69     if len(p) >= 2 and p[1] == ":":
70         drive = p[0].lower()
71         return f"/mnt/{drive}{p[2:]}"
72     return p
73
74
75 @dataclass
76 class TrajectoryPoint:
77     frame: int
78     visible: bool
79     x: float
80     y: float
81
82
83 class TrackNetRunner(WslCommandMixin, DetectorRunner):
84     """Runs TrackNet predict.py in WSL and turns the output CSV into action windows.
85
86     Implements the common DetectorRunner contract so a future Linux-native or
87     alternative trajectory runner can be substituted without touching the hybrids.
88     """
89
90     def __init__(self, cfg: TrackNetConfig):
91         self.cfg = cfg
92
93     def healthcheck(self) -> Tuple[bool, str]:
94         """Verify the WSL env is usable: conda env exists, TF imports, GPU visible.
95
96         Returns (ok, message). Cheap to call before a long run.
97         """
98         cmd = (
99             f"conda activate {self.cfg.conda_env} && "
100             f"python -c \"import tensorflow as tf; "
101             f"print('TF', tf.__version__, 'GPUs', "
102             f"len(tf.config.list_physical_devices('GPU')))\n"
103         )
104         try:
105             res = self._wsl_bash(cmd)
106         except FileNotFoundError:
107             return False, "`wsl` not found ? is this running on Windows with WSL2
108             installed?"
109         except subprocess.TimeoutExpired:
110             return False, "WSL healthcheck timed out."
111         out = (res.stdout or "").strip()
112         if res.returncode != 0:
113             return False, f"WSL env check failed: {(res.stderr or out).strip()}"
114         return True, out
115
116 # ----- inference
117
118 def run_predict(self, video_win_path: str, output_win_dir: str,
119                 video_id: Optional[str] = None) -> Optional[str]:
120     """Run predict.py on a video. Returns the Windows path to the produced CSV, or
121     None.
122
123     ``output_win_dir`` is a Windows directory (under the repo's output/) that
124     WSL writes into via its /mnt view, so the windows process can read the CSV
125     back with no copy. ``video_id`` (file MD5) namespaces the staged video ? and

```

```

123     therefore predict.py's ``<stem>_predict.csv`` output ? so two videos that share
124     a filename cannot collide on the output CSV.
125     """
126     if not self.cfg.weights_path:
127         logger.error(
128             "TrackNet weights_path is not set. Set indexing.tracknet.weights_path "
129             "in config.json to the WSL path of your trained TrackNetV4 weights."
130         )
131         return None
132
133     # Resolve relative dirs BEFORE the /mnt translation ? to_wsl_mnt_path passes
134     # relative paths through unchanged, so WSL would resolve them against the
135     # predict.py cwd instead of the repo (same bug class as wasb_runner).
136     output_win_dir = os.path.abspath(output_win_dir)
137     os.makedirs(output_win_dir, exist_ok=True)
138     # Normalize separators so basename/splitext work when tests (or collector)
139     # pass Windows-style paths on a Linux host.
140     video_win_path = str(video_win_path).replace("\\", "/")
141     base = os.path.basename(video_win_path)
142     stem, ext = os.path.splitext(base)
143
144     # predict.py only accepts .avi/.mp4 and names the CSV "<stem>_predict.csv".
145     if ext.lower() not in (".mp4", ".avi"):
146         logger.error("TrackNet predict.py only supports .mp4/.avi (got %s)", ext)
147         return None
148
149     wsl_out = to_wsl_mnt_path(output_win_dir)
150     # Namespace by video_id so two same-filename videos don't collide on the CSV.
151     # predict.py derives its output name from the (staged) video stem, so staging
152     # under the namespaced name propagates into "<key>_predict.csv".
153     key = f"{stem}_{video_id[:12]}" if video_id else stem
154     expected_csv_win = os.path.join(output_win_dir, f"{key}_predict.csv")
155
156     # Resolve the video path WSL will read.
157     if self.cfg.stage_in_wsl:
158         staged = self._stage_video(video_win_path, f"{key}{ext}")
159         if staged is None:
160             return None
161         wsl_video = staged
162     else:
163         # No staging: predict.py reads /mnt directly and writes
164         # "<stem>_predict.csv".
165         # We cannot rename its output, so fall back to the un-namespaced name.
166         wsl_video = to_wsl_mnt_path(video_win_path)
167         expected_csv_win = os.path.join(output_win_dir, f"{stem}_predict.csv")
168
169     cmd = (
170         f"conda activate {self.cfg.conda_env} && "
171         f"cd {self.cfg.repo_dir}/src && "
172         f"python predict.py "
173         f"--video_path {wsl_tilde_quote(wsl_video)} "
174         f"--model_weights {wsl_tilde_quote(self.cfg.weights_path)} "
175         f"--output_dir {wsl_tilde_quote(wsl_out)} "
176         f"--queue_length {self.cfg.queue_length}"
177     )
178     logger.info("Running TrackNet inference on %s ...", base)
179     try:
180         res = self._wsl_bash(cmd)
181     except subprocess.TimeoutExpired:
182         logger.error("TrackNet inference timed out after %ss.",
183             self.cfg.timeout_sec)
184         return None
185
186     if res.returncode != 0:
187         logger.error("TrackNet predict.py failed:\n%s", (res.stderr or

```

```

res.stdout)[-2000:])
186         return None
187
188         if not os.path.exists(expected_csv_win):
189             logger.error("TrackNet finished but expected CSV not found at %s",
expected_csv_win)
190             return None
191             logger.info("TrackNet trajectory CSV: %s", expected_csv_win)
192             return expected_csv_win
193
194     def _stage_video(self, video_win_path: str, base: str) -> Optional[str]:
195         """Copy the input video into the WSL filesystem (avoids slow /mnt/c reads).
196
197         Returns the WSL path to the staged copy, or None on failure.
198         """
199         src_mnt = to_wsl_mnt_path(video_win_path)
200         dest = f"{self.cfg.wsl_stage_dir}/{base}"
201         cmd = f"mkdir -p {self.cfg.wsl_stage_dir} && cp {shlex.quote(src_mnt)}
{wsl_tilde_quote(dest)} && echo OK"
202         try:
203             res = self._wsl_bash(cmd)
204         except subprocess.TimeoutExpired:
205             logger.error("Staging video into WSL timed out.")
206             return None
207         if res.returncode != 0 or "OK" not in (res.stdout or ""):
208             logger.error("Failed to stage video into WSL: %s", (res.stderr or
res.stdout).strip())
209             return None
210             return dest
211
212     # ----- CSV -> windows
213
214     @staticmethod
215     def parse_trajectory_csv(csv_win_path: str) -> List[TrajectoryPoint]:
216         """Parse a TrackNet predict CSV (columns: Frame,Visibility,X,Y)."""
217         points: List[TrajectoryPoint] = []
218         with open(csv_win_path, newline="") as f:
219             reader = csv.DictReader(f)
220             for row in reader:
221                 try:
222                     points.append(TrajectoryPoint(
223                         frame=int(row["Frame"]),
224                         visible=int(row["Visibility"]) == 1,
225                         x=float(row["X"]),
226                         y=float(row["Y"]),
227                     ))
228                 except (KeyError, ValueError):
229                     continue
230             points.sort(key=lambda p: p.frame)
231             return points
232
233     @staticmethod
234     def trajectory_to_action_windows(
235         points: List[TrajectoryPoint],
236         fps: float,
237         frame_width: float,
238         chunk_start: float = 0.0,
239         velocity_thresh: float = 0.02,
240         merge_gap: float = 2.0,
241         min_window_duration: float = 2.0,
242         chunk_index: Optional[int] = None,
243     ) -> List[Dict[str, Any]]:
244         """Convert a shuttle trajectory into candidate rally windows.
245
246         A frame is "active" when the shuttle is visible AND its inter-frame

```

```

247 displacement (normalised by frame width, per second) exceeds
248 ``velocity_thresh`` ? i.e. the shuttle is in flight. Active frames within
249 ``merge_gap`` seconds of each other are grouped into one window; short
250 windows are dropped. Mirrors HybridYoloSegmenter's windowing so the dict
251 shape feeding the AI-handoff phase is identical.
252 """
253 if fps <= 0:
254     fps = 30.0
255 if frame_width <= 0:
256     frame_width = 1280.0
257
258 # Compute per-frame normalised velocity from the last *visible* point.
259 active = [] # (frame_idx, velocity)
260 prev: Optional[TrajectoryPoint] = None
261 for p in points:
262     if not p.visible:
263         prev = None # break the trajectory; absent shuttle = not in flight
264         continue
265     if prev is not None:
266         dframes = p.frame - prev.frame
267         if dframes > 0:
268             dist = ((p.x - prev.x) ** 2 + (p.y - prev.y) ** 2) ** 0.5
269             velocity = (dist / frame_width) * (fps / dframes)
270             if velocity > velocity_thresh:
271                 active.append((p.frame, velocity))
272     prev = p
273
274 if not active:
275     return []
276
277 max_gap_frames = int(merge_gap * fps)
278
279 def _finalize(group: List[Tuple[int, float]]) -> Dict[str, Any]:
280     vels = [v for _, v in group]
281     return {
282         "start": group[0][0] / fps + chunk_start,
283         "end": group[-1][0] / fps + chunk_start,
284         "active_frame_count": len(group),
285         "peak_velocity": max(vels),
286         "mean_velocity": sum(vels) / len(vels),
287         "track_id_count": 1, # single shuttle; kept for window-shape parity
288         "chunk_index": chunk_index,
289     }
290
291 windows = []
292 group = [active[0]]
293 for af in active[1:]:
294     if af[0] - group[-1][0] <= max_gap_frames:
295         group.append(af)
296     else:
297         windows.append(_finalize(group))
298         group = [af]
299 windows.append(_finalize(group))
300
301 return [w for w in windows if (w["end"] - w["start"]) >= min_window_duration]
302

```

384. user (2026-06-13T14:44:52.410Z)

<https://github.com/avidullu/badminton-highlight-indexer.git>
6de70aa5c840beb91aa50500e6e04b4d812f7429
31069cb fix(detectors): keep ~ expandable in WSL staging/inference commands (#97)

385. assistant (2026-06-13T14:46:13.528Z)

Now I can give you the full picture with exact pointers. Pinning links to the current commit (`6de70aa`) so the line numbers stay stable.

Where `trajectory\_to\_action\_windows` comes from

It's **the indexer's own code** ? not from WASB or TrackNet upstream. It was written as part of the **TrackNet WSL adapter** (ADR-001), and is deliberately **model-agnostic**: it lives as a `@staticmethod` on `TrackNetRunner`, and `WasbRunner` reuses it verbatim because both detectors emit the **same** `Frame, Visibility, X, Y` trajectory CSV:

```
- [tracknet_runner.py::trajectory_to_action_windows](backend/pipeline/detectors/tracknet_runner.py:233) ? the windowing logic ? [GitHub L233?L301](https://github.com/avidullu/badminton-highlight-indexer/blob/6de70aa5c840beb91aa50500e6e04b4d812f7429/backend/pipeline/detectors/tracknet_runner.py#L233-L301)
-
[tracknet_runner.py::parse_trajectory_csv](backend/pipeline/detectors/tracknet_runner.py:214) ? CSV ? points ? [GitHub L214?L231](https://github.com/avidullu/badminton-highlight-indexer/blob/6de70aa5c840beb91aa50500e6e04b4d812f7429/backend/pipeline/detectors/tracknet_runner.py#L214-L231)
- [wasb_runner.py](backend/pipeline/detectors/wasb_runner.py:154) ? WASB inheriting both, the "model-agnostic" seam ? [GitHub L154?L156](https://github.com/avidullu/badminton-highlight-indexer/blob/6de70aa5c840beb91aa50500e6e04b4d812f7429/backend/pipeline/detectors/wasb_runner.py#L154-L156)
```

How the ***trajectory*** itself was detected (the `(x, y, visible)` per frame)

That's WASB, wrapped in `[wasb_infer.py::run](backend/pipeline/detectors/wasb_infer.py:207)` ([GitHub](https://github.com/avidullu/badminton-highlight-indexer/blob/6de70aa5c840beb91aa50500e6e04b4d812f7429/backend/pipeline/detectors/wasb_infer.py#L207-L345)). Two upstream building blocks from **[nttcom/WASB-SBDT](https://github.com/nttcom/WASB-SBDT)** (MIT; "Widely Applicable Strong Baseline for Sports Ball Detection & Tracking", Tarashima et al., BMVC 2023):

1. **Detector** (`[build_detector](backend/pipeline/detectors/wasb_infer.py#L297)` [L297](https://github.com/avidullu/badminton-highlight-indexer/blob/6de70aa5c840beb91aa50500e6e04b4d812f7429/backend/pipeline/detectors/wasb_infer.py#L297)): a heatmap CNN (HRNet-style backbone). It takes a **short stack of consecutive frames** (`frames_in`, here 3 ? that's why 46,080 frames ? 46,078 windows) so it can use motion, and predicts a Gaussian **heatmap** per frame. The peak of the heatmap = the shuttle's pixel location; peak confidence below threshold ? "not visible" that frame.
2. **Online tracker** (`[build_tracker](backend/pipeline/detectors/wasb_infer.py#L338)` [L338](https://github.com/avidullu/badminton-highlight-indexer/blob/6de70aa5c840beb91aa50500e6e04b4d812f7429/backend/pipeline/detectors/wasb_infer.py#L338)): links the per-frame peaks into one coherent track (position/motion association), rejecting spurious blobs and filling `Visibility=0, X=Y=-1` where the shuttle is genuinely lost.

The result is one row per frame ? `Frame, Visibility(0/1), X, Y` ? which is exactly what `parse_trajectory_csv` reads. (Your run: 46,080 rows, 29,879 visible = 64.8%.)

How `trajectory\_to\_action\_windows` turns that into candidate rallies

The core is a **velocity gate + temporal grouping**:

```
```python
```

**per consecutive VISIBLE pair: speed as fraction-of-frame-width per second**

```
dist = ((p.x - prev.x)**2 + (p.y - prev.y)**2) ** 0.5
velocity = (dist / frame_width) * (fps / dframes)
if velocity > velocity_thresh: # shuttle is "in flight"
 active.append((p.frame, velocity))
```

**an INVISIBLE frame resets prev ? absent shuttle = not in flight (a natural break)**

```
```
```

```
```python
```

**group active frames; a gap longer than merge\_gap closes the window**

```

max_gap_frames = int(merge_gap * fps)
if af.frame - group[-1].frame <= max_gap_frames:
 group.append(af) # same rally ? short pause tolerated
else:
 windows.append(_finalize(group)); group = [af] # new rally
...
return [w for w in windows if (w.end - w.start) >= min_window_duration]
` ``

```

Step by step:

1. **Velocity** is normalized by frame width (so it's resolution-independent ? works identically on 1080p) and converted to per-second. Above `velocity_thresh` ? the frame is "active" (shuttle flying).
2. **An invisible shuttle breaks the trajectory** (`prev = None`) ? disappearance is itself a boundary signal.
3. **Grouping** merges active frames within `merge_gap` seconds into one window ? so the brief pauses *between shots within a rally* don't split it, but a long stillness (shuttle down, players resetting) *ends* the window. ? this is the "coarse rally boundary" from the flowchart.
4. Windows shorter than `min_window_duration` are dropped (kills jitter/false blips). Each window carries `start/end`, `peak_velocity`, `mean_velocity`, `active_frame_count`.

These three knobs come from `config.json` ? indexing.wasb`:

- `velocity_threshold` (0.02) ? `velocity_thresh`
- `dead_time_merge_gap` (2.0s) ? `merge_gap`
- `min_action_window_duration` (2.0s) ? `min_window_duration`

So WASB gives you *motion-derived* candidate windows (18 of them on your video); Gemini then refines each into precise rallies with shot counts. The windowing is intentionally the same shape `HybridYoloSegmenter` produces, which is why the AI-handoff + DB stages are detector-agnostic.

---

Meanwhile the *75* reel is still stitching\* (24 clips from 4K, last I saw "Trimming 24 segments") ? I'll confirm it with resolution + duration when it finishes.

**386. user (2026-06-13T14:46:17.653Z)**

<local-command-caveat>Caveat: The messages below were generated by the user while running local commands. DO NOT respond to these messages or otherwise consider them in your response unless the user explicitly asks you to.</local-command-caveat>

**387. user (2026-06-13T14:46:17.653Z)**

```

<command-name>/model</command-name>
 <command-message>model</command-message>
 <command-args>claude-fable-5</command-args>

```

**388. user (2026-06-13T14:46:17.653Z)**

<local-command-stdout>Set model to claude-fable-5</local-command-stdout>

**389. user (2026-06-13T14:49:10.899Z)**

```

<task-notification>
<task-id>bd1qgo1zh</task-id>
<summary>Monitor event: "?5 re-stitch milestones (filter, stitch, done, errors)"</summary>
<event>20:19:10 [backend.pipeline.stitcher.compiler] INFO: Stitching clips together...</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>

```

**390. assistant (2026-06-13T14:49:11.944Z)**

There's an issue with the selected model (claude-fable-5). It may not exist or you may not have access to it.

**391. user (2026-06-13T14:49:13.939Z)**

```
<task-notification>
<task-id>boilngexb</task-id>
<tool-use-id>toolu_01S7LNmPB7cjGuodo87kEQfN</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\boilngexb.output</output-file>
<status>completed</status>
<summary>Background command "Re-stitch ?5-shot reel from 4K using cached rallies" completed (exit code 0)</summary>
</task-notification>
```

**392. assistant (2026-06-13T14:49:14.852Z)**

There's an issue with the selected model (claude-fable-5). It may not exist or you may not have access to it.

**393. user (2026-06-13T14:49:14.870Z)**

```
<task-notification>
<task-id>bd1qgo1zh</task-id>
<summary>Monitor event: "?5 re-stitch milestones (filter, stitch, done, errors)"</summary>
<event>20:19:13 [backend.pipeline.stitcher.compiler] INFO: Dynamic highlight reel saved:
output\GX010125_highlights_min5.mp4 (1987.0 MB)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

**394. assistant (2026-06-13T14:49:15.767Z)**

There's an issue with the selected model (claude-fable-5). It may not exist or you may not have access to it.

**395. user (2026-06-13T14:49:15.778Z)**

```
<task-notification>
<task-id>bd1qgo1zh</task-id>
<summary>Monitor event: "?5 re-stitch milestones (filter, stitch, done, errors)"</summary>
<event>[DONE] C:\Users\avidu\Projects\badminton-highlight-indexer\output\GX010125_highlights_min5.mp4 (24 rallies, 660s)</event>
If this event is something the user would act on now, send a PushNotification. Routine or benign
output doesn't need one.
</task-notification>
```

**396. assistant (2026-06-13T14:49:16.802Z)**

There's an issue with the selected model (claude-fable-5). It may not exist or you may not have access to it.

**397. user (2026-06-13T14:54:07.593Z)**

```
<local-command-caveat>Caveat: The messages below were generated by the user while running local
commands. DO NOT respond to these messages or otherwise consider them in your response unless
the user explicitly asks you to.</local-command-caveat>
```

**398. user (2026-06-13T14:54:07.593Z)**

```
<command-name>/model</command-name>
 <command-message>model</command-message>
 <command-args>claude-opus-4-8</command-args>
```

#### 399. user (2026-06-13T14:54:07.593Z)

<local-command-stdout>Set model to claude-opus-4-8</local-command-stdout>

#### 400. user (2026-06-13T14:56:37.944Z)

can you check what is happening on my gpu? does stictching happen on gpu?

#### 401. assistant (2026-06-13T14:57:04.821Z)

Let me check the GPU right now and confirm how stitching actually encodes:

#### 402. user (2026-06-13T14:57:08.984Z)

```
=== GPU NOW ===
32 %, 2242 MiB, 8192 MiB, 21.63 W
--- compute processes on GPU ---
5684, [Insufficient Permissions], [N/A]
26736, C:\Program Files
(x86)\Microsoft\EdgeWebView\Application\149.0.4022.52\msedgewebview2.exe, [N/A]
9032, C:\Windows\System32\ApplicationFrameHost.exe, [N/A]
22892, C:\Windows\explorer.exe, [N/A]
8876, C:\Program Files\WindowsApps\ReaderNotificationClient_1.0.4.0_x86__e1rzdqpraam7r\AcrobatNo
tificationClient.exe, [N/A]
9436, C:\Program
Files\WindowsApps\5319275A.WhatsAppDesktop_2.2620.102.0_x64__cv1g1gvanyjgm\WhatsApp.Root.exe,
[N/A]
23896, C:\Program
Files\WindowsApps\5319275A.51895FA4EA97F_2.2622.101.0_x64__cv1g1gvanyjgm\WhatsApp.Root.exe,
[N/A]
14804, C:\Windows\SystemApps\ShellExperienceHost_cw5n1h2txyewy\ShellExperienceHost.exe, [N/A]
17388, C:\Windows\SystemApps\Microsoft.LockApp_cw5n1h2txyewy\LockApp.exe, [N/A]
1748, C:\Windows\ImmersiveControlPanel\SystemSettings.exe, [N/A]
24672, C:\Program Files\WindowsApps\Claude_1.12603.1.0_x64__pzs8srxjxfjjc\app\claude.exe, [N/A]
29704, C:\Program Files\WindowsApps\Claude_1.12603.1.0_x64__pzs8srxjxfjjc\app\claude.exe, [N/A]
4, [Insufficient Permissions], [N/A]
35632, C:\Users\avidu\AppData\Local\Programs\Microsoft VS Code\Code.exe, [N/A]
20268, C:\Program
Files\WindowsApps\Microsoft.YourPhone_1.26042.99.0_x64__8wekyb3d8bbwe\PhoneExperienceHost.exe,
[N/A]
11288, C:\Windows\SystemApps\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\TextInputHost.exe, [N/A]
32268, C:\Program Files\NVIDIA Corporation\NVIDIA app\CEF\NVIDIA Overlay.exe, [N/A]
2508, C:\Program Files\NVIDIA Corporation\NVIDIA app\CEF\NVIDIA Overlay.exe, [N/A]
12688, [Insufficient Permissions], [N/A]
13776, C:\Windows\SystemApps\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\SearchHost.exe, [N/A]
15312, C:\Windows\SystemApps\Microsoft.Windows.StartMenuExperienceHost_cw5n1h2txyewy\StartMenuEx
perienceHost.exe, [N/A]
18480, C:\Windows\System32\ShellHost.exe, [N/A]
2856, C:\Program Files\VideoLAN\VLC\vlc.exe, [N/A]
1908, C:\Program Files (x86)\Microsoft\Copilot\Application\mscopilot.exe, [N/A]
11536, [Insufficient Permissions], [N/A]
--- ffmpeg/python running? ---
```

#### 403. user (2026-06-13T14:57:09.504Z)

```
101: watermarking is disabled or has nothing to draw. The concat stage stays -c copy
248: "-c:v", "libx264",
249: "-preset", "superfast",
251: "-c:a", "aac",
258: # whole reel. Clips stay codec-identical either way, so concat -c copy
holds.
332: "-c", "copy",
```

#### 404. user (2026-06-13T14:57:13.883Z)



```
codec_name=h264
width=3840
height=2160
r_frame_rate=60000/1001
duration=247.742933
size_MB=1987
--- watermark failures in min5 stitch? ---
0
```

#### 405. assistant (2026-06-13T14:57:47.965Z)

Stopping the finished min5 monitor, then the answer:

#### 406. user (2026-06-13T14:57:47.980Z)

```
{"message":"Successfully stopped task: bd1qgo1zh (tail -n +1 -f
\"/c/Users/avidu/Projects/badminton-highlight-indexer/output/gx010125_min5_stitch.log\" | grep
-E --line-buffered \"\\[REUSE\\]|\\[FILTER\\]|\\[DONE\\]|Trimming [0-
9]|Stitching|saved|Traceback|ERROR|COMPILER|SystemExit\")\", \"task_id\":\"bd1qgo1zh\", \"task_type\":\"lo
cal_bash\", \"command\":\"tail -n +1 -f \"/c/Users/avidu/Projects/badminton-highlight-
indexer/output/gx010125_min5_stitch.log\" | grep -E --line-buffered
\"\\[REUSE\\]|\\[FILTER\\]|\\[DONE\\]|Trimming
[0-9]|Stitching|saved|Traceback|ERROR|COMPILER|SystemExit\""}]
```

#### 407. assistant (2026-06-13T14:58:03.756Z)

### Your GPU right now: idle of pipeline work

32% util, 2.2 GB, 21 W ? that's just \*\*desktop apps\*\* holding the GPU for compositing/decode (Edge WebView, explorer, WhatsApp, the Claude app, VS Code, and \*\*VLC\*\* ? possibly you previewing the reel). \*\*No compute job is running\*\* ? no `python` or `ffmpeg` process, and no ML/encode load (the 2070 Super pulls ~80 W under real load; 21 W is idle). The ?5 stitch already finished.

### Does stitching use the GPU? \*\*No ? it's CPU.\*\*

The compiler encodes with the \*\*software x264 encoder\*\*, not NVENC:

```
- Per-clip re-encode: [-c:v libx264 -preset
superfast](backend/pipeline/stitcher/compiler.py:248) (CPU)
- Final join: [-c copy](backend/pipeline/stitcher/compiler.py:332) (stream-copy, no encoding
at all)
```

So the ~11-minute ?5 stitch was \*\*CPU-bound\*\* (24 clips of 4K x264 + a watermark drawtext filter, which is also a CPU filter). That's why GPU stayed low while it ran.

**\*\*For contrast, what \*did\* run on the GPU earlier:\*\***

```
- The **4K?1080p proxy transcode** ? `h264_nvenc` (GPU encoder).
- **WASB shuttle inference** ? the detector CNN on CUDA.
```

So in this whole job, the GPU did the \*analysis\* (WASB) and the \*proxy\*, while the \*final stitch\* is CPU.

### ?5 reel is done and verified: `output\GX010125\_highlights\_min5.mp4`

```
- **24 rallies** (shot_count ? 5), **true 4K** 3840x2160, 59.94 fps, **4:08**, 1987 MB, cut from
the original.
- **0 watermark failures** this time (vs 12/12 on the >7 reel) ? so the watermark is now
correctly burned in. That's a clean end-to-end confirmation that the #108 fix works on real 4K
footage.
```

You now have both: `GX010125\_highlights.mp4` (11 rallies, >7 shots, no watermark) and  
`GX010125\_highlights\_min5.mp4` (24 rallies, ?5 shots, watermarked).

---

If the 11-minute CPU stitch felt slow: I can make the stitcher **\*\*GPU-encode with NVENC\*\*** (decode ``-hwaccel cuda` + `-c:v h264_nvenc``), which would cut 4K stitching from ~minutes to ~tens of seconds. It's a legit improvement, but it needs a **\*\*software fallback\*\*** (NVENC isn't present on the Linux CI or non-NVIDIA machines) and a CRF?CQ quality mapping, so I'd do it as a proper PR with the fallback rather than hard-switching. Want me to take that on?

**408. user (2026-06-13T15:05:53.898Z)**

can you do a cleanup of all cached video in the WSL? show me all the files which you will delete

**409. assistant (2026-06-13T15:06:32.369Z)**

Good ? destructive cleanup, so I'll **\*\*inventory first and show you the full list before deleting anything.\*\*** Let me take a complete picture of the WSL staging area:

**410. user (2026-06-13T15:06:40.687Z)**

```
=== ~/clips TOP-LEVEL (size, name), sorted ===
4.0K ./adarsh_avi_full_run.log
4.0K ./adarsh_avi_run.log
4.0K ./gbaaddy_run.log
4.0K ./kushagra_run.log
4.0K ./largetestvideo_run.log
4.0K ./mahadevpura_run.log
4.0K ./multisport_run.log
4.0K ./run_wasb_avi.sh
4.0K ./run_wasb_avi_full.sh
4.0K ./run_wasb_gbaaddy.sh
4.0K ./run_wasb_kushagra.sh
4.0K ./run_wasb_largetestvideo.sh
4.0K ./run_wasb_mahadevpura.sh
4.0K ./run_wasb_multisport.sh
8.0K ./batch_test.csv
8.0K ./wasb_traj_test.csv
96K ./tabletennis_traj.csv
96K ./testlarge_traj.csv
144K ./pickleball_traj.csv
360K ./tabletennis_traj_wasbcache
412K ./mahadevpura_traj.csv
420K ./testlarge_traj_wasbcache
516K ./pickleball_traj_wasbcache
588K ./sample_long_traj.csv
680K ./largetestvideo_traj.csv
1.1M ./adarsh_avi_traj.csv
1.1M ./adarsh_traj.csv
1.2M ./kushagra_traj.csv
1.7M ./gbaaddy_traj.csv
1.7M ./mahadevpura_traj_wasbcache
2.4M ./adarsh_avi_full_traj.csv
2.7M ./largetestvideo_traj_wasbcache
3.9M ./adarsh_traj_wasbcache
4.2M ./adarsh_avi_traj_wasbcache
5.6M ./kushagra_traj_wasbcache
7.2M ./gbaaddy_traj_wasbcache
9.1M ./adarsh_avi_full_traj_wasbcache
87M ./mahadevpura_smoke_180s__05e0e577029d.mp4
659M ./tabletennis_frames
1.5G ./pickleball_frames
2.2G ./GX010125_1080p__89605e80ed01.mp4
2.4G ./mahadevpura_frames
4.5G ./largetestvideo_frames
7.0G ./adarsh_avi_frames
```

```
9.5G ./mahadevpura_smoke_180s__05e0e577029d_frames
11G ./kushagra_frames
16G ./gbaaddy_frames
17G ./adarsh_avi_full_frames
93G ./GX010125_1080p__89605e80ed01_frames
```

```
=== TOTAL in ~/clips ===
163G .
```

```
=== WSL disk free ===
/dev/sdc 1007G 220G 736G 24% /
```

```
=== any other staged videos under ~ (excluding models/weights) ===
/home/avidullu/clips/mahadevpura_smoke_180s__05e0e577029d.mp4
/home/avidullu/clips/GX010125_1080p__89605e80ed01.mp4
```

#### 411. assistant (2026-06-13T15:07:45.296Z)

Here's the complete inventory of `~/clips` (163 GB total). I've categorized it ? **nothing is deleted yet.**

#### ?? Proposed for deletion ? "cached video" (~163 GB)

**Decoded frame caches** (`\*\_frames/` ? the space hogs, all regenerable from the videos):

Dir	Size
GX010125_1080p__89605e80ed01_frames`	93 G
adarsh_avi_full_frames`	17 G
gbaaddy_frames`	16 G
kushagra_frames`	11 G
mahadevpura_smoke_180s__05e0e577029d_frames`	9.5 G
adarsh_avi_frames`	7.0 G
largetestvideo_frames`	4.5 G
mahadevpura_frames`	2.4 G
pickleball_frames`	1.5 G
tabletennis_frames`	659 M

**Staged video copies** (\*.mp4` ? originals exist elsewhere):

File	Size	Original lives at
GX010125_1080p__89605e80ed01.mp4`	2.2 G	Windows `output\GX010125_1080p.mp4`
mahadevpura_smoke_180s__05e0e577029d.mp4`	87 M	golden-set source

#### ? Will be KEPT (not video ? ~50 MB total)

- **Trajectory results** (`\*\_traj.csv`, etc. ? adarsh, kushagra, gbaaddy, mahadevpura, pickleball, tabletennis?) ? these are your **WASB analysis outputs** / golden-set corpus trajectories, GPU-generated and tied to eval work. Tiny, valuable, not video.
- **Detector resume caches** (`\*\_wasbcache/`, ~45 MB) ? not video; deleting would force re-inference.
- Run scripts (\*.sh) and logs (\*.log).

#### Two things to note

- **GX010125's detector cache + trajectory CSV** are NOT in WSL ? they're on the Windows side (`output\wasb\?`), so this WSL cleanup won't touch them. The 52 confirmed rallies are in the SQLite DB, so **re-stitching either reel will still work** after this cleanup (stitch reads the DB + the 4K original, not these frames).
- Everything here is regenerable (frames re-extract from the videos; staged copies re-copy from originals), so this is reclaimable space, not lost work.

How much do you want me to remove?

#### 412. user (2026-06-13T15:08:35.346Z)

Your questions have been answered: "Confirm WSL ~/clips cleanup scope (frame caches + staged videos):"="Keep GX010125, delete rest". You can now continue with these answers in mind.

#### 413. assistant (2026-06-13T15:09:01.206Z)

Confirmed ? keeping GX010125's cache, deleting the 9 older golden-set frame dirs + the mahadevpura smoke clip (~70 GB). Running it now with before/after verification:

#### 414. user (2026-06-13T15:09:10.635Z)

```
Exit code 2
BEFORE:
/dev/sdc 1007G 220G 736G 24% /
AFTER:
/dev/sdc 1007G 152G 804G 16% /
=== remaining *_frames (expect only GX010125) ===
93G GX010125_1080p__89605e80ed01_frames
=== remaining videos (expect only GX010125 proxy) ===
-rwxr-xr-x 1 avidullu avidullu 2.2G Jun 13 08:47 GX010125_1080p__89605e80ed01.mp4
```

#### 415. assistant (2026-06-13T15:09:23.117Z)

Cleanup succeeded ? the `Exit code 2` is just the `ls \*.MP4` glob finding no uppercase matches (harmless), not a deletion error. The reclaim is confirmed: **\*\*804 G free, up from 736 G ? ~68 GB freed\*\***. Let me get a clean final confirmation that the kept items (trajectories, caches) survived:

#### 416. user (2026-06-13T15:09:27.861Z)

```
=== ~/clips total now ===
95G .
=== kept trajectory CSVs ===
11
=== kept wasbcache dirs ===
10
=== anything still >100M (should be only GX010125) ===
2.2G ./GX010125_1080p__89605e80ed01.mp4
93G ./GX010125_1080p__89605e80ed01_frames
```

#### 417. assistant (2026-06-13T15:09:38.160Z)

Cleanup complete and verified. ?

### Result

	Before	After
~/clips` total	163 GB	<b>**95 GB**</b>
WSL disk free	736 GB	<b>**804 GB** (+68 GB)</b>

**\*\*Deleted\*\*** (10 items, ~68 GB): the 9 older golden-set frame caches ? `adarsh\_avi\_full\_frames`, `gbaaddy\_frames`, `kushagra\_frames`, `mahadevpura\_smoke?\_frames`, `adarsh\_avi\_frames`, `largetestvideo\_frames`, `mahadevpura\_frames`, `pickleball\_frames`, `tabletennis\_frames` ? plus the `mahadevpura\_smoke?mp4` staged clip.

**\*\*Kept, as you chose:\*\***

- `GX010125\_1080p\_\_89605e80ed01\_frames` (93 G) + the proxy `?01.mp4` (2.2 G) ? so re-running WASB on this video stays instant.
- **\*\*11 trajectory CSVs\*\*** (your golden-set WASB outputs) and **\*\*10 `\*\_wasbcache` dirs\*\*** ? all survived.
- Run scripts + logs.

Nothing else was touched ? the Windows-side artifacts (the `output\` proxy, the GX010125 detector cache/CSV in `output\wasb\`, and the SQLite DB with the 52 cached rallies) are all intact, so re-stitching either reel still works without re-running anything.

If you later want to reclaim that last 95 GB too (once you're done with this video), just say so and I'll drop the GX010125 frames + proxy as well.

#### 418. user (2026-06-13T15:10:32.516Z)

can you make a plan on how to ensure we keep doing cleanups in a healthy way?

#### 419. assistant (2026-06-13T15:11:50.053Z)

Here's a plan for keeping cache hygiene healthy without it becoming a chore ? built around what we actually saw accumulate.

### The core problem

The bloat is **regenerable intermediates**, dominated by one thing: PNG frame extraction materializes every frame to disk ? **93 GB for a single 12.8-min 4K video**. Across many videos that's unbounded. Staged video copies (2 GB+ each) pile on. Meanwhile the **valuable** outputs are tiny or already archived. So the principle is simple:

> **Separate regenerable intermediates (purge aggressively, automatically) from durable outputs (keep/archive). Make purging default-safe ? dry-run first, the "show me before you delete" pattern.**

### Retention policy ? what lives where, and its lifespan

Artifact	Location	Regenerable?	Policy
---	---	---	---
`*_frames/` (PNG decode)	WSL `~/clips`	? from video	<b>Delete on success</b> (only needed during the detector pass)
Staged `*.mp4` copies	WSL `~/clips`	? from original	<b>Delete on success</b>
`*_wasbcache/` (resume cache)	WSL + `output/wasb`	?	Keep only while a run is in-flight/iterating; drop once CSV exists
`chunks*_handoff/`, `tmp*/`	Windows `output/`	?	Auto-clean per run; sweep stragglers on crash
1080p proxy `_1080p.mp4`	Windows `output/`	? from 4K	Keep while iterating on that video; purge when done
<b>Trajectory CSVs</b>	`~/clips`, `output/wasb`	?? GPU-only	<b>Keep</b> ; archive to `Annotation Setup\Collect\Trajectories`
<b>SQLite DB (rallies)</b>	`output/`	?? Gemini cost	<b>Keep</b> ? this is what makes re-stitch free
<b>Final reels</b>	`output/`	?? expensive	Keep until copied out, then purge

### Tier 1 ? Make the pipeline self-clean (the real fix)

The highest-leverage change: **delete the `\*\_frames/` dir and staged video automatically once the trajectory CSV is written successfully.** Frames are purely intermediate ? after the CSV exists they're dead weight. Done right this caps per-video frame bloat to "during the run only," so it never accumulates again.

- Add `indexing.keep\_frames: false` (default) to config; gate deletion on it so debugging can opt to retain.
- **Only delete on success** (CSV present) ? on failure, keep frames + `wasbcache` so the resume-across-reboots design still works.
- Wire it in `wasb\_runner.run\_predict` / `tracknet\_runner` (they already know the frames path). This is a clean, testable PR.

### Tier 2 ? A dry-run-first cleanup tool

Replace ad-hoc `rm` with `tools/cleanup\_caches.py` (or scratch first):

- Scans **both** WSL `~/clips`` and Windows `output/{wasb,chunks_*_handoff,tmp*}``.
- Classifies each entry (regenerable vs durable) and prints a **sized** report ? dry-run by default (exactly the review-first flow you just used).
- `--apply`` to delete; `--keep-video <id>`` to protect current work; `--older-than 7d`` / `--min-size`` filters.
- One command, repeatable, no per-session re-discovery of paths.

### Tier 3 ? Cadence + guardrails (so it actually happens)

- **Disk preflight in the runner**: check `df`` before extraction; warn (or refuse) if free space < threshold (e.g. 100 GB), optionally offering to GC the oldest frame caches.
- **Cadence tied to the existing checkpoint convention**: run the cleanup tool's dry-run at session wrap and after each completed video; that's when frames are safe to drop (CSV + DB rallies exist).
- **Optional passive hook**: a `Stop`/`SessionEnd`` hook in `settings.json`` that runs the dry-run `*report only*` (never auto-deletes) so you always see accumulation before it's a problem.

### Tier 4 ? Guarantee outputs are durable first

Cleanup is only safe because outputs survive: ensure trajectory CSVs land in `Annotation Setup\Collect\Trajectories`` and reels get copied out, *then* intermediates are always disposable. This is already the convention ? Tier 1/2 just lean on it.

---

**My recommendation**: Tier 1 is the one that actually ends the problem (the pipeline stops hoarding 93 GB/video); Tier 2 makes the occasional manual sweep safe and fast. Tiers 3/4 are policy/cadence, mostly already in your conventions.

Want me to:

#### 420. user (2026-06-13T15:12:50.669Z)

Your questions have been answered: "Which parts of the cleanup plan should I build now?"="Tier 1: self-cleaning pipeline,Tier 2: cleanup tool,Tier 3: disk-guard + hook,Save policy to memory". You can now continue with these answers in mind.

#### 421. assistant (2026-06-13T15:14:09.559Z)

All four ? let me build them. I'll start a new phase, save the policy to memory, and read the exact code I need to modify.

#### 422. user (2026-06-13T15:14:10.023Z)

Chapter marked: "Cache-hygiene automation". Continue your current work.

#### 423. user (2026-06-13T15:14:15.323Z)

```
64:class IndexingConfig(BaseModel):
```

#### 424. user (2026-06-13T15:14:15.705Z)

```
1 """Abstract base for detector runners (TrackNet, WASB, future native, etc.).
2
3 This completes the Detector Runner Abstraction (low-regret item 6 continuation / item 1
4 of the post-observability re-prioritization) to de-risk the WSL/conda dependency for
5 the high-accuracy shuttle trajectory substrate used by tracknet_hybrid / wasb_hybrid.
6
7 The ABC lets us:
8 - Keep the current WSL adapters working unchanged for Windows users.
9 - Swap in (or A/B test) a Linux-native implementation later with zero changes to
10 the hybrid segmenters or the model-agnostic windowing brain.
11 - Provide a single place to evolve the healthcheck / error contract and thread
```

```

12 detector readiness into per-video reports.
13
14 Contract (deliberately matches what the concrete runners + hybrids + tests already did):
15 - run_predict(video_win_path: str, output_win_dir: str) -> Optional[str]
16 Run detector, return the Windows-side path to the produced trajectory CSV
17 (e.g. ".../clip_predict.csv" or ".../clip_wasb.csv"), or None on any failure.
18 The caller (hybrid) is responsible for output dir; impls may stage the video
19 into WSL fs for perf.
20
21 - healthcheck() -> tuple[bool, str]
22 Return (ok, message). Never raises. "ok" decides whether to proceed to run_predict.
23 Message is logged (and can be surfaced to reports or UX in future).
24 Intended to be cheap (env existence, import, GPU visibility, weights present).
25
26 Existing behavior is 100% preserved. The only additions are the inheritance and
27 the explicit common type for future implementers.
28
29 How to add a new runner (for docs / future contributors):
30 1. Subclass DetectorRunner in a new or existing module under pipeline/detectors/.
31 2. Implement run_predict and healthcheck (return (bool, str) tuple).
32 3. If it produces the same CSV schema (Frame,Visibility,X,Y), reuse or re-export
33 TrackNetRunner.parse_trajectory_csv and .trajectory_to_action_windows (they are
34 deliberately static and model-agnostic). If the new detector needs different
35 windowing, it can implement its own or we generalize later.
36 4. Lazy-import heavy native deps inside the methods (so importing the segmenter
37 registry never pulls torch/tf on a machine that doesn't have them).
38 5. Update the two hybrid segmenters (or introduce a detector= choice) to accept
39 DetectorRunner instances (they already do the right thing via duck-typing + the
40 type hints we add here).
41 6. Add unit tests that exercise the contract (isinstance, healthcheck shape, run
42 returns str|None) using mocks for the external bits ? see test_tracknet_runner.py.
43 7. No change to eval/calibration code paths (they consume the CSV + the static
44 windowing helpers directly).
45
46 Optional Linux-native path (the main de-risk goal):
47 A LinuxNativeRunner (or ONNXRunner) would:
48 - Skip all wsl / _stage / conda activate.
49 - Load weights locally (torch or onnx) or call a native binary.
50 - Write the exact same CSV format so trajectory_to_action_windows works verbatim.
51 - healthcheck can just check "weights file exists and torch importable + CUDA?".
52 This removes the WSL requirement for Linux users / future SaaS workers while
53 keeping the exact same candidate quality characteristics (or better, once native
54 weights are available).
55
56 Error surfacing into reports:
57 The healthcheck message is already logged by the hybrids on failure. The reporting
58 harvest sees the outcome as "0 candidates" + detector name. When we wire richer
59 segmenter Results (or pass detector_health into emit_indexer_report context), the
60 (ok, msg) can be recorded explicitly under segmentation.details or a new detector
61 stage. The ABC makes that future change local to one place.
62
63 See also:
64 - backend/pipeline/segmenters/{tracknet_hybrid,wasb_hybrid}.py (the two callers)
65 - tests/test_tracknet_runner.py (contract + pure logic tests; hybrids mock the runner)
66 - docs/TRACKNET_WSL_SETUP.md and CODE_MAP.md (2.2 Detectors section)
67 ""
68
69 import logging
70 import shlex
71 import subprocess
72 from abc import ABC, abstractmethod
73 from typing import Any, Callable, Optional, Tuple
74
75 logger = logging.getLogger(__name__)
76

```

```

77
78 def wsl_tilde_quote(path: str) -> str:
79 """shlex.quote that still lets bash expand a leading ``~/``.
80
81 ``shlex.quote("~/clips/x.mp4")`` single-quotes the whole path, so bash
82 treats ``~`` literally and ``cp``/``mkdir`` fail with "No such file or
83 directory" ? this broke live WSL staging for the default ``~/clips``
84 stage dir (found by the 2026-06-11 wasb_hybrid smoke run; the BACKLOG
85 "shlex-quote breaks ~ expansion" item). Quoting only the part after
86 ``~/`` keeps expansion AND protects spaces/specials in the remainder.
87 """
88 if path == "~":
89 return path
90 if path.startswith("~/"):
91 return "~/\" + shlex.quote(path[2:])
92 return shlex.quote(path)
93
94
95 class WslCommandMixin:
96 """Shared ``wsl -d <distro> bash -c 'source <conda_sh> && <cmd>'`` invocation.
97
98 Lives beside (not on) DetectorRunner: a future Linux-native runner must not
99 inherit WSL plumbing. Requires ``self.cfg`` with ``distro``, ``conda_sh``
100 and ``timeout_sec`` attributes (TrackNetConfig and WasbConfig both qualify).
101 ``timeout_sec == 0`` maps to None = wait forever (configs default 1800s).
102 """
103
104 cfg: Any
105
106 def _wsl_bash(self, inner_cmd: str) -> subprocess.CompletedProcess:
107 full = f"source {self.cfg.conda_sh} && {inner_cmd}"
108 argv = ["wsl", "-d", self.cfg.distro, "bash", "-c", full]
109 logger.debug("WSL exec: %s", full)
110 return subprocess.run(argv, capture_output=True, text=True,
111 timeout=(self.cfg.timeout_sec or None))
112
113
114 class DetectorRunner(ABC):
115 """Common interface for shuttle/ball trajectory detectors (WSL or otherwise)."""
116
117 @abstractmethod
118 def run_predict(self, video_win_path: str, output_win_dir: str,
119 video_id: Optional[str] = None) -> Optional[str]:
120 """Run inference and return path to trajectory CSV (or None on failure).
121
122 ``video_id`` (the file MD5) namespaces all staged/cached/output artifacts so two
123 videos that happen to share a filename cannot reuse each other's frames or CSV.
124 It is optional for back-compat; callers in the live pipeline always pass it.
125 """
126 pass
127
128 @abstractmethod
129 def healthcheck(self) -> Tuple[bool, str]:
130 """Quick check if the runner (env, weights, etc.) is usable.
131
132 Returns (ok, message). Never raises. Cheap to call before a long run.
133 """
134 pass
135
136 # De facto trajectory contract: the trajectory hybrids call these
137 # unconditionally after run_predict. Concrete runners reuse/re-export the
138 # model-agnostic TrackNetRunner staticmethods (same CSV schema + windowing);
139 # a runner with custom windowing supplies its own with the same shapes.
140 parse_trajectory_csv: Callable[..., Any]
141 trajectory_to_action_windows: Callable[..., Any]

```



## 425. user (2026-06-13T15:14:21.056Z)

```

64 class IndexingConfig(BaseModel):
65 default_segmenter: str = "yolo_hybrid"
66 use_gpu: bool = True
67 motion_threshold: float = 0.08
68 min_segment_duration: float = 3.0
69 dead_time_merge_gap: float = 2.0
70 chunk_duration: float = 90.0
71 chunk_overlap: float = 10.0
72 detector_model: str = "fasterrcnn_resnet50_fpn"
73 detector_score_threshold: float = 0.5
74 yolo_velocity_threshold: float = 0.15
75 yolo_frame_skip: int = 3
76 yolo_tracker: str = "bytetrack.yaml"
77 yolo_prefetch_workers: int = 2
78 min_action_window_duration: float = 2.0
79 log_yolo_candidates: bool = True
80 store_yolo_candidates: bool = True
81 ai_handoff_padding: float = 2.0
82 ai_max_workers: int = 5
83 # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
84 # chunks of a high-bitrate (4K) source blow past the provider upload cap
85 # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
86 # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
87 # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
88 # Matches gemini_refine's --clip-scale-width default.
89 ai_chunk_scale_width: int = 1280
90 # Optional debug knob: trim every video to the first N seconds before processing.
91 debug_duration: Optional[float] = None
92
93 tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
94 wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
95
96 @field_validator("default_segmenter")
97 @classmethod
98 def segmenter_must_be_registered(cls, v: str) -> str:
99 # Registry check happens at runtime in main/segmenter selection.
100 # Here we just allow any string for forward compatibility.
101 return v
102
103 @model_validator(mode="after")
104 def _chunk_step_must_be_positive(self) -> "IndexingConfig":
105 # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
106 # positive
107 # step makes `while t < duration: t += step` loop forever, growing memory before
108 # any
109 # GPU work. Reject the bad config at load time instead.
110 if self.chunk_overlap >= self.chunk_duration:
111 raise ValueError(
112 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
113 "
114 f"({self.chunk_duration}); otherwise chunking never advances."
115)
116 return self
117
118 class OutputConfig(BaseModel):
119 default_highlights_name: str = "highlights.mp4"
120 db_name: str = "sports_indexer.db"
121 # Directory where the DB, highlight reels, and processed clips are written.
122 # The CLI's --output-dir overrides this at startup (see backend/main.py::main).
123 output_dir: str = "output"

```

```

122
123
124 class WatermarkConfig(BaseModel):
125 """Branding overlay burned into each highlight clip during the per-clip re-encode
126 (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
127 (under `stitching.watermark` in config.json) to turn it off. The text is fully
128 configurable. The concat stage stays stream-copy (-c copy)."""
129 enabled: bool = True
130 text: str = "Generated by Khelsutra.guru"
131 logo_path: str = "" # optional transparent PNG overlay; "" = no logo
132 font_path: str = "" # optional .ttf; "" = use the fontconfig `font` family
below
133 font: str = "Sans" # fontconfig family used when font_path is empty
134 font_size_ratio: float = Field(default=0.035, gt=0.0, le=0.5) # text height as a
fraction of frame height
135 opacity: float = Field(default=0.85, ge=0.0, le=1.0)
136 box: bool = True # faint backing box behind the text for legibility
137 margin: int = Field(default=24, ge=0) # px inset from the chosen corner
138 position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = "bottom-
right"
139

```

#### 426. user (2026-06-13T15:14:58.254Z)

```

33:class TrackNetConfig(BaseModel):
34- wsl_distro: str = "Ubuntu"
35- repo_dir: str = "~/models/TrackNetV4"
36- conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
37- conda_env: str = "TrackNetV4"
38- weights_path: str = ""
39- queue_length: int = 5
40- stage_in_wsl: bool = True
41- wsl_stage_dir: str = "~/clips"
42- timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
43- velocity_threshold: float = 0.02
44-
45-
46:class WasbIndexConfig(BaseModel):
47- """Typed config for the live WASB shuttle substrate (indexing.wasb).
48-
49- Mirrors backend\pipeline\detectors\wasb_runner.py::WasbConfig.from_indexing_cfg so
50- these knobs actually take effect ? previously this whole block was silently dropped
51- because nested config sections defaulted to extra='ignore'.
52- """
53- wsl_distro: str = "Ubuntu"
54- repo_dir: str = "~/models/WASB-SBDT"
55- conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
56- conda_env: str = "wasb"
57- weights_path: str = "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
58- sport: str = "badminton"
59- wsl_stage_dir: str = "~/clips"
60- timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
61- velocity_threshold: float = 0.02
62-
63-
64-class IndexingConfig(BaseModel):
65- default_segmenter: str = "yolo_hybrid"
66- use_gpu: bool = True
67- motion_threshold: float = 0.08
68- min_segment_duration: float = 3.0

```

#### 427. assistant (2026-06-13T15:15:37.671Z)

Now I have everything I need. Let me implement Tier 1 + Tier 3a (config flags, shared WSL helpers, runner self-clean + disk preflight). Starting with the config models:

**428. user (2026-06-13T15:15:42.939Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**429. user (2026-06-13T15:15:50.733Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\config\models.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**430. assistant (2026-06-13T15:15:54.184Z)**

Now the shared WSL helpers in `base.py`:

**431. user (2026-06-13T15:15:56.134Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**432. user (2026-06-13T15:16:07.608Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\base.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**433. assistant (2026-06-13T15:16:11.689Z)**

Now the WASB runner ? config fields, preflight, and self-clean on success:

**434. user (2026-06-13T15:16:20.360Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb\_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**435. user (2026-06-13T15:16:29.217Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb\_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**436. user (2026-06-13T15:16:37.580Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\wasb\_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**437. assistant (2026-06-13T15:16:43.538Z)**

Now TrackNet for symmetry (staged-video cleanup; it has no frame cache):

**438. user (2026-06-13T15:16:53.207Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\tracknet\_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**439. user (2026-06-13T15:17:01.628Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\tracknet\_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**440. assistant (2026-06-13T15:17:09.369Z)**

Let me add the disk preflight to TrackNet too, for symmetry:

**441. user (2026-06-13T15:17:10.388Z)**

```
158 expected_csv_win = os.path.join(output_win_dir, f"{key}_predict.csv")
159
160 # Resolve the video path WSL will read.
161 if self.cfg.stage_in_wsl:
162 staged = self._stage_video(video_win_path, f"{key}{ext}")
163 if staged is None:
164 return None
165 wsl_video = staged
166 else:
167 # No staging: predict.py reads /mnt directly and writes
168 # We cannot rename its output, so fall back to the un-namespaced name.
169 wsl_video = to_wsl_mnt_path(video_win_path)
170 expected_csv_win = os.path.join(output_win_dir, f"{stem}_predict.csv")
171
172 cmd = (
173 f"conda activate {self.cfg.conda_env} && "
174 f"cd {self.cfg.repo_dir}/src && "
175 f"python predict.py "
176 f"--video_path {wsl_tilde_quote(wsl_video)} "
177 f"--model_weights {wsl_tilde_quote(self.cfg.weights_path)} "
178 f"--output_dir {wsl_tilde_quote(wsl_out)} "
179 f"--queue_length {self.cfg.queue_length}"
180)
```

**442. user (2026-06-13T15:17:17.478Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\backend\pipeline\detectors\tracknet\_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

**443. assistant (2026-06-13T15:18:06.011Z)**

Now Tier 2 ? the cleanup tool. Let me check the repo layout for placement:

**444. user (2026-06-13T15:18:06.399Z)**

No files found

**445. user (2026-06-13T15:18:10.259Z)**

```
backend\tools\rally_slicer.py
backend\tools__init__.py
backend\tools\export.py
```

**446. assistant (2026-06-13T15:18:16.333Z)**

I'll create a top-level `tools/` package for ops tooling. Writing the cleanup tool with pure, testable classification logic:

**447. user (2026-06-13T15:18:17.246Z)**

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tools\\_\_init\_\_.py (file state is current in your context ? no need to Read it back)

**448. assistant (2026-06-13T15:19:00.447Z)**

"""Cache-hygiene tool for the GPU highlight pipeline.

Scans the regenerable intermediates the pipeline leaves behind ? in the WSL stage

dir (`~/clips`) and the Windows `output/` tree ? classifies each entry as  
\*\*purge\*\* (regenerable: frame caches, staged video copies, handoff clips, temp dirs)  
or \*\*keep\*\* (durable: trajectory CSVs, the SQLite DB, final reels) and reports the  
reclaimable space.

DRY-RUN BY DEFAULT ? it only \*shows\* what it would delete. Pass `--apply` to delete.

```
python -m tools.cleanup_caches # report only (safe)
python -m tools.cleanup_caches --apply # delete purge candidates
python -m tools.cleanup_caches --keep-video 89605e80ed01 # protect one video's cache
python -m tools.cleanup_caches --older-than 7 # only purge entries >7 days old
python -m tools.cleanup_caches --include-wasbcache # also drop detector resume caches
```

Pairs with the pipeline's own self-cleaning (indexing.{wasb,tracknet}.keep\_frames=false,  
which deletes frames after a SUCCESSFUL run); this tool reclaims what failed/old runs  
and pre-self-cleaning videos left behind. Stdlib only; safe to run anywhere (the WSL  
scan is skipped automatically when `wsl` is unavailable).

"""

```
from __future__ import annotations
```

```
import argparse
import os
import re
import shutil
import subprocess
import sys
from dataclasses import dataclass
from typing import List, Optional, Tuple
```

**---- classification (pure; unit-tested) ----- #**

**(category, regenerable, default\_purge). default\_purge=True ? deleted unless a flag  
protects it. wasbcache is regenerable but default\_purge=False (it enables resume and  
is tiny) ? opt in with --include-wasbcache.**

```
_KEEP = ("keep", False, False)
```

```
def classify(name: str, is_dir: bool, location: str) -> Tuple[str, bool, bool]:
 """Classify a cache entry by name. Returns (category, regenerable, default_purge)."""
 low = name.lower()
 if is_dir:
 if low.endswith("_frames"):
 return ("frame cache", True, True)
 if low.endswith("_wasbcache"):
 return ("detector resume cache", True, False) # opt-in via --include-wasbcache
 if low.startswith("chunks_") and low.endswith("_handoff"):
 return ("ai handoff clips", True, True)
 if low.startswith("tmp"):
 return ("temp dir", True, True)
 return ("other dir", False, False)
 # files
 if low.endswith((".csv",)):
 return ("trajectory csv", False, False) # durable analysis output
 if low.endswith((".db", ".sqlite", ".sqlite3")):
 return ("database", False, False) # cached rallies ? never purge
 if low.endswith((".log", ".sh", ".txt", ".json", ".md")):
 return ("log/script", False, False)
 if low.endswith((".mp4", ".avi", ".mov", ".mkv")):
 # In the WSL stage dir, every video is a regenerable staged copy.
 if location == "wsl":
 return ("staged video", True, True)
 # In output/, a 1080p/proxy is regenerable; anything else is a produced reel.
```

```

 if "_1080p" in low or "_proxy" in low or low.endswith("_proxy.mp4"):
 return ("proxy video", True, True)
 return ("highlight reel", False, False) # durable output ? keep
return ("other", False, False)

```

```
@dataclass
```

```
class Entry:
```

```

 name: str
 path: str # display path (WSL ~/clips/<name> or windows abs)
 size: int # bytes
 mtime: float # epoch seconds (0 if unknown)
 is_dir: bool
 location: str # "wsl" | "windows"

```

```
@property
```

```

def category(self) -> Tuple[str, bool, bool]:
 return classify(self.name, self.is_dir, self.location)

```

```

def plan_deletions(entries: List[Entry], *, keep_video: Optional[str], older_than_days: float,
 include_wasbcache: bool, now: float) -> Tuple[List[Entry], List[Entry]]:
 """Split entries into (to_delete, kept) per the flags. Pure ? no I/O."""
 to_delete: List[Entry] = []
 kept: List[Entry] = []
 cutoff = older_than_days * 86400.0
 for e in entries:
 cat, _regen, default_purge = e.category
 purge = default_purge or (include_wasbcache and cat == "detector resume cache")
 if purge and keep_video and keep_video in e.name:
 purge = False
 if purge and older_than_days > 0 and e.mtime > 0 and (now - e.mtime) < cutoff:
 purge = False
 (to_delete if purge else kept).append(e)
 return to_delete, kept

```

## ---- scanning (I/O) ----- #

```
def human(n: int) -> str:
```

```

 f = float(n)
 for unit in ("B", "K", "M", "G", "T"):
 if f < 1024 or unit == "T":
 return f"{f:.1f}{unit}" if unit != "B" else f"{int(f)}B"
 f /= 1024
 return f"{f:.1f}T"

```

```
def scan_wsl(distro: str, stage_dir: str) -> List[Entry]:
```

```

 """List ~/clips entries with size+mtime via one `wsl` call. [] if WSL absent."""
 if not shutil.which("wsl"):
 return []
 script = (
 f'cd {stage_dir} 2>/dev/null || exit 0; '
 'for e in *.[!.*]; do [-e "$e"] || continue; '
 'd=0; [-d "$e"] && d=1; '
 'sz=$(du -sb "$e" 2>/dev/null | cut -f1); '
 'mt=$(stat -c %Y "$e" 2>/dev/null); '
 r'printf "%s\t%s\t%s\t%s\n" "$d" "$sz" "$mt" "$e"; done'
)
 try:
 res = subprocess.run(["wsl", "-d", distro, "bash", "-lc", script],
 capture_output=True, text=True, timeout=300)
 except (OSError, subprocess.TimeoutExpired):

```

```

 return []
entries: List[Entry] = []
for line in (res.stdout or "").splitlines():
 parts = line.split("\t")
 if len(parts) != 4:
 continue
 d, sz, mt, name = parts
 try:
 entries.append(Entry(name=name, path=f"{stage_dir}/{name}",
 size=int(sz or 0), mtime=float(mt or 0),
 is_dir=(d == "1"), location="wsl"))
 except ValueError:
 continue
return entries

def _dir_size(path: str) -> int:
 total = 0
 for root, _dirs, files in os.walk(path):
 for fn in files:
 try:
 total += os.path.getsize(os.path.join(root, fn))
 except OSError:
 pass
 return total

def scan_windows(output_dir: str) -> List[Entry]:
 """List output/ entries (top level) plus the output/wasb sub-tree."""
 entries: List[Entry] = []
 for base in (output_dir, os.path.join(output_dir, "wasb")):
 if not os.path.isdir(base):
 continue
 for de in os.scandir(base):
 try:
 is_dir = de.is_dir()
 size = _dir_size(de.path) if is_dir else de.stat().st_size
 mtime = de.stat().st_mtime
 except OSError:
 continue
 entries.append(Entry(name=de.name, path=de.path, size=size,
 mtime=mtime, is_dir=is_dir, location="windows"))
 return entries

def _delete(e: Entry, distro: str) -> bool:
 try:
 if e.location == "wsl":
 subprocess.run(["wsl", "-d", distro, "bash", "-lc", f'rm -rf {_q(e.path)}'],
 capture_output=True, text=True, timeout=600, check=True)
 elif e.is_dir:
 shutil.rmtree(e.path, ignore_errors=True)
 else:
 os.remove(e.path)
 return True
 except (OSError, subprocess.SubprocessError):
 return False

def _q(path: str) -> str:
 if path.startswith("~/"):
 return "~/ " + path[2:].replace("'", "'\\'") + "'"
 return "'" + path.replace("'", "'\\'") + "'"

```

```

def _report(title: str, rows: List[Entry]) -> int:
 print(f"\n{title}")
 if not rows:
 print(" (none)")
 return 0
 total = 0
 for e in sorted(rows, key=lambda x: x.size, reverse=True):
 cat = e.category[0]
 print(f" {human(e.size):>8} {cat:<22} {e.path}")
 total += e.size
 print(f" {'-' * 8}")
 print(f" {human(total):>8} TOTAL ({len(rows)} items)")
 return total

def main(argv: Optional[List[str]] = None) -> int:
 ap = argparse.ArgumentParser(description="Report/clean regenerable pipeline caches (dry-run by default).")
 ap.add_argument("--apply", action="store_true", help="actually delete (default: report only)")
 ap.add_argument("--keep-video", default=None, help="protect entries whose name contains this id/substring")
 ap.add_argument("--older-than", type=float, default=0.0, help="only purge entries older than N days")
 ap.add_argument("--include-wasbcache", action="store_true", help="also purge detector resume caches")
 ap.add_argument("--stage-dir", default="~/clips", help="WSL stage dir (default ~/clips)")
 ap.add_argument("--distro", default="Ubuntu", help="WSL distro (default Ubuntu)")
 ap.add_argument("--output-dir", default="output", help="Windows output dir (default output)")
 args = ap.parse_args(argv)

 import time
 now = time.time()
 entries = scan_wsl(args.distro, args.stage_dir) + scan_windows(args.output_dir)
 to_delete, kept = plan_deletions(
 entries, keep_video=args.keep_video, older_than_days=args.older_than,
 include_wasbcache=args.include_wasbcache, now=now)

 reclaim = _report("PURGE candidates (regenerable):", to_delete)
 _report("KEEP (durable / protected):", kept)
 print(f"\nReclaimable: {human(reclaim)}")

 if not args.apply:
 print("\nDRY RUN ? nothing deleted. Re-run with --apply to delete the PURGE list.")
 return 0
 if not to_delete:
 print("\nNothing to delete.")
 return 0

 print(f"\nDeleting {len(to_delete)} items...")
 freed, failed = 0, 0
 for e in to_delete:
 if _delete(e, args.distro):
 freed += e.size
 else:
 failed += 1
 print(f" FAILED: {e.path}")
 print(f"Freed ~{human(freed)}" + (f" ({failed} failed)" if failed else ""))
 return 0

if __name__ == "__main__":
 sys.exit(main())

```



#### 449. user (2026-06-13T15:19:00.489Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup\_caches.py (file state is current in your context ? no need to Read it back)

#### 450. assistant (2026-06-13T15:19:05.418Z)

Let me test the cleanup tool live (dry-run ? safe) to confirm it scans and classifies correctly on the real machine:

#### 451. user (2026-06-13T15:19:08.405Z)

PURGE candidates (regenerable):

2.2G proxy video output\GX010125\_1080p.mp4

-----

2.2G TOTAL (1 items)

KEEP (durable / protected):

1.9G highlight reel	output\GX010125_highlights_min5.mp4
1.1G highlight reel	output\GX010125_highlights.mp4
153.1M highlight reel	output\kushagra_1280.mp4
119.4M highlight reel	output\MahadevpuraSingles_highlights.mp4
86.0M highlight reel	output\mahadevpura_smoke_180s.mp4
9.0M other dir	output\wasb
6.6M detector resume cache	output\wasb\GX010125_1080p__89605e80ed01_wasb_wasbcache
3.8M highlight reel	output\e2e_reel.mp4
1.6M trajectory csv	output\wasb\GX010125_1080p__89605e80ed01_wasb.csv
730.0K detector resume cache	output\wasb\mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache
168.4K trajectory csv	output\wasb\mahadevpura_smoke_180s__05e0e577029d_wasb.csv
160.0K database	output\sports_indexer.db
138.3K highlight reel	output\ramp_probe.mp4
112.0K database	output\phase3_yolo.db
106.2K log/script	output\gx010125_run.log
46.0K log/script	output\smoke4_log.txt
36.0K database	output\badminton_indexer.db
34.3K log/script	output\gx010125_transcode.log
24.0K database	output\test_indexer.db
8.8K trajectory csv	output\adarsh_avi_full_vs_human.csv
6.6K log/script	output\gx010125_min5_stitch.log
6.0K trajectory csv	output\largetestvideo_vs_human.csv
5.5K trajectory csv	output\gbaaddy_vs_human.csv
3.4K trajectory csv	output\mahadevpura_vs_human.csv
3.1K log/script	output\mahadevpura_smoke_180s.report.json
3.1K trajectory csv	output\kushagra_vs_human.csv
2.9K trajectory csv	output\wasb_tuned_segments_vs_golden.csv
2.8K trajectory csv	output\adarsh_tuned_segments.csv
2.6K log/script	output\kushagra_1280.report.json
2.4K log/script	output\e2e_server_log.txt
2.0K trajectory csv	output\adarsh_baseline_segments.csv
1.9K log/script	output\phase3_probe.json
1.7K log/script	output\human_lovo_manifest.json
1.7K trajectory csv	output\adarsh_baseline_gated.csv
1.6K trajectory csv	output\adarsh_gemini_refined.csv
1.6K trajectory csv	output\adarsh_gemini_fullvideo.csv
1.5K log/script	output\mahadevpura_smoke_180s.report.md
1.5K trajectory csv	output\wasb_tuned_segments.csv
1.4K trajectory csv	output\mahadevpura_gemini_refined.csv
1.3K trajectory csv	output\adarsh_min5s_segments.csv
1.2K trajectory csv	output\wasb_tuned_segments_baseline.csv
1.1K trajectory csv	output\testlarge_vs_human.csv
1.1K log/script	output\kushagra_1280.report.md
662B log/script	output\local_rally_model.json
561B trajectory csv	output\adarsh_avi_vs_human.csv

```

486B log/script output\mahadevpura_smoke_180s.summary.md
480B log/script output\kushagra_1280.summary.md
370B log/script output\local_calib_manifest.json
350B trajectory csv output\testlarge_gemini_fullvideo.csv
270B trajectory csv output\testlarge_gemini_rallies.csv
269B trajectory csv output\adarsh_gemini_smoke.csv
 0B other dir output\uploads

```

3.4G TOTAL (52 items)

Reclaimable: 2.2G

DRY RUN ? nothing deleted. Re-run with --apply to delete the PURGE list.

#### 452. assistant (2026-06-13T15:19:30.067Z)

The Windows scan works, but the \*\*WSL scan returned nothing\*\* ? `~/clips` (with the 93 GB frames, proxy copy, CSVs) didn't show. Let me debug why:

#### 453. user (2026-06-13T15:19:33.938Z)

wsl on PATH: C:\Windows\system32\wsl.EXE  
entries: 0

#### 454. assistant (2026-06-13T15:20:05.490Z)

`wsl` resolves but yields 0 entries. Let me capture the raw output to see if it's the known `wsl.exe` UTF-16 piping quirk:

#### 455. user (2026-06-13T15:20:14.803Z)

Exit code 1

du:

Line |

```
3 | ? e \"\$e\"] || continue; d=0; [-d \"\$e\"] && d=1;
```

```
sz=$(du -sb \"\$?
```

```
|
```

```
~~
```

| The term 'du' is  
not recognized as a name of a cmdlet, function, script file, or executable program.  
Check the spelling of the  
name, or if a path was included, verify that the path is correct and try again.

File "<string>", line 3

```
script = 'cd ~/clips 2>/dev/null || exit 0; for e in *; do [-e \
^
```

SyntaxError: unterminated string literal (detected at line 3)

#### 456. assistant (2026-06-13T15:20:26.436Z)

PowerShell mangled the inline script. Let me use a debug file instead:

#### 457. assistant (2026-06-13T15:20:31.983Z)

```
import os
```

```
import subprocess
```

```
script = (
```

```
'cd ~/clips 2>/dev/null || exit 0; '
```

```
'for e in *; do [-e "$e"] || continue; '
```

```
'd=0; [-d "$e"] && d=1; '
```

```
'sz=$(du -sb "$e" 2>/dev/null | cut -f1); '
```

```
r'printf "%s\t%s\t%s\n" "$d" "$sz" "$e"; done'
```

```
)
```

```

print("--- text=True, default env ---")
r = subprocess.run(["wsl", "-d", "Ubuntu", "bash", "-lc", script],
 capture_output=True, text=True, timeout=120)
print("rc", r.returncode, "stdout chars", len(r.stdout), "stderr", repr(r.stderr[:120]))
print("stdout repr head:", repr(r.stdout[:160]))

print("--- raw bytes head ---")
r2 = subprocess.run(["wsl", "-d", "Ubuntu", "bash", "-lc", script],
 capture_output=True, timeout=120)
print(repr(r2.stdout[:80]))

print("--- WSL_UTF8=1 ---")
r3 = subprocess.run(["wsl", "-d", "Ubuntu", "bash", "-lc", script],
 capture_output=True, text=True, timeout=120,
 env={**os.environ, "WSL_UTF8": "1"})
print("rc", r3.returncode, "lines", len(r3.stdout.splitlines()))
print("head:", repr(r3.stdout[:160]))

```

#### 458. user (2026-06-13T15:20:32.009Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\debug\_wsl.py (file state is current in your context ? no need to Read it back)

#### 459. user (2026-06-13T15:20:39.245Z)

```

--- text=True, default env ---
rc 0 stdout chars 0 stderr ''
stdout repr head: ''
--- raw bytes head ---
b''
--- WSL_UTF8=1 ---
rc 0 lines 0
head: ''

```

#### 460. assistant (2026-06-13T15:21:19.006Z)

All three return empty with rc=0 ? so it's not encoding; the `cd ~/clips || exit 0` is silently bailing. Let me probe `whoami`/`\$HOME`/`pwd` under `bash -c` vs `bash -lc`:

#### 461. assistant (2026-06-13T15:21:22.638Z)

```

import subprocess

probe = 'echo "user=$(whoami) home=$HOME pwd=$(pwd)"; echo "clips_count=$(ls ~/clips 2>/dev/null | wc -l)"; echo "direct=$(ls /home/avidullu/clips 2>/dev/null | wc -l)'"

for label, argv in [
 ("bash -c (no --)", ["wsl", "-d", "Ubuntu", "bash", "-c", probe]),
 ("bash -lc (no --)", ["wsl", "-d", "Ubuntu", "bash", "-lc", probe]),
 ("-- bash -lc", ["wsl", "-d", "Ubuntu", "--", "bash", "-lc", probe]),
]:
 r = subprocess.run(argv, capture_output=True, text=True, timeout=120)
 print(f"=== {label} === rc={r.returncode}")
 print("STDOUT:", repr(r.stdout))
 print("STDERR:", repr(r.stderr[:200]))
 print()

```

#### 462. user (2026-06-13T15:21:22.662Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\debug\_wsl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 463. user (2026-06-13T15:21:28.637Z)

```

=== bash -c (no --) === rc=0
STDOUT: 'user=avidullu home=/home/avidullu pwd=/mnt/c/Users/avidu/Projects/badminton-highlight-indexer\nclips_count=39\nndirect=39\n'
STDERR: ''

```

```

=== bash -lc (no --) === rc=0
STDOUT: 'user=avidullu home=/home/avidullu pwd=/mnt/c/Users/avidu/Projects/badminton-highlight-indexer\nclips_count=39\nndirect=39\n'
STDERR: ''

```

```

=== -- bash -lc === rc=0
STDOUT: 'user=avidullu home=/home/avidullu pwd=/mnt/c/Users/avidu/Projects/badminton-highlight-indexer\nclips_count=39\nndirect=39\n'
STDERR: ''

```

#### 464. assistant (2026-06-13T15:22:14.310Z)

The path resolves fine (39 entries, `home=/home/avidullu`) ? so it's something in my scan \*script\* specifically. Let me isolate the `cd` + loop:

#### 465. assistant (2026-06-13T15:22:18.468Z)

```
import subprocess
```

```

def run(label, script):
 r = subprocess.run(["wsl", "-d", "Ubuntu", "bash", "-lc", script],
 capture_output=True, text=True, timeout=120)
 print(f"=== {label} === rc={r.returncode}")
 print("STDOUT:", repr(r.stdout[:300]))
 print("STDERR:", repr(r.stderr[:200]))
 print()

```

```

run("A: cd + pwd + count", 'cd ~/clips && pwd && ls | wc -l')
run("B: cd guarded + loop echo", 'cd ~/clips 2>/dev/null || exit 0; for e in *; do echo "X$e"; done | head -3')
run("C: full scan loop", (
 'cd ~/clips 2>/dev/null || exit 0; '
 'for e in *; do [-e "$e"] || continue; '
 'd=0; [-d "$e"] && d=1; '
 'sz=$(du -sb "$e" 2>/dev/null | cut -f1); '
 'mt=$(stat -c %Y "$e" 2>/dev/null); '
 r'printf "%s\t%s\t%s\t%s\n" "$d" "$sz" "$mt" "$e"; done | head -3'))

```

#### 466. user (2026-06-13T15:22:18.488Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\debug\_wsl.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 467. user (2026-06-13T15:22:23.826Z)

```

=== A: cd + pwd + count === rc=0
STDOUT: '/home/avidullu/clips\n39\n'
STDERR: ''

```

```

=== B: cd guarded + loop echo === rc=0
STDOUT: 'X\nX\nX\n'
STDERR: ''

```

```

=== C: full scan loop === rc=0
STDOUT: ''
STDERR: ''

```

#### 468. assistant (2026-06-13T15:23:12.134Z)

The loop variable ``$e`` is coming back `**empty**` ? ``[ -e "$e" ]`` then fails and ``continue`` skips everything (that's why C is silent). The glob/loop is misbehaving through the Python?wsl.exe layer. Let me check whether it's Python-subprocess-specific by running the same loop via PowerShell, and test the cleaner ``\\wsl.localhost`` UNC approach:

#### 469. user (2026-06-13T15:23:16.960Z)

```
Exit code 1
ParserError:
Line |
 1 | ? \wsl$\Ubuntu\home\avidullu\clips"; Write-Output "trying
wsl$: $(Test- ?
 |
~~
 | Variable reference
is not valid. '$' was not followed by a valid variable name character. Consider using ${} to
 |
delimit the name.
```

#### 470. assistant (2026-06-13T15:23:25.304Z)

PowerShell choked on the literal ``wsl$``. Let me run those checks cleanly:

#### 471. user (2026-06-13T15:23:27.768Z)

```
X
X
X
```

#### 472. user (2026-06-13T15:23:29.508Z)

```
UNC exists: True
entries: 39
```

#### 473. assistant (2026-06-13T15:24:41.306Z)

Found it ? the ``for e in *`` loop returns empty ``$e`` `**even via PowerShell**`, so it's the user's login-shell environment interfering with the loop (not Python). But two patterns are proven reliable: ``du -sb`` (I used ``du -sh ./`` successfully earlier) and the ``\\wsl.localhost`` UNC path. Let me rebuild ``scan_wsl`` on ``du`` + ``find`` (native-fast, no fragile loop) ? testing the commands first:

#### 474. user (2026-06-13T15:24:45.719Z)

```
=== du -sb (size+path) ===
2352117156 /home/avidullu/clips/GX010125_1080p__89605e80ed01.mp4
98860500712 /home/avidullu/clips/GX010125_1080p__89605e80ed01_frames
404 /home/avidullu/clips/adarsh_avi_full_run.log
2476375 /home/avidullu/clips/adarsh_avi_full_traj.csv
=== find (type+mtime+name) ===
f 1780923256.8006703080 adarsh_avi_full_run.log
f 1780981801.0278589990 mahadevpura_run.log
f 1780907422.4687402170 run_wasb_avi.sh
f 1780743130.8096656640 wasb_traj_test.csv
```

#### 475. assistant (2026-06-13T15:25:12.176Z)

Both work perfectly. Let me rebuild ``scan_wsl`` on ``du`` + ``find`` (no fragile loop):

#### 476. user (2026-06-13T15:25:24.407Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup\_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 477. assistant (2026-06-13T15:25:30.432Z)

Now let me re-test the tool live (dry-run, protecting GX010125):

#### 478. user (2026-06-13T15:25:33.841Z)

PURGE candidates (regenerable):

2.2G	proxy video	output\GX010125_1080p.mp4
------	-------------	---------------------------

-----

2.2G	TOTAL (1 items)
------	-----------------

KEEP (durable / protected):

92.1G	frame cache	~/clips/GX010125_1080p__89605e80ed01_frames
2.2G	staged video	~/clips/GX010125_1080p__89605e80ed01.mp4
1.9G	highlight reel	output\GX010125_highlights_min5.mp4
1.1G	highlight reel	output\GX010125_highlights.mp4
153.1M	highlight reel	output\kushagra_1280.mp4
119.4M	highlight reel	output\MahadevpuraSingles_highlights.mp4
86.0M	highlight reel	output\mahadevpura_smoke_180s.mp4
9.1M	detector resume cache	~/clips/adarsh_avi_full_traj_wasbcache
9.0M	other dir	output\wasb
7.2M	detector resume cache	~/clips/gbaaddy_traj_wasbcache
6.6M	detector resume cache	output\wasb\GX010125_1080p__89605e80ed01_wasb_wasbcache
5.5M	detector resume cache	~/clips/kushagra_traj_wasbcache
4.2M	detector resume cache	~/clips/adarsh_avi_traj_wasbcache
3.9M	detector resume cache	~/clips/adarsh_traj_wasbcache
3.8M	highlight reel	output\e2e_reel.mp4
2.6M	detector resume cache	~/clips/largetestvideo_traj_wasbcache
2.4M	trajectory csv	~/clips/adarsh_avi_full_traj.csv
1.7M	detector resume cache	~/clips/mahadevpura_traj_wasbcache
1.6M	trajectory csv	~/clips/gbaaddy_traj.csv
1.6M	trajectory csv	output\wasb\GX010125_1080p__89605e80ed01_wasb.csv
1.1M	trajectory csv	~/clips/kushagra_traj.csv
1.1M	trajectory csv	~/clips/adarsh_avi_traj.csv
1.1M	trajectory csv	~/clips/adarsh_traj.csv
730.0K	detector resume cache	output\wasb\mahadevpura_smoke_180s__05e0e577029d_wasb_wasbcache
677.6K	trajectory csv	~/clips/largetestvideo_traj.csv
586.4K	trajectory csv	~/clips/sample_long_traj.csv
507.3K	detector resume cache	~/clips/pickleball_traj_wasbcache
408.9K	trajectory csv	~/clips/mahadevpura_traj.csv
408.4K	detector resume cache	~/clips/testlarge_traj_wasbcache
350.5K	detector resume cache	~/clips/tabletennis_traj_wasbcache
168.4K	trajectory csv	output\wasb\mahadevpura_smoke_180s__05e0e577029d_wasb.csv
160.0K	database	output\sports_indexer.db
141.9K	trajectory csv	~/clips/pickleball_traj.csv

#### 479. assistant (2026-06-13T15:26:01.206Z)

The WSL scan works now ? with `--keep-video 89605e80ed01`, GX010125's 92 G frames + staged copy are correctly protected. One cosmetic fix: `output\wasb` is listed both as a dir and via its contents (double-count). Let me skip that redundancy, then write tests:

#### 480. user (2026-06-13T15:26:04.405Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup\_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 481. assistant (2026-06-13T15:26:22.852Z)

""Unit tests for the cache-hygiene tool's pure logic (classify + plan\_deletions).

The I/O (wsl/du/find scan, deletion) is not unit-tested here ? these guard the classification + filtering rules that decide what gets deleted, which is the part where a mistake would be destructive.

"""

```
from tools.cleanup_caches import Entry, classify, plan_deletions
```

```
def test_classify_regenerable_intermediates():
 assert classify("GX010125_1080p__abc_frames", True, "wsl")[2] is True # frame cache ->
 purge
 assert classify("GX010125_1080p__abc.mp4", False, "wsl")[2] is True # staged video
-> purge
 assert classify("GX010125_1080p.mp4", False, "windows")[2] is True # proxy ->
 purge
 assert classify("chunks_gemini_handoff", True, "windows")[2] is True # handoff clips
-> purge
 assert classify("tmp9xk2", True, "windows")[2] is True # temp dir ->
 purge
```

```
def test_classify_durable_kept():
 assert classify("adarsh_traj.csv", False, "wsl")[2] is False # trajectory ->
 keep
 assert classify("sports_indexer.db", False, "windows")[2] is False # DB -> keep
 assert classify("GX010125_highlights.mp4", False, "windows")[2] is False # reel -> keep
 assert classify("run_wasb.sh", False, "wsl")[2] is False # script ->
 keep
 # wasbcache is regenerable but opt-in only (default_purge False)
 cat, regen, default_purge = classify("adarsh_traj_wasbcache", True, "wsl")
 assert cat == "detector resume cache" and regen is True and default_purge is False
```

```
def _e(name, size, is_dir, location="wsl", mtime=0.0):
 return Entry(name=name, path=f"~/clips/{name}", size=size, mtime=mtime,
 is_dir=is_dir, location=location)
```

```
def test_plan_default_purges_frames_keeps_csv_and_wasbcache():
 entries = [
 _e("v_frames", 90_000, True),
 _e("v.mp4", 2_000, False),
 _e("v_traj.csv", 100, False),
 _e("v_traj_wasbcache", 500, True),
]
 to_del, kept = plan_deletions(entries, keep_video=None, older_than_days=0,
 include_wasbcache=False, now=1_000_000.0)
 names = {e.name for e in to_del}
 assert names == {"v_frames", "v.mp4"}
 assert {e.name for e in kept} == {"v_traj.csv", "v_traj_wasbcache"}
```

```
def test_plan_include_wasbcache_opts_it_in():
 entries = [_e("v_traj_wasbcache", 500, True)]
 to_del, _ = plan_deletions(entries, keep_video=None, older_than_days=0,
 include_wasbcache=True, now=1_000_000.0)
 assert [e.name for e in to_del] == ["v_traj_wasbcache"]
```

```
def test_plan_keep_video_protects_matching_substring():
 entries = [_e("GX010125_frames", 90_000, True), _e("kushagra_frames", 5_000, True)]
 to_del, kept = plan_deletions(entries, keep_video="GX010125", older_than_days=0,
 include_wasbcache=False, now=1_000_000.0)
 assert [e.name for e in to_del] == ["kushagra_frames"]
 assert [e.name for e in kept] == ["GX010125_frames"]
```

```

def test_plan_older_than_protects_recent():
 now = 1_000_000.0
 entries = [
 _e("recent_frames", 90_000, True, mtime=now - 3600), # 1h old
 _e("old_frames", 90_000, True, mtime=now - 10 * 86400), # 10d old
]
 to_del, kept = plan_deletions(entries, keep_video=None, older_than_days=7,
 include_wasbcache=False, now=now)
 assert [e.name for e in to_del] == ["old_frames"]
 assert [e.name for e in kept] == ["recent_frames"]

```

#### 482. user (2026-06-13T15:26:22.873Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test\_cleanup\_caches.py (file state is current in your context ? no need to Read it back)

#### 483. assistant (2026-06-13T15:26:31.877Z)

Now tests for the Tier 1 runner self-clean. Let me check the existing runner-test style to match it:

#### 484. user (2026-06-13T15:26:32.240Z)

```

tests\test_calibrate_wasb.py
tests\test_export_wasb_segments.py
tests\test_wasb_runner.py
tests\test_config_wasb.py
tests\test_wasb_cache.py

```

#### 485. user (2026-06-13T15:26:35.832Z)

```

1 """WasbRunner WSL-glue coverage (the LIVE detector path; previously only an ABC-
conformance
2 test existed while the parallel TrackNetRunner had ~21). WSL is mocked."""
3 import types
4 from unittest.mock import patch
5
6 from backend.pipeline.detectors.wasb_runner import WasbRunner, WasbConfig
7 from backend.pipeline.detectors.base import DetectorRunner
8
9
10 def _proc(rc=0, stdout="", stderr=""):
11 return types.SimpleNamespace(returncode=rc, stdout=stdout, stderr=stderr)
12
13
14 def test_is_a_detector_runner():
15 assert isinstance(WasbRunner(WasbConfig()), DetectorRunner)
16
17
18 def test_healthcheck_ok():
19 r = WasbRunner(WasbConfig())
20 with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="torch 2.6 cuda
True")):
21 ok, msg = r.healthcheck()
22 assert ok and "torch" in msg
23
24
25 def test_healthcheck_env_failure():
26 r = WasbRunner(WasbConfig())
27 with patch.object(r, "_wsl_bash", return_value=_proc(1, stderr="ModuleNotFoundError:
torch")):
28 ok, msg = r.healthcheck()
29 assert not ok and "failed" in msg.lower()

```



```

30
31
32 def test_healthcheck_no_wsl_binary():
33 r = WasbRunner(WasbConfig())
34 with patch.object(r, "_wsl_bash", side_effect=FileNotFoundError()):
35 ok, msg = r.healthcheck()
36 assert not ok and "wsl" in msg.lower()
37
38
39 def test_stage_video_success_and_failure():
40 r = WasbRunner(WasbConfig())
41 with patch.object(r, "_wsl_bash", return_value=_proc(0, stdout="OK")):
42 assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") == "~/clips/clip.mp4"
43 with patch.object(r, "_wsl_bash", return_value=_proc(1, stderr="No such file")):
44 assert r._stage_video(r"C:\v\clip.mp4", "clip.mp4") is None
45
46
47 def test_run_predict_returns_none_when_infer_script_stage_fails(tmp_path):
48 r = WasbRunner(WasbConfig())
49 with patch.object(r, "_stage_infer_script", return_value=False):
50 assert r.run_predict(r"C:\v\clip.mp4", str(tmp_path), video_id="abc123abc123")
51 is None
51

```

#### 486. assistant (2026-06-13T15:27:07.805Z)

Now I'll add runner self-clean tests (covering the shared helpers + WASB cleanup behavior):

#### 487. user (2026-06-13T15:27:20.450Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tests\test\_wasb\_runner.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 488. assistant (2026-06-13T15:27:31.536Z)

Let me run the full suite:

#### 489. user (2026-06-13T15:28:15.302Z)

```

..... [51%]
..... [68%]
..... [85%]
..... [100%]
421 passed in 39.24s
=====
RESULT: PASS - all tests green
=====

```

#### 490. assistant (2026-06-13T15:28:26.156Z)

421 passed (+12 new). Now lint/type checks, matching CI. Let me check the exact CI commands and run them:

#### 491. user (2026-06-13T15:28:27.830Z)

```

- name: Ruff check
 run: ruff check . --output-format github

```

```

Type gate (mypy.ini at repo root): lenient green baseline with an explicit
per-module quarantine list ? ratchet by removing entries as modules clean
up. Typed core deps are installed so mypy sees their real signatures
(without them ignore_missing_imports would Any-stub half the checks).
typecheck:
 name: Mypy (lenient baseline)

```

```

 runs-on: ubuntu-latest
 steps:
 - uses: actions/checkout@v6
 --
 - name: Install mypy + typed core deps
 run: python -m pip install mypy pydantic fastapi uvicorn python-dotenv google-genai
numpy
 - name: Mypy
 run: mypy backend training

Fast guard against the 7fbb0 class of breakage: a SyntaxError or
module-level import error shipping while the suite stays green.

```

#### 492. user (2026-06-13T15:28:38.333Z)

```

Exit code 1
tools\cleanup_caches.py:26:8: F401 [*] `re` imported but unused
Found 1 error.
[*] 1 fixable with the `--fix` option.

```

#### 493. user (2026-06-13T15:28:42.701Z)

```

backend\annotations\base.py:85: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\sports__init__.py:7: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs [annotation-unchecked]
backend\pipeline\validators\base.py:36: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
backend\providers__init__.py:7: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs [annotation-unchecked]
Success: no issues found in 76 source files

```

#### 494. assistant (2026-06-13T15:28:47.362Z)

mypy clean. Ruff caught an unused `re` import ? removing it:

#### 495. user (2026-06-13T15:28:48.867Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup\_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 496. user (2026-06-13T15:28:54.947Z)

All checks passed!

#### 497. assistant (2026-06-13T15:29:30.638Z)

Ruff + mypy clean, suite 421 green. For the hook (Tier 3b), I'll add a `--summary` mode to the tool so the hook can emit a one-line nudge only when there's meaningful cruft:

#### 498. user (2026-06-13T15:29:38.835Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\tools\cleanup\_caches.py has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 499. user (2026-06-13T15:29:46.494Z)

All checks passed!

?? 96.5G of regenerable pipeline cache reclaimable ? run `python -m tools.cleanup\_caches` to review, `--apply` to delete.

#### 500. assistant (2026-06-13T15:29:57.357Z)

`--summary` works (silent below threshold). Committing the code PR (Tier 1 + 2 + 3a + tests):

#### 501. user (2026-06-13T15:30:00.923Z)

```
M backend/config/models.py
M backend/pipeline/detectors/base.py
M backend/pipeline/detectors/tracknet_runner.py
M backend/pipeline/detectors/wasb_runner.py
M config.json
M tests/test_wasb_runner.py
Switched to a new branch 'feat/cache-hygiene'
warning: in the working copy of 'tests/test_cleanup_caches.py', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of 'tools/cleanup_caches.py', LF will be replaced by CRLF the next
time Git touches it
M backend/config/models.py
M backend/pipeline/detectors/base.py
M backend/pipeline/detectors/tracknet_runner.py
M backend/pipeline/detectors/wasb_runner.py
M config.json
A tests/test_cleanup_caches.py
M tests/test_wasb_runner.py
A tools/__init__.py
A tools/cleanup_caches.py
?? arial.ttf
?? artifacts/
?? docs/STORAGE_SHARING_MODEL.md
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
?? scratch/debug_wsl.py
?? scratch/keep_awake.ps1
?? scratch/probe_cv2_count.py
?? scratch/probe_gemini.py
?? scratch/repro_drawtext_font.py
?? scratch/repro_movie_overlay.py
?? scratch/run_gx010125.py
?? scratch/stitch_min5.py
```

#### 502. assistant (2026-06-13T15:30:07.392Z)

Exactly the 8 intended files staged (config.json stays local). Committing, pushing, opening PR:

#### 503. user (2026-06-13T15:30:22.222Z)

```
[feat/cache-hygiene fade384] feat: cache hygiene ? self-cleaning pipeline + cleanup tool + disk
guard
8 files changed, 492 insertions(+), 1 deletion(-)
create mode 100644 tests/test_cleanup_caches.py
create mode 100644 tools/__init__.py
create mode 100644 tools/cleanup_caches.py
remote:
remote: Create a pull request for 'feat/cache-hygiene' on GitHub by visiting:
remote: https://github.com/avidullu/badminton-highlight-indexer/pull/new/feat/cache-hygiene
remote:
branch 'feat/cache-hygiene' set up to track 'origin/feat/cache-hygiene'.
To https://github.com/avidullu/badminton-highlight-indexer.git
* [new branch] feat/cache-hygiene -> feat/cache-hygiene
```

#### 504. user (2026-06-13T15:30:40.051Z)

Warning: 18 uncommitted changes  
<https://github.com/avidullu/badminton-highlight-indexer/pull/109>

#### 505. user (2026-06-13T15:32:30.225Z)

```
Syntax sweep + import smoke (no GPU stack) pass 19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27471002466/job/81201765268
Full test suite (offline, mocked) pass 1m31s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27471002466/job/81201765262
Mypy (lenient baseline) pass 26s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27471002466/job/81201765310
Ruff lint pass 10s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27471002466/job/81201765255
Syntax sweep + import smoke (no GPU stack) pass 19s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27471002466/job/81201765268
```

#### 506. assistant (2026-06-13T15:32:33.682Z)

All 4 lanes green. Merging:

#### 507. user (2026-06-13T15:32:44.403Z)

```
["SUCCESS"]
From https://github.com/avidullu/badminton-highlight-indexer
 * branch master -> FETCH_HEAD
 6de70aa..7651687 master -> origin/master
Updating 6de70aa..7651687
Fast-forward
 backend/config/models.py | 14 ++
 backend/pipeline/detectors/base.py | 29 ++-
 backend/pipeline/detectors/tracknet_runner.py | 18 ++
 backend/pipeline/detectors/wasb_runner.py | 23 +++
 docs/VIDEO_LOCALITY_MODEL.md | 147 ++++++
 tests/test_cleanup_caches.py | 71 ++++++
 tests/test_wasb_runner.py | 58 ++++++
 tools/__init__.py | 0
 tools/cleanup_caches.py | 280 ++++++
 9 files changed, 639 insertions(+), 1 deletion(-)
 create mode 100644 docs/VIDEO_LOCALITY_MODEL.md
 create mode 100644 tests/test_cleanup_caches.py
 create mode 100644 tools/__init__.py
 create mode 100644 tools/cleanup_caches.py
M config.json
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
7651687 feat: cache hygiene ? self-cleaning pipeline + cleanup tool + disk guard (#109)
```

#### 508. assistant (2026-06-13T15:32:59.007Z)

PR #109 merged. Now the hook (Tier 3b) ? using the update-config skill, which owns settings.json/hook configuration:

#### 509. user (2026-06-13T15:33:02.219Z)

Launching skill: update-config

#### 510. user (2026-06-13T15:33:02.215Z)

## Update Config Skill

Modify Claude Code configuration by updating settings.json files.

## When Hooks Are Required (Not Memory)

If the user wants something to happen automatically in response to an EVENT, they need a **hook** configured in settings.json. Memory/preferences cannot trigger automated actions.

**These require hooks:**

- "Before compacting, ask me what to preserve" ? PreCompact hook
- "After writing files, run prettier" ? PostToolUse hook with Write|Edit matcher
- "When I run bash commands, log them" ? PreToolUse hook with Bash matcher
- "Always run tests after code changes" ? PostToolUse hook

**Hook events:** PreToolUse, PostToolUse, PreCompact, PostCompact, Stop, Notification, SessionStart

## CRITICAL: Read Before Write

**Always read the existing settings file before making changes.** Merge new settings with existing ones - never replace the entire file.

## CRITICAL: Use AskUserQuestion for Ambiguity

When the user's request is ambiguous, use AskUserQuestion to clarify:

- Which settings file to modify (user/project/local)
- Whether to add to existing arrays or replace them
- Specific values when multiple options exist

## Decision: /config command vs Direct Edit

**Suggest the `/config` slash command** for these simple settings:

- `theme`, `editorMode`, `verbose`, `model`
- `language`, `alwaysThinkingEnabled`
- `permissions.defaultMode`

**Edit settings.json directly** for:

- Hooks (PreToolUse, PostToolUse, etc.)
- Complex permission rules (allow/deny arrays)
- Environment variables
- MCP server configuration
- Plugin configuration

## Workflow

1. **Clarify intent** - Ask if the request is ambiguous
2. **Read existing file** - Use Read tool on the target settings file
3. **Merge carefully** - Preserve existing settings, especially arrays
4. **Edit file** - Use Edit tool (if file doesn't exist, ask user to create it first)
5. **Confirm** - Tell user what was changed

## Merging Arrays (Important!)

When adding to permission arrays or hook arrays, **merge with existing**, don't replace:

**WRONG** (replaces existing permissions):

```
```json
{ "permissions": { "allow": ["Bash(npm *)"] } }
```
```

**RIGHT** (preserves existing + adds new):

```
```json
{
  "permissions": {
    "allow": [
```

```

    "Bash(git *)",      // existing
    "Edit(.claude)",    // existing
    "Bash(npm *)"       // new
  ]
}
}
...

```

Settings File Locations

Choose the appropriate file based on scope:

File	Scope	Git	Use For
~/.claude/settings.json	Global	N/A	Personal preferences for all projects
~/.claude/settings.json	Project	Commit	Team-wide hooks, permissions, plugins
~/.claude/settings.local.json	Project	Gitignore	Personal overrides for this project

Settings load in order: user ? project ? local (later overrides earlier).

Settings Schema Reference

Permissions

```

```json
{
 "permissions": {
 "allow": ["Bash(npm *)", "Edit(.claude)", "Read"],
 "deny": ["Bash(rm -rf *)"],
 "ask": ["Write(/etc/*)"],
 "defaultMode": "default" | "plan" | "acceptEdits" | "dontAsk",
 "additionalDirectories": ["/extra/dir"]
 }
}
...

```

#### \*\*Permission Rule Syntax:\*\*

- Exact match: `"Bash(npm run test)"`
- Prefix wildcard: `"Bash(git *)"` - matches `"git"`, `"git status"`, `"git commit"`, etc.
- Tool only: `"Read"` - allows all Read operations

### Environment Variables

```

```json
{
  "env": {
    "DEBUG": "true",
    "MY_API_KEY": "value"
  }
}
...

```

Model & Agent

```

```json
{
 "model": "sonnet", // or "fable", "opus", "haiku", full model ID
 "agent": "agent-name",
 "alwaysThinkingEnabled": true
}
...

```

### Attribution (Commits & PRs)

```

```json
{
  "attribution": {

```

```

    "commit": "Custom commit trailer text",
    "pr": "Custom PR description text"
  }
}
...

```

Set `commit` or `pr` to empty string `` to hide that attribution.

MCP Server Management

```

```json
{
 "enableAllProjectMcpServers": true,
 "enabledMcpjsonServers": ["server1", "server2"],
 "disabledMcpjsonServers": ["blocked-server"]
}
...

```

## Plugins

```

```json
{
  "enabledPlugins": {
    "formatter@anthropic-tools": true
  }
}
...

```

Plugin syntax: `plugin-name@source` where source is `claude-code-marketplace`, `claude-plugins-official`, or `builtin`.

Other Settings

- `language`: Preferred response language (e.g., "japanese")
- `cleanupPeriodDays`: Days to keep transcripts before automatic cleanup (default: 30; minimum 1)
- `respectGitignore`: Whether to respect .gitignore (default: true)
- `spinnerTipsEnabled`: Show tips in spinner
- `spinnerVerbs`: Customize spinner verbs (`{ "mode": "append" | "replace", "verbs": [...] }`)
- `spinnerTipsOverride`: Override spinner tips (`{ "excludeDefault": true, "tips": ["Custom tip"] }`)
- `syntaxHighlightingDisabled`: Disable diff highlighting

Hooks Configuration

Hooks run commands at specific points in Claude Code's lifecycle.

Hook Structure

```

```json
{
 "hooks": {
 "EVENT_NAME": [
 {
 "matcher": "ToolName|OtherTool",
 "hooks": [
 {
 "type": "command",
 "command": "your-command-here",
 "timeout": 60,
 "statusMessage": "Running..."
 }
]
 }
]
 }
}
...

```

## Hook Events

Event	Matcher	Purpose
PermissionRequest	Tool name	Run before permission prompt
PreToolUse	Tool name	Run before tool, can block
PostToolUse	Tool name	Run after successful tool
PostToolUseFailure	Tool name	Run after tool fails
Notification	Notification type	Run on notifications
Stop	-	Run when Claude stops (including clear, resume, compact)
PreCompact	"manual"/"auto"	Before compaction
PostCompact	"manual"/"auto"	After compaction (receives summary)
UserPromptSubmit	-	When user submits
SessionStart	-	When session starts

**\*\*Common tool matchers:\*\*** `Bash`, `Write`, `Edit`, `Read`, `Glob`, `Grep`

## Hook Types

**\*\*1. Command Hook\*\*** - Runs a shell command:

```
```json
{ "type": "command", "command": "prettier --write $FILE", "timeout": 30 }
```
```

**\*\*2. Prompt Hook\*\*** - Evaluates a condition with LLM:

```
```json
{ "type": "prompt", "prompt": "Is this safe? $ARGUMENTS" }
```
```

Only available for tool events: PreToolUse, PostToolUse, PermissionRequest.

**\*\*3. Agent Hook\*\*** - Runs an agent with tools:

```
```json
{ "type": "agent", "prompt": "Verify tests pass: $ARGUMENTS" }
```
```

Only available for tool events: PreToolUse, PostToolUse, PermissionRequest.

## Hook Input (stdin JSON)

```
```json
{
  "session_id": "abc123",
  "tool_name": "write",
  "tool_input": { "file_path": "/path/to/file.txt", "content": "..."},
  "tool_response": { "success": true } // PostToolUse only
}
```
```

## Hook JSON Output

Hooks can return JSON to control behavior:

```
```json
{
  "systemMessage": "Warning shown to user in UI",
  "continue": false,
  "stopReason": "Message shown when blocking",
  "suppressOutput": false,
  "decision": "block",
  "reason": "Explanation for decision",
  "hookSpecificOutput": {
    "hookEventName": "PostToolUse",
    "additionalContext": "Context injected back to model"
  }
}
```
```



```

Fields:
- `systemMessage` - Display a message to the user (all hooks)
- `continue` - Set to `false` to block/stop (default: true)
- `stopReason` - Message shown when `continue` is false
- `suppressOutput` - Hide stdout from transcript (default: false)
- `decision` - "block" for PostToolUse/Stop/UserPromptSubmit hooks (deprecated for PreToolUse, use hookSpecificOutput.permissionDecision instead)
- `reason` - Explanation for decision
- `hookSpecificOutput` - Event-specific output (must include `hookEventName`):
 - `additionalContext` - Text injected into model context
 - `permissionDecision` - "allow", "deny", or "ask" (PreToolUse only)
 - `permissionDecisionReason` - Reason for the permission decision (PreToolUse only)
 - `updatedInput` - Modified tool input (PreToolUse only)

```

## Common Patterns

**\*\*Auto-format after writes:\*\***

```

```json
{
  "hooks": {
    "PostToolUse": [{
      "matcher": "Write|Edit",
      "hooks": [{
        "type": "command",
        "command": "jq -r '.tool_response.filePath // .tool_input.file_path' | { read -r f;
prettier --write \"$f\"; } 2>/dev/null || true"
      }]
    }]
  }
}
```

```

**\*\*Log all bash commands:\*\***

```

```json
{
  "hooks": {
    "PreToolUse": [{
      "matcher": "Bash",
      "hooks": [{
        "type": "command",
        "command": "jq -r '.tool_input.command' >> ~/.claude/bash-log.txt"
      }]
    }]
  }
}
```

```

**\*\*Stop hook that displays message to user:\*\***

Command must output JSON with `systemMessage` field:

```
```bash
```

Example command that outputs: {"systemMessage": "Session complete!"}

```

echo '{"systemMessage": "Session complete!"}'
```

```

**\*\*Run tests after code changes:\*\***

```

```json
{
  "hooks": {
    "PostToolUse": [{
      "matcher": "Write|Edit",
      "hooks": [{
        "type": "command",
        "command": "jq -r '.tool_input.file_path // .tool_response.filePath' | grep -E

```

```
'\\.(ts|js)$' && npm test || true"
    }}
  }}
}
}
...

```

Constructing a Hook (with verification)

Given an event, matcher, target file, and desired behavior, follow this flow. Each step catches a different failure class ? a hook that silently does nothing is worse than no hook.

1. ****Dedup check.**** Read the target file. If a hook already exists on the same event+matcher, show the existing command and ask: keep it, replace it, or add alongside.
2. ****Construct the command for THIS project ? don't assume.**** The hook receives JSON on stdin. Build a command that:
 - Extracts any needed payload safely ? use ``jq -r`` into a quoted variable or ``{ read -r f; ... "$f"; }``, NOT unquoted ``| xargs`` (splits on spaces)
 - Invokes the underlying tool the way this project runs it (npx/bunx/yarn/pnpm? Makefile target? globally-installed?)
 - Skips inputs the tool doesn't handle (formatters often have ``--ignore-unknown``; if not, guard by extension)
 - Stays RAW for now ? no ``|| true``, no stderr suppression. You'll wrap it after the pipe-test passes.
3. ****Pipe-test the raw command.**** Synthesize the stdin payload the hook will receive and pipe it directly:
 - ``Pre|PostToolUse`` on ``Write|Edit``: ``echo '{"tool_name":"Edit","tool_input":{"file_path":"<a real file from this repo>"}}' | <cmd>``
 - ``Pre|PostToolUse`` on ``Bash``: ``echo '{"tool_name":"Bash","tool_input":{"command":"ls"}}' | <cmd>``
 - ``Stop`/`UserPromptSubmit`/`SessionStart``: most commands don't read stdin, so ``echo '{}`` | `<cmd>`` suffices

Check exit code AND side effect (file actually formatted, test actually ran). If it fails you get a real error ? fix (wrong package manager? tool not installed? jq path wrong?) and retest. Once it works, wrap with ``2>/dev/null || true`` (unless the user wants a blocking check).

4. ****Write the JSON.**** Merge into the target file (schema shape in the "Hook Structure" section above). If this creates ``.claude/settings.local.json`` for the first time, add it to ``.gitignore`` ? the Write tool doesn't auto-gitignore it.

5. ****Validate syntax + schema in one shot:****

```
`jq -e '.hooks.<event>[] | select(.matcher == "<matcher>") | .hooks[] | select(.type == "command") | .command' <target-file>`

```

Exit 0 + prints your command = correct. Exit 4 = matcher doesn't match. Exit 5 = malformed JSON or wrong nesting. A broken settings.json silently disables ALL settings from that file ? fix any pre-existing malformation too.

6. ****Prove the hook fires**** ? only for ``Pre|PostToolUse`` on a matcher you can trigger in-turn (``Write|Edit`` via Edit, ``Bash`` via Bash). ``Stop`/`UserPromptSubmit`/`SessionStart`` fire outside this turn ? skip to step 7.

For a ****formatter**** on ``PostToolUse`/`Write|Edit``: introduce a detectable violation via Edit (two consecutive blank lines, bad indentation, missing semicolon ? something this formatter corrects; NOT trailing whitespace, Edit strips that before writing), re-read, confirm the hook ****fixed**** it. For ****anything else****: temporarily prefix the command in settings.json with ``echo "$(date) hook fired" >> /tmp/claude-hook-check.txt; ``, trigger the matching tool (Edit for ``Write|Edit``, a harmless ``true`` for ``Bash``), read the sentinel file.

****Always clean up**** ? revert the violation, strip the sentinel prefix ? whether the proof passed or failed.

****If proof fails but pipe-test passed and `jq -e` passed****: the settings watcher isn't watching ``.claude/`` ? it only watches directories that had a settings file when this session started. The hook is written correctly. Tell the user to open ``/hooks`` once (reloads config) or restart ? you can't do this yourself; ``/hooks`` is a user UI menu and opening it ends this turn.

7. ****Handoff.**** Tell the user the hook is live (or needs ``/hooks`/restart` per the watcher caveat). Point them at ``/hooks`` to review, edit, or disable it later. The UI only shows "Ran N hooks" if a hook errors or is slow ? silent success is invisible by design.

Example Workflows

Adding a Hook

User: "Format my code after Claude writes it"

```
1. **Clarify**: Which formatter? (prettier, gofmt, etc.)
2. **Read**: `.claude/settings.json` (or create if missing)
3. **Merge**: Add to existing hooks, don't replace
4. **Result**:
```json
{
 "hooks": {
 "PostToolUse": [{
 "matcher": "Write|Edit",
 "hooks": [{
 "type": "command",
 "command": "jq -r '.tool_response.filePath // .tool_input.file_path' | { read -r f;
prettier --write \"$f\"; } 2>/dev/null || true"
 }]
 }
]
}
```

### Adding Permissions

User: "Allow npm commands without prompting"

```
1. **Read**: Existing permissions
2. **Merge**: Add `Bash(npm *)` to allow array
3. **Result**: Combined with existing allows
```

### Environment Variables

User: "Set `DEBUG=true`"

```
1. **Decide**: User settings (global) or project settings?
2. **Read**: Target file
3. **Merge**: Add to env object
```json
{ "env": { "DEBUG": "true" } }
```

Common Mistakes to Avoid

```
1. **Replacing instead of merging** - Always preserve existing settings
2. **Wrong file** - Ask user if scope is unclear
3. **Invalid JSON** - Validate syntax after changes
4. **Forgetting to read first** - Always read before write
```

Troubleshooting Hooks

If a hook isn't running:

1. ****Check the settings file**** - Read ~/.claude/settings.json or .claude/settings.json
2. ****Verify JSON syntax**** - Invalid JSON silently fails
3. ****Check the matcher**** - Does it match the tool name? (e.g., "Bash", "Write", "Edit")
4. ****Check hook type**** - Is it "command", "prompt", or "agent"?
5. ****Test the command**** - Run the hook command manually to see if it works
6. ****Use --debug**** - Run `claude --debug` to see hook execution logs

Full Settings JSON Schema

```
```json
{
 "$schema": "https://json-schema.org/draft/2020-12/schema",
 "type": "object",
 "properties": {
 "$schema": {
 "description": "JSON Schema reference for Claude Code settings",
 "type": "string",
 "const": "https://json.schemastore.org/claude-code-settings.json"
 },
 "apiKeyHelper": {
 "description": "Path to a script that outputs authentication values",
 "type": "string"
 },
 "proxyAuthHelper": {
 "description": "Shell command that outputs a Proxy-Authorization header value (EAP)",
 "type": "string"
 },
 "awsCredentialExport": {
 "description": "Path to a script that exports AWS credentials",
 "type": "string"
 },
 "awsAuthRefresh": {
 "description": "Path to a script that refreshes AWS authentication",
 "type": "string"
 },
 "gcpAuthRefresh": {
 "description": "Command to refresh GCP authentication (e.g., gcloud auth application-
default login)",
 "type": "string"
 },
 "policyHelper": {
 "description": "Executable that computes managed settings at startup. Honored only from
admin-controlled policy sources.",
 "type": "object",
 "properties": {
 "path": {
 "description": "Absolute path to the helper executable",
 "type": "string"
 },
 "timeoutMs": {
 "type": "integer",
 "minimum": 1000,
 "maximum": 9007199254740991
 },
 "refreshIntervalMs": {
 "anyOf": [
 {
 "type": "number",
 "const": 0
 }
]
 }
 }
 }
 }
}
```

```

 {
 "type": "integer",
 "minimum": 60000,
 "maximum": 9007199254740991
 }
]
}
},
"required": [
 "path"
]
},
"fileSuggestion": {
 "description": "Custom file suggestion configuration for @ mentions",
 "type": "object",
 "properties": {
 "type": {
 "type": "string",
 "const": "command"
 },
 "command": {
 "type": "string"
 }
 },
 "required": [
 "type",
 "command"
]
},
"respectGitignore": {
 "description": "Whether file picker should respect .gitignore files (default: true). Note:
.ignore files are always respected.",
 "type": "boolean"
},
"breakReminder": {
 "description": "@internal Opt-in break reminder. When enabled, shows a dismissible nudge
after sustained continuous use. Never blocks ? just a friendly heads-up.",
 "type": "object",
 "properties": {
 "enabled": {
 "description": "Show a friendly nudge after sustained continuous use (default false).
Must be true for the reminder to fire.",
 "type": "boolean"
 },
 "intervalMinutes": {
 "description": "Minutes of continuous use before the reminder fires (default 120). Re-
fires every interval until you take a break.",
 "type": "integer",
 "exclusiveMinimum": 0,
 "maximum": 9007199254740991
 },
 "breakThresholdMinutes": {
 "description": "Minutes of inactivity that count as a break and reset the timer
(default 15)",
 "type": "integer",
 "exclusiveMinimum": 0,
 "maximum": 9007199254740991
 },
 "message": {
 "description": "Custom reminder text. Leave unset for a rotating set of friendly
nudges.",
 "type": "string"
 }
 }
}
},

```

```

"quietHours": {
 "description": "@internal Opt-in quiet hours. When enabled, shows a single soft nudge per
session while inside the configured local-time window. Never blocks.",
 "type": "object",
 "properties": {
 "enabled": {
 "description": "Show a one-time nudge when you start or keep using the CLI inside your
quiet-hours window (default false).",
 "type": "boolean"
 },
 "start": {
 "description": "Start of the quiet-hours window, 24-hour local time \"HH:MM\".",
 "type": "string",
 "pattern": "^[01]?\\d|2[0-3]):[0-5]\\d$"
 },
 "end": {
 "description": "End of the quiet-hours window, 24-hour local time \"HH:MM\". May be
earlier than start for an overnight range.",
 "type": "string",
 "pattern": "^[01]?\\d|2[0-3]):[0-5]\\d$"
 }
 }
},
"cleanupPeriodDays": {
 "description": "Number of days to retain chat transcripts before automatic cleanup
(default: 30). Minimum 1. Use a large value for long retention; use --no-session-persistence to
disable transcript writes entirely.",
 "type": "integer",
 "exclusiveMinimum": 0,
 "maximum": 9007199254740991
},
"skillListingMaxDescChars": {
 "description": "Per-skill description character cap in the skill listing sent to Claude
(default: 1536). Descriptions longer than this are truncated. Raise to opt in to higher per-turn
context cost.",
 "type": "integer",
 "exclusiveMinimum": 0,
 "maximum": 9007199254740991
},
"skillListingBudgetFraction": {
 "description": "Fraction of the context window (in characters) reserved for the skill
listing sent to Claude (default: 0.01 = 1%). When the listing exceeds this, descriptions are
shortened to fit. Raise to opt in to higher per-turn context cost.",
 "type": "number",
 "exclusiveMinimum": 0,
 "maximum": 1
},
"wslInheritsWindowsSettings": {
 "description": "When set to true in either admin-only Windows source ? the HKLM
SOFTWARE/Policies/ClaudeCode registry key or C:/Program Files/ClaudeCode/managed-settings.json ?
WSL reads managed settings from the full Windows policy chain (HKLM, C:/Program Files/ClaudeCode
via DrvFs, HKCU) in addition to /etc/claude-code. Windows sources take priority. The flag is
also required in HKCU itself for HKCU policy to apply on WSL (double opt-in: admin enables the
chain, user confirms HKCU). On native Windows the flag has no effect.",
 "type": "boolean"
},
"env": {
 "description": "Environment variables to set for Claude Code sessions",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "type": "string"
 }
}

```

```

},
"attribution": {
 "description": "Customize attribution text for commits and PRs. Each field defaults to the
standard Claude Code attribution if not set.",
 "type": "object",
 "properties": {
 "commit": {
 "description": "Attribution text for git commits, including any trailers. Empty string
hides attribution.",
 "type": "string"
 },
 "pr": {
 "description": "Attribution text for pull request descriptions. Empty string hides
attribution.",
 "type": "string"
 }
 }
},
"includeCoAuthoredBy": {
 "description": "Deprecated: Use attribution instead. Whether to include Claude's co-
authored by attribution in commits and PRs (defaults to true)",
 "type": "boolean"
},
"includeGitInstructions": {
 "description": "Include built-in commit and PR workflow instructions in Claude's system
prompt (default: true)",
 "type": "boolean"
},
"permissions": {
 "description": "Tool usage permissions configuration",
 "type": "object",
 "properties": {
 "allow": {
 "description": "List of permission rules for allowed operations",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "deny": {
 "description": "List of permission rules for denied operations",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "ask": {
 "description": "List of permission rules that should always prompt for confirmation",
 "type": "array",
 "items": {
 "type": "string"
 }
 }
 }
},
"defaultMode": {
 "description": "Default permission mode when Claude Code needs access",
 "type": "string",
 "enum": [
 "acceptEdits",
 "auto",
 "bypassPermissions",
 "default",
 "dontAsk",
 "plan"
]
},
},

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 "disableBypassPermissionsMode": {
 "description": "Disable the ability to bypass permission prompts",
 "type": "string",
 "enum": [
 "disable"
]
 },
 "disableAutoMode": {
 "description": "Disable auto mode",
 "type": "string",
 "enum": [
 "disable"
]
 },
 "additionalDirectories": {
 "description": "Additional directories to include in the permission scope",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "additionalProperties": {}
 },
 "model": {
 "description": "Override the default model used by Claude Code",
 "type": "string"
 },
 "fallbackModel": {
 "description": "Fallback model(s) tried in order when the primary model is overloaded or
unavailable. Each element accepts a model name or alias; \"default\" expands to the default
model. CLI --fallback-model takes precedence.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "availableModels": {
 "description": "Allowlist of models that users can select. Accepts family aliases
(\"opus\" allows any opus version), version prefixes (\"opus-4-5\" allows only that version),
and full model IDs. If undefined, all models are available. If empty array, only the default
model is available. Typically set in managed settings by enterprise administrators.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "modelOverrides": {
 "description": "Override mapping from Anthropic model ID (e.g. \"claude-opus-4-6\") to
provider-specific model ID (e.g. a Bedrock inference profile ARN). Typically set in managed
settings by enterprise administrators.",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "type": "string"
 }
 },
 "enableAllProjectMcpServers": {
 "description": "Whether to automatically approve all MCP servers in the project",
 "type": "boolean"
 },
 "enabledMcpjsonServers": {
 "description": "List of approved MCP servers from .mcp.json",

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 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "disabledMcpjsonServers": {
 "description": "List of rejected MCP servers from .mcp.json",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "skillOverrides": {
 "description": "Per-skill listing overrides keyed by skill name. \"name-only\" lists the skill without its description; \"user-invocable-only\" hides it from the model but keeps /name; \"off\" hides it from both. Absent = on.",
 "type": "object",
 "propertyNames": {
 "type": "string"
 }
 },
 "additionalProperties": {
 "type": "string",
 "enum": [
 "on",
 "name-only",
 "user-invocable-only",
 "off"
]
 }
},
"disableBundledSkills": {
 "description": "Disable the skills and workflows that ship with Claude Code: bundled skills and workflows are removed entirely; built-in slash commands stay typable but are hidden from the model. Plugins, .claude/skills/, and .claude/commands/ are unaffected. Equivalent to CLAUDE_CODE_DISABLE_BUNDLED_SKILLS=1.",
 "type": "boolean"
},
"allowedMcpServers": {
 "description": "Enterprise allowlist of MCP servers that can be used. Applies to all scopes including enterprise servers from managed-mcp.json. If undefined, all servers are allowed. If empty array, no servers are allowed. Denylist takes precedence - if a server is on both lists, it is denied.",
 "type": "array",
 "items": {
 "type": "object",
 "properties": {
 "serverName": {
 "description": "Name of the MCP server that users are allowed to configure",
 "type": "string",
 "pattern": "^[a-zA-Z0-9_-]+$"
 },
 "serverCommand": {
 "description": "Command array [command, ...args] to match exactly for allowed stdio servers",
 "minItems": 1,
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "serverUrl": {
 "description": "URL pattern with wildcard support (e.g., \"https://*.example.com/*\") for allowed remote MCP servers",
 "type": "string"
 }
 }
 }
}

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 }
 },
 "deniedMcpServers": {
 "description": "Enterprise denylist of MCP servers that are explicitly blocked. If a
server is on the denylist, it will be blocked across all scopes including enterprise. Denylist
takes precedence over allowlist - if a server is on both lists, it is denied.",
 "type": "array",
 "items": {
 "type": "object",
 "properties": {
 "serverName": {
 "description": "Name of the MCP server that is explicitly blocked",
 "type": "string",
 "pattern": "^[a-zA-Z0-9_-]+$"
 },
 "serverCommand": {
 "description": "Command array [command, ...args] to match exactly for blocked stdio
servers",
 "minItems": 1,
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "serverUrl": {
 "description": "URL pattern with wildcard support (e.g.,
\"https://*.example.com/*\") for blocked remote MCP servers",
 "type": "string"
 }
 }
 }
 },
 "hooks": {
 "description": "Custom commands to run before/after tool executions",
 "type": "object",
 "propertyNames": {
 "anyOf": [
 {
 "type": "string",
 "enum": [
 "PreToolUse",
 "PostToolUse",
 "PostToolUseFailure",
 "PostToolBatch",
 "Notification",
 "UserPromptSubmit",
 "UserPromptExpansion",
 "SessionStart",
 "SessionEnd",
 "Stop",
 "StopFailure",
 "SubagentStart",
 "SubagentStop",
 "PreCompact",
 "PostCompact",
 "PermissionRequest",
 "PermissionDenied",
 "Setup",
 "TeammateIdle",
 "TaskCreated",
 "TaskCompleted",
 "Elicitation",
 "ElicitationResult",
 "ConfigChange",

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 "WorktreeCreate",
 "WorktreeRemove",
 "InstructionsLoaded",
 "CwdChanged",
 "FileChanged",
 "MessageDisplay"
]
},
{
 "not": {}
}
]
},
"additionalProperties": {
 "type": "array",
 "items": {
 "type": "object",
 "properties": {
 "matcher": {
 "description": "String pattern to match (e.g. tool names like \"Write\")",
 "type": "string"
 },
 "hooks": {
 "description": "List of hooks to execute when the matcher matches",
 "type": "array",
 "items": {
 "anyOf": [
 {
 "type": "object",
 "properties": {
 "type": {
 "description": "Shell command hook type",
 "type": "string",
 "const": "command"
 },
 "command": {
 "description": "Shell command to execute",
 "type": "string"
 },
 "args": {
 "description": "Argument list for exec form. When present, `command` is
resolved as an executable and spawned directly with these arguments ? no shell. Path
placeholders like ${CLAUDE_PLUGIN_ROOT} are substituted per-element as plain strings, so paths
with quotes, $, or backticks never reach a shell parser. When absent, `command` runs through a
shell (bash on POSIX, PowerShell on Windows without Git Bash).",
 "type": "array",
 "items": {
 "type": "string"
 }
 }
 }
 },
 {
 "description": "Permission rule syntax to filter when this hook runs
(e.g., \"Bash(git *)\"). Only runs if the tool call matches the pattern. Avoids spawning hooks
for non-matching commands.",
 "type": "string"
 }
]
 },
 "shell": {
 "description": "Shell interpreter. 'bash' uses your $SHELL
(bash/zsh/sh); 'powershell' uses pwsh. Defaults to bash (powershell on Windows without Git
Bash).",
 "type": "string",
 "enum": [
 "bash",
 "powershell"
]
 }
 }
 }
 }
}

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 },
 "timeout": {
 "description": "Timeout in seconds for this specific command",
 "type": "number",
 "exclusiveMinimum": 0
 },
 },
 "statusMessage": {
 "description": "Custom status message to display in spinner while hook
runs",
 "type": "string"
 },
 },
 "once": {
 "description": "If true, hook runs once and is removed after execution",
 "type": "boolean"
 },
 },
 "async": {
 "description": "If true, hook runs in background without blocking",
 "type": "boolean"
 },
 },
 "asyncRewake": {
 "description": "If true, hook runs in background and wakes the model on
exit code 2 (blocking error). Implies async.",
 "type": "boolean"
 },
 },
 "rewakeMessage": {
 "description": "@internal Custom prefix for the system-reminder shown to
the model when an asyncRewake hook exits with code 2. The hook output is appended after this
prefix.",
 "type": "string",
 "minLength": 1
 },
 },
 "rewakeSummary": {
 "description": "@internal One-line summary shown to the user in the
terminal when an asyncRewake hook exits with code 2. Defaults to \"Stop hook feedback\".",
 "type": "string",
 "minLength": 1
 }
 },
 "required": [
 "type",
 "command"
]
},
{
 "type": "object",
 "properties": {
 "type": {
 "description": "LLM prompt hook type",
 "type": "string",
 "const": "prompt"
 },
 },
 "prompt": {
 "description": "Prompt to evaluate with LLM. Use $ARGUMENTS placeholder
for hook input JSON.",
 "type": "string"
 },
 },
 "if": {
 "description": "Permission rule syntax to filter when this hook runs
(e.g., \"Bash(git *)\"). Only runs if the tool call matches the pattern. Avoids spawning hooks
for non-matching commands.",
 "type": "string"
 },
 },
 "timeout": {
 "description": "Timeout in seconds for this specific prompt evaluation",
 "type": "number",

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 "exclusiveMinimum": 0
 },
 "model": {
 "description": "Model to use for this prompt hook (e.g., \"claude-sonnet-4-6\"). If not specified, uses the default small fast model.",
 "type": "string"
 },
 "continueOnBlock": {
 "description": "Sets the continue value for the decision:\\"block\\" produced when ok is false. Default false (turn ends). Whether continue:true lets the turn proceed depends on the event's decision:\\"block\\" semantics. On PostToolUse, the reason is fed back to Claude and the turn continues.",
 "type": "boolean"
 },
 "statusMessage": {
 "description": "Custom status message to display in spinner while hook runs",
 "type": "string"
 },
 "once": {
 "description": "If true, hook runs once and is removed after execution",
 "type": "boolean"
 }
},
"required": [
 "type",
 "prompt"
]
},
{
 "type": "object",
 "properties": {
 "type": {
 "description": "Agentic verifier hook type",
 "type": "string",
 "const": "agent"
 },
 "prompt": {
 "description": "Prompt describing what to verify (e.g. \"Verify that unit tests ran and passed.\"). Use $ARGUMENTS placeholder for hook input JSON.",
 "type": "string"
 },
 "if": {
 "description": "Permission rule syntax to filter when this hook runs (e.g., \"Bash(git *)\"). Only runs if the tool call matches the pattern. Avoids spawning hooks for non-matching commands.",
 "type": "string"
 },
 "timeout": {
 "description": "Timeout in seconds for agent execution (default 60)",
 "type": "number",
 "exclusiveMinimum": 0
 },
 "model": {
 "description": "Model to use for this agent hook (e.g., \"claude-sonnet-4-6\"). If not specified, uses Haiku.",
 "type": "string"
 },
 "statusMessage": {
 "description": "Custom status message to display in spinner while hook runs",
 "type": "string"
 },
 "once": {
 "description": "If true, hook runs once and is removed after execution",

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 "type": "boolean"
 }
},
"required": [
 "type",
 "prompt"
]
},
{
 "type": "object",
 "properties": {
 "type": {
 "description": "HTTP hook type",
 "type": "string",
 "const": "http"
 },
 "url": {
 "description": "URL to POST the hook input JSON to",
 "type": "string",
 "format": "uri"
 },
 "if": {
 "description": "Permission rule syntax to filter when this hook runs
(e.g., \"Bash(git *)\"). Only runs if the tool call matches the pattern. Avoids spawning hooks
for non-matching commands.",
 "type": "string"
 },
 "timeout": {
 "description": "Timeout in seconds for this specific request",
 "type": "number",
 "exclusiveMinimum": 0
 },
 "headers": {
 "description": "Additional headers to include in the request. Values may
reference environment variables using $VAR_NAME or ${VAR_NAME} syntax (e.g., \"Authorization\":
\"Bearer $MY_TOKEN\"). Only variables listed in allowedEnvVars will be interpolated.",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "type": "string"
 }
 },
 "allowedEnvVars": {
 "description": "Explicit list of environment variable names that may be
interpolated in header values. Only variables listed here will be resolved; all other $VAR
references are left as empty strings. Required for env var interpolation to work.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "statusMessage": {
 "description": "Custom status message to display in spinner while hook
runs",
 "type": "string"
 },
 "once": {
 "description": "If true, hook runs once and is removed after execution",
 "type": "boolean"
 }
 },
 "required": [
 "type",

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 "url"
]
},
{
 "type": "object",
 "properties": {
 "type": {
 "description": "MCP tool hook type",
 "type": "string",
 "const": "mcp_tool"
 },
 "server": {
 "description": "Name of an already-configured MCP server to invoke",
 "type": "string"
 },
 "tool": {
 "description": "Name of the tool on that server to call",
 "type": "string"
 },
 "input": {
 "description": "Arguments passed to the MCP tool. String values support
${path} interpolation from the hook input JSON (e.g. \"${tool_input.file_path}\").",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {}
 },
 "if": {
 "description": "Permission rule syntax to filter when this hook runs
(e.g., \"Bash(git *)\"). Only runs if the tool call matches the pattern. Avoids spawning hooks
for non-matching commands.",
 "type": "string"
 },
 "timeout": {
 "description": "Timeout in seconds for this specific tool call",
 "type": "number",
 "exclusiveMinimum": 0
 },
 "statusMessage": {
 "description": "Custom status message to display in spinner while hook
runs",
 "type": "string"
 },
 "once": {
 "description": "If true, hook runs once and is removed after execution",
 "type": "boolean"
 }
 },
 "required": [
 "type",
 "server",
 "tool"
]
}
]
}
},
"required": [
 "hooks"
]
}
},
},

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"worktree": {
 "description": "Git worktree configuration for --worktree flag.",
 "type": "object",
 "properties": {
 "symlinkDirectories": {
 "description": "Directories to symlink from main repository to worktrees to avoid disk
bloat. Must be explicitly configured - no directories are symlinked by default. Common examples:
\"node_modules\", \".cache\", \".bin\"",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "sparsePaths": {
 "description": "Directories to include when creating worktrees, via git sparse-
checkout (cone mode). Dramatically faster in large monorepos ? only the listed paths are written
to disk.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "baseRef": {
 "description": "Which ref new worktrees branch from. 'fresh' (default) branches from
origin/<default-branch> for a clean tree. 'head' branches from your current local HEAD so
unpushed commits and feature-branch state are present. Applies to --worktree, EnterWorktree, and
agent isolation.",
 "type": "string",
 "enum": [
 "fresh",
 "head"
]
 },
 "bgIsolation": {
 "description": "Isolation mode for background sessions in this repo. 'worktree'
(default) blocks Edit/Write in the main checkout until EnterWorktree is called. 'none' lets
background jobs edit the working copy directly.",
 "type": "string",
 "enum": [
 "worktree",
 "none"
]
 }
 },
 "disableAllHooks": {
 "description": "Disable all hooks and statusLine execution",
 "type": "boolean"
 },
 "disableAgentView": {
 "description": "Disable agent view (`claude agents`, `--bg`, /background, the on-demand
daemon). Typically set in managed settings. Equivalent to CLAUDE_CODE_DISABLE_AGENT_VIEW=1.",
 "type": "boolean"
 },
 "disableRemoteControl": {
 "description": "Disable Remote Control (claude.ai/code, `claude remote-control`,
`--remote-control`/`--rc`, auto-start, and the in-session toggle). Typically set in managed
settings.",
 "type": "boolean"
 },
 "disableWorkflows": {
 "description": "Disable the Workflows feature (also via CLAUDE_CODE_DISABLE_WORKFLOWS).",
 "type": "boolean"
 },
 "enableWorkflows": {

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 "description": "Enable or disable the Workflows feature for this user. Unset = default by
plan once the feature is available.",
 "type": "boolean"
 },
 "workflowKeywordTriggerEnabled": {
 "description": "Enable the \"ultracode\" keyword trigger: including the keyword in a
prompt opts that turn into the Workflow tool. Set to false to disable the trigger. Default:
true.",
 "type": "boolean"
 },
 "disableSkillShellExecution": {
 "description": "Disable inline shell execution in skills and custom slash commands from
user, project, or plugin sources. Commands are replaced with a placeholder instead of being
run.",
 "type": "boolean"
 },
 "defaultShell": {
 "description": "Default shell for input-box ! commands. Defaults to 'bash' on all
platforms (no Windows auto-flip).",
 "type": "string",
 "enum": [
 "bash",
 "powershell"
]
 },
 "allowManagedHooksOnly": {
 "description": "When true (and set in managed settings), only hooks from managed settings
run. User, project, and local hooks are ignored.",
 "type": "boolean"
 },
 "allowedHttpHookUrls": {
 "description": "Allowlist of URL patterns that HTTP hooks may target. Supports * as a
wildcard (e.g. \"https://hooks.example.com/*\"). When set, HTTP hooks with non-matching URLs are
blocked. If undefined, all URLs are allowed. If empty array, no HTTP hooks are allowed. Arrays
merge across settings sources (same semantics as allowedMcpServers).",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "httpHookAllowedEnvVars": {
 "description": "Allowlist of environment variable names HTTP hooks may interpolate into
headers. When set, each hook's effective allowedEnvVars is the intersection with this list. If
undefined, no restriction is applied. Arrays merge across settings sources (same semantics as
allowedMcpServers).",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "allowManagedPermissionRulesOnly": {
 "description": "When true (and set in managed settings), only permission rules
(allow/deny/ask) from managed settings are respected. User, project, local, and CLI argument
permission rules are ignored.",
 "type": "boolean"
 },
 "allowManagedMcpServersOnly": {
 "description": "When true (and set in managed settings), allowedMcpServers is only read
from managed settings. deniedMcpServers still merges from all sources, so users can deny servers
for themselves. Users can still add their own MCP servers, but only the admin-defined allowlist
applies.",
 "type": "boolean"
 },
 "allowAllClaudeAiMcps": {
 "description": "When true (and set in managed settings), claude.ai cloud MCP connectors

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load alongside managed-mcp.json instead of being suppressed by its exclusive-control lockdown. Default off preserves the lockdown. Read from managed settings only.",

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"type": "boolean"
},
"strictPluginOnlyCustomization": {
 "description": "When set in managed settings, blocks non-plugin customization sources for
the listed surfaces. Array form locks specific surfaces (e.g. [\"skills\", \"hooks\"]); `true`
locks all four; `false` is an explicit no-op. Blocked: ~/.claude/{surface}/, .claude/{surface}/
(project), settings.json hooks, .mcp.json. NOT blocked: managed (policySettings) sources,
plugin-provided customizations. Composes with strictKnownMarketplaces for end-to-end admin
control ? plugins gated by marketplace allowlist, everything else blocked here.",
 "anyOf": [
 {
 "type": "boolean"
 },
 {
 "type": "array",
 "items": {
 "type": "string",
 "enum": [
 "skills",
 "agents",
 "hooks",
 "mcp"
]
 }
 }
]
},
"statusLine": {
 "description": "Custom status line display configuration",
 "type": "object",
 "properties": {
 "type": {
 "type": "string",
 "const": "command"
 },
 "command": {
 "type": "string"
 },
 "padding": {
 "type": "number"
 },
 "refreshInterval": {
 "description": "Re-run the status line command every N seconds in addition to event-
driven updates",
 "type": "number",
 "minimum": 1
 },
 "hideVimModeIndicator": {
 "description": "Hide the built-in `-- INSERT --` / `-- VISUAL --` indicator below the
prompt. Use this when your status line script renders `vim.mode` itself.",
 "type": "boolean"
 }
 },
 "required": [
 "type",
 "command"
]
},
"prUrlTemplate": {
 "description": "URL template for PR links in the footer badge and inline messages.
Placeholders: {host} {owner} {repo} {number} {url}. Example:
`https://reviews.example.com/{owner}/{repo}/pull/{number}`",
 "type": "string"
}
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 },
 "subagentStatusLine": {
 "description": "Custom per-subagent status line shown in the agent panel; receives row
context as JSON on stdin",
 "type": "object",
 "properties": {
 "type": {
 "type": "string",
 "const": "command"
 },
 "command": {
 "type": "string"
 }
 },
 "required": [
 "type",
 "command"
]
 },
 "enabledPlugins": {
 "description": "Enabled plugins using plugin-id@marketplace-id format. Example: {
\"formatter@anthropic-tools\": true }. Also supports extended format with version constraints.
Settings precedence is user < project < local < flag < policy, so to disable a plugin that
project settings enable, set it to false in .claude/settings.local.json ? setting false in
~/.claude/settings.json is overridden by the project.",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "anyOf": [
 {
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 {
 "type": "boolean"
 },
 {
 "not": {}
 }
]
 }
 },
 "extraKnownMarketplaces": {
 "description": "Additional marketplaces to make available for this repository. Typically
used in repository .claude/settings.json to ensure team members have required plugin sources.",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "type": "object",
 "properties": {
 "source": {
 "description": "Where to fetch the marketplace from",
 "anyOf": [
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "url"
 }
 }
 }
]
 }
 }
 }
 }
 }
}

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 },
 "url": {
 "description": "Direct URL to marketplace.json file",
 "type": "string",
 "format": "uri"
 },
 "headers": {
 "description": "Custom HTTP headers (e.g., for authentication)",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "type": "string"
 }
 },
 "required": [
 "source",
 "url"
]
 },
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "github"
 },
 "repo": {
 "description": "GitHub repository in owner/repo format",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\", \"v1.0.0\").
Defaults to repository default branch.",
 "type": "string"
 },
 "path": {
 "description": "Path to marketplace.json within repo (defaults to .claude-
plugin/marketplace.json)",
 "type": "string"
 },
 "sparsePaths": {
 "description": "Directories to include via git sparse-checkout (cone mode).
Use for monorepos where the marketplace lives in a subdirectory. Example: [\".claude-plugin\",
\"plugins\"]. If omitted, the full repository is cloned.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "skipLfs": {
 "description": "Skip Git LFS smudge during clone and update (sets
GIT_LFS_SKIP_SMUDGE=1) so LFS pointer files stay as pointers instead of downloading their
content. Use for marketplaces hosted in repos with large LFS objects.",
 "type": "boolean"
 }
 },
 "required": [
 "source",
 "repo"
]
 },
 {

```

```

 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "git"
 },
 "url": {
 "description": "Full git repository URL",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\", \"v1.0.0\").
Defaults to repository default branch.",
 "type": "string"
 },
 "path": {
 "description": "Path to marketplace.json within repo (defaults to .claude-
plugin/marketplace.json)",
 "type": "string"
 },
 "sparsePaths": {
 "description": "Directories to include via git sparse-checkout (cone mode).
Use for monorepos where the marketplace lives in a subdirectory. Example: [\".claude-plugin\",
\"plugins\"]. If omitted, the full repository is cloned.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "skipLfs": {
 "description": "Skip Git LFS smudge during clone and update (sets
GIT_LFS_SKIP_SMUDGE=1) so LFS pointer files stay as pointers instead of downloading their
content. Use for marketplaces hosted in repos with large LFS objects.",
 "type": "boolean"
 }
 },
 "required": [
 "source",
 "url"
]
 },
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "npm"
 },
 "package": {
 "description": "NPM package containing marketplace.json",
 "type": "string"
 }
 },
 "required": [
 "source",
 "package"
]
 },
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "file"
 }
 }
 },

```

```

 "path": {
 "description": "Local file path to marketplace.json",
 "type": "string"
 },
 },
 "required": [
 "source",
 "path"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "directory"
 },
 "path": {
 "description": "Local directory containing .claude-plugin/marketplace.json",
 "type": "string"
 }
 },
 "required": [
 "source",
 "path"
]
},
{
 "description": "Policy-list sentinel for the ~/.claude/skills/ auto-load
(@skills-dir plugins). In strictKnownMarketplaces: opt the scan back IN (by default any
allowlist blocks it). In blockedMarketplaces: turn the scan OFF without otherwise restricting
marketplaces. Only meaningful in those two managed-settings lists
(areLocalPluginDirsAllowedByPolicy); known_marketplaces.json / marketplace add etc. ignore it.",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "skills-dir"
 }
 },
 "required": [
 "source"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "hostPattern"
 },
 "hostPattern": {
 "description": "Regex pattern to match the host/domain extracted from any
marketplace source type. For github sources, matches against \"github.com\". For git sources
(SSSH or HTTPS), extracts the hostname from the URL. Use in strictKnownMarketplaces to allow all
marketplaces from a specific host (e.g., \"^github\\.mycompany\\.com$\"),",
 "type": "string"
 }
 },
 "required": [
 "source",
 "hostPattern"
]
},
{

```

```

 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "pathPattern"
 },
 "pathPattern": {
 "description": "Regex pattern matched against the .path field of file and
directory sources. Use in strictKnownMarketplaces to allow filesystem-based marketplaces
alongside hostPattern restrictions for network sources. Use \".*\\" to allow all filesystem
paths, or a narrower pattern (e.g., \"/opt/approved/") to restrict to specific directories.",
 "type": "string"
 }
 },
 "required": [
 "source",
 "pathPattern"
]
 },
 {
 "description": "Inline marketplace manifest defined directly in settings.json.
The reconciler writes a synthetic marketplace.json to the cache; diffMarketplaces detects edits
via isEqual on the stored source (the plugins array is inside this object, so edits surface as
sourceChanged).",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "settings"
 },
 "name": {
 "description": "Marketplace name. Must match the extraKnownMarketplaces key
(enforced); the synthetic manifest is written under this name. Same validation as
PluginMarketplaceSchema plus reserved-name rejection ? validateOfficialNameSource runs after the
disk write, too late to clean up.",
 "type": "string",
 "minLength": 1
 },
 "plugins": {
 "description": "Plugin entries declared inline in settings.json",
 "type": "array",
 "items": {
 "type": "object",
 "properties": {
 "name": {
 "description": "Plugin name as it appears in the target repository",
 "type": "string",
 "minLength": 1
 },
 "source": {
 "description": "Where to fetch the plugin from. Must be a remote
source ? relative paths have no marketplace repository to resolve against.",
 "anyOf": [
 {
 "description": "Path to the plugin root, relative to the
marketplace root (the directory containing .claude-plugin/, not .claude-plugin/ itself)",
 "type": "string",
 "pattern": "^\\.\\.\\.\\.\\.\\.*"
 },
 {
 "description": "NPM package as plugin source",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",

```

```

 "const": "npm"
 },
 "package": {
 "description": "Package name (or url, or local path, or
anything else that can be passed to `npm` as a package)",
 "anyOf": [
 {
 "type": "string"
 },
 {
 "type": "string"
 }
]
 },
 "version": {
 "description": "Specific version or version range (e.g.,
^1.0.0, ~2.1.0)",
 "type": "string"
 },
 "registry": {
 "description": "Custom NPM registry URL (defaults to using
system default, likely npmjs.org)",
 "type": "string",
 "format": "uri"
 },
 "required": [
 "source",
 "package"
],
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "url"
 },
 "url": {
 "description": "Full git repository URL (https:// or git@)",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\"v1.0.0\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "^[a-f0-9]{40}$"
 }
 },
 "required": [
 "source",
 "url"
],
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",

```



```

 "const": "github"
 },
 "repo": {
 "description": "GitHub repository in owner/repo format",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\\\"v1.0.0\\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "^[a-f0-9]{40}$"
 }
 },
 "required": [
 "source",
 "repo"
]
 },
 {
 "description": "Plugin located in a subdirectory of a larger
repository (monorepo). Only the specified subdirectory is materialized; the rest of the repo is
not downloaded.",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "git-subdir"
 },
 "url": {
 "description": "Git repository: GitHub owner/repo shorthand,
https://, or git@ URL",
 "type": "string"
 },
 "path": {
 "description": "Subdirectory within the repo containing the
plugin (e.g., \"tools/claude-plugin\"). Cloned sparsely using partial clone (--filter=tree:0) to
minimize bandwidth for monorepos.",
 "type": "string",
 "minLength": 1
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\\\"v1.0.0\\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "^[a-f0-9]{40}$"
 }
 },
 "required": [
 "source",
 "url",
 "path"
]
 },
},

```

{  
 "description": "Placeholder for source types this Claude Code  
version does not recognize. Never authored by hand ? PluginMarketplaceSchema rewrites  
unparseable sources to this so the entry remains in marketplace.plugins (detectDelistedPlugins  
must not see it as removed). Install attempts fail at cachePlugin with a clear \"update Claude  
Code\" message.",

```
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "unsupported"
 }
 },
 "required": [
 "source"
]
 }
],
 "description": {
 "type": "string"
 },
 "version": {
 "type": "string"
 },
 "strict": {
 "type": "boolean"
 }
},
"required": [
 "name",
 "source"
]
}
},
"owner": {
 "type": "object",
 "properties": {
 "name": {
 "description": "Display name of the plugin author or organization",
 "type": "string",
 "minLength": 1
 },
 "email": {
 "description": "Contact email for support or feedback",
 "type": "string"
 },
 "url": {
 "description": "Website, GitHub profile, or organization URL",
 "type": "string"
 }
 },
 "required": [
 "name"
]
}
},
"required": [
 "source",
 "name",
 "plugins"
]
}
],
},
```

```

 "installLocation": {
 "description": "Local cache path where marketplace manifest is stored (auto-
generated if not provided)",
 "type": "string"
 },
 "autoUpdate": {
 "description": "Whether to automatically update this marketplace and its installed
plugins on startup",
 "type": "boolean"
 }
 },
 "required": [
 "source"
]
}
},
"strictKnownMarketplaces": {
 "description": "Enterprise strict list of allowed marketplace sources. When set in managed
settings, ONLY these exact sources can be added as marketplaces. The check happens BEFORE
downloading, so blocked sources never touch the filesystem. Note: this is a policy gate only ?
it does NOT register marketplaces. To pre-register allowed marketplaces for users, also set
extraKnownMarketplaces.",
 "type": "array",
 "items": {
 "anyOf": [
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "url"
 },
 "url": {
 "description": "Direct URL to marketplace.json file",
 "type": "string",
 "format": "uri"
 },
 "headers": {
 "description": "Custom HTTP headers (e.g., for authentication)",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "type": "string"
 }
 }
 }
 },
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "github"
 },
 "repo": {
 "description": "GitHub repository in owner/repo format",
 "type": "string"
 },
 "ref": {

```

```

 "description": "Git branch or tag to use (e.g., \"main\", \"v1.0.0\"). Defaults
to repository default branch.",
 "type": "string"
 },
 "path": {
 "description": "Path to marketplace.json within repo (defaults to .claude-
plugin/marketplace.json)",
 "type": "string"
 },
 "sparsePaths": {
 "description": "Directories to include via git sparse-checkout (cone mode). Use
for monorepos where the marketplace lives in a subdirectory. Example: [\".claude-plugin\",
\"plugins\"]. If omitted, the full repository is cloned.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "skipLfs": {
 "description": "Skip Git LFS smudge during clone and update (sets
GIT_LFS_SKIP_SMUDGE=1) so LFS pointer files stay as pointers instead of downloading their
content. Use for marketplaces hosted in repos with large LFS objects.",
 "type": "boolean"
 }
},
"required": [
 "source",
 "repo"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "git"
 },
 "url": {
 "description": "Full git repository URL",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\", \"v1.0.0\"). Defaults
to repository default branch.",
 "type": "string"
 },
 "path": {
 "description": "Path to marketplace.json within repo (defaults to .claude-
plugin/marketplace.json)",
 "type": "string"
 },
 "sparsePaths": {
 "description": "Directories to include via git sparse-checkout (cone mode). Use
for monorepos where the marketplace lives in a subdirectory. Example: [\".claude-plugin\",
\"plugins\"]. If omitted, the full repository is cloned.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "skipLfs": {
 "description": "Skip Git LFS smudge during clone and update (sets
GIT_LFS_SKIP_SMUDGE=1) so LFS pointer files stay as pointers instead of downloading their
content. Use for marketplaces hosted in repos with large LFS objects.",
 "type": "boolean"
 }
 }
}

```

```

 }
 },
 "required": [
 "source",
 "url"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "npm"
 },
 "package": {
 "description": "NPM package containing marketplace.json",
 "type": "string"
 }
 },
 "required": [
 "source",
 "package"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "file"
 },
 "path": {
 "description": "Local file path to marketplace.json",
 "type": "string"
 }
 },
 "required": [
 "source",
 "path"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "directory"
 },
 "path": {
 "description": "Local directory containing .claude-plugin/marketplace.json",
 "type": "string"
 }
 },
 "required": [
 "source",
 "path"
]
},
{
 "description": "Policy-list sentinel for the ~/.claude/skills/ auto-load (@skills-
dir plugins). In strictKnownMarketplaces: opt the scan back IN (by default any allowlist blocks
it). In blockedMarketplaces: turn the scan OFF without otherwise restricting marketplaces. Only
meaningful in those two managed-settings lists (areLocalPluginDirsAllowedByPolicy);
known_marketplaces.json / marketplace add etc. ignore it.",
 "type": "object",

```

```

 "properties": {
 "source": {
 "type": "string",
 "const": "skills-dir"
 }
 },
 "required": [
 "source"
]
 },
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "hostPattern"
 },
 "hostPattern": {
 "description": "Regex pattern to match the host/domain extracted from any
marketplace source type. For github sources, matches against \"github.com\". For git sources
(SSSH or HTTPS), extracts the hostname from the URL. Use in strictKnownMarketplaces to allow all
marketplaces from a specific host (e.g., \"^github\\.mycompany\\.com$\"),",
 "type": "string"
 }
 },
 "required": [
 "source",
 "hostPattern"
]
 },
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "pathPattern"
 },
 "pathPattern": {
 "description": "Regex pattern matched against the .path field of file and
directory sources. Use in strictKnownMarketplaces to allow filesystem-based marketplaces
alongside hostPattern restrictions for network sources. Use \".*\" to allow all filesystem
paths, or a narrower pattern (e.g., \"^/opt/approved/\") to restrict to specific directories.",
 "type": "string"
 }
 },
 "required": [
 "source",
 "pathPattern"
]
 },
 {
 "description": "Inline marketplace manifest defined directly in settings.json. The
reconciler writes a synthetic marketplace.json to the cache; diffMarketplaces detects edits via
isEqual on the stored source (the plugins array is inside this object, so edits surface as
sourceChanged).",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "settings"
 },
 "name": {
 "description": "Marketplace name. Must match the extraKnownMarketplaces key
(enforced); the synthetic manifest is written under this name. Same validation as
PluginMarketplaceSchema plus reserved-name rejection ? validateOfficialNameSource runs after the

```

```

disk write, too late to clean up.",
 "type": "string",
 "minLength": 1
},
"plugins": {
 "description": "Plugin entries declared inline in settings.json",
 "type": "array",
 "items": {
 "type": "object",
 "properties": {
 "name": {
 "description": "Plugin name as it appears in the target repository",
 "type": "string",
 "minLength": 1
 },
 "source": {
 "description": "Where to fetch the plugin from. Must be a remote source ?
relative paths have no marketplace repository to resolve against.",
 "anyOf": [
 {
 "description": "Path to the plugin root, relative to the marketplace
root (the directory containing .claude-plugin/, not .claude-plugin/ itself)",
 "type": "string",
 "pattern": "^\\.\\.\\.\\.\\.\\.*"
 },
 {
 "description": "NPM package as plugin source",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "npm"
 },
 "package": {
 "description": "Package name (or url, or local path, or anything
else that can be passed to `npm` as a package)",
 "anyOf": [
 {
 "type": "string"
 },
 {
 "type": "string"
 }
]
 }
 }
 }
],
 "version": {
 "description": "Specific version or version range (e.g., ^1.0.0,
~2.1.0)",
 "type": "string"
 },
 "registry": {
 "description": "Custom NPM registry URL (defaults to using system
default, likely npmjs.org)",
 "type": "string",
 "format": "uri"
 }
 },
 "required": [
 "source",
 "package"
]
 },
 {
 "type": "object",
 "properties": {

```

```

 "source": {
 "type": "string",
 "const": "url"
 },
 "url": {
 "description": "Full git repository URL (https:// or git@)",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\\\"v1.0.0\\\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "[a-f0-9]{40}$"
 }
 },
 "required": [
 "source",
 "url"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "github"
 },
 "repo": {
 "description": "GitHub repository in owner/repo format",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\\\"v1.0.0\\\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "[a-f0-9]{40}$"
 }
 },
 "required": [
 "source",
 "repo"
]
},
{
 "description": "Plugin located in a subdirectory of a larger
repository (monorepo). Only the specified subdirectory is materialized; the rest of the repo is
not downloaded.",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "git-subdir"
 },
 },

```



```

 "url": {
 "description": "Git repository: GitHub owner/repo shorthand,
https://, or git@ URL",
 "type": "string"
 },
 "path": {
 "description": "Subdirectory within the repo containing the plugin
(e.g., \"tools/claude-plugin\"). Cloned sparsely using partial clone (--filter=tree:0) to
minimize bandwidth for monorepos.",
 "type": "string",
 "minLength": 1
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\"v1.0.0\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "^[a-f0-9]{40}$"
 }
 },
 "required": [
 "source",
 "url",
 "path"
]
},
{
 "description": "Placeholder for source types this Claude Code version
does not recognize. Never authored by hand ? PluginMarketplaceSchema rewrites unparseable
sources to this so the entry remains in marketplace.plugins (detectDelistedPlugins must not see
it as removed). Install attempts fail at cachePlugin with a clear \"update Claude Code\"
message.",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "unsupported"
 }
 },
 "required": [
 "source"
]
}
],
},
"description": {
 "type": "string"
},
"version": {
 "type": "string"
},
"strict": {
 "type": "boolean"
}
},
"required": [
 "name",
 "source"
]
}

```

```

 },
 "owner": {
 "type": "object",
 "properties": {
 "name": {
 "description": "Display name of the plugin author or organization",
 "type": "string",
 "minLength": 1
 },
 "email": {
 "description": "Contact email for support or feedback",
 "type": "string"
 },
 "url": {
 "description": "Website, GitHub profile, or organization URL",
 "type": "string"
 }
 },
 "required": [
 "name"
]
 },
 "required": [
 "source",
 "name",
 "plugins"
]
 }
],
}
},
"blockedMarketplaces": {
 "description": "Enterprise blacklist of marketplace sources. When set in managed settings,
these exact sources are blocked from being added as marketplaces. The check happens BEFORE
downloading, so blocked sources never touch the filesystem.",
 "type": "array",
 "items": {
 "anyOf": [
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "url"
 },
 "url": {
 "description": "Direct URL to marketplace.json file",
 "type": "string",
 "format": "uri"
 }
 },
 "headers": {
 "description": "Custom HTTP headers (e.g., for authentication)",
 "type": "object",
 "propertyNames": {
 "type": "string"
 }
 },
 "additionalProperties": {
 "type": "string"
 }
 }
]
 },
 "required": [
 "source",
 "url"
]
}

```

```

]
 },
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "github"
 },
 "repo": {
 "description": "GitHub repository in owner/repo format",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\", \"v1.0.0\"). Defaults
to repository default branch.",
 "type": "string"
 },
 "path": {
 "description": "Path to marketplace.json within repo (defaults to .claude-
plugin/marketplace.json)",
 "type": "string"
 },
 "sparsePaths": {
 "description": "Directories to include via git sparse-checkout (cone mode). Use
for monorepos where the marketplace lives in a subdirectory. Example: [\".claude-plugin\",
\"plugins\"]. If omitted, the full repository is cloned.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "skipLfs": {
 "description": "Skip Git LFS smudge during clone and update (sets
GIT_LFS_SKIP_SMUDGE=1) so LFS pointer files stay as pointers instead of downloading their
content. Use for marketplaces hosted in repos with large LFS objects.",
 "type": "boolean"
 }
 },
 "required": [
 "source",
 "repo"
]
 },
 {
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "git"
 },
 "url": {
 "description": "Full git repository URL",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\", \"v1.0.0\"). Defaults
to repository default branch.",
 "type": "string"
 },
 "path": {
 "description": "Path to marketplace.json within repo (defaults to .claude-
plugin/marketplace.json)",
 "type": "string"
 }
 },

```

```

 "sparsePaths": {
 "description": "Directories to include via git sparse-checkout (cone mode). Use
for monorepos where the marketplace lives in a subdirectory. Example: [\".claude-plugin\",
\"plugins\"]. If omitted, the full repository is cloned.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "skipLfs": {
 "description": "Skip Git LFS smudge during clone and update (sets
GIT_LFS_SKIP_SMUDGE=1) so LFS pointer files stay as pointers instead of downloading their
content. Use for marketplaces hosted in repos with large LFS objects.",
 "type": "boolean"
 }
 },
 "required": [
 "source",
 "url"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "npm"
 },
 "package": {
 "description": "NPM package containing marketplace.json",
 "type": "string"
 }
 },
 "required": [
 "source",
 "package"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "file"
 },
 "path": {
 "description": "Local file path to marketplace.json",
 "type": "string"
 }
 },
 "required": [
 "source",
 "path"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "directory"
 },
 "path": {
 "description": "Local directory containing .claude-plugin/marketplace.json",
 "type": "string"
 }
 }
}

```

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 }
 },
 "required": [
 "source",
 "path"
]
},
{
 "description": "Policy-list sentinel for the ~/.claude/skills/ auto-load (@skills-dir plugins). In strictKnownMarketplaces: opt the scan back IN (by default any allowlist blocks it). In blockedMarketplaces: turn the scan OFF without otherwise restricting marketplaces. Only meaningful in those two managed-settings lists (areLocalPluginDirsAllowedByPolicy); known_marketplaces.json / marketplace add etc. ignore it.",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "skills-dir"
 }
 },
 "required": [
 "source"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "hostPattern"
 },
 "hostPattern": {
 "description": "Regex pattern to match the host/domain extracted from any marketplace source type. For github sources, matches against \"github.com\". For git sources (SSH or HTTPS), extracts the hostname from the URL. Use in strictKnownMarketplaces to allow all marketplaces from a specific host (e.g., \"^github\\.mycompany\\.com$\"),",
 "type": "string"
 }
 },
 "required": [
 "source",
 "hostPattern"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "pathPattern"
 },
 "pathPattern": {
 "description": "Regex pattern matched against the .path field of file and directory sources. Use in strictKnownMarketplaces to allow filesystem-based marketplaces alongside hostPattern restrictions for network sources. Use \".*\" to allow all filesystem paths, or a narrower pattern (e.g., \"^/opt/approved/\") to restrict to specific directories.",
 "type": "string"
 }
 },
 "required": [
 "source",
 "pathPattern"
]
},
{

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 "description": "Inline marketplace manifest defined directly in settings.json. The reconciler writes a synthetic marketplace.json to the cache; diffMarketplaces detects edits via isEqual on the stored source (the plugins array is inside this object, so edits surface as sourceChanged).",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "settings"
 },
 "name": {
 "description": "Marketplace name. Must match the extraKnownMarketplaces key (enforced); the synthetic manifest is written under this name. Same validation as PluginMarketplaceSchema plus reserved-name rejection ? validateOfficialNameSource runs after the disk write, too late to clean up.",
 "type": "string",
 "minLength": 1
 },
 "plugins": {
 "description": "Plugin entries declared inline in settings.json",
 "type": "array",
 "items": {
 "type": "object",
 "properties": {
 "name": {
 "description": "Plugin name as it appears in the target repository",
 "type": "string",
 "minLength": 1
 },
 "source": {
 "description": "Where to fetch the plugin from. Must be a remote source ? relative paths have no marketplace repository to resolve against.",
 "anyOf": [
 {
 "description": "Path to the plugin root, relative to the marketplace root (the directory containing .claude-plugin/, not .claude-plugin/ itself)",
 "type": "string",
 "pattern": "^\\.\\.\\.\\.\\.\\/*"
 },
 {
 "description": "NPM package as plugin source",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "npm"
 },
 "package": {
 "description": "Package name (or url, or local path, or anything else that can be passed to `npm` as a package)",
 "anyOf": [
 {
 "type": "string"
 },
 {
 "type": "string"
 }
]
 }
 }
 }
]
 },
 "version": {
 "description": "Specific version or version range (e.g., ^1.0.0, ~2.1.0)",
 "type": "string"
 },
 "registry": {

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 "description": "Custom NPM registry URL (defaults to using system
default, likely npmjs.org)",
 "type": "string",
 "format": "uri"
 },
 "required": [
 "source",
 "package"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "url"
 },
 "url": {
 "description": "Full git repository URL (https:// or git@)",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\"v1.0.0\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "^[a-f0-9]{40}$"
 }
 },
 "required": [
 "source",
 "url"
]
},
{
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "github"
 },
 "repo": {
 "description": "GitHub repository in owner/repo format",
 "type": "string"
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\"v1.0.0\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "^[a-f0-9]{40}$"
 }
 },
 "required": [

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 "source",
 "repo"
]
},
{
 "description": "Plugin located in a subdirectory of a larger
repository (monorepo). Only the specified subdirectory is materialized; the rest of the repo is
not downloaded.",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "git-subdir"
 },
 "url": {
 "description": "Git repository: GitHub owner/repo shorthand,
https://, or git@ URL",
 "type": "string"
 },
 "path": {
 "description": "Subdirectory within the repo containing the plugin
(e.g., \"tools/claude-plugin\"). Cloned sparsely using partial clone (--filter=tree:0) to
minimize bandwidth for monorepos.",
 "type": "string",
 "minLength": 1
 },
 "ref": {
 "description": "Git branch or tag to use (e.g., \"main\",
\"v1.0.0\"). Defaults to repository default branch.",
 "type": "string"
 },
 "sha": {
 "description": "Specific commit SHA to use",
 "type": "string",
 "minLength": 40,
 "maxLength": 40,
 "pattern": "^[a-f0-9]{40}$"
 }
 },
 "required": [
 "source",
 "url",
 "path"
]
},
{
 "description": "Placeholder for source types this Claude Code version
does not recognize. Never authored by hand ? PluginMarketplaceSchema rewrites unparseable
sources to this so the entry remains in marketplace.plugins (detectDelistedPlugins must not see
it as removed). Install attempts fail at cachePlugin with a clear \"update Claude Code\"
message.",
 "type": "object",
 "properties": {
 "source": {
 "type": "string",
 "const": "unsupported"
 }
 },
 "required": [
 "source"
]
}
],
},
"description": {

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 "type": "string"
 },
 "version": {
 "type": "string"
 },
 "strict": {
 "type": "boolean"
 }
],
 "required": [
 "name",
 "source"
]
 },
 "owner": {
 "type": "object",
 "properties": {
 "name": {
 "description": "Display name of the plugin author or organization",
 "type": "string",
 "minLength": 1
 },
 "email": {
 "description": "Contact email for support or feedback",
 "type": "string"
 },
 "url": {
 "description": "Website, GitHub profile, or organization URL",
 "type": "string"
 }
 },
 "required": [
 "name"
]
 },
 "required": [
 "source",
 "name",
 "plugins"
]
},
"pluginSuggestionMarketplaces": {
 "description": "Marketplace names whose plugins may surface as contextual install suggestions (relevance-based tips). No marketplace-declared suggestions surface without this allowlist; the built-in first-party frontend-design tip is unaffected. Only honored when set in managed settings (policy scope); the key is ignored in user, project, and local settings. A name only takes effect when the marketplace is registered on the machine AND its registered source is also declared in managed settings, either as the extraKnownMarketplaces entry for that name or as an entry of strictKnownMarketplaces. A marketplace registered from a different source under an allowlisted name is ignored. The official marketplace is exempt from the source requirement: allowlisting its name alone suffices, since that name can only register from the official Anthropic source.",
 "type": "array",
 "items": {
 "type": "string"
 }
},
"forceLoginMethod": {
 "description": "Force a specific login method: \"claudeai\" for Claude Pro/Max, \"console\" for Console billing, \"gateway\" for the Cloud gateway OIDC device flow",

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 "type": "string",
 "enum": [
 "claudeai",
 "console",
 "gateway"
]
},
"forceLoginGatewayUrl": {
 "description": "@internal Cloud gateway URL to pre-fill and auto-connect to during login. Typically set in local managed settings alongside forceLoginMethod: \"gateway\" so users never type the URL. Hidden from public SDK types until Cloud gateway is documented.",
 "type": "string",
 "format": "uri"
},
"parentSettingsBehavior": {
 "description": "Controls whether the SDK parent tier (Options.managedSettings / --managed-settings) layers under this admin tier. \"first-wins\" (default): parent is dropped ? admin tiers are the only policy source. \"merge\": parent's restrictive-only-filtered settings union under the admin winner. Has no effect when no admin tier exists (parent applies as the sole policy tier, still filtered restrictive-only).",
 "type": "string",
 "enum": [
 "first-wins",
 "merge"
]
},
"forceLoginOrgUUID": {
 "description": "Organization UUID to require for OAuth login. Accepts a single UUID string or an array of UUIDs (any one is permitted). When set in managed settings, login fails if the authenticated account does not belong to a listed organization.",
 "anyOf": [
 {
 "type": "string"
 },
 {
 "type": "array",
 "items": {
 "type": "string"
 }
 }
]
},
"forceRemoteSettingsRefresh": {
 "description": "When set in managed settings, the CLI blocks startup until remote managed settings are freshly fetched, and exits if the fetch fails",
 "type": "boolean"
},
"otelHeadersHelper": {
 "description": "Path to a script that outputs OpenTelemetry headers",
 "type": "string"
},
"outputStyle": {
 "description": "Controls the output style for assistant responses",
 "type": "string"
},
"viewMode": {
 "description": "Default transcript view mode on startup",
 "type": "string",
 "enum": [
 "default",
 "verbose",
 "focus"
]
},
"language": {

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 "description": "Preferred language for Claude responses and voice dictation (e.g.,
\"japanese\", \"spanish\")",
 "type": "string"
 },
 "skipWebFetchPreflight": {
 "description": "Skip the WebFetch blocklist check for enterprise environments with
restrictive security policies",
 "type": "boolean"
 },
 "sandbox": {
 "type": "object",
 "properties": {
 "enabled": {
 "type": "boolean"
 },
 "failIfUnavailable": {
 "description": "Exit with an error at startup if sandbox.enabled is true but the
sandbox cannot start (missing dependencies or unsupported platform). When false (default), a
warning is shown and commands run unsandboxed. Intended for managed-settings deployments that
require sandboxing as a hard gate.",
 "type": "boolean"
 },
 "autoAllowBashIfSandboxed": {
 "type": "boolean"
 },
 "allowUnsandboxedCommands": {
 "description": "Allow commands to run outside the sandbox via the
dangerouslyDisableSandbox parameter. When false, the dangerouslyDisableSandbox parameter is
completely ignored and all commands must run sandboxed. Default: true.",
 "type": "boolean"
 },
 "network": {
 "type": "object",
 "properties": {
 "allowedDomains": {
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "deniedDomains": {
 "description": "Domains that are always blocked, even if matched by
allowedDomains. Supports the same wildcard syntax as allowedDomains. Merged from all settings
sources regardless of allowManagedDomainsOnly.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "allowManagedDomainsOnly": {
 "description": "When true (and set in managed settings), only allowedDomains and
WebFetch(domain:...) allow rules from managed settings are respected. User, project, local, and
flag settings domains are ignored. Denied domains are still respected from all sources.",
 "type": "boolean"
 },
 "allowUnixSockets": {
 "description": "macOS only: Unix socket paths to allow. Ignored on Linux (seccomp
cannot filter by path).",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "allowAllUnixSockets": {
 "description": "If true, allow all Unix sockets (disables blocking on both

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platforms).",
 "type": "boolean"
 },
 "allowLocalBinding": {
 "type": "boolean"
 },
 "allowMachLookup": {
 "description": "macOS only: Additional XPC/Mach service names to allow looking up.
Supports trailing-wildcard prefix matching (e.g., \"com.apple.coresimulator.*\"). Needed for
tools that communicate via XPC such as the iOS Simulator or Playwright.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "httpProxyPort": {
 "type": "number"
 },
 "socksProxyPort": {
 "type": "number"
 },
 "tlsTerminate": {
 "description": "[EXPERIMENTAL] Enable in-process TLS termination so the per-
request filter can see HTTPS request bodies. Provide a CA cert+key, or omit both to have
sandbox-runtime generate an ephemeral one for the session.",
 "type": "object",
 "properties": {
 "caCertPath": {
 "type": "string",
 "minLength": 1
 },
 "caKeyPath": {
 "type": "string",
 "minLength": 1
 }
 }
 }
},
"filesystem": {
 "type": "object",
 "properties": {
 "allowWrite": {
 "description": "Additional paths to allow writing within the sandbox. Merged with
paths from Edit(...) allow permission rules.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "denyWrite": {
 "description": "Additional paths to deny writing within the sandbox. Merged with
paths from Edit(...) deny permission rules.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "denyRead": {
 "description": "Additional paths to deny reading within the sandbox. Merged with
paths from Read(...) deny permission rules.",
 "type": "array",
 "items": {
 "type": "string"
 }
 }
 }
}

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 },
 "allowRead": {
 "description": "Paths to re-allow reading within denyRead regions. Takes
precedence over denyRead for matching paths.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "allowManagedReadPathsOnly": {
 "description": "When true (set in managed settings), only allowRead paths from
policySettings are used.",
 "type": "boolean"
 }
 },
 "ignoreViolations": {
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "type": "array",
 "items": {
 "type": "string"
 }
 }
 },
 "enableWeakerNestedSandbox": {
 "type": "boolean"
 },
 "enableWeakerNetworkIsolation": {
 "description": "macOS only: Allow access to com.apple.trustd.agent in the sandbox.
Needed for Go-based CLI tools (gh, gcloud, terraform, etc.) to verify TLS certificates when
using httpProxyPort with a MITM proxy and custom CA. **Reduces security** ? opens a potential
data exfiltration vector through the trustd service. Default: false",
 "type": "boolean"
 },
 "excludedCommands": {
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "ripgrep": {
 "description": "Custom ripgrep configuration for bundled ripgrep support",
 "type": "object",
 "properties": {
 "command": {
 "type": "string"
 },
 "args": {
 "type": "array",
 "items": {
 "type": "string"
 }
 }
 }
 },
 "required": [
 "command"
],
 "bwrapPath": {
 "description": "Linux/WSL only: Absolute path to the bwrap (bubblewrap) binary.
Overrides auto-detection via PATH. Only honored from admin-controlled managed settings.",

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 "type": "string"
 },
 "socatPath": {
 "description": "Linux/WSL only: Absolute path to the socat binary used for the sandbox
network proxy. Overrides auto-detection via PATH. Only honored from admin-controlled managed
settings.",
 "type": "string"
 }
},
"additionalProperties": {}
},
"feedbackSurveyRate": {
 "description": "Probability (0?1) that the session quality survey appears when eligible.
0.05 is a reasonable starting point.",
 "type": "number",
 "minimum": 0,
 "maximum": 1
},
"spinnerTipsEnabled": {
 "description": "Whether to show tips in the spinner",
 "type": "boolean"
},
"spinnerVerbs": {
 "description": "Customize spinner verbs. mode: \"append\" adds verbs to defaults,
\"replace\" uses only your verbs.",
 "type": "object",
 "properties": {
 "mode": {
 "type": "string",
 "enum": [
 "append",
 "replace"
]
 },
 "verbs": {
 "type": "array",
 "items": {
 "type": "string"
 }
 }
 },
 "required": [
 "mode",
 "verbs"
]
},
"spinnerTipsOverride": {
 "description": "Override spinner tips. tips: array of tip strings. excludeDefault: if
true, only show custom tips (default: false).",
 "type": "object",
 "properties": {
 "excludeDefault": {
 "type": "boolean"
 },
 "tips": {
 "type": "array",
 "items": {
 "type": "string"
 }
 }
 },
 "required": [
 "tips"
]
},

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"syntaxHighlightingDisabled": {
 "description": "Whether to disable syntax highlighting in diffs",
 "type": "boolean"
},
"terminalTitleFromRename": {
 "description": "Whether /rename updates the terminal tab title (defaults to true). Set to
false to keep auto-generated topic titles.",
 "type": "boolean"
},
"alwaysThinkingEnabled": {
 "description": "When false, thinking is disabled. When absent or true, thinking is enabled
automatically for supported models.",
 "type": "boolean"
},
"effortLevel": {
 "description": "Persisted effort level for supported models.",
 "type": "string",
 "enum": [
 "low",
 "medium",
 "high",
 "xhigh"
]
},
"ultracode": {
 "description": "Enable ultracode for the session: xhigh effort plus standing dynamic-
workflow orchestration. Session-scoped ? typically provided via --settings or the
apply_flag_settings control request; interactive toggles never persist it. Requires workflows to
be enabled and an xhigh-capable model.",
 "type": "boolean"
},
"autoCompactWindow": {
 "description": "Auto-compact window size",
 "type": "integer",
 "minimum": 100000,
 "maximum": 1000000
},
"advisorModel": {
 "description": "Advisor model for the server-side advisor tool.",
 "type": "string"
},
"fastMode": {
 "description": "When true, fast mode is enabled. When absent or false, fast mode is off.",
 "type": "boolean"
},
"fastModePerSessionOptIn": {
 "description": "When true, fast mode does not persist across sessions. Each session starts
with fast mode off.",
 "type": "boolean"
},
"promptSuggestionEnabled": {
 "description": "When false, prompt suggestions are disabled. When absent or true, prompt
suggestions are enabled.",
 "type": "boolean"
},
"awaySummaryEnabled": {
 "description": "@internal When false, the session recap (shown when you return after being
away for 5+ minutes) is disabled. When absent or true, recap is enabled. Hidden from public SDK
types until external launch.",
 "type": "boolean"
},
"showClearContextOnPlanAccept": {
 "description": "When true, the plan-approval dialog offers a \"clear context\" option.
Defaults to false.",
 "type": "boolean"
}

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 },
 "agent": {
 "description": "Name of an agent (built-in or custom) to use for the main thread. Applies
the agent's system prompt, tool restrictions, and model.",
 "type": "string"
 },
 "companyAnnouncements": {
 "description": "Company announcements to display at startup (one will be randomly selected
if multiple are provided)",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "pluginConfigs": {
 "description": "Per-plugin configuration including MCP server user configs, keyed by
plugin ID (plugin@marketplace format)",
 "type": "object",
 "propertyNames": {
 "type": "string"
 }
 },
 "additionalProperties": {
 "type": "object",
 "properties": {
 "mcpServers": {
 "description": "User configuration values for MCP servers keyed by server name",
 "type": "object",
 "propertyNames": {
 "type": "string"
 }
 },
 "additionalProperties": {
 "type": "object",
 "propertyNames": {
 "type": "string"
 }
 },
 "additionalProperties": {
 "anyOf": [
 {
 "type": "string"
 },
 {
 "type": "number"
 },
 {
 "type": "boolean"
 },
 {
 "type": "array",
 "items": {
 "type": "string"
 }
 }
]
 }
 }
 },
 "options": {
 "description": "Non-sensitive option values from plugin manifest userConfig, keyed
by option name. Sensitive values go to secure storage instead.",
 "type": "object",
 "propertyNames": {
 "type": "string"
 },
 "additionalProperties": {
 "anyOf": [

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 {
 "type": "string"
 },
 {
 "type": "number"
 },
 {
 "type": "boolean"
 },
 {
 "type": "array",
 "items": {
 "type": "string"
 }
 }
]
}
}
},
"remote": {
 "description": "Cloud session configuration",
 "type": "object",
 "properties": {
 "defaultEnvironmentId": {
 "description": "Default environment ID to use for cloud sessions",
 "type": "string"
 }
 }
},
"autoUpdatesChannel": {
 "description": "Release channel for auto-updates (latest or stable)",
 "type": "string",
 "enum": [
 "latest",
 "stable",
 "rc"
]
},
"minimumVersion": {
 "description": "Minimum version to stay on - prevents downgrades when switching to stable channel",
 "type": "string"
},
"requiredMinimumVersion": {
 "description": "Minimum Claude Code version required to start. If the running version is older, Claude Code exits at startup with instructions to update. Only enforced from managed (policy) settings.",
 "type": "string"
},
"requiredMaximumVersion": {
 "description": "Maximum Claude Code version allowed to start. If the running version is newer, Claude Code exits at startup with instructions to install an approved version. Only enforced from managed (policy) settings.",
 "type": "string"
},
"plansDirectory": {
 "description": "Custom directory for plan files, relative to project root. If not set, defaults to ~/.claude/plans/",
 "type": "string"
},
"tui": {
 "description": "Terminal UI renderer. \"fullscreen\" uses the flicker-free alt-screen renderer with virtualized scrollbar (equivalent to CLAUDE_CODE_NO_FLICKER=1). \"default\" uses

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the classic main-screen renderer.",
 "type": "string",
 "enum": [
 "default",
 "fullscreen"
]
},
"voice": {
 "description": "Voice mode settings (hold-to-talk / tap-to-toggle dictation)",
 "type": "object",
 "properties": {
 "enabled": {
 "type": "boolean"
 },
 "mode": {
 "description": "'hold' (default): hold to talk. 'tap': tap to start, tap to
stop+submit.",
 "type": "string",
 "enum": [
 "hold",
 "tap"
]
 },
 "autoSubmit": {
 "description": "Submit the prompt when hold-to-talk is released (hold mode only)",
 "type": "boolean"
 }
 }
},
"channelsEnabled": {
 "description": "Managed-org opt-in for channel notifications (MCP servers with the
claude/channel capability pushing inbound messages). claude.ai Teams/Enterprise: default off.
Console: default on unless managed settings exist. Set true to allow; users then select servers
via --channels.",
 "type": "boolean"
},
"allowedChannelPlugins": {
 "description": "Managed-org allowlist of channel plugins. When set, replaces the default
Anthropic allowlist ? admins decide which plugins may push inbound messages. Undefined falls
back to the default. Requires channelsEnabled: true.",
 "type": "array",
 "items": {
 "type": "object",
 "properties": {
 "marketplace": {
 "type": "string"
 },
 "plugin": {
 "type": "string"
 }
 }
 },
 "required": [
 "marketplace",
 "plugin"
]
},
"prefersReducedMotion": {
 "description": "Reduce or disable animations for accessibility (spinner shimmer, flash
effects, etc.)",
 "type": "boolean"
},
"doneMeansMerged": {
 "description": "@internal When true, Claude keeps working until the PR is ready for you to
merge, a cron/Monitor is armed to resume later, or it hands you a self-contained next step.",

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 "type": "boolean"
 },
 "autoMemoryEnabled": {
 "description": "Enable auto-memory for this project. When false, Claude will not read from
or write to the auto-memory directory.",
 "type": "boolean"
 },
 "autoMemoryDirectory": {
 "description": "Custom directory path for auto-memory storage. Supports ~/ prefix for home
directory expansion. Ignored if set in projectSettings (checked-in .claude/settings.json) for
security. When unset, defaults to ~/.claude/projects/<sanitized-cwd>/memory/.",
 "type": "string"
 },
 "autoDreamEnabled": {
 "description": "Enable background memory consolidation (auto-dream). When set, overrides
the server-side default.",
 "type": "boolean"
 },
 "showThinkingSummaries": {
 "description": "Request API-side thinking summaries and show them in the conversation and
in the transcript view (ctrl+o). Set explicitly to override the default for your install.",
 "type": "boolean"
 },
 "skipDangerousModePermissionPrompt": {
 "description": "Whether the user has accepted the bypass permissions mode dialog",
 "type": "boolean"
 },
 "skipWorkflowUsageWarning": {
 "description": "@internal Whether the user has accepted the multi-agent workflow usage
warning. Until set, auto permission mode prompts before running a workflow.",
 "type": "boolean"
 },
 "disableAutoMode": {
 "description": "Disable auto mode",
 "type": "string",
 "enum": [
 "disable"
]
 },
 "sshConfigs": {
 "description": "SSH connection configurations for remote environments. Typically set in
managed settings by enterprise administrators to pre-configure SSH connections for team
members.",
 "type": "array",
 "items": {
 "type": "object",
 "properties": {
 "id": {
 "description": "Unique identifier for this SSH config. Used to match configs across
settings sources.",
 "type": "string"
 },
 "name": {
 "description": "Display name for the SSH connection",
 "type": "string"
 },
 "sshHost": {
 "description": "SSH host in format \"user@hostname\" or \"hostname\", or a host
alias from ~/.ssh/config",
 "type": "string"
 },
 "sshPort": {
 "description": "SSH port (default: 22)",
 "type": "integer",
 "minimum": -9007199254740991,

```

```

 "maximum": 9007199254740991
 },
 "sshIdentityFile": {
 "description": "Path to SSH identity file (private key)",
 "type": "string"
 },
 "startDirectory": {
 "description": "Default working directory on the remote host. Supports tilde
expansion (e.g. ~/projects). If not specified, defaults to the remote user home directory. Can
be overridden by the [dir] positional argument in `claude ssh <config> [dir]`.",
 "type": "string"
 }
},
"required": [
 "id",
 "name",
 "sshHost"
]
}
},
"claudeMd": {
 "description": "CLAUDE.md-style instructions injected as organization-managed memory. Only
honored from managed/policy settings.",
 "type": "string"
},
"claudeMdExcludes": {
 "description": "Glob patterns or absolute paths of CLAUDE.md files to exclude from
loading. Patterns are matched against absolute file paths using picomatch. Only applies to User,
Project, and Local memory types (Managed/policy files cannot be excluded). Examples:
`~/home/user/monorepo/CLAUDE.md`, `**/code/CLAUDE.md`, `**/some-dir/.claude/rules/**`,
 "type": "array",
 "items": {
 "type": "string"
 }
},
"pluginTrustMessage": {
 "description": "Custom message to append to the plugin trust warning shown before
installation. Only read from policy settings (managed-settings.json / MDM). Useful for
enterprise administrators to add organization-specific context (e.g., `All plugins from our
internal marketplace are vetted and approved.`).",
 "type": "string"
},
"theme": {
 "description": "Color theme for the UI",
 "anyOf": [
 {
 "type": "string",
 "enum": [
 "auto",
 "dark",
 "light",
 "light-daltonized",
 "dark-daltonized",
 "light-ansi",
 "dark-ansi"
]
 },
 {
 "type": "string",
 "pattern": "^custom:.*"
 }
]
},
"editorMode": {
 "description": "Key binding mode for the prompt input",

```

```

 "type": "string",
 "enum": [
 "normal",
 "vim"
]
},
"verbose": {
 "description": "Show full tool output instead of truncated summaries",
 "type": "boolean"
},
"preferredNotifChannel": {
 "description": "Preferred OS notification channel",
 "type": "string",
 "enum": [
 "auto",
 "iterm2",
 "iterm2_with_bell",
 "terminal_bell",
 "kitty",
 "ghostty",
 "notifications_disabled"
]
},
"autoCompactEnabled": {
 "description": "Automatically compact conversation when context fills",
 "type": "boolean"
},
"switchModelsOnFlag": {
 "description": "When safety measures flag a message, automatically switch to a different
model to keep chatting. When off, your session will pause instead.",
 "type": "boolean"
},
"autoScrollEnabled": {
 "description": "Auto-scroll the conversation view to bottom (fullscreen mode only)",
 "type": "boolean"
},
"fileCheckpointingEnabled": {
 "description": "Snapshot files before edits so /rewind can restore them",
 "type": "boolean"
},
"showTurnDuration": {
 "description": "Show \"Cooked for Nm Ns\" after each assistant turn",
 "type": "boolean"
},
"showMessageTimestamps": {
 "description": "Stamp each assistant message with its arrival time",
 "type": "boolean"
},
"terminalProgressBarEnabled": {
 "description": "Emit OSC 9;4 progress sequences during long operations",
 "type": "boolean"
},
"todoFeatureEnabled": {
 "description": "Enable the todo / task tracking panel",
 "type": "boolean"
},
"teammateMode": {
 "description": "How spawned teammates execute (tmux, in-process, auto)",
 "type": "string",
 "enum": [
 "auto",
 "tmux",
 "in-process"
]
},

```

```

"remoteControlAtStartup": {
 "description": "Start Remote Control bridge automatically each session",
 "type": "boolean"
},
"isolatePeerMachines": {
 "description": "Require explicit approval before SendMessage can reach a peer session on
another machine via Remote Control",
 "type": "boolean"
},
"daemonColdStart": {
 "description": "When no background service is running: 'transient' spawns one for this
login session; 'ask' offers to install it persistently",
 "type": "string",
 "enum": [
 "transient",
 "ask"
]
},
"autoUploadSessions": {
 "description": "Mirror local sessions to claude.ai as view-only (no remote control)",
 "type": "boolean"
},
"inputNeededNotifEnabled": {
 "description": "Push to mobile when a permission prompt or question is waiting",
 "type": "boolean"
},
"agentPushNotifEnabled": {
 "description": "Allow Claude to push proactive mobile notifications",
 "type": "boolean"
},
"skipAutoPermissionPrompt": {
 "description": "Whether the user has accepted the auto mode opt-in dialog",
 "type": "boolean"
},
"useAutoModeDuringPlan": {
 "description": "Whether plan mode uses auto mode semantics when auto mode is available
(default: true)",
 "type": "boolean"
},
"autoMode": {
 "description": "Auto mode classifier prompt customization",
 "type": "object",
 "properties": {
 "allow": {
 "description": "Rules for the auto mode classifier allow section. Include the literal
string \"$defaults\" to inherit the built-in rules at that position.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "soft_deny": {
 "description": "Rules for the auto mode classifier SOFT BLOCK section ?
destructive/irreversible actions that user intent can clear. Include the literal string
\"$defaults\" to inherit the built-in rules at that position.",
 "type": "array",
 "items": {
 "type": "string"
 }
 },
 "hard_deny": {
 "description": "Rules for the auto mode classifier HARD BLOCK section ? security
boundaries that user intent does NOT clear. Include the literal string \"$defaults\" to inherit
the built-in rules at that position.",
 "type": "array",

```

```

 "items": {
 "type": "string"
 }
 },
 "environment": {
 "description": "Entries for the auto mode classifier environment section. Include the
literal string \"$defaults\" to inherit the built-in entries at that position.",
 "type": "array",
 "items": {
 "type": "string"
 }
 }
},
"disableDeepLinkRegistration": {
 "description": "Prevent claude-cli:// protocol handler registration with the OS",
 "type": "string",
 "enum": [
 "disable"
]
},
"voiceEnabled": {
 "description": "Enable voice mode (hold-to-talk dictation)",
 "type": "boolean"
},
"defaultView": {
 "description": "Default transcript view: chat (SendUserMessage checkpoints only) or
transcript (full)",
 "type": "string",
 "enum": [
 "chat",
 "transcript"
]
}
},
"additionalProperties": {}
}
...

```

## User Request

Add a SessionStart hook (project-local settings, .claude/settings.local.json ? not committed) that runs the cache-hygiene nudge for this project. Command: ``python -m tools.cleanup_caches --summary`` run from the project root (C:\Users\avidu\Projects\badminton-highlight-indexer). It prints a one-line reminder to stdout ONLY when >5 GB of regenerable pipeline cache is reclaimable (silent otherwise), so its stdout should surface as session-start context. Goal: passively remind me at the start of a session if WSL/output caches have accumulated, without nagging when there's little to clean.

### 511. assistant (2026-06-13T15:33:33.466Z)

Now I'll configure the SessionStart hook in ``.claude/settings.local.json``. First, checking existing settings and testing the exact command:

### 512. user (2026-06-13T15:33:36.122Z)

```

--- existing .claude settings files ---
-rw-r--r-- 1 avidu 197609 10909 Jun 2 17:12 settings.local.json
--- settings.local.json content (if any) ---
{
 "permissions": {
 "allow": [
 "WebSearch",
 "mcp__7a624b0a-c38a-49de-a788-e6773a216a2c__create_file",

```

```

 "Bash(python -c ' *)",
 "Bash(python -m pytest
tests/test_tracknet_hybrid.py::test_pipeline_no_api_key_returns_empty -q)",
 "Bash(git commit -m 'feat: add TrackNet shuttle-trajectory segmenter \\\(WSL2\\\) + WASB
setup *)",
 "Bash(gh pr create --base master --head feat/tracknet-wasb-shuttle-detector --title 'feat:
TrackNet shuttle-trajectory segmenter \\\(WSL2\\\) + WASB setup' --body ' *)",
 "Bash(gh pr *)",
 "Bash(git merge *)",
 "Bash(python -m pytest tests/test_eval_metrics.py -q)",
 "Bash(python -m pytest tests/test_eval_annotations.py -q)",
 "Bash(python -m pytest tests/test_eval_harness.py -q)",
 "Bash(python -m pytest tests/ --cov=backend --cov=run_eval --cov-report=term-missing -q)",
 "Bash(ffprobe -v error -show_entries stream=width,height -show_entries format=duration -of
csv=p=0:s=x data/videos/___TRACKED_VAR__.mp4)",
 "Bash(PYTHONPATH=. python -m src.cli.curate export --out-dir data/eval_export)",
 "Bash(PYTHONPATH=. python -m src.cli.curate export --out-dir data/eval_export_all
--include-staging)",
 "Bash(PYTHONPATH=. python -c ' *)",
 "Bash(PYTHONPATH=. python -m pytest tests/ -q)",
 "Bash(git init *)",
 "Bash(git add *)",
 "Bash(git commit -q -m ' *)",
 "Bash(timeout 90 yt-dlp -f \"bv*+ba/b\" -S \"res,fps,ext:mp4:m4a\" --no-playlist
--simulate --print \"selected: %\\(resolution\\)s @ %\\(fps\\)sfps ext=%\\(ext\\)s\"
\"https://www.youtube.com/watch?v=0669aZhH0LI\")",
 "Bash(PYTHONPATH=. python -m src.cli.curate --help)",
 "Bash(PYTHONPATH=. python -m src.cli.curate fetch --help)",
 "PowerShell($ErrorActionPreference='Stop'; try { irm https://deno.land/install.ps1 | iex }
catch { Write-Output \"installer error: $_\" }; $deno = Join-Path $env:USERPROFILE
\".deno\\\\bin\\\\deno.exe\"; if \"(Test-Path $deno)\" { Write-Output \"deno installed: $\\(&
$deno --version | Select-Object -First 1\\)\" } else { Write-Output \"deno.exe not found at
$deno\" } }",
 "Bash(PYTHONPATH=. python -m pytest tests/test_downloader.py -q)",
 "Bash(PYTHONPATH=. python -u -m src.cli.curate fetch --video-id shuttleset_1 --force)",
 "Bash(tasklist)",
 "Bash(tasklist //FI \"IMAGENAME eq chrome.exe\")",
 "Bash(PYTHONPATH=. python -u -m src.cli.curate fetch --video-id shuttleset_1 --force
--cookies-from-browser firefox)",
 "Bash(time PYTHONPATH=. python -u -m src.cli.curate fetch --video-id shuttleset_1 --force
--cookies-from-browser firefox --player-client default)",
 "Bash(ffprobe -v error -select_streams v:0 -show_entries stream=width,height,codec_name
-of csv=p=0 data/videos/shuttleset_1.mp4)",
 "Bash(awk '{print $5, $NF}')",
 "Bash(time PYTHONPATH=. python -u -m src.cli.curate fetch --video-id shuttleset_2 --force
--cookies-from-browser firefox)",
 "Bash(time PYTHONPATH=. python -u -m src.cli.curate fetch --cookies-from-browser firefox
--sleep 8 --retries 3 --retry-backoff 30)",
 "Bash(python scratch_verify.py)",
 "Bash(PYTHONPATH=. python -m src.cli.curate export --out-dir data/eval_export --include-
staging)",
 "Bash(PYTHONPATH=. python -m pytest tests/test_eval_batch.py -q)",
 "Bash(PYTHONPATH=. python run_phase3_eval.py --help)",
 "Bash(PYTHONPATH=. python run_phase3_eval.py --manifest
'C:\\\\Users\\\\\\\\avidu\\\\\\\\Projects\\\\\\\\sports-data-
collector\\\\\\\\data\\\\\\\\eval_export\\\\\\\\manifest.json' --segmenter motion --only shuttleset_1 --max-
frames 600)",
 "Bash(PYTHONPATH=. python -u run_phase3_eval.py --manifest
'C:\\\\Users\\\\\\\\avidu\\\\\\\\Projects\\\\\\\\sports-data-
collector\\\\\\\\data\\\\\\\\eval_export\\\\\\\\manifest.json' --segmenter motion --only shuttleset_1
--reprocess)",
 "Bash(PYTHONPATH=. python -u run_phase3_eval.py --manifest
'C:\\\\Users\\\\\\\\avidu\\\\\\\\Projects\\\\\\\\sports-data-
collector\\\\\\\\data\\\\\\\\eval_export\\\\\\\\manifest.json' --segmenter yolo_hybrid --only shuttleset_8
--db output/phase3_yolo.db --json output/phase3_probe.json)",

```



```

"Bash(PYTHONPATH=. python -m pytest tests/test_eval_training.py tests/test_eval_batch.py
-q)",
"Bash(PYTHONPATH=. python -m pytest tests/test_eval_classifier.py -q)",
"Bash(ps -w)",
"Bash(PYTHONPATH=. python scratch_diag.py)",
"Bash(PYTHONPATH=. python scratch_merge.py)",
"Bash(PYTHONPATH=. python -c \"import backend.indexer.motion\")",
"WebFetch(domain:ai.google.dev)",
"WebFetch(domain:api.github.com)",
"WebFetch(domain:raw.githubusercontent.com)",
"WebFetch(domain:www.ffmpeg.org)",
"WebFetch(domain:www.via-la.com)",
"WebFetch(domain:en.wikipedia.org)",
"WebFetch(domain:www.streamingmedia.com)",
"WebFetch(domain:developer.nvidia.com)",
"WebFetch(domain:www.anthropic.com)",
"WebFetch(domain:learnopencv.com)",
"WebFetch(domain:docs.ultralytics.com)",
"WebFetch(domain:blog.roboflow.com)",
"WebFetch(domain:www.ultralytics.com)",
"WebFetch(domain:nvidia.custhelp.com)",
"Bash(curl -s \"https://api.github.com/repos/qaz812345/TrackNetV3\")",
"Bash(curl -s -o /dev/null -w \"%{http_code}\")",
\"https://raw.githubusercontent.com/qaz812345/TrackNetV3/main/LICENSE\"",
"Bash(curl -s -o /dev/null -w \"%{http_code}\")",
\"https://raw.githubusercontent.com/qaz812345/TrackNetV3/master/LICENSE\"",
"Bash(curl -s -o /dev/null -w \"%{http_code}\\\\n'
https://raw.githubusercontent.com/qaz812345/TrackNetV3/__TRACKED_VAR__/LICENSE)",
"Bash(curl -s \"https://api.github.com/repos/qaz812345/TrackNetV3/contents/\")",
"Bash(curl -s -L -H \"Accept: application/vnd.github+json\"",
https://api.github.com/repos/alenzenx/TrackNetV3)",
"WebFetch(domain:gitlab.com)",
"Bash(curl -s \"https://raw.githubusercontent.com/qaz812345/TrackNetV3/main/README.md\"",
"Bash(curl -s \"https://raw.githubusercontent.com/qaz812345/TrackNetV3/master/LICENSE\"",
"Bash(curl -s
\"https://raw.githubusercontent.com/qaz812345/TrackNetV3/master/README.md\"",
"Bash(curl -s \"https://raw.githubusercontent.com/qaz812345/TrackNetV3/master/LICENSE\" -o
/tmp/tn3_license.txt)",
"Bash(echo \"exit=$?\")",
"Read(//tmp/**)",
"Bash(curl -s \"https://raw.githubusercontent.com/qaz812345/TrackNetV3/master/README.md\"
-o /tmp/tn3_readme.md)",
"Bash(curl -s https://api.github.com/repos/nttcom/WASB-SBDT/license)",
"WebFetch(domain:www.gnu.org)",
"Bash(curl -s https://www.gnu.org/licenses/agpl-3.0.txt)",
"WebFetch(domain:opensource.org)",
"Bash(curl -s -L
\"https://raw.githubusercontent.com/ultralytics/ultralytics/main/LICENSE\"",
"WebFetch(domain:choosealicense.com)",
"Bash(curl -s -L --max-time 30
https://raw.githubusercontent.com/ultralytics/ultralytics/main/LICENSE)",
"Bash(curl -s -L --max-time 40
https://raw.githubusercontent.com/ultralytics/ultralytics/main/LICENSE)",
"Bash(curl -s -L --max-time 40
\"https://api.github.com/repos/ultralytics/ultralytics/license\"",
"Bash(curl -sS -L --max-time 60 -A \"Mozilla/5.0\"
\"https://api.github.com/repos/ultralytics/ultralytics/license\" -o /tmp/ul_license.json)",
"Bash(curl -sS -L --max-time 60 -A \"Mozilla/5.0\"
\"https://api.github.com/repos/ultralytics/ultralytics/license\" -o
C:/Users/avidu/ul_license.json)",
"Bash(curl -sS -L --max-time 60
\"https://raw.githubusercontent.com/ultralytics/ultralytics/main/LICENSE\" -o
C:/Users/avidu/ul_license.txt)",
"Bash(curl -s -L \"https://docs.ultralytics.com/help/license/\" -A \"Mozilla/5.0\")",
"Bash(curl -s -L -H \"Accept: application/vnd.github+json\"

```

```

\"https://api.github.com/repos/ultralytics/ultralytics\"),
 \"Bash(curl -s -L \"https://docs.ultralytics.com/help/license/\" -A \"Mozilla/5.0
\\(Windows NT 10.0; Win64; x64\\)\" -o /tmp/ultra_license.html)\",
 \"Bash(python3 -c \"import sys,html,re; t=sys.stdin.read\\(\\);
t=re.sub\\(r'<script.*?</script>',' ',t,flags=re.S\\);
t=re.sub\\(r'<style.*?</style>',' ',t,flags=re.S\\); t=re.sub\\(r'<[^>]+>',' ',t\\);
t=html.unescape\\(t\\); t=re.sub\\(r'[\\\\t]+' , ' ',t\\); print\\(t\\)\\)\",
 \"PowerShell(try { $r = Invoke-WebRequest -Uri
\"https://raw.githubusercontent.com/ultralytics/ultralytics/main/LICENSE\" -UseBasicParsing
-TimeoutSec 60; $r.Content.Substring\\(0, [Math]::Min\\(300, $r.Content.Length\\)\\) } catch {
\\\"ERR: $\\(\\$_Exception.Message\\)\\\" })\",
 \"Bash(curl -s -o /dev/null -w \"%{http_code}\\\"
\"https://docs.ultralytics.com/help/license/\" --max-time 20)\",
 \"Bash(curl -sL https://raw.githubusercontent.com/ultralytics/ultralytics/main/LICENSE)\",
 \"Read(/c/Users/avidu/**)\",
 \"Bash(curl -sL https://api.github.com/repos/ultralytics/ultralytics/license)\",
 \"Bash(curl -s --max-time 25
https://raw.githubusercontent.com/ultralytics/ultralytics/main/LICENSE)\",
 \"Bash(curl -sSL -w \"HTTP_STATUS:%{http_code}\\\"\\n\\n\"
https://raw.githubusercontent.com/ultralytics/ultralytics/main/LICENSE)\",
 \"Bash(curl -s -L \"https://docs.ultralytics.com/help/license/\\)\",
 \"WebFetch(domain:r.jina.ai)\",
 \"PowerShell(2+2)\",
 \"Bash(curl -sL --max-time 60 \"https://docs.ultralytics.com/help/license/\" -A
\\\"Mozilla/5.0\\)\",
 \"Bash(sed 's/<[^>]*>//g')\",
 \"Bash(curl -sL --max-time 60 \"https://docs.ultralytics.com/help/license/\" -H \"User-
Agent: Mozilla/5.0 \\(Windows NT 10.0; Win64; x64\\) AppleWebKit/537.36 \\(KHTML, like Gecko\\)
Chrome/120.0 Safari/537.36\\)\",
 \"Bash(python3 -c \"import sys,re; t=sys.stdin.read\\(\\);
t=re.sub\\(r'<script.*?</script>',' ',t,flags=re.S\\);
t=re.sub\\(r'<style.*?</style>',' ',t,flags=re.S\\); t=re.sub\\(r'<[^>]+>',' ',t\\);
t=re.sub\\(r'&','&',t\\); t=re.sub\\(r'\\\\s+', ' ',t\\); print\\(t\\)\\)\",
 \"PowerShell($r = Invoke-WebRequest -Uri
\"https://raw.githubusercontent.com/ultralytics/ultralytics/main/pyproject.toml\"
-UseBasicParsing; \\($r.Content -split \"\\n\\n\\n\\n\" | Where-Object { $_ -match
\\\"license|AGPL|Affero\\\" } | Select-Object -First 8)\",
 \"WebFetch(domain:medium.com)\",
 \"WebFetch(domain:webcache.googleusercontent.com)\",
 \"WebFetch(domain:opensource.stackexchange.com)\",
 \"WebFetch(domain:pypi.org)\",
 \"Bash(pip show *)\",
 \"Bash(python -c \"import importlib.metadata as m; print\\(m.metadata\\('google-
genai'\\).get\\('License'\\)\\); print\\('CLASSIFIERS:'\\); [print\\(c\\) for c in
m.metadata\\('google-genai'\\).get_all\\('Classifier'\\) if 'License' in c]\\)\",
 \"Bash(python -c \"import importlib.metadata as m; files=m.files\\('google-genai'\\);
[print\\(f\\) for f in files if 'LICENSE' in str\\(f\\).upper\\(\\) or 'COPYING' in
str\\(f\\).upper\\(\\)\\)\\)\",
 \"Bash(python -c \"import importlib.metadata as m; d=m.distribution\\('google-genai'\\);
print\\(d._path\\)\\)\",
 \"WebFetch(domain:www.rooragrp.com)\"
]
}
}
--- gitignore has settings.local? ---
(not in .gitignore)

```

### 513. user (2026-06-13T15:33:38.043Z)

?? 96.5G of regenerable pipeline cache reclaimable ? run `python -m tools.cleanup\_caches` to review, `--apply` to delete.  
exit=0

### 514. assistant (2026-06-13T15:34:08.022Z)

Command works (exit 0, prints the nudge). The existing `settings.local.json` only has `permissions` ? I'll merge the hook in without touching that. Let me Read it (Edit requires it) and check git tracking:

**515. user (2026-06-13T15:34:08.972Z)**

<system-reminder>Warning: the file exists but is shorter than the provided offset (130). The file has 122 lines.</system-reminder>

**516. user (2026-06-13T15:34:11.277Z)**

```
tracked by git?:
NO-untracked
git check-ignore:
.claude/settings.local.json
(ignored)
```

**517. assistant (2026-06-13T15:34:18.645Z)**

Good ? it's untracked and git-ignored, safe to edit. Let me read the tail for the exact closing:

**518. user (2026-06-13T15:34:19.372Z)**

```
116 "Bash(python -c \"import importlib.metadata as m; files=m.files\\('google-
genai'\"); [print\\(f\\) for f in files if 'LICENSE' in str\\(f\\).upper\\(\\) or 'COPYING' in
str\\(f\\).upper\\(\\)]\\)\",
117 "Bash(python -c \"import importlib.metadata as m; d=m.distribution\\('google-
genai'\"); print\\(d._path\\)\\)\",
118 "WebFetch(domain:www.rooragrp.com)"
119]
120 }
121 }
122
```

**519. user (2026-06-13T15:34:28.894Z)**

The file C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\settings.local.json has been updated successfully. (file state is current in your context ? no need to Read it back)

**520. assistant (2026-06-13T15:34:33.523Z)**

Now validate the JSON syntax and confirm the hook is correctly registered:

**521. user (2026-06-13T15:34:36.482Z)**

```
Exit code 127
--- JSON valid + permissions preserved ---
permissions.allow entries: 115
hook command: python -m tools.cleanup_caches --summary 2>/dev/null || true
--- jq schema-path check ---
/usr/bin/bash: line 1: jq: command not found
```

**522. assistant (2026-06-13T15:34:56.046Z)**

Validated ? `json.load` confirms valid JSON, permissions preserved (115 entries), and the hook is correctly nested (`python -m tools.cleanup\_caches --summary 2>/dev/null || true`). The SessionStart hook fires next session start, so I can't trigger it in this turn, but the config + the underlying command are both verified working. (`jq` isn't installed here ? the Python check is the equivalent validation.)

Now the last selected item ? save the policy to memory:

**523. assistant (2026-06-13T15:35:12.012Z)**

```

name: cache-hygiene-policy
description: "Retention policy + tooling for the GPU pipeline's regenerable caches
(frames/staged videos) ? what's purgeable vs durable, the self-cleaning runners,
tools/cleanup_caches.py, disk guard, and the SessionStart nudge hook."
metadata:
 type: project

```

How we keep pipeline caches from accumulating (designed + shipped 2026-06-13, PR #109).  
Pairs with [[gotcha-long-runs-gpu-wsl]] (why frames balloon: ~93 GB/4K video).

**\*\*Principle:\*\*** separate **\*\*regenerable intermediates\*\*** (purge aggressively, automatically)  
from **\*\*durable outputs\*\*** (keep/archive). Purging is default-safe ? dry-run first.

**\*\*Retention table:\*\***

- PURGE (regenerable): ``*_frames/`` (PNG decode), staged ``*.mp4`` copies in ``~/clips``,  
``*_1080p/proxy`` videos in ``output/``, ``chunks*_handoff/``, ``tmp*/``.
- KEEP (durable): trajectory CSVs (WASB outputs / golden corpus), the SQLite rally DB,  
final highlight reels, logs/scripts.
- OPT-IN purge: ``*_wasbcache/`` (detector resume cache ? small, aids resume; only via  
``--include-wasbcache``).

**\*\*Mechanisms (all live on master):\*\***

1. **\*\*Self-cleaning runners (the root fix):\*\*** ``indexing.{wasb,tracknet}.keep_frames``  
(default **\*\*false\*\***) ? after a SUCCESSFUL run (CSV written) the WASB runner deletes the  
frame cache + staged video; TrackNet deletes its staged video. On FAILURE nothing is  
deleted (resume intact). So per-video frame bloat no longer accumulates by default.
2. **\*\*`tools/cleanup\_caches.py`\*\*** ? dry-run-first ops tool. ``python -m tools.cleanup_caches``  
(report), ``--apply`` (delete), ``--keep-video <id|stem>`` (protect; pass the STEM e.g.  
``GX010125`` to also catch the Windows proxy, the hash only matches WSL frames),  
``--older-than N``, ``--include-wasbcache``, ``--summary`` (one-line nudge). Scans WSL ``~/clips``  
+ Windows ``output/``. ? WSL scan uses ``du`+`find`` arg-globbing on purpose ? the owner's  
login shell mangles ``for e in *`` loops (loop var comes back EMPTY); ``du -sh */`` /  
``find -printf`` are reliable. wsl.exe output decodes fine with text=True here (not UTF-16).
3. **\*\*Disk guard:\*\*** ``indexing.{wasb,tracknet}.min_free_gb`` (default 0=off) ? warns before a  
run if WSL free space is low.
4. **\*\*SessionStart hook\*\*** (``./claude/settings.local.json``, local+gitignored): runs  
``python -m tools.cleanup_caches --summary`` ? one-line nudge only when >5 GB reclaimable.

**\*\*Manual cadence:\*\*** review the dry-run at session wrap / after each completed video; delete  
frame caches once the trajectory CSV + DB rallies exist. Keep ``keep_frames=false`` (the default)  
unless debugging a specific run.

#### 524. user (2026-06-13T15:35:12.043Z)

File created successfully at: C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\cache-hygiene-policy.md (file state is current in your context ? no need to Read it back)

#### 525. user (2026-06-13T15:35:21.030Z)

The file C:\Users\avidu\claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

#### 526. user (2026-06-13T15:35:38.851Z)

<tool\_use\_error>File has not been read yet. Read it first before writing to it.</tool\_use\_error>

#### 527. user (2026-06-13T15:35:47.026Z)

```

1 ---
2 name: current-state

```

```

3 description: "Live micro-checkpoint state for Claude ? updated every checkpoint (not
only on request). What just landed, what's next, open decisions."
4 metadata:
5 node_type: memory
6 type: project
7 originSessionId: b13673cc-f087-43b1-9527-35dc0b67b301
8 ---
9
10 # Current state ? updated continuously (see [[working-agreement-checkpoints]])
11
12 **Checkpoint:** 2026-06-13 (SYNC + HANDOFF REFRESH ? read-only reconciliation by a ramp-
up session; no new code from
13 me this pass). `git pull --ff-only` = already up to date at **master `6de70aa` (#108)**.
Reconciled remote vs my last
14 session (#100): seven PRs landed from OTHER slates (NOT my work ? facts from git/gh):
#101 configurable branding

```

## 528. user (2026-06-13T15:36:01.876Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

## 529. assistant (2026-06-13T15:36:25.224Z)

All four pieces are shipped. Here's what now keeps your caches healthy, automatically.

## What landed ? [PR #109](https://github.com/avidullu/badminton-highlight-indexer/pull/109) (merged, CI green)

**\*\*Tier 1 ? self-cleaning pipeline (the root fix).\*\*** After a *\*successful\** run (trajectory CSV written), the WASB runner now deletes the frame cache + staged video; TrackNet deletes its staged copy. Gated by ``indexing.{wasb,tracknet}.keep_frames`` (default **\*\*false\*\***). On *\*failure\** nothing is deleted, so a re-run still resumes. This is what actually ends the 93 GB/video accumulation ? it never builds up again by default.

**\*\*Tier 2 ? `tools/cleanup\_caches.py` (dry-run-first).\*\***

...

```

python -m tools.cleanup_caches # report only (safe)
python -m tools.cleanup_caches --apply # delete the purge list
python -m tools.cleanup_caches --keep-video GX010125 # protect one video
python -m tools.cleanup_caches --older-than 7 # only >7-day-old cruft

```

Scans WSL ``~/clips`` + Windows ``output/``, classifies regenerable (frames, staged/proxy videos, handoff/temp) vs durable (trajectory CSVs, DB, reels), reports reclaimable space. ``--apply``, ``--keep-video``, ``--older-than``, ``--include-wasbcache`` all supported.

**\*\*Tier 3 ? disk guard + hook.\*\***

- ``indexing.{wasb,tracknet}.min_free_gb`` (default 0=off): warns before a run if WSL space is low.
- A **\*\*SessionStart hook\*\*** in ``.claude/settings.local.json`` (local, gitignored) runs ``--summary`` and surfaces a one-line nudge **\*\*only when >5 GB is reclaimable\*\*** ? silent otherwise, so no nagging. It activates next session start (right now it would report ~96 GB, since GX010125's 93 GB frames are still on disk by your choice).

**\*\*Policy saved to memory\*\*** ? ``cache-hygiene-policy.md`` (retention table + tooling + the ``for e`` in ``WSL footgun I hit``), indexed in MEMORY.md, with a checkpoint in current-state.

## Two honest notes

- I verified the hook config (valid JSON, permissions preserved, command pipe-tested) but **\*\*couldn't fire a SessionStart hook mid-turn\*\*** ? it'll run at your next session start. If it doesn't, opening ``/hooks`` once reloads the watcher.
- A real footgun surfaced while building the tool: your **\*\*login shell mangles `for e in` loops\*\*** (the loop variable comes back empty), which silently broke my first WSL scanner. The shipped tool uses ``du`+`find`` arg-globbing instead, and I documented it in memory so it doesn't

bite a future session.

Net effect: you shouldn't have to think about this again ? the pipeline self-cleans, the guard warns you before disk pressure, and the session nudge catches anything that slips through. The one-time ``python -m tools.cleanup_caches --apply`` whenever you want will sweep the pre-existing backlog (including GX010125's frames once you're done with it).

**530. user (2026-06-13T15:36:27.677Z)**

please follow the diligence of docs updates, tests and CIs health and go on merging the PRs

**531. assistant (2026-06-13T15:36:51.317Z)**

Let me first see what's actually open across both repos (and sync to head per convention) before applying the docs/tests/CI diligence:

**532. user (2026-06-13T15:37:00.886Z)**

```
Exit code 1
=== INDEXER open PRs ===
failed to parse jq expression (line 1, column 7)
 .[] | \
 ^ unexpected token "\\\"
```

**533. user (2026-06-13T15:37:09.685Z)**

```
=== INDEXER open PRs ===

=== COLLECTOR open PRs ===
```

**534. assistant (2026-06-13T15:37:32.094Z)**

Both repos have **\*\*zero open PRs\*\*** ? everything I opened this session is already merged. Let me confirm the collector docstring PR actually merged (not left dangling), then audit whether my merged PRs left docs gaps ? that's the diligence to close:

**535. user (2026-06-13T15:37:36.941Z)**

```
=== collector #20 state ===
MERGED merged=2026-06-12T10:15:31Z ? docs(cvat): update detect_issues padding rationale to
config-driven values
=== recent indexer merges (mine this session) ===
#109 feat: cache hygiene ? self-cleaning pipeline + cleanup tool + disk guard
#108 fix(stitcher): escape drive-letter colon in ffmpeg filtergraph paths
#107 feat(api): enforce stitching.min_shot_count on /api/compile
#106 docs(platform): video-locality mental model ? the 10GB question
#105 docs: internationalization (i18n) design plan ? Kannada-first, by design
#104 docs(pmf): C13 field kit ? recruiting flyer, visit playbook, answer sheet, pre-registered
signals
```

**536. assistant (2026-06-13T15:37:51.436Z)**

Collector #20 merged, all PRs merged, nothing open. So the diligence to close is **\*\*docs\*\***: my three feature PRs (#107 API min\_shot\_count, #108 watermark fix, #109 cache hygiene) shipped user-facing config knobs + a new ops tool without updating README/CODE\_MAP. Let me audit exactly what's documented vs missing:

**537. user (2026-06-13T15:37:52.390Z)**

```
1 # Badminton Highlight & Rally Indexer
2
3 A local, AI-powered web application and CLI utility designed to process long-form
badminton recordings (e.g., from GoPro), run comprehensive pluggable validations (quality,
```

integrity, and relevance checks), index all rallies into a relational SQLite database, and dynamically compile custom highlight reels using high-speed FFmpeg stitching.

```
4
5 The pipeline is built around three pluggable registries ? **validators**,
segmenters, and **AI providers** ? plus a **sport profile** registry so the same engine can
target other racket/court sports in the future. Badminton is the default profile.
6
7 > **Where this is heading:** the project is evolving from an AI-API-dependent prototype
into a self-improving tool that learns rally detection *offline* from a commercially-licensed,
annotated corpus. See **docs/ROADMAP.md** for the four-phase plan and
docs/archives/decisions/DECISIONS.md (historical
ADRs) for the rationale.
8
9 ---
10
11 ## ? Key Features
12
13 * **Pluggable Video Validation Framework**: Easily extendable base-class architecture
that automatically runs checks on:
14 * *Static Format & Resolution*: Verifies container is MP4, resolution >= 720p, and
framerate >= 30 FPS.
15 * *PTS Timeline Continuity*: Scans keyframe timestamps for timeline gaps, jumps,
or editing splices to detect tampering or splices.
16 * *Scene Cut Density Check*: Measures sub-sampled HSV frame histogram correlation
to verify visual cuts. A continuous raw camera stream should have near 0 cuts.
17 * *Blurriness Check*: Calculates average frame Laplacian variance to reject out-
of-focus footage.
18 * *Relevance (Is it Badminton?)*: Uses high-performance HSV color court matching
to identify court layouts.
19 * **Five Pluggable Rally Segmenters** (`backend/pipeline/segmenters/` ? `motion`,
`gemini`, `yolo_hybrid`, `tracknet_hybrid`, `wasb_hybrid`). The three core ones:
20 * `yolo_hybrid` ? two-stage pipeline. Stage 1 runs **torchvision person
detection** (Faster R-CNN, BSD-licensed) with **ByteTrack** tracking (supervision, MIT-licensed)
to cheaply filter the video down to a small set of high-motion candidate windows. Stage 2 hands
each candidate (with a small padding) to the AI provider for high-precision semantic boundaries
and shot counts. Massively cheaper than running the LLM on the full video. *(The registry key
remains `yolo_hybrid` for back-compat; the AGPL-licensed YOLOv8 was replaced with permissive
components ? see docs/archives/MONETIZATION_AUDIT.md and
ADR-005.)*
21 * `gemini` ? sends the entire video (or a `chunk_duration` slice) to Google Gemini
directly. Highest precision per call, highest cost.
22 * `motion` ? zero-dependency OpenCV pixel-difference heuristic. No API key
required. Useful as a baseline or for offline/air-gapped use.
23 * **Pluggable AI Provider Registry** (`backend/providers/`): The Gemini provider ships
in-tree; new providers (e.g. an OpenAI or local-model backend) can be registered without
touching the segmenters.
24 * **Pluggable Sport Profile Registry** (`backend/sports/`): The per-sport LLM prompt
and relevance config live in a `SportProfile`. Add a new sport by subclassing `SportProfile` and
registering it.
25 * **SQLite Indexing**: Each processed video is keyed by MD5 hash. Play segments
(start/end/duration/server/receiver/metadata) and per-segment events are persisted in
`output/sports_indexer.db`. Re-running on the same file is a cache hit.
26 * **Query System**: Filter indexed play segments by `server`, `receiver`,
`duration_min`, and `event_type` (the schema supports rich event types like
smashes/serves/fouls; today the shipped segmenters mainly populate timing + estimated
`shot_count`).
27 * **Zero-Dependency Local UI**: Served statically directly by FastAPI (Vite + Node.js
NOT required!) featuring a glassmorphic dashboard, responsive filtering, and immediate clip
stitching.
28 * **High-Speed FFmpeg Stitching** with configurable `begin_padding`, `end_padding`,
and `min_shot_count` filtering, plus an optional **branding watermark** burned into every reel
(**on by default** ? "Generated by Khelsutra.guru", fully configurable, with an off switch).
29 * **Detailed CLI Debugging Tools**: `--debug-clip <t>` inspects frame metrics,
validator scores, and motion values around a timestamp. `--trim-start/--trim-end` extracts an
arbitrary segment via stream-copy ffmpeg for fast iteration.
```

```

30
31 ---
32
33 ## ?? Setup & Diagnostics
34
35 ### 1. External System Requirements
36 The indexer depends on FFmpeg and FFprobe to parse media containers, extract
keyframes, and stitch clips.
37
38 * On Windows (PowerShell as Admin):
39 ```powershell
40 winget install Gyan.FFmpeg
41 ```
42 *Restart your terminal afterwards to register the path changes.*
43
44 ### 2. Auto-Installer & Diagnostics Checker
45 Run the diagnostics checker script in the root directory to verify your dependencies,
initialize default configs, and install PIP packages:
46 ```bash
47 python setup.py
48 ```
49 This script will:
50 * Validate `ffmpeg` and `ffprobe` are present in your PATH.
51 * Initialize a default `config.json` (if one does not exist) with thresholds for blur,
court HSV bounds, motion signals, YOLO velocity, chunking, and stitching.
52 * Detect whether an NVIDIA GPU is available (via `nvidia-smi`) and select the matching
PyTorch wheel index automatically (`cu124` if GPU present, `cpu` otherwise).
53 * Install everything from `requirements.txt`: `fastapi`, `uvicorn`, `opencv-python`,
`numpy`, `pydantic`, `jinja2`, `python-dotenv`, `google-genai`, `torch`, `torchvision`,
`supervision`.
54
55 > The default `yolo_hybrid` segmenter uses torchvision's COCO-pretrained Faster
R-CNN, whose weights are downloaded and cached automatically by torchvision on first run ?
nothing is bundled in the repo. (Pick a lighter/heavier detector via `indexing.detector_model`:
`fasterrcnn_resnet50_fpn` (default), `fasterrcnn_mobilenet_v3_large_fpn`, or
`retinanet_resnet50_fpn`.)
56
57 ---
58
59 ## ? Running the Application
60
61 ### 1. Launching the Local Web App Dashboard
62 To start the FastAPI server and open the web dashboard:
63 ```bash
64 python -m backend.main --server --port 8000
65 ```
66 Open your browser and navigate to http://localhost:8000.
67 *Simply input your local raw video file path in the sidebar to process, validate, index,
and compile highlights instantly.*
68
69 ### 2. Running in CLI Processing Mode
70 To process and index a video completely via terminal commands:
71 ```bash
72 python -m backend.main --video-path "C:/path/to/my_video.mp4"
73 ```
74 This outputs a validation summary and indexes all rallies directly into the local SQLite
database, using whichever segmenter is set as `default_segmenter` in `config.json`. The
shipped `config.json` sets `gemini` (needs `GEMINI_API_KEY`; without it the run silently
produces 0 segments and the observability report flags `silent_provider_noop`). Set it to
`yolo_hybrid`/`wasb_hybrid`/`motion` for the local paths.
75
76 To explicitly pick a segmenter for one run:
77 ```bash
78 python -m backend.main --video-path "C:/path/to/my_video.mp4" --segmenter yolo_hybrid
79 python -m backend.main --video-path "C:/path/to/my_video.mp4" --segmenter gemini

```



```

80 python -m backend.main --video-path "C:/path/to/my_video.mp4" --segmenter motion
81 ```
82
83 ### 3. Rich CLI Debugging (`--debug-clip`)
84 To diagnose exactly why a video segment was registered as active/dead play, or inspect
85 detailed focus/continuity metrics at a particular time in seconds:
86 ```bash
87 python -m backend.main --video-path "C:/path/to/my_video.mp4" --debug-clip 125.5
88 ```
89 *Outputs granular validation check logs and motion ratios within a +/- 3 second frame
90 window around the timestamp.*
91
92 ### 4. Extracting a Trimmed Segment (`--trim-start` / `--trim-end`) ? also the quickest
93 watermark preview
94 Extract a slice of a recording to disk (re-encoded; and, with the default-on branding
95 watermark, stamped "Generated by Khelsutra.guru" in the corner):
96 ```bash
97 python -m backend.main --video-path "C:/path/to/my_video.mp4" \
98 --trim-start 600 --trim-end 660 --trim-output slice_10m_to_11m.mp4
99 ```
100 Output lands in `--output-dir` (default `output/`) unless `--trim-output` is an absolute
101 path. This is the fastest way to see the watermark on a real clip ? set
102 `stitching.watermark.enabled: false` in `config.json` to turn it off.
103
104 > To preview the exact overlay on any file without the pipeline:
105 > ```bash
106 > ffmpeg -i in.mp4 -vf "drawtext=text='Generated by Khelsutra.guru':font='DejaVu Sans':f
107 ontcolor=white@0.85:fontsize=h/28:x=w-tw-24:y=h-th-24:box=1:boxcolor=black@0.4:boxborderw=10"
108 -c:a copy out.mp4
109 > ```
110
111 ### 5. End-to-End "Process + Stitch" Script
112 `run_process_and_stitch.py` is an interactive convenience runner: it MD5-hashes a
113 hardcoded video path, reuses cached indexed segments if available (offering to re-index from
114 scratch), and then prompts for stitching parameters (`begin_padding`, `end_padding`,
115 `min_shot_count`) before compiling the final highlight reel. Useful when iterating on stitching
116 settings against a fixed large recording without re-running expensive segmentation.
117
118 ---
119
120 ## ?? Threshold Tuning (`config.json`)
121 You can adjust the parameters of your validators, segmenters, AI provider, and stitcher
122 by editing `config.json` in the root folder. The default config emitted by `setup.py`:
123
124 ```json
125 {
126 "sport": "badminton",
127 "ai_provider": "gemini",
128 "ai_model": "gemini-2.5-flash",
129 "validation": {
130 "min_resolution_width": 1280,
131 "min_resolution_height": 720,
132 "min_fps": 29.9,
133 "min_blur_threshold": 20.0,
134 "max_scene_cuts_per_minute": 0.5,
135 "relevance": {
136 "court_green_lower": [35, 40, 40],
137 "court_green_upper": [85, 255, 255],
138 "court_pixels_min_ratio": 0.05
139 }
140 },
141 "indexing": {
142 "default_segmenter": "gemini",
143 "use_gpu": true,
144 "motion_threshold": 0.08,
145 }
146 }
147 ```

```

```

132 "min_segment_duration": 3.0,
133 "dead_time_merge_gap": 2.0,
134 "chunk_duration": 90,
135 "chunk_overlap": 10,
136 "yolo_velocity_threshold": 0.15,
137 "yolo_frame_skip": 3,
138 "yolo_tracker": "bytetrack.yaml",
139 "yolo_prefetch_workers": 2,
140 "min_action_window_duration": 2.0,
141 "ai_handoff_padding": 2.0,
142 "ai_max_workers": 5
143 },
144 "output": {
145 "default_highlights_name": "highlights.mp4",
146 "db_name": "sports_indexer.db"
147 },
148 "stitching": {
149 "begin_padding": 0.5,
150 "end_padding": 0.5,
151 "use_cache": false,
152 "min_shot_count": 1,
153 "watermark": {
154 "enabled": true,
155 "text": "Generated by Khelsutra.guru",
156 "position": "bottom-right",
157 "opacity": 0.85,
158 "logo_path": "",
159 "font_path": ""
160 }
161 }
162 }
163 ...
164
165 Notable knobs:
166 * `validation.min_blur_threshold` ? defaults to `20.0` (deliberately permissive for
noisy indoor-arena GoPro footage; see `docs/archives/TEST_RUN_SUMMARY.md` for why).
167 * `indexing.default_segmenter` ? `yolo_hybrid` | `gemini` | `motion`.
168 * `indexing.use_gpu` ? when true and CUDA is available, YOLO runs on GPU.
169 * `indexing.yolo_velocity_threshold` ? fraction of frame width per second; lower
values catch slower play, higher values are stricter about what counts as "action".
170 * `indexing.yolo_frame_skip` ? process every Nth frame during YOLO tracking. `3` at 30
fps ? 10 samples/sec.
171 * `indexing.chunk_duration` / `chunk_overlap` ? how to split long videos for the YOLO
phase.
172 * `indexing.ai_handoff_padding` ? seconds of slack added around each YOLO candidate
window before sending to the AI provider.
173 * `indexing.ai_max_workers` ? parallel calls to the AI provider during phase 3.
174 * `stitching.min_shot_count` ? drop play segments whose estimated shot count is below
this threshold when compiling highlights.
175 * `stitching.watermark` ? branding overlay burned into every highlight clip (**on by
default**). `enabled: false` turns it off; `text` is the fully-configurable string; `position` ?
{`bottom-right`, `bottom-left`, `top-right`, `top-left`}; `opacity` 0?1; optional `logo_path`
(transparent PNG). It ships with a **bundled Apache-2.0 font (Roboto Slab)** so the text renders
out-of-the-box **without a system fontconfig setup** (which Windows ffmpeg often lacks); set
font_path to a different `.ttf` (your own, or another in `backend/assets/fonts/`) to
change the typeface. If the watermark encode still fails, the clip is re-encoded without it (so
the reel is never lost) and the ffmpeg reason is logged.
176
177 ---
178
179 ## ? Frequently Asked Questions (FAQ)
180
181 ### Q1: The server complains that `ffprobe` is not found.
182 Ensure FFmpeg is installed and added to your system environment variables. You can check
this by typing `ffmpeg -version` or `ffprobe -version` in your terminal. If running `setup.py`

```

```

outputs a `[FAIL]`, re-install FFmpeg or manually copy the binary files to a PATH folder.
183
184 ### Q2: Why does my video fail the Relevance check?
185 Our default relevance check scans for the typical indoor green badminton court floor
color space (HSV). If your court is blue, or uses standard wood grain flooring, simply adjust
the HSV values inside your `config.json` under `validation.relevance.court_green_lower` and
`court_green_upper` to match your court, or lower the `court_pixels_min_ratio` to `0.0` to
bypass.
186
187 ### Q3: How do I add a new custom validator to the pipeline?
188 1. Create a new python file in `backend/pipeline/validators/` (e.g. `my_check.py`).
189 2. Subclass `VideoValidator` and implement `validate(self, video_path, config,
context)`.
190 3. Register your class inside `backend/pipeline/validators/__init__.py` using
`registry.register(MyCheckValidator())`.
191 It's that simple! The app will automatically run it, display it in the CLI diagnostics
summary, and render it in the frontend checklist.
192
193 ### Q4: Why does the CLI or server output freeze or take a long time on large videos?
194 * **Print Buffering**: Standard Python buffers console output when stdout is
redirected or running in background shells on Windows. To get real-time unbuffered log
statements immediately, run commands with Python's unbuffered flag:
195 ```bash
196 python -u -m backend.main --video-path "C:/path/to/video.mp4"
197 ```
198 * **Video Seeking Overhead**: Seeking to random frames in high-bitrate long H.265/HEVC
videos using OpenCV's `cap.set(cv2.CAP_PROP_POS_FRAMES)` incurs massive I/O decoding delays. The
`motion` segmenter reads sequentially and skips frame decodes using container-level
`cap.grab()`, which delivers a **100x speedup** on multi-gigabyte raw files. `yolo_hybrid`
sidesteps the problem entirely by extracting fixed chunks via ffmpeg first.
199
200 ### Q5: Why did the video fail with "No keyframes / packet timestamps could be
extracted"?
201 Some H.265/HEVC streams (e.g. GoPro footage) encode frames as GDR or CRA keyframes.
Standard FFmpeg frame-level skipping (`-skip_frame nokey`) fails to identify them, returning
empty logs. We resolved this by extracting packet indexes (`packet=pts_time,flags`) at the
container demuxer level instead, which is incredibly robust and executes instantly without
decoding a single pixel.
202
203 ### Q6: How can I quickly test the indexing without waiting to parse an entire multi-GB
video file?
204 You can pass the `--max-frames <N>` argument in CLI or `max_frames` in the POST API
payload. This restricts the segmenter frame loop to process only the first N sub-sampled
frames, enabling you to test motion filters and court relevance checks in seconds:
205 ```bash
206 python -m backend.main --video-path "C:\Users\avidu\Videos\LargeTestVideo.MP4" --max-
frames 300
207 ```
208 *(At 5 sub-sampled frames per second, `--max-frames 300` analyzes exactly the first 60
seconds of video).*
209
210 ### Q7: How do I configure and run an AI-powered segmenter (Gemini / YOLO-hybrid)?
211 Both `yolo_hybrid` (default) and `gemini` need an API key for the AI provider. To use
Google Gemini:
212 1. Create a local `.env` file in the project root (this is already in `.gitignore` to
prevent secret leaks).
213 2. Set your Google AI Studio API key in the file:
214 ```env
215 GEMINI_API_KEY=your_key_here
216 ```
217 3. Run with whichever segmenter you want:
218 ```bash
219 # Default: cheap candidate filtering with YOLO, then precise boundaries via Gemini
220 python -m backend.main --video-path "C:/path/to/video.mp4" --segmenter yolo_hybrid
221

```

```

222 # Full-video Gemini analysis (more expensive, highest semantic precision per call)
223 python -m backend.main --video-path "C:/path/to/video.mp4" --segmenter gemini
224
225 # Pure CV baseline ? no API key needed
226 python -m backend.main --video-path "C:/path/to/video.mp4" --segmenter motion
227 ```
228 *Or set `"default_segmenter": "yolo_hybrid"` (or `"gemini"`/`"motion"`) under
 `"indexing"` in `config.json`.
229
230 The AI provider used by `yolo_hybrid` and `gemini` is controlled by `ai_provider` +
 `ai_model` at the top level of `config.json`. Today only `gemini` is registered; new providers
 can be added under `backend/providers/`.
231
232 **Trajectory-based segmenters (`wasb_hybrid`, `tracknet_hybrid`).** The registry also
 includes two shuttle-trajectory variants. Like `yolo_hybrid`, they cheaply filter the video down
 to candidate windows *locally* and then route each window to the AI provider (`gemini`) for
 precise boundaries ? so they need the same `GEMINI_API_KEY`. **If an observability report shows
 `segmenter_actual: wasb_hybrid` (or `tracknet_hybrid`), that is a valid, registered segmenter ?
 not an error.** Pick one with `--segmenter wasb_hybrid` or via `default_segmenter`.
 (`wasb_hybrid` additionally requires the WSL WASB setup ? see
 docs/TRACKNET_WSL_SETUP.md.)
233
234 ### Q8: How does the `yolo_hybrid` pipeline actually save money?
235 Sending a multi-gigabyte video to a hosted LLM is slow and expensive. `yolo_hybrid` runs
 in four phases:
236
237 1. **Chunking** ? splits the video into overlapping windows (`chunk_duration`,
 `chunk_overlap`) so we can process incrementally.
238 2. **Detection + tracking** ? torchvision person detection (Faster R-CNN) + ByteTrack
 runs on each chunk locally (GPU-accelerated when available), measuring per-track velocity.
 Frames whose normalised player velocity exceeds `yolo_velocity_threshold` are kept; the rest are
 dropped. Adjacent active frames are merged into candidate windows, with
 sub-`min_action_window_duration` windows discarded.
239 3. **AI handoff** ? only the surviving candidate windows (padded by `ai_handoff_padding`
 seconds on each side) are extracted and sent to the AI provider for precise rally boundaries and
 shot-count estimation. These calls run in parallel up to `ai_max_workers`.
240 4. **DB population** ? confirmed segments are written to `play_segments` along with
 `shot_count`, `shot_count_confidence`, and a `detector` tag like `yolo-hybrid-gemini-2.5-flash`.
241
242 In practice this typically cuts AI-provider runtime/cost by an order of magnitude versus
 uploading the full video to `gemini` directly, while preserving the LLM's strength at semantic
 boundary detection.
243
244 ### Q9: How can I debug/test the Gemini Segmenter on a small portion of a video without
 uploading a huge multi-GB file?
245 Ingesting and uploading multi-gigabyte video files to Google Cloud can be slow and
 expensive. We built a native **fast local trimming** developer debugging feature:
246 1. Open your `config.json` file.
247 2. Under the `"indexing"` configuration block, add a `"debug_duration"` key (in
 seconds):
248 ```json
249 "indexing": {
250 "default_segmenter": "gemini",
251 "debug_duration": 15.0,
252 ...
253 }
254 ```
255 3. When `"debug_duration"` is set, the Gemini segmenter will instantly run a zero-re-
 encode, sub-second `ffmpeg` slice to create a tiny temporary clip of the first 15 seconds,
 upload only that tiny clip to Google AI Studio (which takes <1 second to upload and <5 seconds
 to process), run the model query, and clean up the temporary files automatically. This allows
 you to iterate on model prompts, evaluate validation overlaps, and verify rally boundaries in
 under 10 seconds!
256
257 ### Q10: Can I trust the per-segment metadata from the `motion` segmenter?

```

258     **\*\*No.\*\*** The zero-dependency ``motion`` segmenter detects *timing* via pixel-difference, but its per-segment **metadata** is heuristic/synthetic ? ``intensity``, ``avg_speed_kmh``, the `serve/smash/foul/winner` **events**, and ``server`/`receiver`` are generated values, **not measured**. They exist so the schema is populated for demos; do not treat them as ground truth. For real shot counts and boundaries use ``yolo_hybrid`` (default) or ``gemini``. If your observability report shows a ``synthetic_motion_metadata`` warning, this is why.

259

260     **### Q11:** I re-ran a video and my previous results changed/disappeared. Why?

261     Videos are keyed by the **MD5** of the file's bytes. Re-processing the *same* file is treated as a fresh ingest: ``add_video`` is ``INSERT OR REPLACE``, and the foreign-key cascade **deletes** the previous play/candidate segments before writing the new ones. So the earlier results are intentionally replaced, not merged. (A ``reingest_replaced_prior`` note in the report flags this.) Note also that re-encoding or re-downloading a video changes its bytes ? a new MD5 ? a separate record.

262

263     **### Q12:** What is the ``*.report.json`` / ``*.report.md`` / ``*.summary.md`` file next to my video?

264     Every processing run now writes a **per-video observability report** next to the video (or to ``report.output_dir`` if set in ``config.json``):

265     - ``<name>.report.json`` ? machine-readable: every stage, metric, and finding. The source of truth; diff two runs to see what changed.

266     - ``<name>.report.md`` ? the **technical** report: validation results, processing stages, and **findings/warnings**. Each warning names the problem and cites the FAQ entry to read. **Start here** when a run looks wrong.

267     - ``<name>.summary.md`` ? a **customer-friendly** summary: just the video metadata and highlights, no system internals.

268

269     To read the technical report: scan **Verdict** first (PASS / PASS-with-warnings / FAIL), then **Findings & warnings** ? each finding has a plain-English message, evidence, and a ? see `README#Qn`` pointer. To turn reports off or relocate them, set ``"report": {"enabled": false}`` or ``"report": {"output_dir": "...}"`` in ``config.json``.

270

271     **### Q13:** A validation check failed and indexing was halted ? what do I do?

272     Open the report's **Processing stages ? validation** block (or the ``VALIDATION SUMMARY`` printed to the console). Each failed check states the exact reason. Common cases: **relevance** (the court color didn't match ? see [Q2](#q2-why-does-my-video-fail-the-relevance-check)); **resolution/fps** (below the configured minimums ? adjust ``validation.min_`` in ``config.json``); **format** (not an MP4 container); **blur** (footage too soft ? improve sharpness/lighting or lower ``validation.min_blur_threshold``; a reading of ``0.0`` often means undecodable frames, so re-encode with ``ffmpeg -i in.mp4 -c:v libx264 out.mp4`` before lowering the threshold); **keyframes/PTS** (HEVC/GoPro keyframe quirks ? see [Q5](#q5-why-did-the-video-fail-with-no-keyframes--packet-timestamps-could-be-extracted)). Fix the underlying cause or relax the relevant threshold in ``config.json``, then re-run.

273

274     **### Q14:** The report says fps was a "fallback" (not probed) ? what does that mean?

275     The ``Video`` section shows ``FPS | 30 (fallback)`` and a ``fps_fallback_used`` warning when the pipeline could **not probe** the real frame rate from the file and fell back to a hardcoded default (30 fps). Because rally detection converts frames/seconds and scales motion thresholds by fps, a wrong fps **miscalibrates** timing and motion sensitivity ? segment boundaries and which motion counts as "play" can be off.

276

277     This usually means validation didn't run (it's the step that probes the file via ``ffprobe``), or ``ffprobe`` couldn't read the stream's ``r_frame_rate``. To fix: confirm the real fps ?

278     

```
```bash
```

279

```
ffprobe -v error -select_streams v:0 -show_entries stream=r_frame_rate -of csv=p=0
```

280

```
"C:/path/to/video.mp4"
```

281 ? make sure ``ffprobe`` is on your PATH (see [Q1](#q1-the-server-complains-that-ffprobe-is-not-found)) and that the run validates the file, then re-run so the real fps is used. (When fps shows ``(probed)`` instead, this is all fine.)

282

283 ---

284

285 **## ? Annotation & Evaluation**

```

286
287     Once a video is processed, you can measure how well the pipeline detected its
rallies against hand-annotated ground truth. You annotate each rally's [start, end) time
range in a small CSV; the harness scores the detected rallies straight from the SQLite DB ? no
API calls, fully deterministic.
288
289     ```bash
290     python run_eval.py --scaffold --annotations rallies.csv          # make an empty
template
291     # ...fill in start,end rows, then:
292     python run_eval.py --video "C:/videos/match.mp4" --annotations rallies.csv --show-
failures
293     ```
294
295     It reports precision/recall/F1, mean IoU, and boundary error for two stages ? raw
detector 'candidates' vs. AI-confirmed 'final' segments ? which pinpoints whether a weak result
is a detection problem or a segmentation problem.
296
297     > Rally timestamps evaluate and tune segmentation; they do not train the shuttle
detector (that needs per-frame coordinate labels). See
[docs/EVALUATION.md](docs/EVALUATION.md) for the annotation format, CLI flags, and how to
read the output.
298
299     ---
300
301     ## ? Running the Test Suite
302     The test suite uses 'pytest' and lives in 'tests/'. From the project root:
303
304     ```bash
305     pip install pytest pytest-cov --user
306
307     # One-liner full suite (recommended before opening a PR) ? discoverable wrapper that
308     # runs the whole suite, reports a clear PASS/FAIL, and returns pytest's exit code:
309     python run_all_tests.py
310     python run_all_tests.py -k yolo -v          # any args pass straight through to pytest
311
312     # ...or call pytest directly:
313     pytest                                     # run everything
314     pytest tests/test_yolo_hybrid.py          # run a single file
315     pytest -k yolo --maxfail=1 -v             # filter by keyword, fail fast, verbose
316     pytest --cov=backend --cov-report=term-missing # coverage report (uses .coveragerc)
317     ```
318
319     > The optional 'obsreport' package powers the per-video observability reports; its tests
320     > skip cleanly if it isn't installed. To exercise them: 'pip install -e ../sports-
obsreport --no-build-isolation'.
321
322     The suite mocks external dependencies (the Gemini client, ffprobe/ffmpeg, OpenCV
'VideoCapture', the YOLO model) so it runs offline and without a GPU. The tests cover:
323
324     * 'tests/test_base.py' ? segmenter registry
325     * 'tests/test_database.py' ? SQLite schema, video/segment insert/query
326     * 'tests/test_geometry.py' ? bbox center / distance / velocity / IoU helpers
327     * 'tests/test_video.py' ? ffprobe duration + ffmpeg chunk extraction helpers
328     * 'tests/test_providers.py' + 'tests/test_provider_gemini.py' ? AI provider registry
and the Gemini provider
329     * 'tests/test_sport_profiles.py' ? sport profile registry + the badminton prompt
330     * 'tests/test_gemini.py' ? 'GeminiPlaySegmenter' (single-file and chunked paths)
331     * 'tests/test_yolo_hybrid.py' ? 'HybridYoloSegmenter' happy path, missing-API-key
bailout, and timestamp-validation drops
332     * 'tests/test_tracknet_runner.py' + 'tests/test_tracknet_hybrid.py' ? TrackNet WSL
runner (mocked subprocess) and the 'tracknet_hybrid' segmenter
333     * 'tests/test_eval_metrics.py' + 'tests/test_eval_annotations.py' +
'tests/test_eval_harness.py' ? rally evaluation harness (interval matching, CSV loading, DB
scoring, CLI)

```

```

334
335     ---
336
337     ## ? Project Structure (high level)
338
339     ```
340     backend/
341         main.py                # FastAPI app + CLI orchestrator
342         database.py            # SQLite schema and queries
343         validators/            # Pluggable video validators (format, blur, scene-cut,
relevance, ...)
344         indexer/                # Pluggable rally segmenters: motion, gemini, yolo_hybrid,
tracknet_hybrid
345         detectors/            # TrackNet WSL2 runner (shuttle-trajectory subprocess adapter)
346         providers/            # Pluggable AI providers (gemini today)
347         sports/                # Pluggable sport profiles (badminton today)
348         eval/                  # Rally evaluation harness: metrics, annotations, scoring
349         stitcher/compiler.py    # FFmpeg-based highlight compiler
350         utils/                 # ffprobe/ffmpeg + bbox geometry helpers
351         static/                # Zero-build frontend served by FastAPI
352     docs/                      # Live docs (CODE_MAP, COMMERCIALIZATION, BACKLOG, ROADMAP,
guidelines, golden-set). Historical items in docs/archives/
353     tests/                     # pytest suite (mocked, runs offline)
354     run_process_and_stitch.py   # Interactive end-to-end runner
355     run_eval.py                # Rally evaluation CLI (see "Annotation & Evaluation" above)
356     setup.py                   # Dependency & FFmpeg diagnostics + default config init
357     config.json                # Runtime configuration (see "Threshold Tuning" above)
358     ```
359

```

538. assistant (2026-06-13T15:38:32.182Z)

I have the README. The clear docs gap is **##109** (cache hygiene) ? the new ``tools/cleanup_caches.py`` tool and the ``keep_frames`/`min_free_gb`` config knobs are undocumented; **##107** (min_shot_count now applies to the API too) needs a one-line clarification. Let me read the CODE_MAP anchors I'll touch, then make surgical edits:

539. user (2026-06-13T15:38:34.240Z)

```

7:[Omitted long matching line]
53: -> WasbRunner.healthcheck() (gate; not ok -> return [])
@backend/pipeline/detectors/wasb_runner.py
57: -> TrackNetRunner.parse_trajectory_csv(csv)
@backend/pipeline/detectors/tracknet_runner.py (shared, re-exported by WasbRunner)
120:### 2.2 Detectors & windowing (WSL-quarantined; ABC abstraction complete)
122:- `@backend/pipeline/detectors/tracknet_runner.py`
126:[Omitted long matching line]
273:| Tune candidate windowing (velocity/merge/min-dur) |
`@backend/pipeline/detectors/tracknet_runner.py::trajectory_to_action_windows`; cfg `config.json
indexing.{wasb,tracknet}.velocity_threshold`, `dead_time_merge_gap`,
`min_action_window_duration` |
307:| `test_tracknet_runner.py` | `pipeline/detectors/tracknet_runner` (`to_wsl_mnt_path`,
`from_indexing_cfg`, parse + `trajectory_to_action_windows` shared brain, healthcheck,
run_predict, stage) |
334:| `test_providers.py` / `test_provider_gemini.py` | `providers` registry + GeminiProvider
lifecycle (`MAX_UPLOAD_MB` guard refuses LOUDLY w/ `oversize_skipped` metric, non-list JSON ->
[], cleanup delete) |
336:| (no dedicated test) | individual `pipeline/validators/*` impls,
`pipeline/stitcher.compiler`, `main.py`, `run_process_and_stitch.py`,
`pipeline/detectors/wasb_runner`, `pipeline/segmenters/{wasb_hybrid,motion}`,
`eval/audio/e1_probe` (diagnostic) ?? exercised only indirectly |

```

540. user (2026-06-13T15:38:38.237Z)

```

120     ### 2.2 Detectors & windowing (WSL-quarantined; ABC abstraction complete)
121     - `@backend/pipeline/detectors/base.py` ? DetectorRunner ABC (healthcheck() ->
Tuple[bool,str] never-raises; run_predict(...) -> Optional[str] CSV path). Contract deliberately
matches the pre-existing tuple returns and call sites in the hybrids. Rich module docstring
includes "how to add a new runner", Linux-native path notes (the de-risk goal), and guidance on
threading health into reports. Re-exported from `detectors/__init__.py`. This finishes the WSL
de-risk seam (low-regret 6 / review item 1).
122     - `@backend/pipeline/detectors/tracknet_runner.py`
123     - `::TrackNetConfig` (+`.from_indexing_cfg`; ? key drift `wsl_distro` JSON -> `distro`
field), `::TrackNetRunner` ? runs TrackNet `predict.py` in WSL.
124     - `::to_wsl_mnt_path` (`C:\x`->`/mnt/c/x`; ? POSIX/~` passthrough, UNC unhandled),
`::TrajectoryPoint`(frame,visible,x,y).
125     - `*:::parse_trajectory_csv` @staticmethod* + `*:::trajectory_to_action_windows`
@staticmethod* = MODEL-AGNOSTIC brain, re-exported verbatim by WasbRunner. ? `weights_path`
defaults `` -> `run_predict` returns None; predict.py only `.mp4/.avi` + hard-names
`<stem>_predict.csv`; `visible==False` resets prev (occlusion gap breaks track); `fps<=0->30`,
`frame_width<=0->1280` silent; `track_id_count` hardcoded 1; `timeout_sec=0` -> NO timeout
(hangs forever). `::stage_video` "OK" sentinel.
126     - `@backend/pipeline/detectors/wasb_runner.py::WasbRunner` (inherits DetectorRunner)
(+`::WasbConfig`, from `indexing.wasb`; ? `weights_path` HAS a real default
`wasb_badminton_best.pth.tar` ? no required guard) ? stages `wasb_infer.py` (`::INFER_WIN`) +
video into WSL, runs in `wasb` env. `::parse_trajectory_csv`/`::trajectory_to_action_windows` =
`staticmethod(TrackNetRunner.*)` (explicit rebind = the plugin-equivalence point). Output hard-
named `<stem>_wasb.csv`. ? `wasb_infer.py` re-staged each run (editing Windows copy inert until
staged); `wsl_bash` now shared via `detectors/base.py::WslCommandMixin` (2026-06-11 dedupe).
Both runners now satisfy the DetectorRunner ABC for future swappability.
127     - `@backend/pipeline/detectors/wasb_infer.py` (WSL) ? `::run` core. ? imports WASB-SBDT
repo modules + torch ? NOT importable on Windows. Reboot-resumable detector cache:
`::DET_FILE='det_raw.jsonl'`, `::MANIFEST_FILE='manifest.json'`, `::CACHE_VERSION=1`,
`::load_and_clean_cache` (reads longest CONTIGUOUS prefix where `w==line index`, rewrites to
heal crash-truncated tail), `::manifest_compatible` (? `frames_dir` abspath'd but `weights`
compared raw), `::atomic_write_text/_atomic_write_trajectory`, `::dets_to_tracker` (xy MUST be
np.array), `::build_cfg` (`runner.gpus=[0]` hardcoded), `::extract_frames` (cv2-PNG SLOW),
`::default_cache_dir` (`<stem>_wasbcache` next to out). ? GPU pass resumable; CPU tracker
STATEFUL -> always fully re-run; `--fresh` ignores cache.
128     - `@backend/pipeline/detectors/rally_gate.py` ? Gate A net-crossing post-filter (pure,
no I/O/model). `::estimate_play_axis` (larger-spread axis; ? ties -> 'x'),
`::count_net_crossings` (sign changes vs median net; deadband jitter zone),
`::gate_windows`/`::apply_gate` (`min_crossings=2` default; kills single-toss service feeds).
`::GatedWindow`(start,end,crossings,visible_points,kept). ? window contract here is
`(start,end)` tuples NOT the rich window dict ? caller must adapt. Consumed by calibrate_local /
distill_local / export_wasb_segments.
129
130     ### 2.3 Eval & calibration (all pure core; I/O in CLI drivers)
131     - `@backend/eval/metrics.py` ? scoring primitive (THE spine). `::interval_iou`,
`::match_intervals` (greedy, deterministic), `::score`->`::Score`(`.as_dict()` = wire shape),
`::pr_curve`, `::Match`, `::MatchResult`. ? greedy not Hungarian; empty matches -> 0.0 (not NaN)
for mean_iou/boundary errors.

```

541. assistant (2026-06-13T15:39:25.263Z)

Now I'll make surgical docs edits. CODE_MAP first ? the detector self-clean, the new tool, and a common-task row:

542. user (2026-06-13T15:39:30.270Z)

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543. user (2026-06-13T15:39:43.373Z)

<tool_use_error>String to replace not found in file.

String: - `@backend/tools/rally\_slicer.py` ? standalone CLI. `::compute\_clip\_windows` (pure, unit-tested), `::load\_rally\_labels` (golden schema; ? silently skips degenerate rows),


```
`::slice_rallies`, `::ClipWindow`(rally:str), `::BEGIN_PADDING=0.5`/`::END_PADDING=1.0` (?
duplicate compiler's by copy). `_read_time` -> backend/eval/annotations.parse_timestamp`.
(note: Edit also tried swapping \uxxxx escapes and their characters; neither form matched, so
the mismatch is likely elsewhere in old_string. Re-read the file and copy the exact surrounding
text.)</tool_use_error>
```

544. user (2026-06-13T15:39:49.069Z)

```
152     ### 2.4 Stitcher / tools / utils
153     - `@backend/pipeline/stitcher/compiler.py::VideoCompiler` ? `compile_highlights(raw,
segments:[(start,end)], out_name)`. Per-clip re-encode -> `-f concat -c copy`. `::parse_time`
(mirrors gemini.parse_time; ? unparseable -> 0.0), `::get_video_duration`. ? if duration probe
fails (0.0) NO boundary validation; `begin_padding`/`end_padding` ctor defaults 0.5/1.0 but
every live call site (`/api/compile`, CLI trim, `run_process_and_stitch.py`) threads
`config.json` stitching.{begin_padding,end_padding}` (typed default 0.5/0.5); temp clips in a
TemporaryDirectory under output_dir; raises ValueError/FileNotFoundError/RuntimeError; uses
`print()`. ? two-stage codec contract: clips re-encoded with IDENTICAL settings precisely so
final concat can stream-copy ? change a preset and concat may break. ? optional **branding
watermark** (`stitching.watermark`, default-on): `::watermark_filter` builds a `-vf` (drawtext
via `textfile` + optional logo `movie/overlay`) applied to the per-clip re-encode only (concat
stays `-c copy`); a failed watermark encode retries the clip WITHOUT it (never nukes the reel,
and logs the ffmpeg stderr reason). Pass `watermark=` to `VideoCompiler(...)` (a
`WatermarkConfig` model or dict; `None` = off). Default font = bundled Apache-2.0
`::BUNDLED_FONT` (`backend/assets/fonts/RobotoSlab-Regular.ttf`) so it renders without
fontconfig; `font_path` overrides.
154     - `@backend/tools/rally_slicer.py` ? standalone CLI. `::compute_clip_windows` (pure,
unit-tested), `::load_rally_labels` (golden schema; ? silently skips degenerate rows),
`::slice_rallies`, `::ClipWindow`(rally:str), `::BEGIN_PADDING=0.5`/`::END_PADDING=1.0` (?
duplicate compiler's by copy). `_read_time` -> backend/eval/annotations.parse_timestamp`.
155     - `@backend/utils/video.py` ? `::get_video_duration` (ffprobe; ? 0.0 on failure =
"unknown"), `::extract_video_chunk(...,reencode=False,scale_width=None)`. ? copy mode cuts at
nearest keyframe (drift) ? pass `reencode=True` for short/accurate clips; `scale_width` requires
`reencode=True`.
156     - `@backend/utils/geometry.py` ? `::get_velocity_normalised` (fraction of frame-width/s;
? raw-px fallback if width<=0; used by yolo_hybrid), `::calculate_iou` (? +1 px convention
inflates IoU), `::get_center/get_distance/get_velocity`.
```

545. assistant (2026-06-13T15:39:55.717Z)

Path uses forward slashes. Retrying:

546. user (2026-06-13T15:40:03.210Z)

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547. assistant (2026-06-13T15:40:07.222Z)

Now the common-tasks table row:

548. user (2026-06-13T15:40:07.995Z)

```
272     | Change a config default / add a config field | `@backend/config/models.py` (single
source of truth) -> regenerate via `setup.py`/`save_default_config`; ? config.json may override
(default_segmenter) |
273     | Read config in new code | `from backend.config import load_config, AppConfig`
(`@backend/config/__init__.py`); attribute access, NOT raw `json.load` |
274     | Tune candidate windowing (velocity/merge/min-dur) |
`@backend/pipeline/detectors/tracknet_runner.py::trajectory_to_action_windows`; cfg `config.json`
indexing.{wasb,tracknet}.velocity_threshold`, `dead_time_merge_gap`,
`min_action_window_duration` |
275     | Add a segmenter | subclass `@backend/pipeline/segmenters/base.py::BasePlaySegmenter`;
`segmenter_registry.register(...)` at module bottom; import in
`@backend/pipeline/segmenters/__init__.py`; if AI-based add to
```

```
`@backend/reporting/harvest.py::_AI_SEGMENTERS` |
276 | Change stitching padding | `config.json stitching.{begin_padding,end_padding}`; impl
`@backend/pipeline/stitcher/compiler.py::VideoCompiler.compile_highlights`; wiring
`@backend/main.py` (`/api/compile` + CLI trim); mirror
`@backend/tools/rally_slicer.py::BEGIN_PADDING/END_PADDING` |
277 | Add a sport | subclass `@backend/sports/base.py::SportProfile`;
`sport_registry.register(...)` in `@backend/sports/__init__.py`; define prompt+HSV in new file |
278 | Add an AI provider | subclass `@backend/providers/base.py::AIVideoProvider`;
`provider_registry.register('name', Cls)` |
279 | Add an annotation/GT tier | subclass
`@backend/annotations/base.py::AnnotationProvider`; register + import in
`@backend/annotations/__init__.py` |
```

549. user (2026-06-13T15:40:17.241Z)

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550. assistant (2026-06-13T15:40:25.352Z)

Now README. The min_shot_count clarification + the new cache-hygiene knobs:

551. user (2026-06-13T15:40:30.389Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

552. user (2026-06-13T15:40:49.000Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

553. user (2026-06-13T15:40:55.784Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\README.md has been updated successfully. (file state is current in your context ? no need to Read it back)

554. assistant (2026-06-13T15:41:07.346Z)

Docs updated. Now the diligence pass ? branch, run the full suite (docs-only, but honoring the tests/CI check), then PR:

555. user (2026-06-13T15:41:38.724Z)

```
M      README.md
M      config.json
M      docs/CODE_MAP.md
Switched to a new branch 'docs/cache-hygiene-and-api-min-shots'
M  README.md
  M config.json
M  docs/CODE_MAP.md
?? arial.ttf
?? artifacts/
?? docs/STORAGE_SHARING_MODEL.md
?? scratch/HIGH_RISK_REMEDIATION_PLAN.md
?? scratch/PR1_BODY.md
?? scratch/PR1_HANDOFF.md
?? scratch/PR2_HANDOFF.md
?? scratch/PR2_deferred_hardening_status.patch
?? scratch/PR88_FIX.md
?? scratch/debug_wsl.py
?? scratch/keep_awake.ps1
?? scratch/probe_cv2_count.py
?? scratch/probe_gemini.py
```

```
?? scratch/repro_drawtext_font.py
?? scratch/repro_movie_overlay.py
?? scratch/run_gx010125.py
?? scratch/stitch_min5.py
421 passed in 26.66s
```

```
=====
RESULT: PASS - all tests green
=====
```

556. assistant (2026-06-13T15:41:44.188Z)

Suite green (421, docs-only). Committing, pushing, opening the PR:

557. user (2026-06-13T15:42:03.721Z)

```
[docs/cache-hygiene-and-api-min-shots 8270a5b] docs: cache-hygiene tooling/knobs +
min_shot_count API parity (README + CODE_MAP)
 2 files changed, 21 insertions(+), 2 deletions(-)
remote:
remote: Create a pull request for 'docs/cache-hygiene-and-api-min-shots' on GitHub by visiting:
remote:   https://github.com/avidullu/badminton-highlight-indexer/pull/new/docs/cache-
hygiene-and-api-min-shots
remote:
branch 'docs/cache-hygiene-and-api-min-shots' set up to track 'origin/docs/cache-hygiene-and-
api-min-shots'.
To https://github.com/avidullu/badminton-highlight-indexer.git
 * [new branch]      docs/cache-hygiene-and-api-min-shots -> docs/cache-hygiene-and-api-min-
shots
Warning: 18 uncommitted changes
https://github.com/avidullu/badminton-highlight-indexer/pull/110
```

558. user (2026-06-13T15:43:54.211Z)

```
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27471284838/job/81202540268
Full test suite (offline, mocked)                pass    1m27s https://github.com/avidullu/badminton-
highlight-indexer/actions/runs/27471284838/job/81202540230
Mypy (lenient baseline) pass    28s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27471284838/job/81202540208
Ruff lint      pass    10s https://github.com/avidullu/badminton-highlight-
indexer/actions/runs/27471284838/job/81202540194
Syntax sweep + import smoke (no GPU stack)      pass    21s
https://github.com/avidullu/badminton-highlight-indexer/actions/runs/27471284838/job/81202540268
```

559. assistant (2026-06-13T15:43:58.236Z)

All 4 lanes green. Merging and syncing:

560. user (2026-06-13T15:44:09.559Z)

```
["SUCCESS"]
From https://github.com/avidullu/badminton-highlight-indexer
 * branch          master      -> FETCH_HEAD
   7651687..15e4e0e master      -> origin/master
Updating 7651687..15e4e0e
Fast-forward
 README.md          | 18 ++++++
 docs/CODE_MAP.md   |  5 +++-
 2 files changed, 21 insertions(+), 2 deletions(-)
M       config.json
Your branch is up to date with 'origin/master'.
Already on 'master'
Already up to date.
15e4e0e docs: cache-hygiene tooling/knobs + min_shot_count API parity (README + CODE_MAP) (#110)
```

7651687 feat: cache hygiene ? self-cleaning pipeline + cleanup tool + disk guard (#109)
1c9be4c docs(platform): video-locality mental model ? the 10GB question (#106)

561. assistant (2026-06-13T15:44:22.699Z)

PR #110 merged (master `15e4e0e`). Let me confirm both repos are fully clean and update the checkpoint:

562. user (2026-06-13T15:44:27.079Z)

```
=== open PRs indexer ===  
=== open PRs collector ===  
=== done ===
```

563. user (2026-06-13T15:44:28.793Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\current-state.md has been updated successfully. (file state is current in your context ? no need to Read it back)

564. assistant (2026-06-13T15:44:44.242Z)

Done ? diligence applied and the merge queue is clear.

What I found and did

When I checked, **both repos had zero open PRs** ? everything from this session (#102, collector #20, #107, #108, #109) was already merged. So "the PRs to merge" had effectively become a **docs-diligence gap**: my three feature merges (#107 API min_shot_count, #108 watermark fix, #109 cache hygiene) shipped working code + tests + green CI, but the **user-facing docs** hadn't caught up.

I closed that gap with **[PR #110](https://github.com/avidullu/badminton-highlight-indexer/pull/110)** (merged, master `15e4e0e`):

- **README** ? new "Cache Hygiene & Disk Cleanup" section (self-clean + `tools/cleanup\_caches.py` usage), documented the `keep\_frames` / `min\_free\_gb` knobs, clarified `min\_shot\_count` now applies to `/api/compile` too, and added `tools/` to the project structure.
- **CODE_MAP** ? detector self-clean behavior (§2.2), the `tools/cleanup\_caches.py` entry with the `for`-loop WSL footgun note (§2.4), and two common-task rows ("Reclaim disk / clean caches", "Filter highlights by shot count").

The diligence gates, each verified

- **Docs**: README + CODE_MAP brought in sync with the merged features (the actual gap).
- **Tests**: `python run\_all\_tests.py` ? **421 passed** (docs-only, unchanged).
- **CI health**: all **4 lanes green** (smoke / full-suite / ruff / mypy) on a verified-**SUCCESS** rollup before squash-merging.

State now

- **Indexer + collector**: 0 open PRs. master = `15e4e0e`, synced.
- Memory checkpoint updated.
- The only outstanding local item is unchanged and intentional: `config.json` still carries this run's settings (`wasb\_hybrid` / `min\_shot\_count=8` / `timeout\_sec=7200`) ? uncommitted, yours to keep or revert.

One honest note for the record: ideally those docs would have ridden *inside* each feature PR rather than as a follow-up ? I'll fold doc updates into the feature PRs going forward so "diligence" is built in rather than swept up after. Nothing else is open or pending.

565. user (2026-06-13T16:53:52.308Z)

Can you run an ffmpeg command and make a copy of "C:\Users\avidu\Projects\badminton-highlight-indexer\output\GX010125_highlights_min5.mp4" in the same directory but trim the first 70 seconds

of this video because its garbage

566. assistant (2026-06-13T16:54:55.045Z)

I'll trim the first 70s into a new file (original untouched), using GPU NVENC for a fast, exact cut:

567. user (2026-06-13T16:55:00.818Z)

```
Exit code 187
[h264_nvenc @ 0000020ca3a100c0] No capable devices found
[vost#0:0/h264_nvenc @ 0000020ca1972240] [enc:h264_nvenc @ 0000020ca3bb7d40] Error while opening
encoder - maybe incorrect parameters such as bit_rate, rate, width or height.
[vf#0:0 @ 0000020ca3a10440] Error sending frames to consumers: Generic error in an external
library
[vf#0:0 @ 0000020ca3a10440] Task finished with error code: -542398533 (Generic error in an
external library)
[vf#0:0 @ 0000020ca3a10440] Terminating thread with return code -542398533 (Generic error in an
external library)
[vost#0:0/h264_nvenc @ 0000020ca1972240] [enc:h264_nvenc @ 0000020ca3bb7d40] Could not open
encoder before EOF
[vost#0:0/h264_nvenc @ 0000020ca1972240] Task finished with error code: -22 (Invalid argument)
[vost#0:0/h264_nvenc @ 0000020ca1972240] Terminating thread with return code -22 (Invalid
argument)
[out#0/mp4 @ 0000020ca196fe40] Nothing was written into output file, because at least one of its
streams received no packets.
frame=    0 fps=0.0 q=0.0 Lsize=           0KiB time=N/A bitrate=N/A speed=N/A elapsed=0:00:01.39
```

568. user (2026-06-13T16:55:04.486Z)

[Request interrupted by user]

569. user (2026-06-13T16:55:30.733Z)

I mean trim off the first 70 seconds and leave the rest

570. assistant (2026-06-13T16:55:52.120Z)

Right ? ``-ss 70`` does keep everything *after* 70s (drops the intro), so the intent was correct; the failure was just NVENC ("No capable devices found" ? the GPU encoder couldn't open this time, even though it worked for the proxy earlier). No need to re-encode at all for a trim ? ****stream copy**** is instant and lossless:

571. user (2026-06-13T16:55:59.172Z)

```
OK
codec_name=h264
width=3840
height=2160
duration=179.152750
size=1510356547
size_MB=1440.4
```

572. assistant (2026-06-13T16:56:46.345Z)

Stream copy worked and is lossless, but note the catch: it can only cut at a keyframe, so it actually trimmed ****68.6s**** (output is 179.15s = 247.74 ? 68.6), leaving ~1.4s of the garbage intro at the start. Since you specifically want the first 70s *gone*, let me redo it as an ****exact**** cut at 70.0s. NVENC isn't available right now, so I'll use the CPU encoder (a few minutes for 4K):

573. user (2026-06-13T16:56:50.228Z)

Command running in background with ID: brsyiht1z. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\47ecf23a-6312-4118-b2bd-c1af33f2eaa7\tasks\brsyiht1z.output. You will be notified when it completes. To check interim output, use Read on that file path.

574. assistant (2026-06-13T16:57:01.524Z)

The exact-cut re-encode is running in the background (CPU libx264, ~3?5 min for 179s of 4K). It writes to `output\GX010125\_highlights\_min5\_trimmed.mp4`, leaving your original `GX010125\_highlights\_min5.mp4` untouched. I'll confirm the result (exact 177.7s duration, 4K) when it finishes.

(Also: the SessionStart cache-hygiene hook fired at the top of this session ? "?? 96.5G reclaimable" ? so that piece is confirmed working. That 96.5G is mostly GX010125's frame cache you chose to keep; `python -m tools.cleanup\_caches --apply` clears it whenever you're done.)

575. user (2026-06-13T16:57:04.521Z)

the first 70s are not useful

576. assistant (2026-06-13T16:57:12.202Z)

Right ? that's exactly what the running trim does (drops 0?70s, keeps 70s?end). Let me check whether it's finished:

577. user (2026-06-13T16:57:21.495Z)

```
Exit code 1
frame= 502 fps= 18 q=39.0 size= 56064KiB time=00:00:08.34 bitrate=55058.1kbits/s speed=0.292x
elapsed=0:00:28.56
frame= 511 fps= 18 q=37.0 size= 56832KiB time=00:00:08.49 bitrate=54825.5kbits/s speed=0.292x
elapsed=0:00:29.09
--- output state ---
[mov,mp4,m4a,3gp,3g2,mj2 @ 00000212051e4b00] moov atom not found
C:\Users\avidu\Projects\badminton-highlight-indexer\output\GX010125_highlights_min5_trimmed.mp4:
Invalid data found when processing input
size_MB=57.3
lastWrite=06/13/2026 22:27:20
```

578. assistant (2026-06-13T16:58:19.793Z)

It's running but slow ? CPU libx264 at 4K is only **0.29x realtime** (~10 min for this clip), and the `moov atom` error just means the file's still being written. That's too slow and risks the timeout. Let me stop it, check if the GPU encoder is usable now, and pick the fastest exact-cut path:

579. user (2026-06-13T16:58:27.706Z)

```
{"message":"Successfully stopped task: brsyiht1z ($in=\"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\output\\GX010125_highlights_min5.mp4\"; $out=\"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\output\\GX010125_highlights_min5_trimmed.mp4\"; ffmpeg -y -hide_banner -loglevel error -stats -ss 70 -i $in -c:v libx264 -preset superfast -crf 22 -c:a aac -b:a 128k $out 2>&1 | Out-File -Encoding utf8 C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\output\\trim70.log) \", \"task_id\":\"brsyiht1z\", \"task_type\":\"local_bash\", \"command\":\"$in=\"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\output\\GX010125_highlights_min5.mp4\"; $out=\"C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\output\\GX010125_highlights_min5_trimmed.mp4\"; ffmpeg -y -hide_banner -loglevel error -stats -ss 70 -i $in -c:v libx264 -preset superfast -crf 22 -c:a aac -b:a 128k $out 2>&1 | Out-File -Encoding utf8 C:\\Users\\avidu\\Projects\\badminton-highlight-indexer\\output\\trim70.log\"}
```

580. assistant (2026-06-13T16:58:49.145Z)

Now let me check the GPU state and retry NVENC (it's fast if the device is available again):

581. user (2026-06-13T16:58:51.746Z)

GPU 0: NVIDIA GeForce RTX 2070 Super with Max-Q Design (UUID: GPU-bde8e710-b1e3-9905-edee-d7cc100c5460)
23 %, 1865 MiB, 8192 MiB, 0

582. assistant (2026-06-13T16:59:00.495Z)

GPU is free (0 NVENC sessions, 6 GB available) ? the earlier failure was transient. Retrying NVENC for a fast exact cut:

583. user (2026-06-13T16:59:06.577Z)

```
Exit code 187
[h264_nvenc @ 00000228ff3e3880] No capable devices found
[vost#0:0/h264_nvenc @ 00000228ff3e3640] [enc:h264_nvenc @ 00000228fb585280] Error while opening
encoder - maybe incorrect parameters such as bit_rate, rate, width or height.
[vf#0:0 @ 00000228fb03e0c0] Error sending frames to consumers: Generic error in an external
library
[vf#0:0 @ 00000228fb03e0c0] Task finished with error code: -542398533 (Generic error in an
external library)
[vf#0:0 @ 00000228fb03e0c0] Terminating thread with return code -542398533 (Generic error in an
external library)
[vost#0:0/h264_nvenc @ 00000228ff3e3640] [enc:h264_nvenc @ 00000228fb585280] Could not open
encoder before EOF
[vost#0:0/h264_nvenc @ 00000228ff3e3640] Task finished with error code: -22 (Invalid argument)
[vost#0:0/h264_nvenc @ 00000228ff3e3640] Terminating thread with return code -22 (Invalid
argument)
[out#0/mp4 @ 00000228ff3e13c0] Nothing was written into output file, because at least one of its
streams received no packets.
frame=    0 fps=0.0 q=0.0 Lsize=           0KiB time=N/A bitrate=N/A speed=N/A elapsed=0:00:01.13
exit=-542398533
```

584. assistant (2026-06-13T16:59:39.652Z)

NVENC is genuinely unavailable in this context now (device error despite the GPU being free) ?
I'll stop retrying it. Before committing to a slow CPU encode, let me check where the keyframes
actually are near 70s ? if there's one right after 70s, a ****lossless instant**** stream-copy from
it is the better cut (no quality loss, no garbage):

585. user (2026-06-13T16:59:43.503Z)

```
keyframe pts_times in [66,82]:
64.467350
68.638183
70.289833,
74.460667
78.631500
```

586. assistant (2026-06-13T16:59:56.890Z)

A keyframe sits at ****70.29s**** ? just 0.29s past the mark. So a lossless stream-copy from there
drops the ***entire*** first 70s with no quality loss and no slow encode. That's the ideal cut:

587. user (2026-06-13T17:00:04.584Z)

```
OK
codec_name=h264
width=3840
height=2160
r_frame_rate=60000/1001
duration=177.489100
```

size_MB=1427.1

588. assistant (2026-06-13T17:00:15.321Z)

Done ? clean, lossless, and the full first 70s is gone.

`output\GX010125\_highlights\_min5\_trimmed.mp4`

- ****Starts at 70.29s**** of the original (the first keyframe past the 70s mark), so the entire not-useful intro is dropped ? no garbage left over.
- ****177.5s**** long (247.74 ? 70.29), ****true 4K**** 3840×2160, h264, 59.94 fps, 1.43 GB.
- ****Lossless**** ? stream copy, zero re-encode, so identical quality to the source (and instant).
- Your original `GX010125\_highlights\_min5.mp4` is untouched.

Why this beat the alternatives: NVENC threw "No capable devices found" (transient GPU-encoder issue, even with the card free), and CPU libx264 was crawling at 0.29× realtime (~10 min) for the exact 70.000s cut. Since you said the first 70s isn't useful, cutting at the keyframe 0.29s later gives a perfect lossless result with no quality loss and no wait ? strictly better than re-encoding here.

One cleanup note: I left a stray `output\trim70.log` and a 57 MB partial from the aborted CPU encode was overwritten by the final file, so nothing dangling there except that small log ? want me to delete `trim70.log`?

589. user (2026-06-13T17:00:17.290Z)

I think it can be a good tool to check in scratch as well

590. assistant (2026-06-13T17:01:22.811Z)

That's a good idea ? the keyframe-aware lossless trim (with the re-encode fallback) generalizes well beyond this one clip. Let me write it as a reusable `scratch/trim\_video.py`:

591. assistant (2026-06-13T17:01:51.898Z)

"""Trim seconds off the front (and optionally the tail) of a video ? fast + lossless by default.

The smart bit, learned the hard way on GX010125_highlights_min5.mp4: a plain `ffmpeg -ss N -c copy` can only cut at a keyframe, and it snaps to the keyframe **before** N ? leaving a sliver of the unwanted intro. And re-encoding a 4K clip on CPU is ~0.3x realtime (minutes), while NVENC is flaky in some shells ("No capable devices found").

So the default here is ****keyframe-aware lossless****: probe the keyframes, pick the first one at/after the requested start, and stream-copy from it. The entire [0, start) region is dropped (you lose at most one GOP of **wanted** content, usually a fraction of a second), with zero re-encode and zero quality loss ? instant.

Use `--exact` only when you need the cut at the precise frame (re-encodes: tries NVENC, falls back to CPU libx264).

Examples:

```
python scratch/trim_video.py "C:/path/in.mp4" --start 70
python scratch/trim_video.py "in.mp4" --start 70 --end 240 -o out.mp4
python scratch/trim_video.py "in.mp4" --start 70 --exact      # frame-exact, re-encodes
python scratch/trim_video.py "in.mp4" --start 70 --dry-run    # show the plan only
"""

import argparse
import os
import subprocess
import sys
```

```
def _run(cmd):
    return subprocess.run(cmd, capture_output=True, text=True)
```



```

def probe_duration(path):
    r = _run(["ffprobe", "-v", "error", "-show_entries", "format=duration",
              "-of", "default=noprint_wrappers=1:nokey=1", path])
    try:
        return float(r.stdout.strip())
    except ValueError:
        return 0.0

def first_keyframe_at_or_after(path, t, window=15.0, max_window=240.0):
    """Smallest video keyframe timestamp >= t. Expands the probe window until found."""
    while window <= max_window:
        r = _run(["ffprobe", "-v", "error", "-select_streams", "v:0", "-skip_frame", "nokey",
                  "-show_entries", "frame=pts_time", "-of", "csv=p=0",
                  "-read_intervals", f"{max(0.0, t)}%{t + window}", path])
        times = []
        for tok in r.stdout.replace(",", " ").split():
            try:
                times.append(float(tok))
            except ValueError:
                pass
        cands = sorted(x for x in times if x >= t - 1e-3)
        if cands:
            return cands[0]
        window *= 2
    return None

def trim_lossless(inp, out, start, end):
    kf = first_keyframe_at_or_after(inp, start)
    if kf is None:
        raise SystemExit(f"no keyframe found at/after {start}s ? try --exact")
    cmd = ["ffmpeg", "-y", "-hide_banner", "-loglevel", "error", "-ss", f"{kf:.6f}"]
    if end is not None:
        cmd += ["-to", f"{end:.6f}"] # both before -i = original-timeline absolute
    times
    cmd += ["-i", inp, "-c", "copy", "-avoid_negative_ts", "make_zero", out]
    return kf, _run(cmd)

def trim_exact(inp, out, start, end):
    """Frame-exact: re-encode [start, end). Tries NVENC, falls back to CPU libx264."""
    dur_args = ["-t", f"{end - start:.6f}"] if end is not None else []
    base = ["ffmpeg", "-y", "-hide_banner", "-loglevel", "error", "-ss", f"{start:.6f}",
            "-i", inp, *dur_args, "-c:a", "aac", "-b:a", "128k"]
    nvenc = base[:]; nvenc[-4:-1] = [] # (no-op; keep clear below)
    nvenc = base + []
    nvenc_cmd = ["ffmpeg", "-y", "-hide_banner", "-loglevel", "error", "-ss", f"{start:.6f}",
                 "-i", inp, *dur_args, "-c:v", "h264_nvenc", "-preset", "p5", "-cq", "21",
                 "-c:a", "aac", "-b:a", "128k", out]
    r = _run(nvenc_cmd)
    if r.returncode == 0:
        return "h264_nvenc", r
    x264_cmd = ["ffmpeg", "-y", "-hide_banner", "-loglevel", "error", "-ss", f"{start:.6f}",
                "-i", inp, *dur_args, "-c:v", "libx264", "-preset", "veryfast", "-crf", "22",
                "-c:a", "aac", "-b:a", "128k", out]
    return "libx264", _run(x264_cmd)

def main(argv=None):
    ap = argparse.ArgumentParser(description=__doc__,
    formatter_class=argparse.RawDescriptionHelpFormatter)
    ap.add_argument("input")
    ap.add_argument("--start", type=float, required=True, help="seconds to drop from the front")
    ap.add_argument("--end", type=float, default=None, help="absolute end time (seconds) in the

```

```

source")
    ap.add_argument("-o", "--output", default=None, help="output path (default:
<stem>_trimmed<ext> beside input)")
    ap.add_argument("--exact", action="store_true", help="frame-exact cut (re-encodes;
NVENC->libx264)")
    ap.add_argument("--dry-run", action="store_true", help="print the plan, don't run ffmpeg")
    args = ap.parse_args(argv)

    if not os.path.exists(args.input):
        raise SystemExit(f"input not found: {args.input}")
    out = args.output or (lambda s, e: f"{s}_trimmed{e}")(*os.path.splitext(args.input))
    dur = probe_duration(args.input)
    print(f"input: {args.input} ({dur:.2f}s)")
    print(f"mode: {'EXACT re-encode' if args.exact else 'lossless keyframe-copy'} | "
          f"drop first {args.start}s" + (f", end at {args.end}s" if args.end else "") + f" ->
{out}")

    if args.exact:
        if args.dry_run:
            print(f"[dry-run] would re-encode [{args.start}, {args.end or dur}] (frame-exact).")
            return 0
        enc, r = trim_exact(args.input, out, args.start, args.end)
        if r.returncode != 0:
            print(r.stderr[-600:]); raise SystemExit(f"exact trim failed (encoder={enc})")
        print(f"[done] {enc} exact cut -> {out} ({probe_duration(out):.2f}s, "
              f"{os.path.getsize(out)/1024/1024:.1f} MB)")
        return 0

    kf = first_keyframe_at_or_after(args.input, args.start)
    if kf is None:
        raise SystemExit(f"no keyframe at/after {args.start}s ? use --exact")
    print(f"keyframe at/after {args.start}s = {kf:.3f}s (drops {kf:.3f}s; "
          f"{kf - args.start:.3f}s more than requested, lossless)")
    if args.dry_run:
        print(f"[dry-run] would stream-copy [{kf:.3f}, {args.end or dur}] -> {out}")
        return 0
    _, r = trim_lossless(args.input, out, args.start, args.end)
    if r.returncode != 0:
        print(r.stderr[-600:]); raise SystemExit("lossless trim failed ? try --exact")
    print(f"[done] lossless -> {out} ({probe_duration(out):.2f}s, "
          f"{os.path.getsize(out)/1024/1024:.1f} MB)")
    return 0

if __name__ == "__main__":
    sys.exit(main())

```

592. user (2026-06-13T17:01:51.935Z)

File created successfully at: C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\trim_video.py (file state is current in your context ? no need to Read it back)

593. assistant (2026-06-13T17:01:57.493Z)

Let me clean up some dead lines I left in `trim\_exact`:

594. user (2026-06-13T17:02:02.735Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\trim_video.py has been updated successfully. (file state is current in your context ? no need to Read it back)

595. assistant (2026-06-13T17:02:06.777Z)

Now let me verify the tool works ? `--dry-run` (proves the keyframe logic), then a real run to a

throwaway output to confirm it reproduces the correct cut:

596. user (2026-06-13T17:02:16.753Z)

```
=== dry-run ===
input: output\GX010125_highlights_min5.mp4 (247.74s)
mode: lossless keyframe-copy | drop first 70.0s -> output\GX010125_highlights_min5_trimmed.mp4
keyframe at/after 70.0s = 70.290s (drops 70.290s; 0.290s more than requested, lossless)
[dry-run] would stream-copy [70.290, 247.742933] -> output\GX010125_highlights_min5_trimmed.mp4
=== real run to throwaway ===
input: output\GX010125_highlights_min5.mp4 (247.74s)
mode: lossless keyframe-copy | drop first 70.0s -> output\_trim_selftest.mp4
keyframe at/after 70.0s = 70.290s (drops 70.290s; 0.290s more than requested, lossless)
[done] lossless -> output\_trim_selftest.mp4 (177.49s, 1427.1 MB)
=== probe + cleanup throwaway ===
177.489100
(throwaway removed)
```

597. assistant (2026-06-13T17:02:40.879Z)

The tool works ? verified end to end. Dry-run shows the plan, and a real run reproduced the exact same cut (177.49s, 1427.1 MB, lossless) before I removed the throwaway.

`scratch/trim\_video.py`

A reusable, keyframe-aware video trimmer that bakes in everything we learned this session:

```
```bash
python scratch/trim_video.py "in.mp4" --start 70 # lossless, instant (default)
python scratch/trim_video.py "in.mp4" --start 70 --end 240 # also trim the tail
python scratch/trim_video.py "in.mp4" --start 70 --exact # frame-exact (re-encodes)
python scratch/trim_video.py "in.mp4" --start 70 --dry-run # show the plan only
```
```

What it encodes as reusable knowledge:

- **Default = lossless keyframe-copy** ? probes keyframes, snaps to the first one ***at/after*** **`--start`** so the whole unwanted region is gone (no leftover sliver), then stream-copies. Instant, zero quality loss.
- *****--exact***** re-encodes for a frame-precise cut, trying ****NVENC** first and falling back to CPU **libx264** ? so it survives the "No capable devices found" flakiness we hit.
- Reports exactly what it did (e.g. "keyframe at/after 70.0s = 70.290s; drops 0.290s more than requested, lossless").
- Defaults output to **<stem>_trimmed<ext>** beside the input; never touches the original.

It's in **`scratch/`** (untracked, like your other helpers). Two optional follow-ups if you want:

- ****Promote it to `tools/`**** as a committed, tested tool (like **`cleanup\_caches.py`**) ? say the word and I'll add it with unit tests + a README line via a PR.
- ****Tidy scratch****: this session left a few one-off debug artifacts (**`debug\_wsl.py`**, **`probe\_cv2\_count.py`**, **`probe\_gemini.py`**, **`repro\_drawtext\_font.py`**, **`repro\_movie\_overlay.py`**) and **`output/trim70.log`** ? want me to delete those?

598. user (2026-06-13T17:02:42.866Z)

and also for the memory of this model to reuse this tool

599. assistant (2026-06-13T17:03:01.080Z)

I'll save a memory so future sessions reach for this tool instead of rebuilding the trim logic (and re-learning the keyframe/NVENC gotchas):

600. assistant (2026-06-13T17:03:11.824Z)

```
---
name: tool-trim-video
```

description: "Reusable scratch tool scratch/trim_video.py ? keyframe-aware LOSSLESS video trim (drop first/last N sec) with NVENC->libx264 exact-mode fallback. Use it instead of hand-rolling ffmpeg trims."

metadata:

type: reference

``scratch/trim\_video.py`` (untracked scratch helper, built 2026-06-13) ? trim seconds off the front (and optionally tail) of a video. **``Reach for this instead of re-deriving ffmpeg trim flags.``**

...

```
python scratch/trim_video.py "in.mp4" --start 70          # lossless, instant (default)
python scratch/trim_video.py "in.mp4" --start 70 --end 240 # also cut the tail
python scratch/trim_video.py "in.mp4" --start 70 --exact   # frame-exact (re-encodes)
python scratch/trim_video.py "in.mp4" --start 70 --dry-run  # show the plan only
```

...

Default output: `<stem>_trimmed<ext>` beside the input; original untouched.

``Why it exists (gotchas it bakes in ? discovered trimming GX010125\_highlights\_min5.mp4):``

- Plain `ffmpeg -ss N -c copy` snaps to the keyframe *before* N ? leaves a sliver of the unwanted

- intro. This tool snaps to the first keyframe *at/after* N (probes keyframes via `ffprobe -skip_frame nokey -read_intervals`), so the whole $[0,N)$ region is gone, still lossless/instant.

- CPU `libx264` on 4K ? 0.3x realtime (minutes) ? avoid unless `--exact` is truly needed.

- **``NVENC is flaky in this shell context``** ? `h264_nvenc` throws "No capable devices found" even with the GPU free/0 sessions (worked for the proxy transcode earlier same session, failed later).

- So `--exact` tries NVENC then falls back to libx264 automatically. See `[[gotcha-long-runs-gpu-wsl]]`.

Default (lossless keyframe-copy) is the right call `--always` for "drop the garbage intro/outro"; only use `--exact` when the cut must land on a precise frame. Could be promoted to a committed `tools/` tool with tests if it earns its keep (owner offered, not yet done).

601. user (2026-06-13T17:03:12.007Z)

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602. user (2026-06-13T17:03:21.024Z)

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603. assistant (2026-06-13T17:03:29.639Z)

Saved to memory. Future sessions will now know to reuse the tool rather than rebuild it.

- **``tool-trim-video.md``** ? points at `scratch/trim_video.py` with usage, and crucially captures *why* it exists: the keyframe-snap-before-N gotcha (leaves a sliver), the slow 4K CPU encode, and the flaky NVENC "No capable devices found" ? so the next session inherits the lessons, not just the script.

- **``MEMORY.md``** index updated with a one-line pointer under "Start here", linked to `[[gotcha-long-runs-gpu-wsl]]`.

So the trim tool is now: written (`scratch/trim_video.py`), verified (dry-run + real run reproduce the exact lossless cut), and remembered (auto-memory entry + index).

Still open from earlier, whenever you want them ? both optional:

- Promote `trim_video.py` to a committed `tools/` tool with tests (like `cleanup_caches.py`).

- Tidy the one-off scratch debug artifacts (``debug_wsl.py``, ``probe_*.py``, ``repro_*.py``) and ``output/trim70.log``.